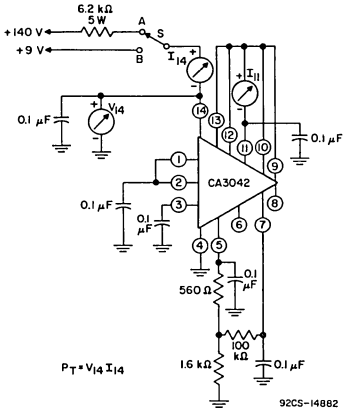


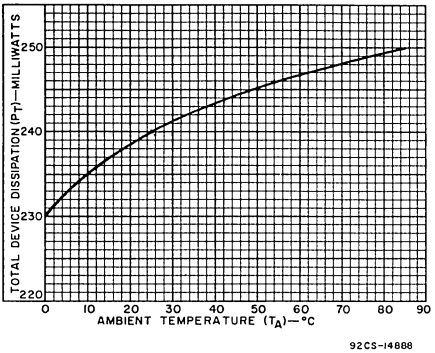


Fig.4 - Typical dissipation characteristic.

PROCEDURES:

Total Device Dissipation:

1. Set switch S in position A
2. Measure and record V_{14} and I_{14}.
3. Determine Total Device Dissipation from $P_T = V_{14} I_{14}$

Quiescent Operating Current into Terminal 11:

1. Turn switch S to position B
2. Measure I_{11} and record as Quiescent Operating Current into Terminal 11.

9-Volt Current Drain:

1. Set switch S in position B
2. Measure I_{14} and record as 9-Volt Current Drain.

Fig.3 - Test setup for measurement of total device dissipation, quiescent current into terminal No.11, and 9-volt current drain.

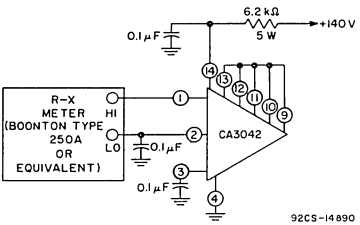
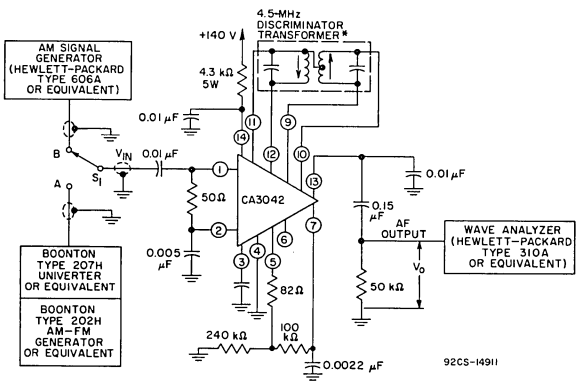


Fig.5 - Test setup for measurement of input-impedance components.



PROCEDURES:

1. Set FM Signal Generator as follows:
   Output Frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Deviation = ±25 kHz
   Output level for V_{in} = 100 mV rms
2. Set AM Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1000 Hz
   Per cent modulation = 30
   Output level for V_{in} = 10 mV rms
3. With S_1 in Position A measure AF Output Voltage and record as $V_{o(FM)}$.
4. With S_1 in Position B measure AF Output Voltage and record as $V_{o(AM)}$.
5. Determine AM Rejection from AMR = $\frac{V_{o(FM)}}{V_{o(AM)}}$

* TRW Electronics, Des Plaines, Illinois. Part No. EO23874, or equivalent.

Fig.6 - Test setup for measurement of AM rejection.

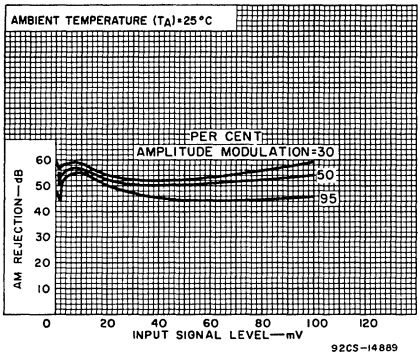
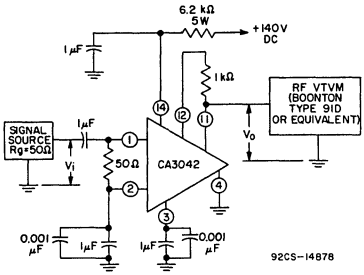


Fig.7 - Typical AM rejection characteristics.



PROCEDURE Voltage Gain:

1. Set input frequency at desired value, v_i = 100 μV rms.
2. Record v_o.
3. Calculate Voltage Gain A from A = 20 $\log_{10} v_o/v_i$.
4. Repeat Steps 1, 2, and 3 for each frequency and/or for temperature desired.

Fig.8 - Test setup for measurement of IF amplifier voltage gain.

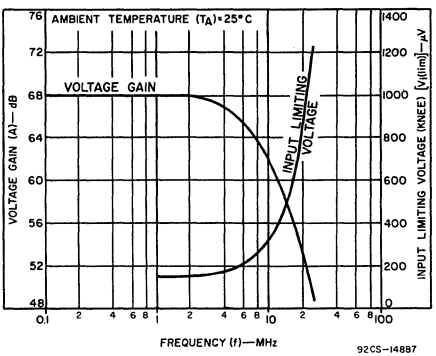


Fig.9 - Typical IF amplifier voltage gain and input limiting voltage (knee) characteristics.

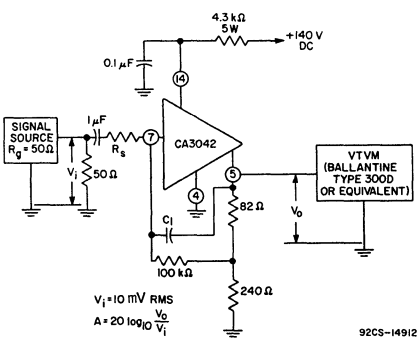


Fig.10 - Test setup for measurement of AF amplifier voltage gain.

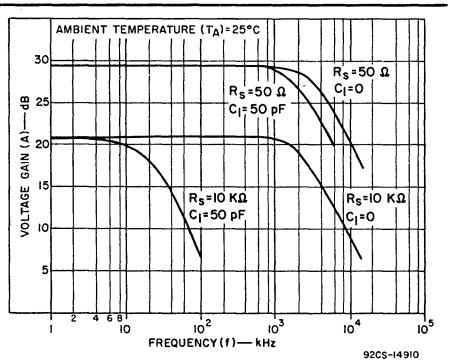
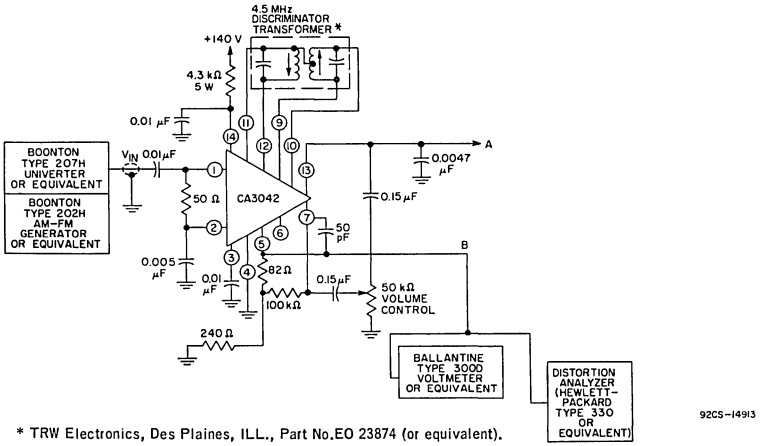


Fig.11 - Typical AF amplifier voltage gain characteristics.



* TRW Electronics, Des Plaines, ILL., Part No.EO 23874 (or equivalent).

PROCEDURES:

Recovered AF Voltage:

1. Set Input Signal Generator as follows:
   Output frequency = 4.5 MHz
   Modulating frequency = 1 kHz
   Deviation = ±25 kHz
   Output level for V_{in} = 100 mV rms
2. Set volume control for maximum af output
3. Measure af output voltage and record as Recovered AF Voltage.

Total Harmonic Distortion:

1. Adjust volume control for an af output voltage of 500 mV rms.
2. Measure Total Harmonic Distortion of the output signal in accordance with the Operating Instructions for the Distortion Analyzer.

Input Limiting Voltage (Knee):

1. Decrease V_{in} until the af output voltage is 3 dB less than the value set in Step 1 of the procedure for measurement of Total Harmonic Distortion (500mV - 3 dB = 350 mV)
2. Measure resulting value of V_{in} and record as Input Limiting Voltage (Knee).

Fig.12 - Test setup for measurement of input limiting voltage (knee), recovered AF voltage, and total harmonic distortion.

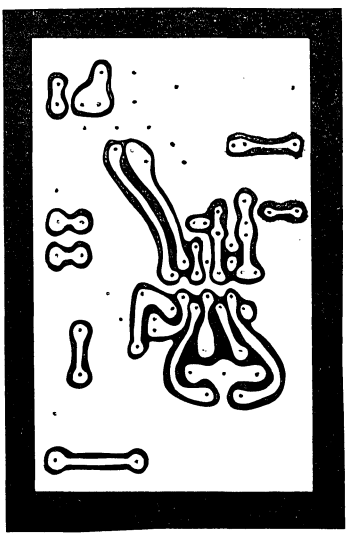


Fig.13 - Recommended layout of printed-circuit board for TV-receiver sound strip utilizing RCA-CA3042. (Bottom View)

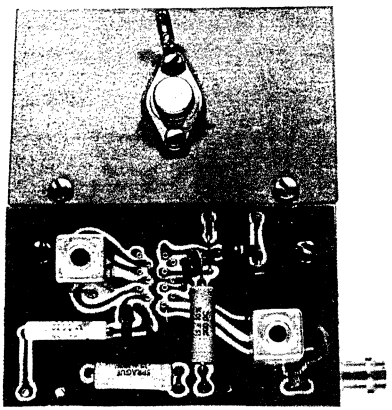


Fig.14 - Recommended parts layout for TV-receiver sound strip utilizing RCA-CA3042. (Top View)

RCA
Solid State
Division

Linear Integrated Circuits

CA3043

Special-Function Sub-System

Monolithic Silicon



HIGH-GAIN IF AMPLIFIER, LIMITER, FM DETECTOR, AND AF PREAMPLIFIER/DRIVER

For FM IF Amplifier Applications in Communications Receivers and High-Fidelity FM Receivers up to 20 MHz

RCA Integrated Circuit Type CA3043 provides in a single monolithic silicon chip, a major sub-system for the IF sections of Communications and high-fidelity FM receivers. As shown in the Schematic Diagram (Fig.2) and the FM Receiver Block Diagram (Fig.1), the CA3043 contains a multistage if-amplifier/limiter section, an FM-detector stage, a Zener-diode regulated power-supply section, and an af-amplifier section. In FM receivers, the CA3043 can be used to provide if amplification and limiting, FM detection, and af preamplification. The CA3043 provides exceptional versatility of circuit design because the if-amplifier/limiter section, FM detector section, and af-preamplifier/driver section can be used independently of each other.

The four stage emitter-follower-coupled if amplifier section provides 80-dB voltage gain at 10.7 MHz, and features an output stage with exceptionally good limiting characteristics because of its transistor constant-current sink.

The FM detector section is distinguished by circuitry which provides forward bias to the detector diodes, D2 and D3, and also provides a reference voltage for AFC.

The audio amplifier provides a low-impedance drive for subsequent audio amplifiers.

The power supply section provides zener-regulated, decoupled voltages for the IF amplifier, detector, and audio amplifier sections.

FEATURES

- high sensitivity – – input limiting voltage (knee) 50 μV typ. at 10.7 MHz
- excellent AM rejection – – 58 dB typ. at 10.7 MHz
- inherent high stability – – internally shielded
- internal Zener-diode regulated voltage supply
- low harmonic radiation
- wide frequency capability – – <100 kHz to >20 MHz
- low harmonic distortion

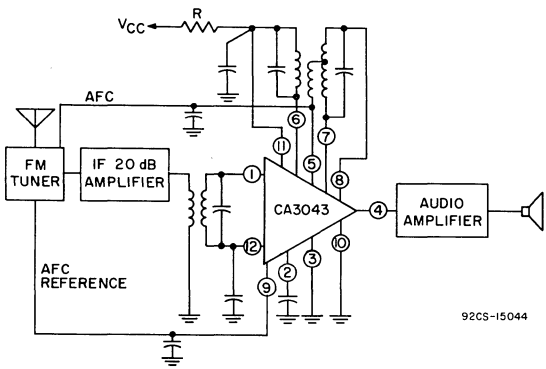


Fig.1 - Typical application of the CA3043 as a high-gain limiter, amplifier-detector in an FM receiver.

ABSOLUTE-MAXIMUM RATINGS at T_A = 25°C

DISSIPATION:
At T_A = 25°C to T_A = 85°C . . . . . .450 mW
Above T_A = 85°C . . . . . Derate linearly 5 mW/°C

TEMPERATURE RANGE:
Operating . . . . . -55°C to + 125°C
Storage . . . . . -65°C to + 150°C

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 min)
from case for 10 seconds max . . . . . + 265°C

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed horizontally with respect to the terminals listed vertically. For example, the voltage range between horizontal terminal 5 and vertical terminal 3 is +6 to 0 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12
1		+4 -4	0 -5	*	*	*	*	*	*	0 -5	*	Note(1)
2			0 -3	*	*	*	*	*	*	0 -3	*	*
3				+6 0	+6 0	+15 +2	+6 0	+6 0	+6 0	0	Note(2)	+3 0
4					+2 -4	*	*	*	*	0 -6	*	*
5						*	*	*	*	0 -6	+6 0	*
6							*	*	*	-2 -15	*	*
7								Note(1)	*	0 -6	*	*
8									*	0 -6	*	*
9										0 -6	*	*
10											Note(2) 0	+3 0
11												*
12												

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	-	-
2	-	-
3	0.1	40
4	-	20
5	-	-
6	-	-
7	-	-
8	-	-
9	-	20
10	0.1	40
11	40	0.1
12	-	-

Note 1: These terminals should be connected through a dc resistance to any terminal which does not exceed 100 ohms.

Note 2: Pin 11 may be connected to any positive voltage source through a suitable resistor provided its current rating is not exceeded.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	TEST CIRCUIT AND PROCEDURE	LIMITS TYPE CA3043			UNITS
			Fig.	Min.	Typ.	Max.	
STATIC CHARACTERISTICS							
Current Drain at 6V into Pin No.11	I_{11}	V_{CC} = +6V	3	10	16	20	mA
Regulator Voltage Pin No.11	V_{11}	V_{CC} = +30V, R_S = 750 Ω	3	6.9	7.4	8	V
Total Device Dissipation	P_T		3	200	225	260	mW
Quiescent Operating Current into Pin No.6	I_6		3	-	0.65	-	mA
DYNAMIC CHARACTERISTICS at V_{CC} = +30V, R_S = 750 Ω, f = 10.7 MHz							
Voltage Gain	A_v		4	72	80	-	dB
Input Limiting Voltage (knee)	v_i(lim)	v_o(af) at -3 dB point	6	-	50	-	μV (RMS)
Limiting Current from Pin No.6	I_6(lim)		4	-	0.42	-	mA (RMS)
Recovered AF Voltage	v_o(af)	v_i = 1 mV (RMS) f (modulating) = 1 kHz Deviation = ± 75 kHz	6	75	110	150	mV (RMS)
Amplitude-Modulation Rejection	AMR	v_i = 10 mV f (modulating) = 1 kHz % modulation = 50%	8	-	58	-	dB
Total Harmonic Distortion	THD	v_i = 1 mV (RMS)	6	-	0.3	-	%
Input Impedance Components:							
Parallel Input Resistance	R_I		-	-	7	-	kΩ
Parallel Input Capacitance	C_I		-	-	5	-	pF

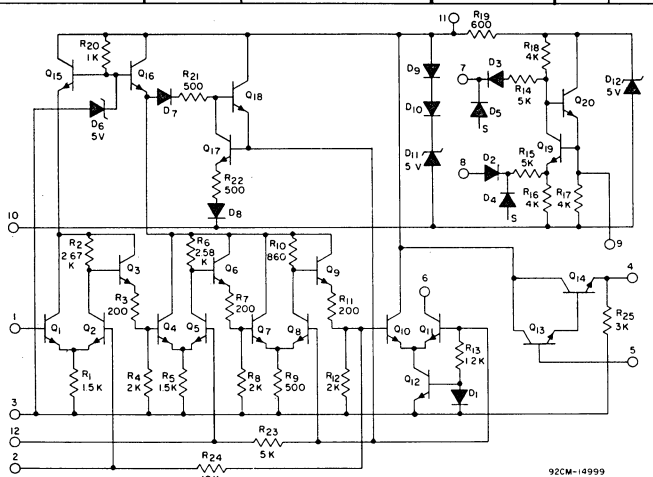


Fig.2 - Schematic diagram.

Notes:

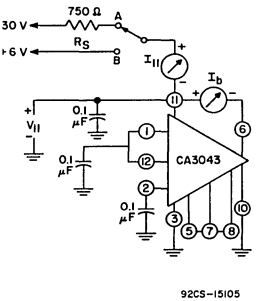
S = Substrate

Terminal No.3 wire-connected to the case.

Terminal No.10 connected to the case through the substrate.

Terminals No.3 and 10 which are connected to the substrate should be connected to the most negative point in the circuit.

Diodes D_4 and D_5, act as capacitors and are used to balance the detector substrate capacitances.

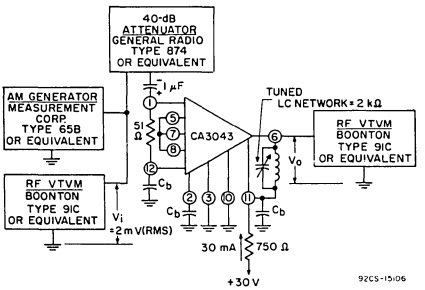


Switch in Position A for:

Regulator-Voltage, Quiescent-Operating-Current, and Device Dissipation Test

Switch in Position B for Current into Pin No.11

Fig.3 - Regulator voltage, device dissipation, quiescent operating current, and current at 6 volts into Pin No. 11.



$$\text{Voltage Gain} = 20 \log_{10} 100 \frac{V_o}{V_i}$$

C_b - Bypass Capacitor, 0.1 μF electrolytic in parallel with 0.01 μF

$I_6(1im) = \frac{V_o}{2K\Omega}$, V_i = 100 mV(RMS)

* Output circuit should be completely shielded from the input circuit at the socket.

Fig.4 - Voltage gain test circuit.

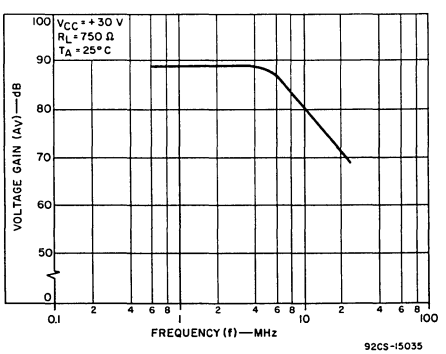
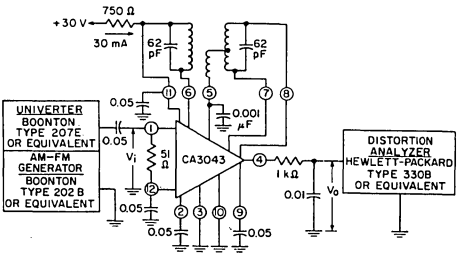


Fig.5 - Voltage gain vs frequency.



PROCEDURE:

1. Recovered Audio Voltage $v_{o(af)}$ —
   Set input frequency to 10.7 MHz,
   v_i = 1 mV(RMS), modulating frequency = 1 kHz
   Deviation = ±75 kHz
   Record v_o as measured on the Distortion Analyzer meter scale.
   This is the recovered Audio Voltage $v_{o(af)}$
2. 3 dB Limiting Sensitivity $v_{i(1im)}$ —
   Reduce v_i until $v_{o(af)}$ drops 3 dB.
   Record this value of v_i as $v_{i(1im)}$
3. Total Harmonic Distortion THD —
   Reset v_i to 1 mV(RMS) and operate Distortion Analyzer per manufacturer's instructions to measure THD.

* See Fig.9 for details on Discriminator Transformer.

Fig.6 - Input limiting voltage (knee), recovered AF voltage, and total harmonic distortion test circuit.

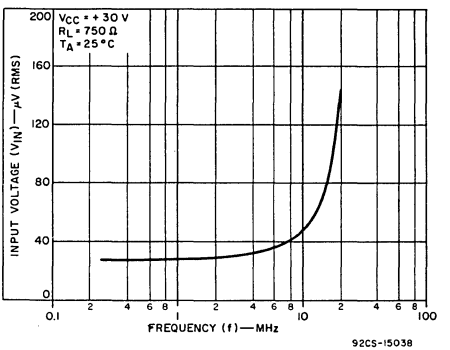
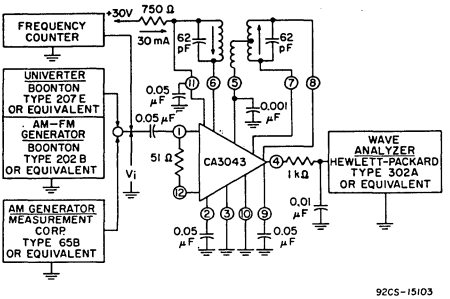


Fig.7 - Input limiting voltage (knee) at -3 dB point vs frequency.



PROCEDURE:

A. Connect FM Generator to CA3043 input.

Set frequency to 10.7 MHz, v_i = 10 mV, modulating frequency = 1 kHz

Deviation = ±75 kHz.

Tune Wave Analyzer to peak reading at 1 kHz and record recovered Audio Voltage $v_{o(af)FM}$.

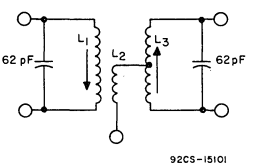
B. Disconnect FM Generator and Connect AM Generator to CA3043 input.

Set frequency to 10.7 MHz, v_i = 10 mV, modulating frequency = 1 kHz, percent modulation = 50%.

Tune Wave Analyzer to peak reading and record recovered audio voltage $v_{o(af)AM}$

Amplitude Modulation Rejection Ratio = $20 \log_{10} \frac{v_{o(af)FM}}{v_{o(af)AM}}$

Fig.8 - Amplitude modulation rejection test circuit.



Coil Form, Outside Diameter = 7/32"

Can = 1/2" square X 1-1/8" long

Slugs - Radio Industries Type MP34/MP100 Material

L_1 & L_3 = 20 Turns 5-44 litz wire universal wound

L_2 = 10 Turns 5-44 litz wire wound bifilar with L_1

L_1 & L_3 coupling adjusted to 520 kHz peak to peak separation on S curve when operated in circuit shown in Fig.6.

Fig.9 - 10.7-MHz discriminator transformer for CA3043.

RCA
Solid State
Division

Linear Integrated Circuits

CA3044
CA3044V1

Special-Function Sub-System

Monolithic Silicon

The RCA CA3044 and CA3044V1 represent a second generation of integrated circuits designed primarily for AFC (Automatic-Frequency-Control) applications.

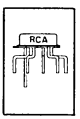
The CA3044V1 is electrically identical to the CA3044 but is supplied with formed leads for easier PC board design and construction.

CA3044



10-LEAD
TO-5

CA3044V1



FORMED
10-LEAD
TO-5

WIDE-BAND AMPLIFIER/PHASE DETECTOR WITH ZENER DIODE VOLTAGE REGULATOR

For AFC (Automatic Frequency Control) Applications

FEATURES

- Primarily intended for AFC (automatic frequency control) Applications
- Internal Zener Diode Voltage Regulator
- Differential Input Amplifier/Limiter
- Full-Wave Diode Bridge Detector
- Differential Output Voltage Amplifier
- Available in Two Electrically Identical Versions, CA3044 With Straight Leads; CA3044V1 With Formed Leads
- Wide Operating Temperature Range; -55 to +125°C

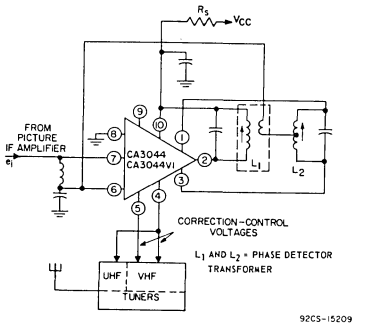


Fig.1 - Block diagram of Typical Automatic Fine Tuning (AFT) Application using CA3044 or CA3044V1 in Color-TV Receiver.

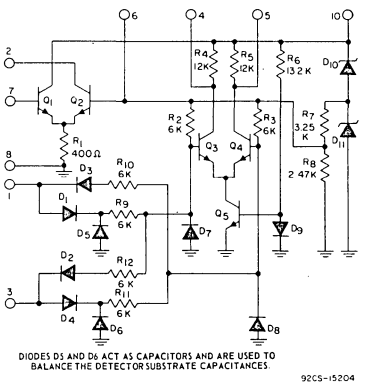


Fig.2 - Schematic diagram CA3044, CA3044V1

ABSOLUTE-MAXIMUM RATINGS

DISSIPATION:
At T_A = 25°C . . . . . 830 mW
Above T_A = 25°C . . . . . Derate linearly 5.6 mW/°C

TEMPERATURE RANGE:
Operating . . . . . -55°C to +125°C
Storage . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. . . . . + 265C

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 6 is +20 to 0 volts.

TERMINAL No.	9	10	1	2	3	4	5	6	7	8
9	NO INTERNAL CONNECTION									
10			+20 0	+20 -10	+20 0	+20 0	+20 0	+20 0	+20 0	▲
1				*	+12 -12	*	*	+6 -6	*	+6 0
2					*	*	*	+20 0	*	+20 0
3						*	*	+6 -6	*	+6 0
4							*	*	*	+12 0
5								*	*	+12 0
6									+5 -5	+5 0
7										+8 -5
8										REF. SUB-STRATE

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
9	-	-
10	50	50
1	5	5
2	20	20
3	5	5
4	5	5
5	5	5
6	5	5
7	5	5
8	50	50

▲ Terminal No. 10 may be connected to any positive voltage source through a suitable dropping resistor—provided the dissipation rating is not exceeded.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Unless Otherwise Specified

CHARACTERISTICS	SYMBOLS	TEST CIRCUITS FIG.	TEST CONDITIONS	LIMITS CA3044 and CA3044V1 MIN.	TYP.	MAX.	UNITS	CHARACTERISTIC CURVES FIG.
STATIC CHARACTERISTICS								
Device Dissipation	P_T	3	V_{CC} = 30 V, R_S = 1.5 kΩ, T_A = -55°C	90	120	150	mW	-
Device Dissipation	P_T	3	V_{CC} = 30 V, R_S = 1.5 kΩ, T_A = 25°C	110	140	170	mW	-
Device Dissipation	P_T	3	V_{CC} = 30 V, R_S = 1.5 kΩ, T_A = +125°C	130	160	190	mW	-
9-Volt Current Drain	I_T	3	V_{10} = 9 V	2.5	4	5.5	mA	-
Zener Regulating Voltage - DC Supply Voltage at Terminal 10	V_{10}	3	V_{CC} = 30 V, R_S = 1.5 kΩ	10.5	11.2	11.9	V	-
Quiescent Operating Current into Terminal 2	I_2	3		1	2	4	mA	-
Quiescent Operating Voltage at Terminal 4	V_4	-		5.0	6.5	8.0	V	-
Quiescent Operating Voltage at Terminal 5	V_5	-		5.0	6.5	8.0	V	-
Output Offset Voltage between Terminals 4 and 5	V_{4-5}	-		-1.5	0	1.5	V	-
DYNAMIC CHARACTERISTICS (AS RF AMPLIFIER)								
Input Limiting Voltage (Knee)	V_i Limiting	4	f=45.75 MHz	-	75	-	mV	-
Input Admittance	y_{11}	-	f = 45.75 MHz, V_{CC} = 30 V, R_S = 1.5 kΩ	-	0.5+j1.1	-	mmho	-
Reverse Transfer Admittance	y_{12}	-		-	3.8+j3.4	-	μmho	-
Forward Transfer Admittance	y_{21}	-		-	-11.7 +10.1	-	mmho	-
Output Admittance	y_{22}	-		-	0.077+j0.9	-	mmho	-
OUTPUT vs FREQUENCY DEVIATION - AFC								
Correction-Control Voltage at Terminal 4	V corr. (4)	5	V_{CC} = +30 V, V_{in} = 200 mV RMS, f_o = MHz as indicated	% of V_{10}		% of V_{10}		
			45.750 - 0.025	85	-	-	V	6,7
			45.750 + 0.025	-	-	33	V	
			45.750 - 0.900	75	-	-	V	7
			45.750 + 0.900	-	-	43	V	
			45.750 - 1.500	-	-	85	V	
			45.750 + 1.500	33	-	-	V	
Correction-Control Voltage at Terminal 5	V corr. (5)	5	45.750 - 0.025	-	-	33	V	6,7
			45.750 + 0.025	85	-	-	V	
			45.750 - 0.900	-	-	43	V	7
			45.750 + 0.900	75	-	-	V	
			45.750 - 1.500	33	-	-	V	
			45.750 + 1.500	-	-	85	V	

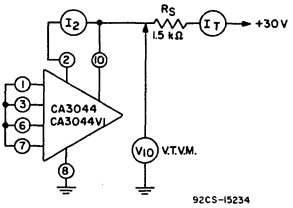


Fig.3 - Test setup: Measurement of total device dissipation, Zener regulating voltage, quiescent operating current (terminal 2).

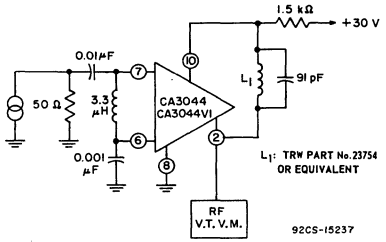


Fig.4 - Input limiting sensitivity test circuit.

DYNAMIC CONTROL VOLTAGE CHARACTERISTICS

The CA3044 and CA3044VI are specifically intended for use in the AFT system of color television receivers. Each device is tested so that the control voltages generated by the circuit meet the critical requirements of the system. Figure 5 is the schematic diagram of the test circuit.

Figure 6 and 7 show the control voltages generated at terminals 4 and 5 of the Integrated Circuit as a function of the frequency deviation from the nominal center frequency. Figure 6 shows the region within 25 KHz of the center frequency while Figure 7 covers the entire bandwidth of the system. The horizontal reference lines on the figures are generated by a voltage divider connected between the power supply voltage on Terminal 10 and ground. The dynamic control voltages are compared with these references according to the Output vs Frequency Deviation Table. For example: when the frequency deviation is -25 KHz the control voltage at Terminal 4 is greater than the reference A voltage; the control voltage at Terminal 5 is less than the reference B voltage.

The shape of the correction voltage characteristics is dependent to a large degree upon transformer characteristics and the parts layout. In order to closely duplicate the curves shown, the printed circuit board shown in Figure 8 and the parts layout shown in Figure 9 should be followed as closely as possible.

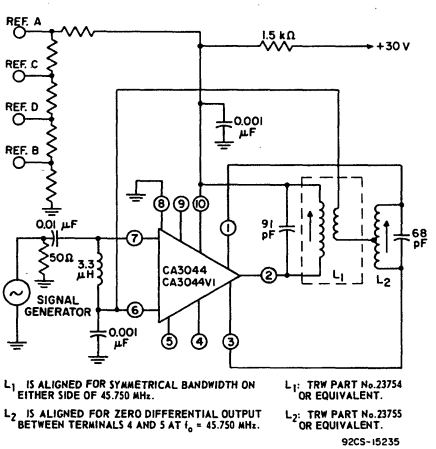


Fig.5 - Correction voltage test circuit for CA3044 and CA3044VI.

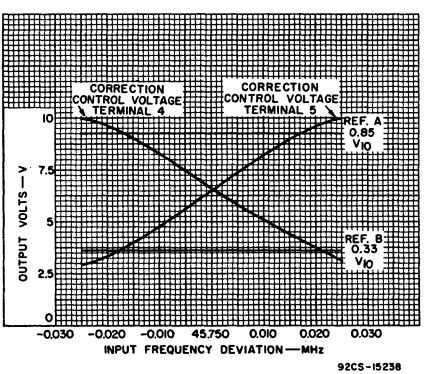


Fig.6 - Typical narrow-band dynamic control voltage characteristics.

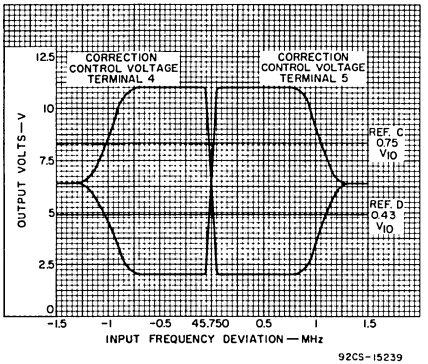


Fig.7 - Typical wide-band dynamic control voltage characteristics.

DEFINITIONS OF TERMS

Input Limiting Voltage (Knee) [$v_i(lim)$]

The input signal voltage which will cause the output signal to decrease 3 dB from its maximum level.

Total Device Dissipation (P_T)

The total power drain of the device with no signal applied and no external load current.

Quiescent Operating Voltage

The dc voltage at the output terminal, with respect to ground, with no signal applied.

Quiescent Operating Current

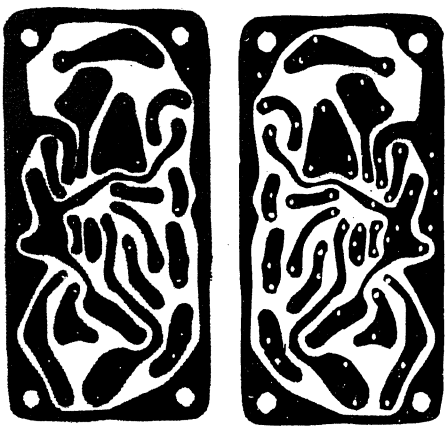
The average (dc) value of the current in either output, terminal, with no signal applied.

Output Offset Voltage

The dc voltage between output terminals with no signal applied.

Control Voltage

The dc voltage at either output terminal with respect to ground with an RF signal of specified frequency applied.



a) Top view b) Bottom view

Fig.8 - Printed Circuit Board for Test Circuit

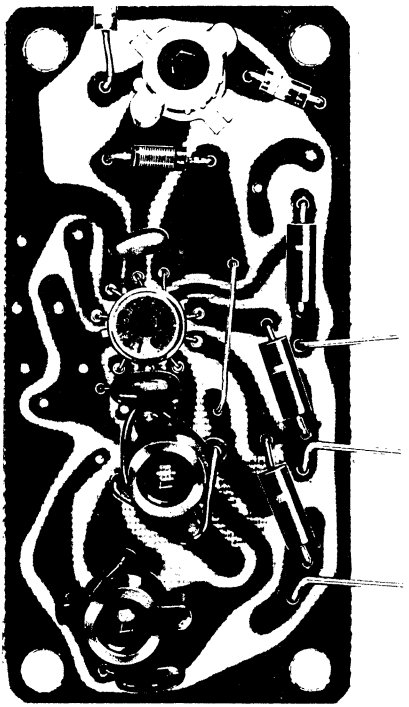


Fig.9 - Top view of wired test board.

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Monolithic Silicon

CA3045, CA3045F, CA3046

14-Lead Dual-In-Line Ceramic



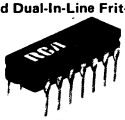
H-1553

14-Lead Dual-In-Line Plastic



H-1517

14-Lead Dual-In-Line Frit-Seal



H-1806

General-Purpose Transistor Arrays

THREE ISOLATED TRANSISTORS AND ONE DIFFERENTIALLY-CONNECTED TRANSISTOR PAIR

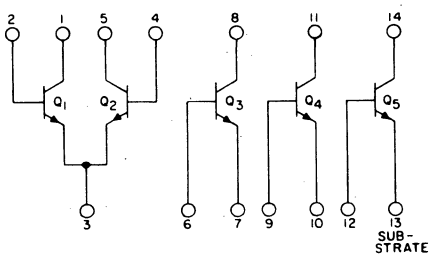
For Low-Power Applications at Frequencies from DC through the VHF Range

The CA3045 and CA3046 each consist of five general-purpose silicon n-p-n transistors on a common monolithic substrate. Two of the transistors are internally connected to form a differentially-connected pair.

The transistors of the CA3045 and CA3046 are well suited to a wide variety of applications in low power systems in the DC through VHF range. They may be used as discrete transistors in conventional circuits. However, in addition, they provide the very significant inherent integrated circuit advantages of close electrical and thermal matching.

The CA3045 is supplied in a 14-lead dual-in-line hermetic (welded-seal) ceramic package and the CA3045F in a 14-lead dual-in-line hermetic (frit-seal) ceramic package.

The CA3046 is electrically identical to the CA3045 but is supplied in a dual-in-line plastic package for applications requiring only a limited temperature range.



92CS-15206

Fig.1 - Schematic diagram.

FEATURES

- Two matched pairs of transistors
  V_{BE} matched ±5 mV
  Input offset current 2 μA max. at I_C = 1 mA
- 5 general purpose monolithic transistors
- Operation from DC to 120 MHz
- Wide operating current range
- Low noise figure - - 3.2 dB typ. at 1 kHz
- Full military temperature range for CA3045
  -55 to +125°C

APPLICATIONS

- General use in all types of signal processing systems operating anywhere in the frequency range from DC to VHF
- Custom designed differential amplifiers
- Temperature compensated amplifiers
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Integrated-Circuit Transistor Array" for suggested applications.

ABSOLUTE MAXIMUM RATINGS AT $T_A = 25^oC$

	CA3045		CA3045F, CA3046		
	Each Transistor	Total Package	Each Transistor	Total Package	
Power Dissipation:					
T_A up to 55°C	–	–	300	750	mW
T_A > 55°C	–	–	Derate at 6.67		mW/°C
T_A up to 75°C	300	750	–	–	mW
T_A > 75°C	Derate at 8		–	–	mW/°C
Collector-to-Emitter Voltage, V_{CEO}	15	–	15	–	V
Collector-to-Base Voltage, V_{CBO}	20	–	20	–	V
Collector-to-Substrate Voltage, V_{CIO}*	20	–	20	–	V
Emitter-to-Base Voltage, V_{EBO}	5	–	5	–	V
Temperature Range:					
Operating	–55 to +125		–55 to +125		°C
Storage	–65 to +150		–65 to +150		°C
Lead Temperature (During Soldering): At distance 1/16 ±1/32" (1.59 ±0.79 mm) from case for 10 seconds max.	+265		+265		°C

* The collector of each transistor of the CA3045 and CA3046 is isolated from the substrate by an integral diode. *The substrate (terminal 13) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.*

ELECTRICAL CHARACTERISTICS, at $T_A = 25^oC$

Characteristics apply for each transistor in the CA3045 and CA3046 as specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS Type CA3045 Type CA3046			UNITS
			MIN.	TYP.	MAX.	
STATIC CHARACTERISTICS						
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	20	60	-	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	15	24	-	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10μA, I_{CI} = 0	20	60	-	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	5	7	-	V
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	-	0.002	40	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10 V, I_B = 0	-	See curve	0.5	μA
Static Forward Current-Transfer Ratio (Static Beta)	h_{FE}	V_{CE} = 3 V, I_C = 10 mA	-	100	-	-
		V_{CE} = 3 V, I_C = 1 mA	40	100	-	-
		V_{CE} = 3 V, I_C = 10 μA	-	54	-	-
Input Offset Current for Matched Pair Q_1 and Q_2. $\lvert I_{IO_1} - I_{IO_2} \rvert$		V_{CE} = 3 V, I_C = 1 mA	-	0.3	2	μA
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3 V, I_E = 1 mA	-	0.715	-	V
		V_{CE} = 3 V, I_E = 10 mA	-	0.800	-	
Magnitude of Input Offset Voltage for Differential Pair $\lvert V_{BE_1} - V_{BE_2} \rvert$		V_{CE} = 3 V, I_C = 1 mA	-	0.45	5	mV
Magnitude of Input Offset Voltage for Isolated Transistors $\lvert V_{BE_3} - V_{BE_4} \rvert$, $\lvert V_{BE_4} - V_{BE_5} \rvert$, $\lvert V_{BE_5} - V_{BE_3} \rvert$		V_{CE} = 3 V, I_C = 1 mA	-	0.45	5	mV
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 3 V, I_C = 1 mA	-	-1.9	-	mV °C
Collector-to-Emitter Saturation Voltage	V_{CES}	I_B = 1 mA, I_C = 10 mA	-	0.23	-	V
Temperature Coefficient: Magnitude of Input-Offset Voltage	$\frac{\lvert \Delta V_{IO} \rvert}{\Delta T}$	V_{CE} 3 V, I_C 1 mA	-	1.1	-	μV°C

ELECTRICAL CHARACTERISTICS, at T_A = 25°C

Characteristics apply for each transistor in the CA3045 and CA3046 as specified.

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS Type CA3045 Type CA3046			UNITS
			MIN.	TYP.	MAX.	
DYNAMIC CHARACTERISTICS						
Low-Frequency Noise Figure	NF	f = 1 kHz, V_{CE} = 3V, I_C = 100 μA Source Resistance = 1 kΩ	-	3.25	-	dB
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:						
Forward Current-Transfer Ratio	h_{fe}	f = 1 kHz, V_{CE} = 3 V, I_C = 1 mA	-	110	-	-
Short-Circuit Input Impedance	h_{ie}		-	3.5	-	kΩ
Open-Circuit Output Impedance	h_{oe}		-	15.6	-	μmho
Open-Circuit Reverse Voltage-Transfer Ratio	h_{re}		-	1.8×10^{-4}	-	-
Admittance Characteristics:						
Forward Transfer Admittance	Y_{fe}	f = 1 MHz, V_{CE} = 3 V, I_C = 1 mA	-	31-j1.5	-	-
Input Admittance	Y_{ie}		-	0.3+j0.04	-	-
Output Admittance	Y_{oe}		-	0.001+j0.03	-	-
Reverse Transfer Admittance	Y_{re}		-	See curve	-	-
Gain-Bandwidth Product	f_T	V_{CE} = 3 V, I_C = 3 mA	300	550	-	-
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 3 V, I_E = 0	-	0.6	-	pF
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 3 V, I_C = 0	-	0.58	-	pF
Collector-to-Substrate Capacitance	C_{CI}	V_{CS} = 3 V, I_C = 0	-	2.8	-	pF

STATIC CHARACTERISTICS

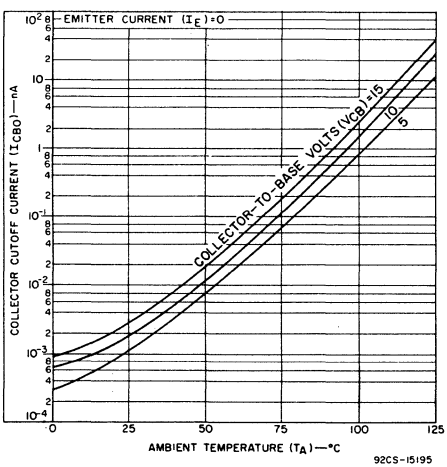


Fig.2 - Typical collector-to-base cutoff current vs ambient temperature for each transistor.

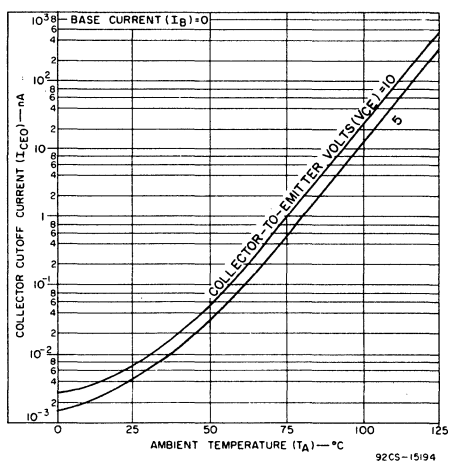


Fig.3 - Typical collector-to-emitter cutoff current vs ambient temperature for each transistor.

STATIC CHARACTERISTICS

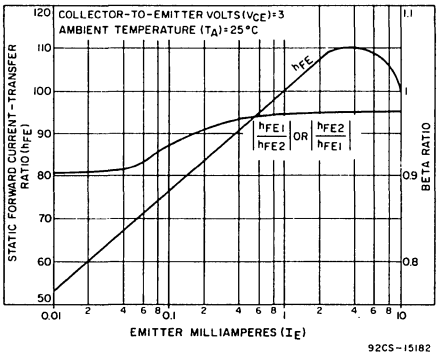


Fig.4 - Typical static forward current-transfer ratio and beta ratio for transistors Q_1 and Q_2 vs emitter current.

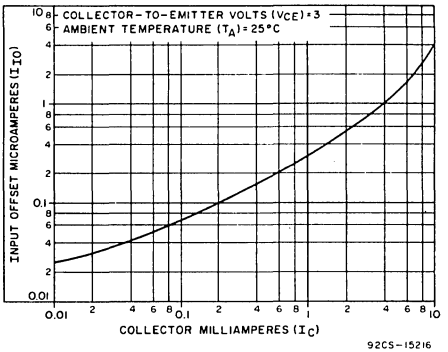


Fig.5 - Typical input offset current for matched transistor pair Q_1Q_2 vs collector current.

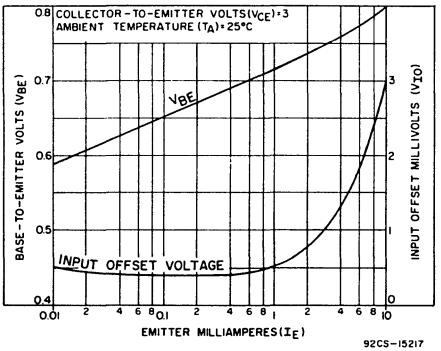


Fig.6 - Typical static base-to-emitter voltage characteristic and input offset voltage for differential pair and paired isolated transistors vs emitter current.

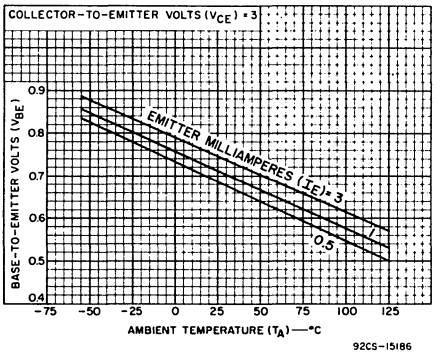


Fig.7 - Typical base-to-emitter voltage characteristic vs ambient temperature for each transistor.

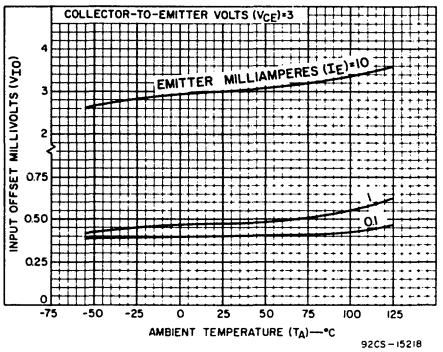


Fig.8 - Typical input offset voltage characteristics for differential pair and paired isolated transistors vs ambient temperature.

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

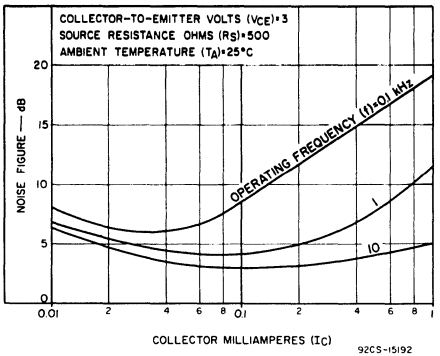


Fig.9(a) - Typical noise figure vs collector current.

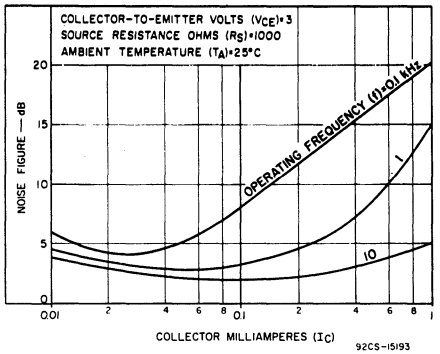


Fig.9(b) - Typical noise figure vs collector current.

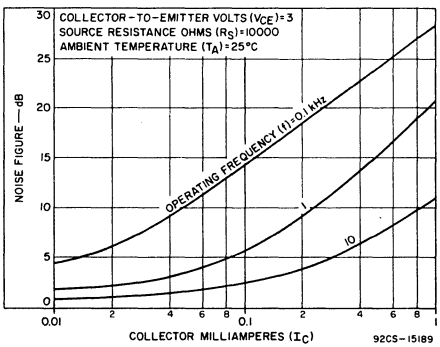


Fig.9(c) - Typical noise figure vs collector current.

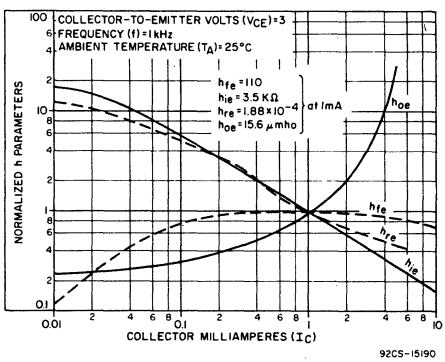


Fig.10 - Typical normalized forward current-transfer ratio, short-circuit input impedance, open-circuit output impedance, and open-circuit reverse voltage-transfer ratio vs collector current.

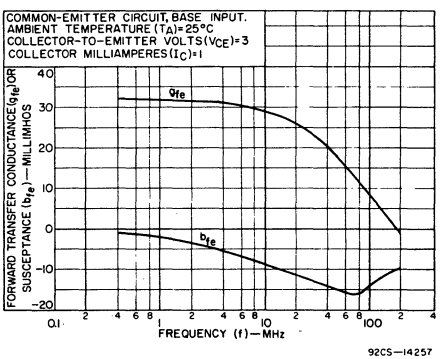


Fig.11 - Typical forward transfer admittance vs frequency.

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

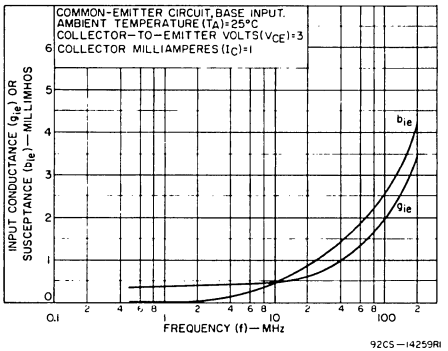


Fig.12 - Typical input admittance vs frequency.

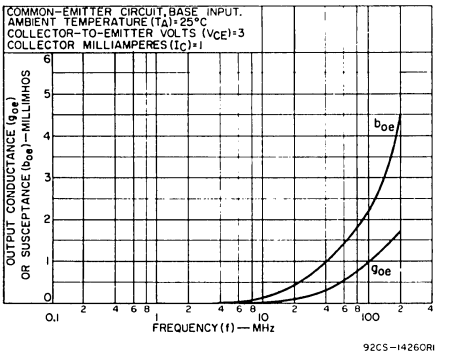


Fig.13 - Typical output admittance vs frequency.

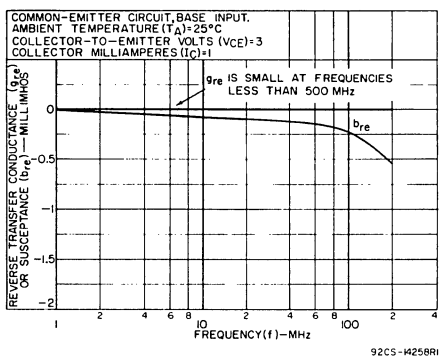


Fig.14 - Typical reverse transfer admittance vs frequency.

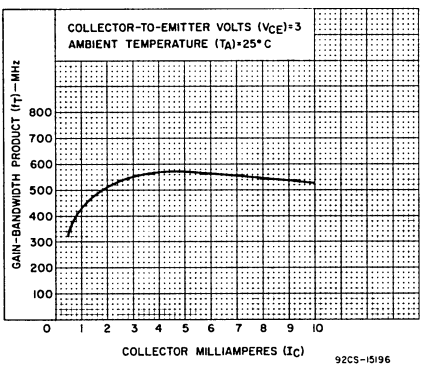


Fig.15 - Typical gain-bandwidth product vs collector current.

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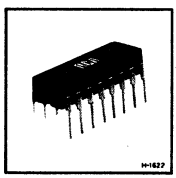
CA3048

Amplifier Array

Monolithic Silicon

FOUR INDEPENDENT AC AMPLIFIERS

For Low-Noise and General AC Applications In Industrial Service



CA3048

The RCA CA3048 is a silicon monolithic integrated circuit consisting of four independent identical AC amplifiers which can operate from a single-ended power supply.

The amplifiers include internal DC bias and feedback to provide temperature-stabilized operation. They may be used in a wide variety of AC applications in which operational amplifiers have previously been used.

Each high gain amplifier has a high impedance non-inverting input, and a lower impedance inverting input for the application of feedback. Two power-supply terminals and two ground terminals are provided to reduce internal and external coupling between amplifiers.

The CA3048 is supplied in a 16-lead dual-in-line plastic package.

APPLICATIONS

- Multi-channel or cascade operation
- Low-level preamplifiers
- Equalizers
- Linear signal mixers
- Tone generators
- Multivibrators
- AC integrators

FEATURES

- Four AC amplifiers on a common substrate
- Independently accessible inputs and outputs
- Operates from single-ended supply

EACH AMPLIFIER

- Noise figure at 1 kHz ..... 2 dB typ.
- High voltage gain ..... 53 dB min.
- High input resistance ..... 90 kΩ typ.
- Undistorted output voltage ..... 2 V rms min.
- Output Impedance ..... 1 kΩ typ.
- Open-loop bandwidth ..... 300 kHz typ.

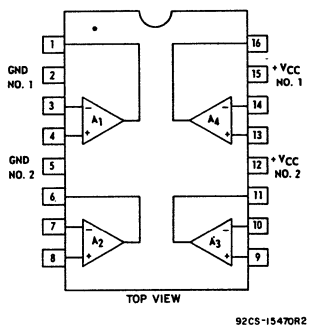


Fig.1 - Block diagram for CA3048.

ABSOLUTE-MAXIMUM RATINGS at T_A = 25°C:

DISSIPATION:

At T_A = 55°C . . . . . 750 mW

Above T_A = 55°C. . . . . Derate linearly at 7.7 mW/°C

TEMPERATURE RANGE:

Operating . . . . . -40°C to +85°C

Storage . . . . . -65°C to +150°C

LEAD TEMPERATURE (During Soldering)

At distance 1/16 ± 1/32 inch (1.59 ±0.79mm)

from case for 10 seconds max. . . . . +265°C

POWER SUPPLY VOLTAGE . . . . . +16 V

AC INPUT VOLTAGE . . . . . 0.5 V rms

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 4 is +2 to -3.6 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		+16 0	*	*	*	*	*	*	*	*	*	*	*	*	0 -16	*
2			*	+2 -3.6	0	*	*	+2 -3.6	-3.6	*	*	+16 0	+2 -3.6	*	+16 0	0 -16
3				+5 -5	*	*	*	*	*	*	*	*	*	*	*	*
4					+3.6 -2	*	*	*	*	*	*	*	*	*	*	*
5						0 -16	*	+2 -3.6	+2 -3.6	*	0 -16	+16 0	+2 -3.6	*	+16 0	*
6							*	*	*	*	*	*	0 -16	*	*	*
7								+5 -5	*	*	*	*	*	*	*	*
8									*	*	*	*	*	*	*	*
9										+5 -5	*	*	*	*		*
10											*	*	*	*	*	*
11												*	*	*	*	*
12													0 -16	*	*	*
13														+5 -5	*	*
14															*	*
15																+16 0
16																

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIR-CUIT	LIMITS CA3048			UNITS	TYPICAL CHARAC-TERISTICS CURVES
			FIG.	MIN.	TYP.	MAX.		FIG.
STATIC								
Current drain per amplifier pair	I_{12} or I_{15}	V_{CC} = +12V	3	9.5	13.5	17.5	mA	4,5
DC Voltage at Output Terminals	V_1, V_6, V_{11}, V_{16}	V_{CC} = +12V	3	6.1	6.9	8.1	V	-
DC Voltage at Feedback Terminals	V_3, V_7, V_{10}, V_{14}	V_{CC} = +12V	3	1.7	2.0	2.3	V	-
DC Voltage at Input Terminals	V_4, V_8, V_9, V_{13}	V_{CC} = +12V	3	2.2	2.5	2.8	V	-
DYNAMIC (Characteristics given are for each amplifier with no AC feedback)								
Open-Loop Gain	A_{OL}	V_{CC} = +12V, E_{IN} = 2mV, f = 10kHz	6	53	58	-	dB	7,8
Output Voltage Swing	V_O(rms)	V_{CC} = +12V, f = 1kHz, THD = 5%	6	2.0	2.4	-	V	-
Open-Loop -3dB Bandwidth	BW	V_{CC} = +12V, E_{IN} = 2mV	6	250	300	-	kHz	9
Total Harmonic Distortion	THD	V_{CC} = +12V, f = 1kHz, E_{OUT} = 2V rms	6	-	0.65	-	%	10
Input Resistance	R_{IN}	OPEN LOOP Terminals 3, 7, 10, and 14 are by-passed to ground, f = 1kHz	-	-	90	-	kΩ	-
Input Capacitance	C_{IN}	f = 1MHz	-	-	9	-	pF	-
Output Resistance	R_{OUT}	Terminals 3, 7, 10 and 14 are by-passed to ground	-	-	1	-	kΩ	-
Output Capacitance	C_{OUT}	f = 1MHz	-	-	18	-	pF	-
Feedback Capacitance (Output to non-inverting Input)	C_{FB}	V_{CC} = +12V, f = 1MHz	-	-	<0.1	-	pF	-
Broad-Band Output Noise Voltage	E_N	V_{CC} = +12V, R_S = 10kΩ, A = 40dB, Equivalent Noise BW = 50kHz	11	-	0.3	1	mV	-
Output Noise Voltage "Weighted"	E_N(WT)		12	-	0.5	2.2	mV	-
Noise Figure	NF (R_S = 5 kΩ)	f = 10Hz	-	-	10	-	dB	-
		f = 100Hz	-	-	5.8	-	dB	
		f = 1kHz	-	-	2	-	dB	
		f = 10kHz	-	-	1.1	-	dB	
		f = 100kHz	-	-	0.6	-	dB	
Inter-Amplifier Audio Separation "Cross Talk"		V_{CC} = +12V, f = 1kHz, 0dB = 0.78V	13	-	<-45	-	dB	-
Inter-Amplifier Capacitance (Any amplifier output to any *other* amplifier input)	C	V_{CC} = +12V, f = 1MHz	-	-	<0.02	-	pF	-

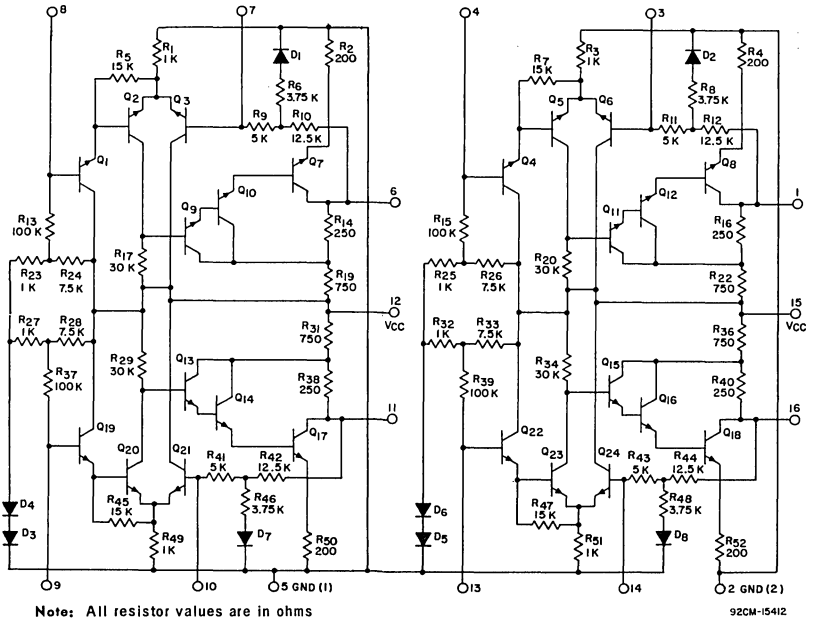


Fig.2 - Schematic diagram for CA3048.

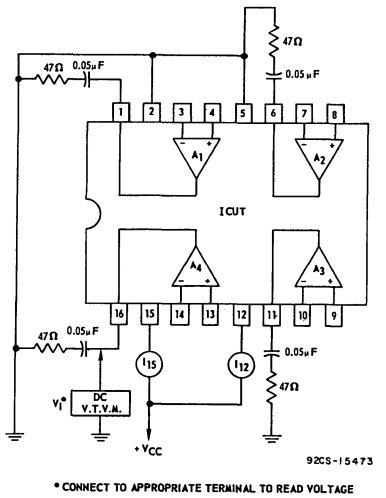


Fig.3 - Test circuit for measurement of collector supply voltage and currents.

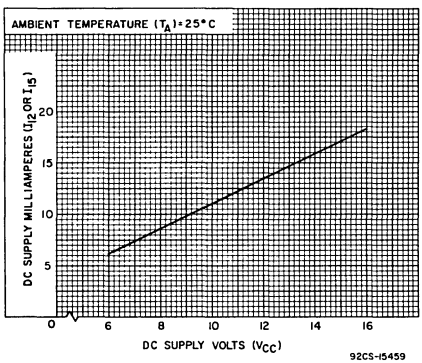


Fig.4 - Typical DC supply current vs supply voltage.

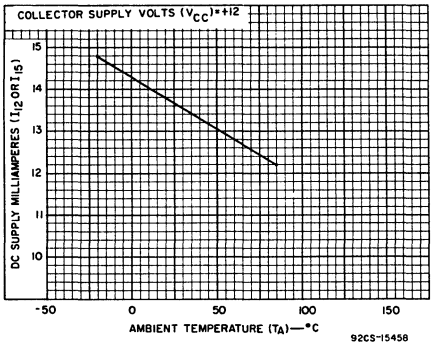


Fig.5 - Typical DC supply current vs ambient temperature.

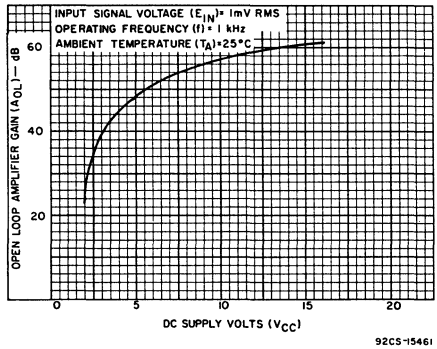
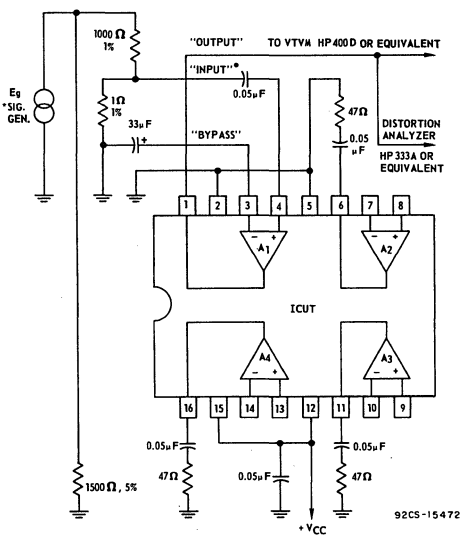


Fig.7 - Typical amplifier gain vs DC supply voltage.



* Sig Gen should be a low distortion type (0.2% THD or less) HP206A or equivalent.

● Adjustment of E_g to 2 volts will make E_S = 2mV.

Test Circuit shows Amplifier #1 under test, to test Amplifiers 2, 3, or 4; Connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig.6 - Test circuit for measurement of distortion, open-loop gain and bandwidth characteristics.

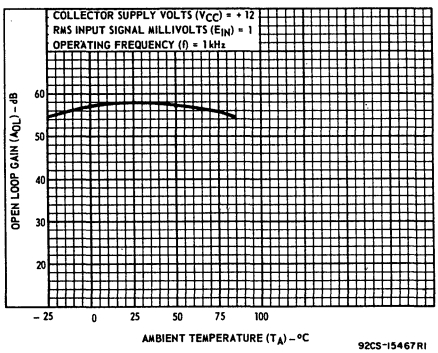


Fig.8 - Typical open-loop gain vs ambient temperature.

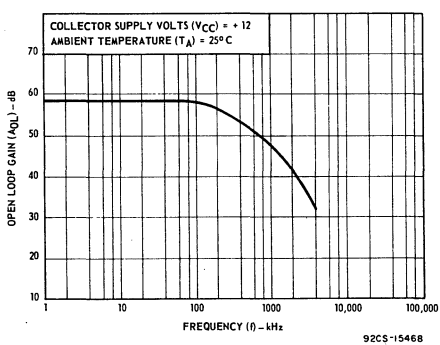


Fig.9 - Typical open-loop gain vs frequency.

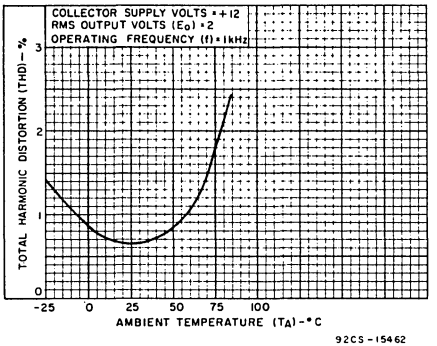
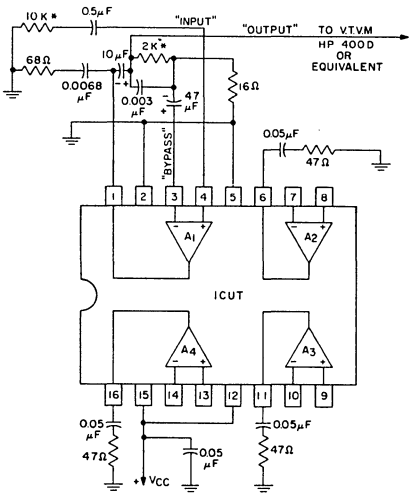


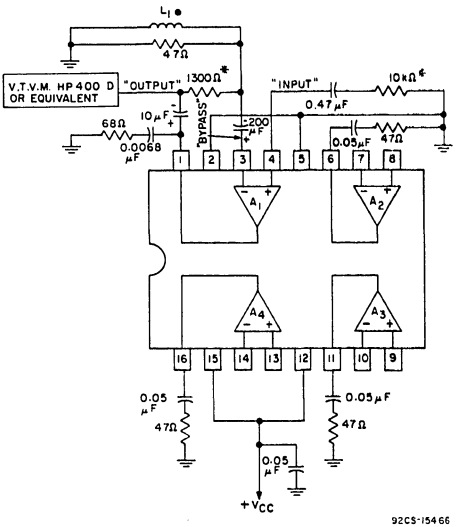
Fig.10 - Typical total harmonic distortion vs ambient temperature.



To test Amplifiers 1, 2, 3, or 4, connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig.11 - Test circuit for measurement of broadband noise characteristic.

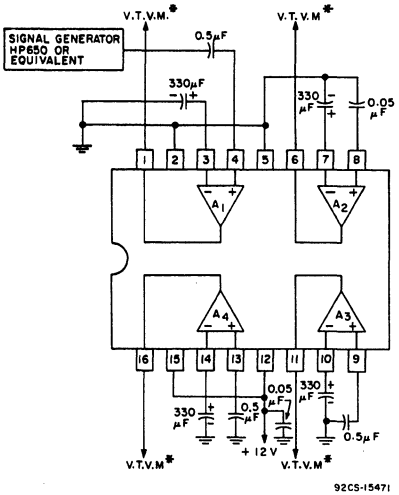


● L_1 - 2.5 millihenry inductor, dc resistance 0.3 ohms or less.

* Resistors metal film type, 1%. To test amplifiers, connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig.12 - Test circuit for measurement of "weighted" output noise voltage characteristic.



* V.T.V.M. - Hewlett-Packard Model 400D or equivalent.

Procedure:

1. Adjust Signal Generator for 0 dB output at reference terminal.
2. Read voltage at other output terminals (Figure shows terminal #1 used as reference).

Fig.13 - Test circuit for measurement of inter-amplifier audio separation "cross talk" characteristic.

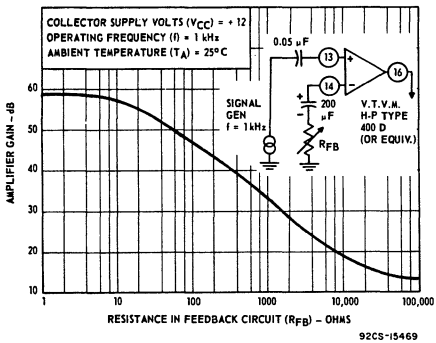


Fig.14 - Typical amplifier gain vs feedback resistance.

OPERATING CONSIDERATIONS

Economical Gain Control

The CA3048 is designed to permit flexibility in the methods by which amplifier gain can be controlled. Fig.14 shows a curve of the gain of an amplifier when the internal resistive feedback of the device is used in conjunction with an external resistor. Although measured gain of various amplifiers will not be uniform, because of tolerances of internal resistances, this method is very economical and easy to apply.

Stability

The CA3048, as in other devices having high gain-bandwidth product, requires some attention to circuit layout, design, and construction to achieve stability.

Should the CA3048 be left unterminated, socket capacitance alone will provide sufficient feedback to cause high frequency oscillations; therefore, all test circuits in this data bulletin include loading networks that provide stability under all conditions.

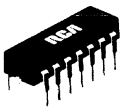
RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3049T CA3102E

14-Lead DIP
CA3102E



12-Lead TO-5



CA3049T

DUAL HIGH-FREQUENCY DIFFERENTIAL AMPLIFIERS

For Low-Power Applications at Frequencies up to 500 MHz

Features:

- **Power Gain 23 dB (typ.) at 200 MHz**
- **Noise Figure 4.6 dB (typ.) at 200 MHz**
- **Two differential amplifiers on a common substrate**
- **Independently accessible inputs and outputs**
- **Full military-temperature-range capability- (–55°C to + 125°C) for the CA3102E and for the CA3049T**

RCA-CA3049T and CA3102E* consist of two independent differential amplifiers with associated constant-current transistors on a common monolithic substrate. The six transistors which comprise the amplifiers are general-purpose devices which exhibit low I/f noise and a value of f_T in excess of 1 GHz. These features make the CA3049T and CA3102E useful from dc to 500 MHz. Bias and load resistors have been omitted to provide maximum application flexibility.

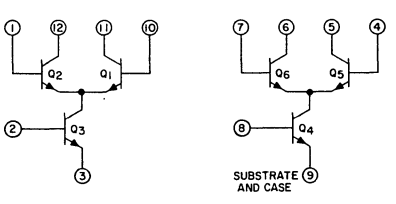
The monolithic construction of the CA3049T and CA3102E provides close electrical and thermal matching of the amplifiers. This feature makes these devices particularly useful in dual-channel applications where matched performance of the two channels is required.

The CA3102E is like the CA3049T except that it has a separate substrate connection for greater design flexibility. The CA3049T is supplied in the 12-lead TO-5 package; the CA3102E, in the 14-lead plastic dual-in-line package.

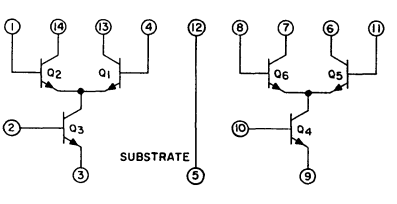
Applications

- **VHF amplifiers**
- **VHF mixers**
- **Multifunction combinations — RF/Mixer/Oscillator; Converter/IF**
- **IF amplifiers (differential and/or cascode)**
- **Product detectors**
- **Doubly balanced modulators and demodulators**
- **Balanced quadrature detectors**
- **Cascade limiters**
- **Synchronous detectors**
- **Balanced mixers**
- **Synthesizers**
- **Balanced (push-pull) cascode amplifiers**
- **Sense amplifiers**

*Formerly Developmental No. TA6228.



Schematic Diagram for CA3049T



Schematic Diagram for CA3102E

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		TEST CIR-CUIT FIG.	CA3102E LIMITS MIN.	TYP.	MAX.	UNITS	TYPICAL CHARAC-TERISTICS CURVES FIG.
STATIC CHARACTERISTICS									
For Each Differential Amplifier									
Input Offset Voltage	V_{IO}			1	---	0.25	5	mV	-4
Input Offset Current	I_{IO}	$I_3 = I_9 = 2$ mA		1	---	0.3	3	μA	---
Input Bias Current	I_{IB}			1	---	13.5	33	μA	5
Temperature Coefficient Magnitude of Input-Offset Voltage	$\left\|\frac{\Delta V_{IO}}{\Delta T}\right\|$			1	---	1.1	---	μV/°C	4
For Each Transistor									
DC Forward Base-to-Emitter Voltage	V_{BE}	$V_{CE} = 6$ V, $I_C = 1$ mA		---	674	774	874	mV	6
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	$V_{CE} = 6$ V, $I_C = 1$ mA		---	---	-0.9	---	mV/°C	6
Collector-Cutoff Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$		---	---	0.0013	100	nA	7
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$		---	15	24	---	V	---
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10$ μA, $I_E = 0$		---	20	60	---	V	---
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_C = 10$ μA, $I_B = 0$, $I_E = 0$		---	20	60	---	V	---
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 10$ μA, $I_C = 0$		---	5	7	---	V	---
DYNAMIC CHARACTERISTICS									
1/f Noise Figure (For Single Transistor)	NF	f = 100 KH_3, $R_S = 500\ \Omega$ $I_C = 1$ mA		---	---	1.5	---	dB	12
Gain-Bandwidth Product (For Single Transistor)	f_T	$V_{CE} = 6$ V, $I_C = 5$ mA		---	---	1.35	---	GH_z	11
Collector-Base Capacitance	C_{CB}	$I_C = 0$ $V_{CB} = 5$V	*	---	---	0.28	---	pF	8
			**	---	---	0.15	---	pF	
Collector-Substrate Capacitance	C_{CI}	$I_C = 0$ $V_{CI} = 5$V			---	1.65	---	pF	8
For Each Differential Amplifier									
Common-Mode Rejection Ratio	CMR	$I_3 = I_9 = 2$ mA		---	---	100	---	dB	---
AGC Range, One Stage	AGC	Bias Voltage = -6V		2	---	75	---	dB	---
Voltage Gain, Single-Ended Output	A	Bias Voltage = -4.2V f = 10 MHz		2	18	22	---	dB	9, 10
Insertion Power Gain	G_p	f = 200 MHz	Cascode	3	---	23	---	dB	---
Noise Figure	NF	$V_{CC} = 12$V	Cascode	3	---	4.6	---	dB	---
Input Admittance	Y_{11}	For Cascode Configuration $I_3 = I_9 = 2$ mA	Cascode	---	---	1.5 + j 2.45	---	mmho	14, 16, 18
			Diff.Amp.	---	---	0.878 + j 1.3	---		15, 17, 19
Reverse Transfer Admittance	Y_{12}	For Diff. Amplifier Configuration $I_3 = I_9 = 4$mA (each collector $I_C \simeq 2$mA)	Cascode	---	---	0 - j 0.008	---	mmho	---
			Diff.Amp.	---	---	0 - j 0.013	---		---
Forward Transfer Admittance	Y_{21}		Cascode	---	---	17.9 - j 30.7	---	mmho	26, 28, 30
			Diff. Amp.	---	---	- 10.5 + j 13	---		27, 29, 31
Output Admittance	Y_{22}		Cascode	---	---	- 0.503 - j 15	---	mmho	20, 22, 24
			Diff.Amp.	---	---	0.071 + j 0.62	---		21, 23, 25

*Terminals 1 & 14, or 7 & 8. (CA3102E) 1 & 12 or 6 & 7 (CA3049T)
**Terminals 13 & 4, or 6 & 11. (CA3102E) 10 & 11 or 4 & 5 (CA3049T)

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		TEST CIRCUIT FIG.	CA3049T LIMITS MIN.	TYP.	MAX.	UNITS	TYPICAL CHARACTERISTICS CURVES FIG.
STATIC CHARACTERISTICS									
For Each Differential Amplifier									
Input Offset Voltage	V_{IO}			1	---	0.25	---	mV	-4
Input Offset Current	I_{IO}	$I_3 = I_9 = 2$ mA		1	---	0.3	---	μA	---
Input Bias Current	I_{IB}			1		13.5	33	μA	5
Temperature Coefficient Magnitude of Input-Offset Voltage	$\frac{\lvert\Delta V_{IO}\rvert}{\Delta T}$			1	---	1.1	---	μV/°C	4
For Each Transistor									
DC Forward Base-to-Emitter Voltage	V_{BE}	$V_{CE} = 6$ V, $I_C = 1$ mA		---	---	774	---	mV	6
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	$V_{CE} = 6$ V, $I_C = 1$ mA		---	---	-0.9	---	mV/°C	6
Collector-Cutoff Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$		---	---	0.0013	100	nA	7
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$		---	15	24	---	V	---
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10$ μA, $I_E = 0$		---	20	60	---	V	---
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_C = 10$ μA, $I_B = 0$, $I_E = 0$		---	20	60	---	V	---
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 10$ μA, $I_C = 0$		---	5	7	---	V	---
DYNAMIC CHARACTERISTICS									
1/f Noise Figure (For Single Transistor)	NF	f = 100 KH$_3$, $R_S = 500\ \Omega$, $I_C = 1$ mA		---	---	1.5	---	dB	12
Gain-Bandwidth Product (For Single Transistor)	f_T	$V_{CE} = 6$ V, $I_C = 5$ mA		---	---	1.35	---	GH$_z$	11
Collector-Base Capacitance	C_{CB}	$I_C = 0$ $V_{CB} = 5$V	*	---	---	0.28	---	pF	8
			**	---	---	0.28	---	pF	
Collector-Substrate Capacitance	C_{CI}	$I_C = 0$ $V_{CI} = 5$V			---	1.65	---	pF	8
For Each Differential Amplifier									
Common-Mode Rejection Ratio	CMR	$I_3 = I_9 = 2$ mA		---	---	100	---	dB	---
AGC Range, One Stage	AGC	Bias Voltage = -6V		2	---	75	---	dB	---
Voltage Gain, Single-Ended Output	A	Bias Voltage = -4.2V, f = 10 MHz		2	---	22	---	dB	9, 10
Insertion Power Gain	G_p	f = 200 MHz	Cascode	3	---	23	---	dB	---
Noise Figure	NF	$V_{CC} = 12$V	Cascode	3	---	4.6		dB	---
Input Admittance	Y_{11}	For Cascode Configuration, $I_3 = I_9 = 2$ mA	Cascode	---	---	1.5 + j 2.45	---	mmho	14, 16, 18
			Diff.Amp.	---	---	0.878 + j 1.3	---		15, 17, 19
Reverse Transfer Admittance	Y_{12}	For Diff. Amplifier Configuration, $I_3 = I_9 = 4$mA (each collector $I_C \simeq 2$mA)	Cascode	---	---	0 - j 0.008	---	mmho	---
			Diff.Amp.	---	---	0 - j 0.013	---		---
Forward Transfer Admittance	Y_{21}		Cascode	---	---	17.9 - j 30.7	---	mmho	26, 28, 30
			Diff. Amp.	---	---	- 10.5 + j 13	---		27, 29, 31
Output Admittance	Y_{22}		Cascode	---	---	- 0.503 - j 15	---	mmho	20, 22, 24
			Diff.Amp.	---	---	0.071 + j 0.62	---		21, 23, 25

*Terminals 1 & 14, or 7 & 8. (CA3102E) 1 & 12 or 6 & 7 (CA3049T)
**Terminals 13 & 4, or 6 & 11. (CA3102E) 10 & 11 or 4 & 5 (CA3049T)

MAXIMUM RATINGS, *ABSOLUTE-MAXIMUM VALUES, AT $T_A = 25°C$*

Power Dissipation, P:	CA3049T	CA3102E	
Any one transistor	300	300	mW
Total package	600	750	mW
For $T_A > 55°C$ Derate at:	5	6.67	mW/°C
Temperature Range:			
Operating	−55 to + 125	−55 to + 125	°C
Storage	−65 to + 150	−65 to + 150	°C

Lead Temperature (During Soldering):
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm)
from case for 10 seconds max. ..... +265°C

The following ratings apply for each transistor in the devices

Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

*The collector of each transistor of the CA3049T and CA3102E is isolated from the substrate by an integral diode. The substrate (terminal 9) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action.

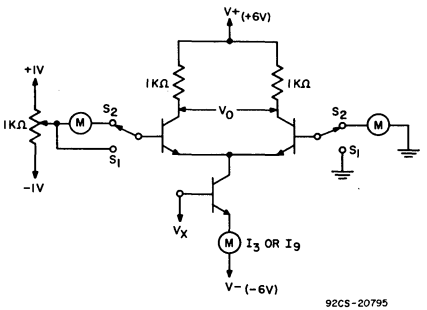


Fig.1—Static characteristics test circuit for CA3102E.

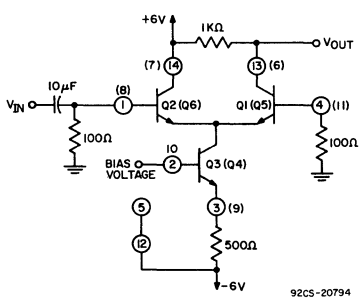
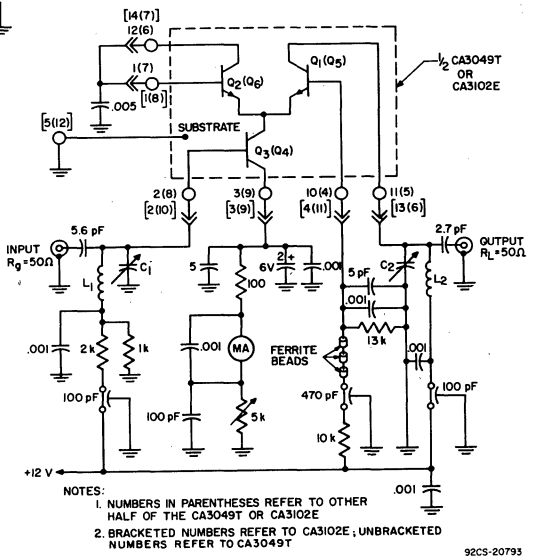


Fig.2—AGC range and voltage gain test circuit for CA3102E.



L_1, L_2 – Approx. 1/2 Turn #18 Tinned Copper Wire, 5/8" Dia.

C_1, C_2 – 15 pF Variable Capacitors (Hammarlund, MAC-15; or Equivalent)

All Capacitors in μF Unless Otherwise Indicated

All Resistors in Ohms Unless Otherwise Indicated

Fig.3—200 MHz cascode power gain and noise figure test circuit.

Typical Characteristics for CA3049T and CA3102E

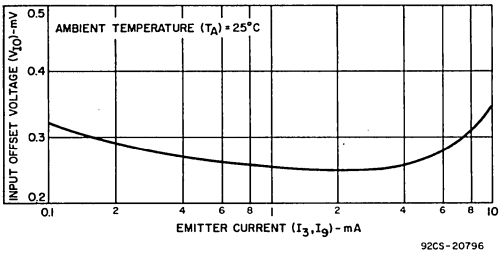


Fig. 4—Input offset voltage vs. emitter current.

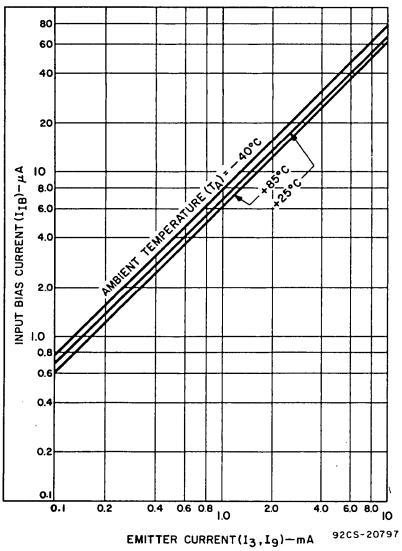


Fig. 5—Input bias current vs. emitter current.

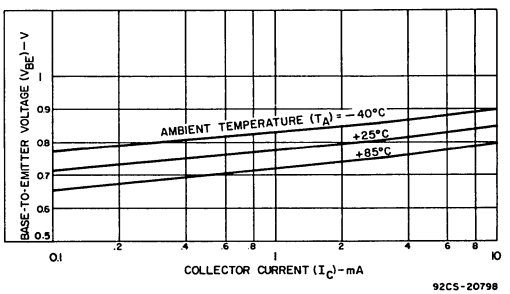


Fig. 6—Base-to-emitter voltage vs. collector current.

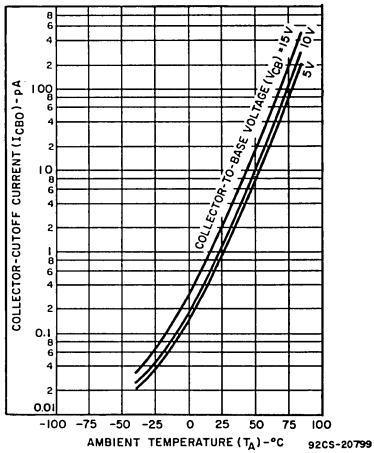


Fig. 7—Collector-cutoff current vs. temperature.

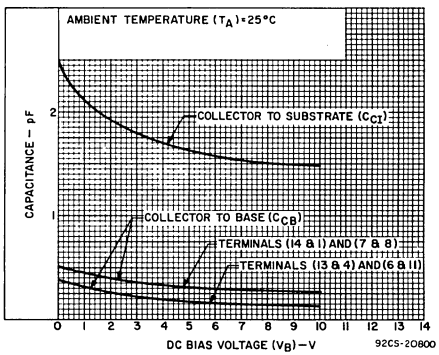


Fig. 8—Capacitance vs. dc bias voltage.

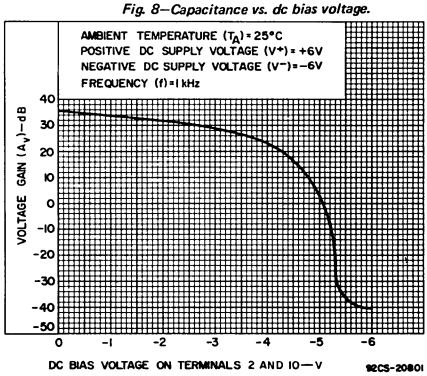


Fig. 9—Voltage gain vs. dc bias voltage.

Typical Characteristics for CA3049T and CA3102E (cont'd)

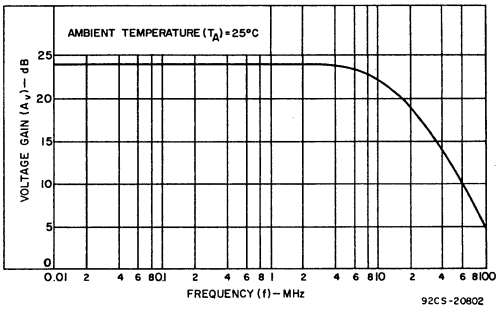


Fig. 10—Voltage gain vs. frequency.

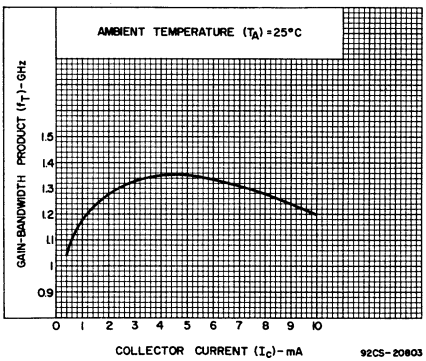


Fig. 11—Gain-bandwidth product vs. collector current.

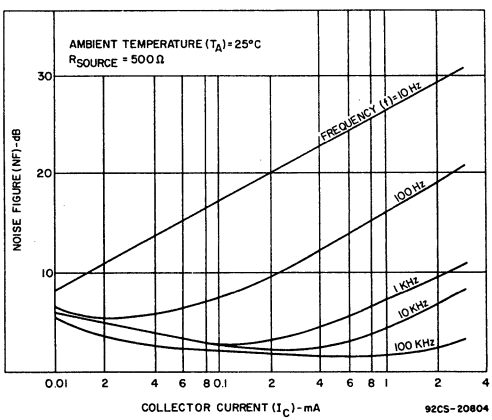


Fig. 12—1/f noise figure vs. collector current.

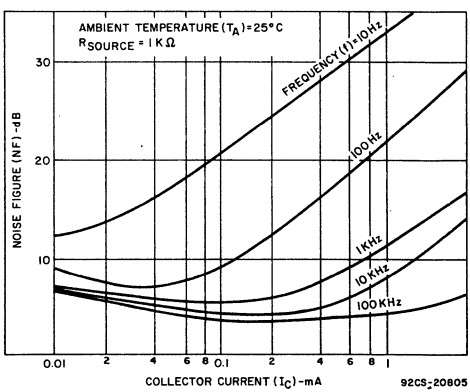


Fig. 13—1/f noise figure vs. collector current.

Typical Input Admittance Characteristics for CA3049T and CA3102E

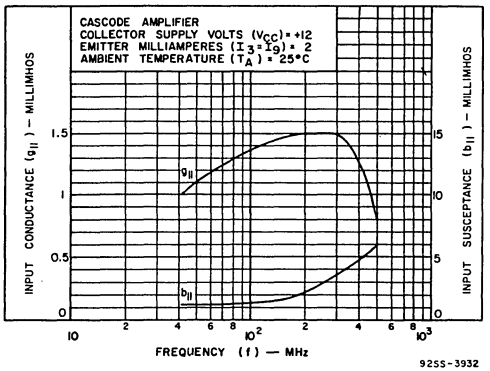


Fig. 14—Input admittance (Y_{11}) vs. frequency.

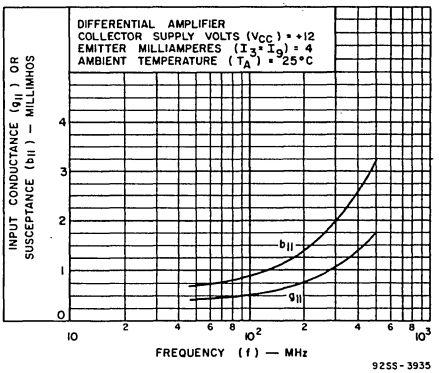


Fig. 15—Input admittance (Y_{11}) vs. frequency.

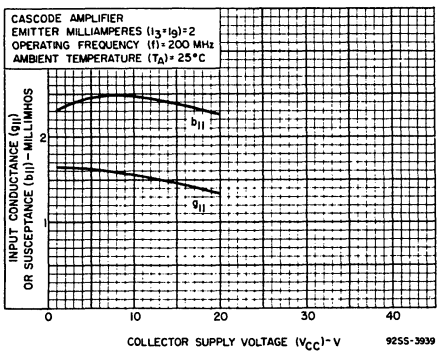


Fig. 16—Input admittance (Y_{11}) vs. collector supply voltage.

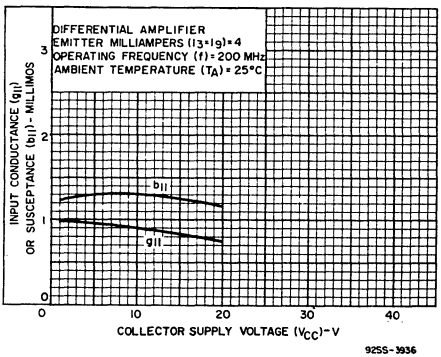


Fig. 17—Input admittance (Y_{11}) vs. collector supply voltage.

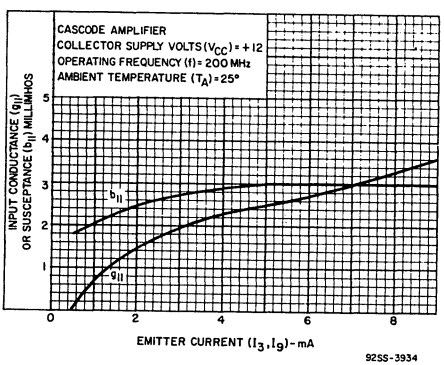


Fig. 18—Input admittance (Y_{11}) vs. emitter current.

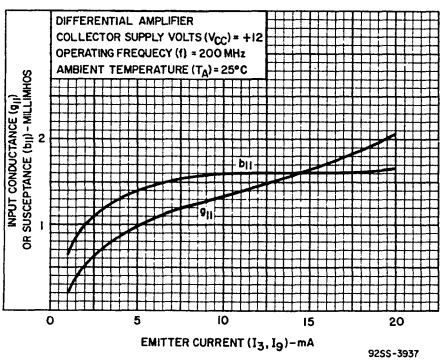


Fig. 19—Input admittance (Y_{11}) vs. emitter current.

Typical Output Admittance Characteristics for CA3049T and CA3102E

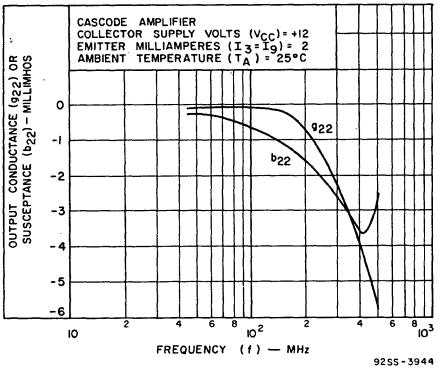


Fig. 20—Output admittance (Y_{22}) vs. frequency.

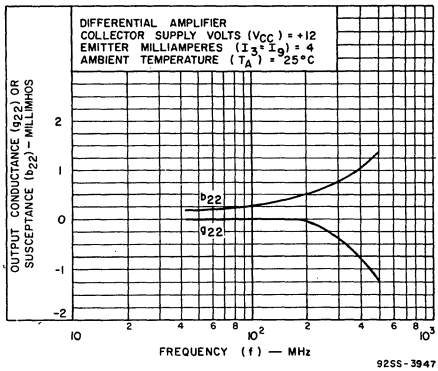


Fig. 21—Output admittance (Y_{22}) vs. frequency.

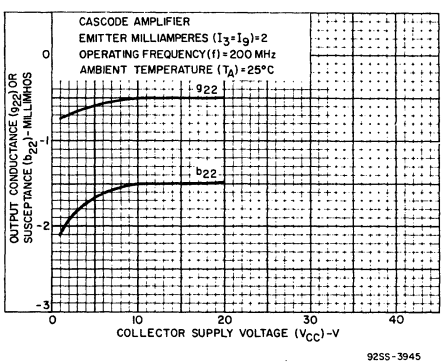


Fig. 22—Output admittance (Y_{22}) vs. collector supply voltage.

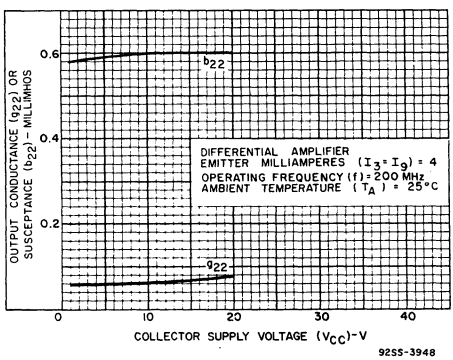


Fig. 23—Output admittance (Y_{22}) vs. collector supply voltage.

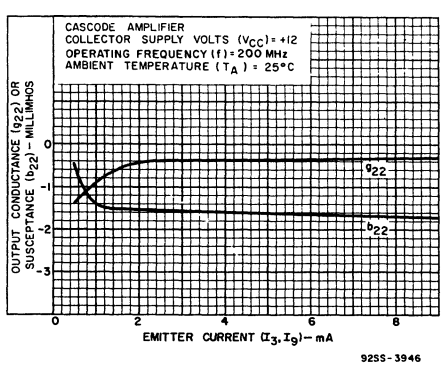


Fig. 24—Output admittance (Y_{22}) vs. emitter current.

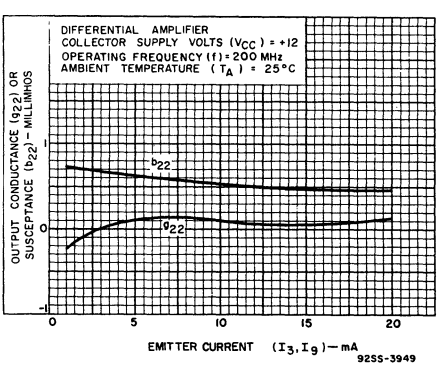


Fig. 25—Output admittance (Y_{22}) vs. emitter current.

Typical Forward Transfer Characteristics for CA3049T and CA3102E

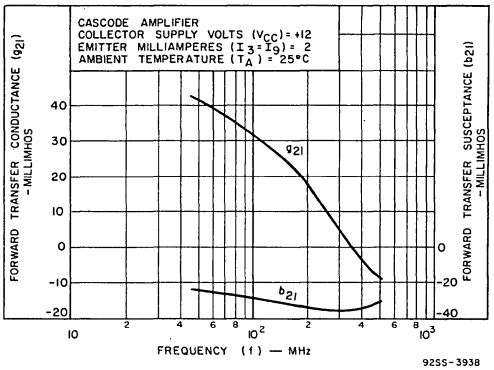


Fig. 26—Forward transfer admittance (Y_{21}) vs. frequency.

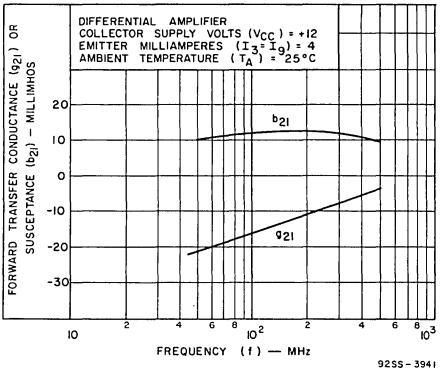


Fig. 27—Forward transfer admittance (Y_{21}) vs. frequency.

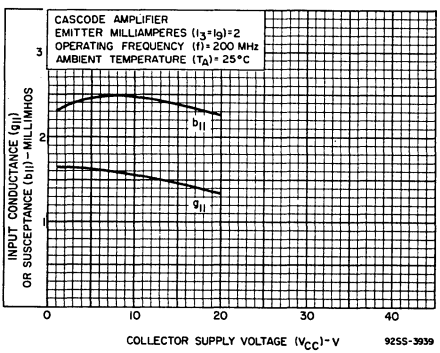


Fig. 28—Forward transfer admittance (Y_{21}) vs. collector supply voltage.

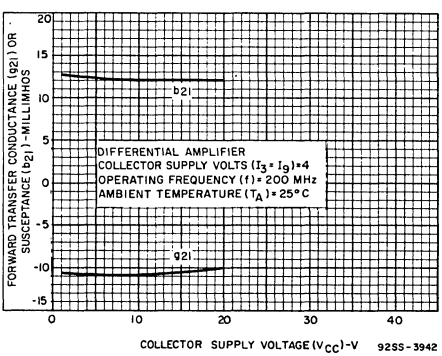


Fig. 29—Forward transfer admittance (Y_{21}) vs. collector supply voltage.

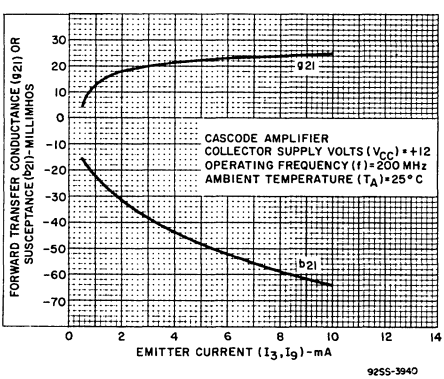


Fig. 30—Forward transfer admittance (Y_{21}) vs. emitter current.

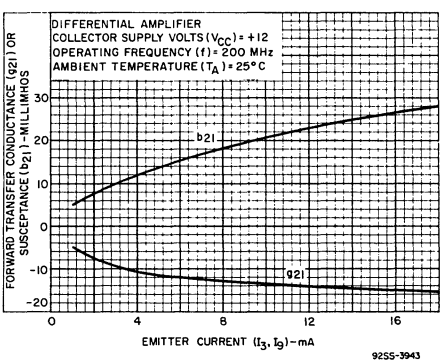


Fig. 31—Forward transfer admittance (Y_{21}) vs. emitter current.

RCA
Solid State
Division

Linear Integrated Circuits

CA3050
CA3051

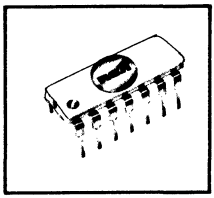
Dual Differential Amplifiers

Monolithic Silicon

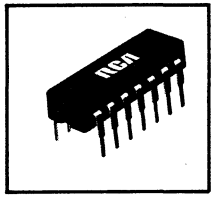
The CA3050 and CA3051 each consists of two differential amplifiers with associated constant current transistors on a common substrate. Each amplifier is driven by Darlington-connected emitter follower inputs to provide high input impedance, low bias current, and low offset current. A string of diodes is included to provide temperature-compensated bias to the constant current transistors and a low impedance bias point for the inputs to the differential amplifiers when a single power supply is used.

TWO DARLINGTON-CONNECTED DIFFERENTIAL AMPLIFIERS WITH DIODE BIAS STRING

For Low-Power Applications at Frequencies from DC to 20 MHz



CA3050 14-Lead Dual-in-Line Ceramic Package



CA3051 14-Lead Dual-in-Line Plastic Package

FEATURES

- Input offset current . . . . . 70 nA max.
- Input bias current . . . . . 500 nA max.
- Input offset voltage . . . . . 5 mV max.
- Input impedance . . . . . 460 kΩ typ.
- Independently accessible inputs and outputs

APPLICATIONS

- Matched dual amplifiers
- Dual sense amplifiers
- Dual Schmitt triggers
- Dual multivibrators
- Doubly balanced detectors and modulators
- Balanced quadrature detectors
- Synthesizer mixers
- Product detectors

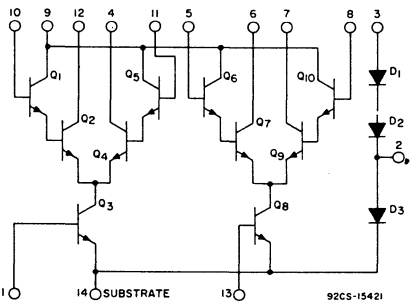


Fig.1 - Schematic diagram.

MAXIMUM RATINGS, ABSOLUTE-MAXIMUM VALUES, AT T_A = 25°C

	CA3050	CA3051	
Power Dissipation, P:			
Any one transistor	150	150	mW
Total package	900	750	mW
For $T_A > 55°C$, Derate at	8	6.67	mW/°C
Temperature Range:			
Operating	-55 to +125		°C
Storage	-65 to +150		°C
LEAD TEMPERATURE (During Soldering) At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265		°C

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage, V_{CEO}	15	V
Collector-to-Base Voltage, V_{CBO}	20	V
Collector-to-Substrate Voltage, V_{CIO}*	20	V
Emitter-to-Base Voltage, V_{EBO}	5	V
Collector Current, I_C	50	mA

* The collector of each transistor of the CA3050 and CA3051 is isolated from the substrate by an integral diode. *The substrate (terminal 14) must be more negative than all collectors to maintain isolation between transistors and to provide for normal transistor action.*

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 3 is +5 to −2 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	–	*	*	*	*	*	*	*	*	*	*	*	*	+1 -5
2			+5 -2	*	*	*	*	*	*	*	*	*	*	+1 -1
3				*	*	*	*	*	*	*	*	*	*	+3 -1
4					*	*	*	*	*	+14 -2.5 Note 3	+14 -2.5 Note 4	*	*	+20 -1
5						+2.5 -14 Note 1	+2.5 -14 Note 1	+10 -10	+1 -20	*	*	*	*	+16 -1
6							*	+14 -2.5 Note 2	*	*	*	*	*	+20 -1
7								+14 -2.5 Note 2	*	*	*	*	*	+20 -1
8									+1 20	*	*	*	*	+16 -1
9										+20 -1	+20 -1	*	*	+20 -1
10											+10 -10	+2.5 -14 Note 3	*	+16 -1
11												+2.5 -14 Note 4	*	+16 -1
12														+20 -1
13														+1 -5
14														Ref. Substrate

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
1	5	0.1
2	50	50
3	50	1
4	50	1
5	5	0.1
6	50	1
7	50	1
8	5	0.1
9	50	1
10	5	0.1
11	5	0.1
12	50	1
13	5	0.1
14	100	5

NOTE 1: This rating is important only when terminal 5 is more positive than terminal 8.

NOTE 2: This rating is important only when terminal 8 is more positive than terminal 5.

NOTE 3: This rating is important only when terminal 10 is more positive than terminal 11.

NOTE 4: This rating is important only when terminal 11 is more positive than terminal 10.

*Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIR-CUIT	LIMITS CA3050/CA3051			UNITS	TYPICAL CHARAC-TERISTICS CURVES
			FIG.	MIN.	TYP.	MAX.		FIG.
STATIC								
Amplifier Characteristics								
Input Offset Voltage	V_{IO}		–	–	1.5	5	mV	2a,b
Input Offset Current	I_{IO}		–	–	7	70	nA	3a,b
Input Bias Current	I_{IB}		–	–	200	500	nA	4a,b
Quiescent Operating Current Ratio	$\frac{(I_4+I_{12})}{I_3}$ or $\frac{(I_6+I_7)}{I_3}$	V_{CC} = + 6 V, I_3 = 2 mA	–	0.9	1.00	1.13	–	5a,b
DC Forward Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3 V, I_C = 50 μA	–	–	0.645	0.700	V	6
		V_{CE} = 3 V, I_C = 1 mA	–	–	0.725	0.800		
		V_{CE} = 3 V, I_C = 3 mA	–	–	0.760	0.850		
		V_{CE} = 3 V, I_C = 10 mA	–	–	0.805	0.900		
Temperature Coefficient of Base-to-Emitter Voltage	$\frac{\Delta V_{BE}}{\Delta T}$	V_{CE} = 3 V, I_C = 1 mA	–	–	-1.9	–	mV/°C	7
Transistor Characteristics								
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	–	–	0.002	100	nA	8
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	–	15	24	–	V	–
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	–	20	60	–	V	–
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	–	20	60	–	V	–
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	–	5	7	–	V	–
DYNAMIC								
Transistor Characteristics								
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 3 V, I_E = 0	–	–	0.78	–	pF	9
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 3 V, I_C = 0	–	–	0.47	–	pF	9
Collector-to-Substrate Capacitance	C_{CI}	V_{CS} = 3 V, I_C = 0	–	–	1.92	–	pF	9
Amplifier Characteristics								
Gain-Bandwidth Product (For Single Transistor)	f_T	V_{CE} = 5 V, I_C = 3 mA	–	–	600	–	MHz	10
Forward Transadmittance (With single-ended input and output)	$\lvert y_{21} \rvert$	V_{CC} = 10 V, I_3 = 2 mA f = 1 MHz	11	7	9	11	mmho	11
Bandwidth at -3 dB Point	BW	V_{CC} = 10 V, I_3 = 2 mA	11	–	4.3	–	MHz	11
Input Impedance	Z_I	V_{CC} = 10 V, I_3 = 2 mA f = 1 KHz	12	–	460	–	kΩ	12
Output Impedance	Z_O	I_3 = 2 mA, f = 1 KHz	13	–	170	–	kΩ	13
Common-Mode Rejection Ratio	CMR	I_3 = 2 mA, f = 1 KHz	–	–	65	–	dB	–
AGC Range	AGC	I_3 = 2 mA, f = 1 KHz Terminal No.3 Grounded	11	–	60	–	dB	–

TYPICAL STATIC CHARACTERISTICS

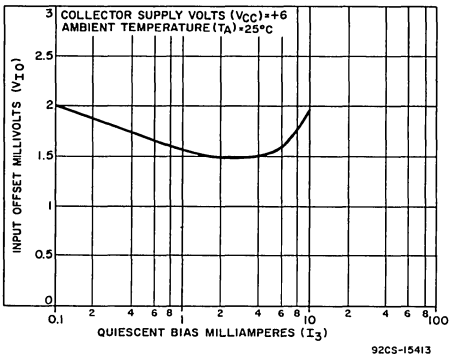


Fig.2(a) - Typical input offset voltage vs quiescent bias current.

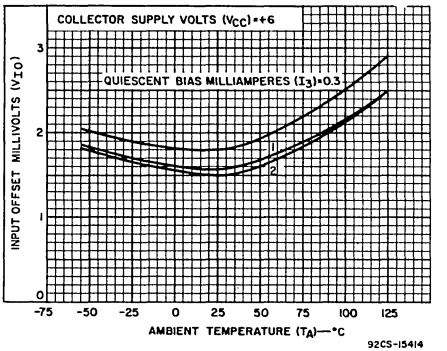


Fig.2(b) - Typical input offset voltage vs ambient temperature.

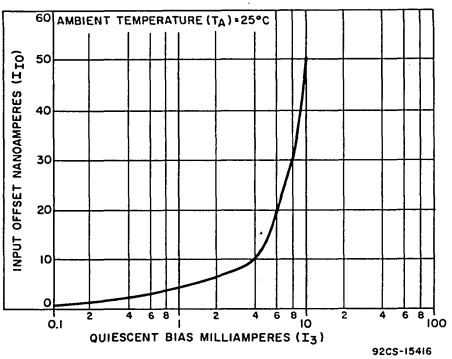


Fig.3(a) - Typical input offset current vs quiescent bias current.

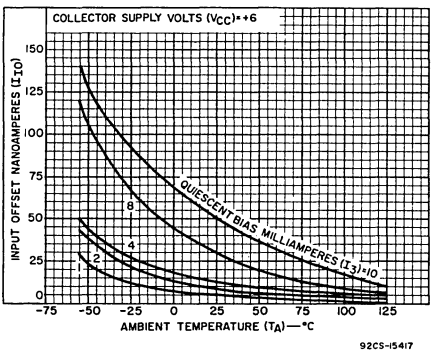


Fig.3(b) - Typical input offset current vs ambient temperature.

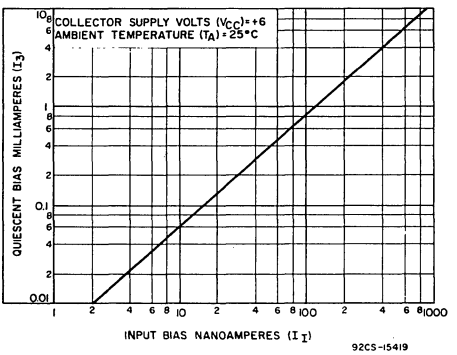


Fig.4(a) - Typical quiescent bias current vs input bias current.

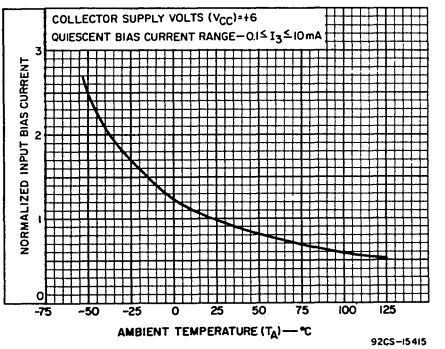


Fig.4(b) - Typical normalized input bias current vs ambient temperature.

STATIC CHARACTERISTICS

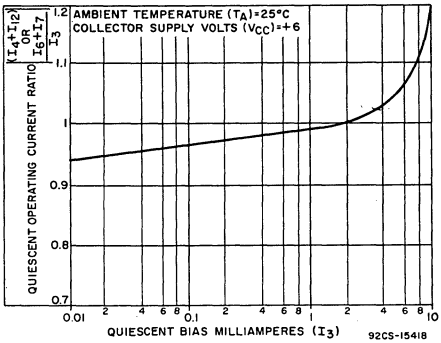


Fig.5(a) - Typical quiescent operating current ratio vs quiescent bias current.

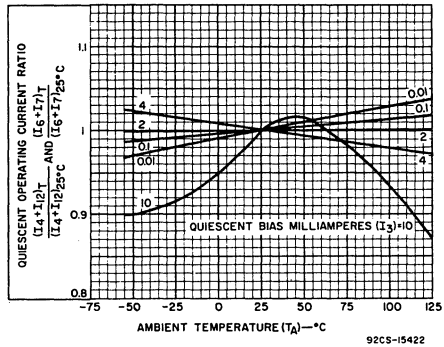


Fig.5(b) - Typical quiescent operating current ratio vs ambient temperature.

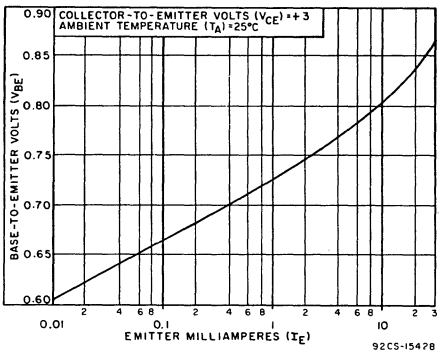


Fig.6 - Typical static base-to-emitter voltage characteristic vs emitter current for all transistors and forward diode voltage drops.

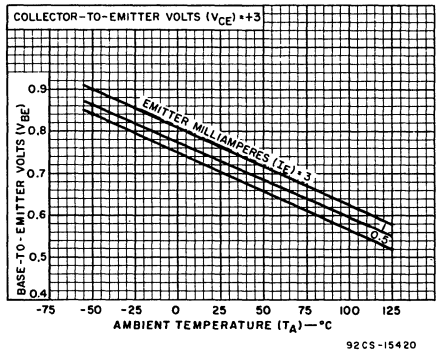


Fig.7 - Typical base-to-emitter voltage characteristic vs ambient temperature for each transistor.

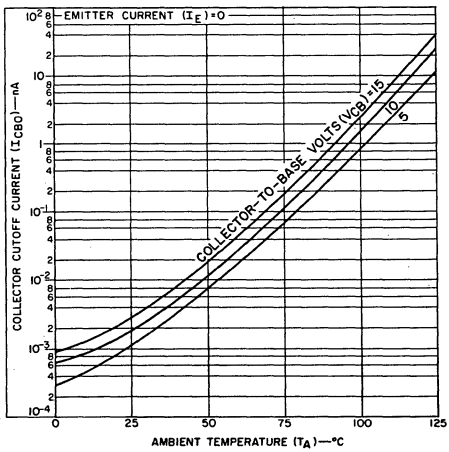


Fig.8 - Typical collector-to-base cutoff current vs ambient temperature for each transistor.

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

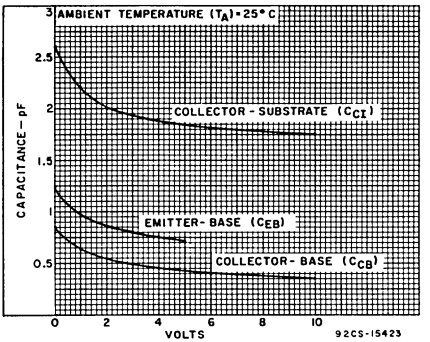


Fig.9 - Typical capacitance for each transistor.

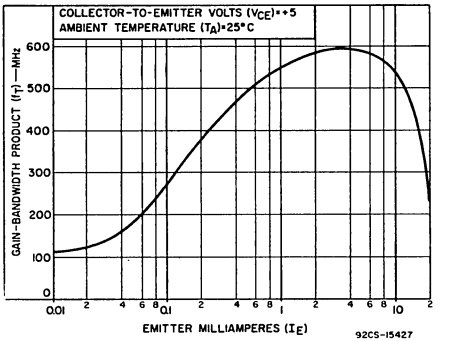


Fig.10 - Typical gain-bandwidth product (f_T) for each transistor vs emitter current.

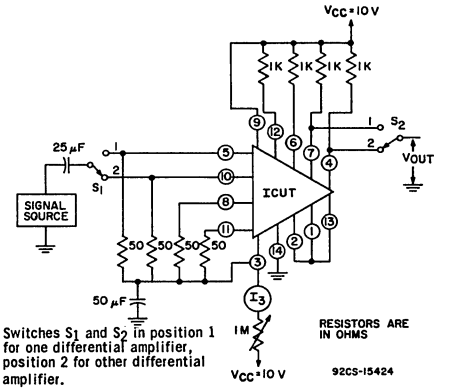


Fig.11(a) - Test circuit for forward transadmittance, -3 dB bandwidth, and AGC range.

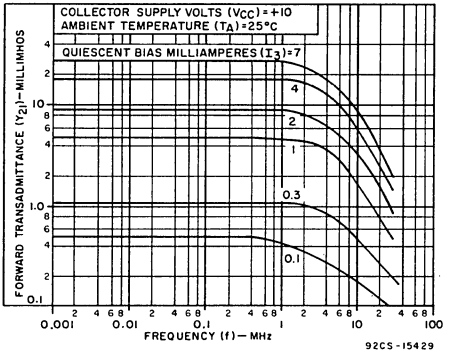


Fig.11(b) - Typical differential amplifier forward transadmittance with single-ended output vs frequency.

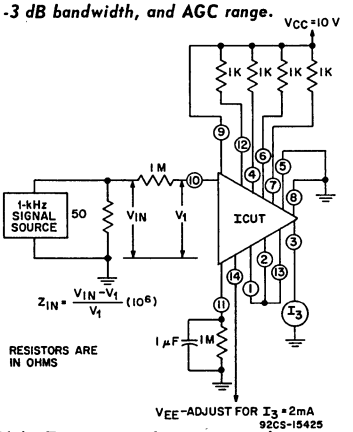


Fig.12(a) - Test circuit for input impedance.

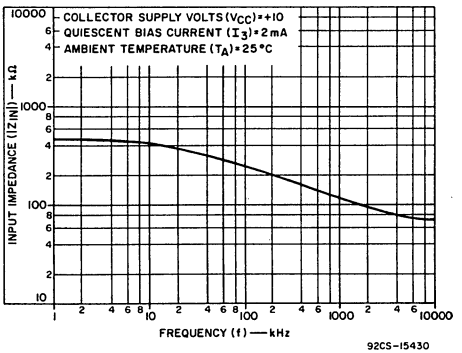
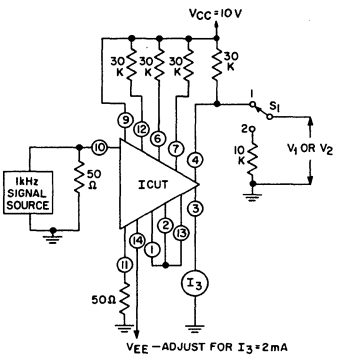


Fig.12(b) - Typical input impedance vs frequency with output short-circuited.

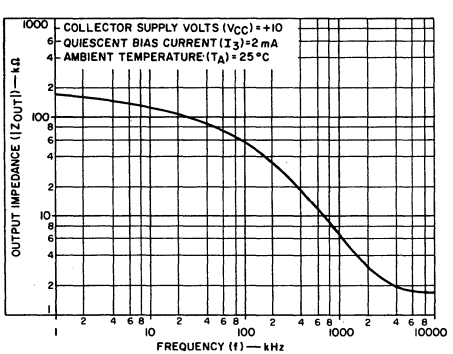
DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR



$$Z_{OUT} = \frac{(30K \times 10K)\frac{V_2}{V_1}}{\frac{V_2}{V_1}(30K+10K)-10K}$$

92CS-15426

Fig.13(a) - Test circuit for output impedance.



92CS-15431

Fig.13(b) - Typical output impedance vs frequency with input short-circuited.

RCA
Solid State
Division

Linear Integrated Circuits

CA3052

Special-Function Sub-System Stereo Preamplifier

The RCA CA3052 is a silicon monolithic integrated circuit designed specifically for stereo preamplifier service. The circuit consists of four independent AC amplifiers which can operate from a single-ended supply.

The CA3052 can operate as an equalizer amplifier in tape recorders, magnetic cartridge phonograph applications, and tone control amplifiers. The CA3052 can provide all of the amplification necessary for a full-function stereo preamplifier.

The CA3052 is supplied in a 16-lead dual-in-line plastic package.

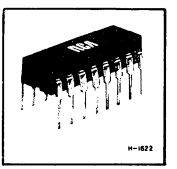
APPLICATIONS

- Full-function stereo preamplifiers
- Tape recorder and playback preamplifiers
- Tone Generators

RCA CA3048 Amplifier Array (File No.377) is schematically identical with the CA3052. Each amplifier of the CA3048 is tightly specified for equivalent output noise under a variety of test methods. The CA3052 is specified using RIAA test methods for equivalent input noise using one test method for amplifiers 1 and 4, and an appropriately different method for amplifiers 2 and 3.

FOUR INDEPENDENT AC AMPLIFIERS

For Stereo Preamplifiers, Magnetic Pickups, Tape Heads, etc.



CA3052

FEATURES

- Four AC amplifiers on a common substrate
- Independently accessible inputs and outputs
- Operates from single-ended supply

EACH AMPLIFIER

- High voltage gain . . . . . 53 dB min.
- High input resistance . . . . . 90 k Ω typ.
- Undistorted output voltage . . . . . 2 V rms min.
- Output Impedance . . . . . 1 k Ω typ.
- Open-loop bandwidth . . . . . 300 kHz typ.

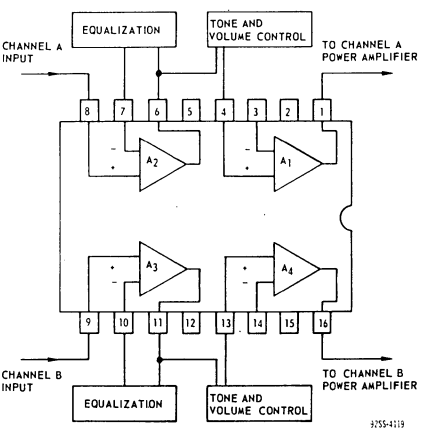


Fig.1 — Block diagram of stereo preamplifier using CA3052.

ABSOLUTE-MAXIMUM RATINGS at T_A = 25°C:

DISSIPATION:

Up to T_A = 55°C . . . . . 750 mW
Above T_A = 55°C . . . . . Derate linearly at 7.7 mW/°C

TEMPERATURE RANGE:

Operating . . . . . −40°C to +85°C
Storage . . . . . −65°C to +150°C

LEAD TEMPERATURE (During Soldering):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max. . . . . +265°C

POWER SUPPLY VOLTAGE . . . . . +16 V

AC INPUT VOLTAGE . . . . . 0.5 V rms

MAXIMUM VOLTAGE RATINGS

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 4 is +2 to -3.6 volts.

TERMINAL No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1		+16 0	*	*	*	*	*	*	*	*	*	*	*	*	0 −16	*
2			*	+2 −3.6	0	*	*	+2 −3.6	+2 −3.6	*	*	+16 0	+2 −3.6	*	+16 0	0 −16
3				+5 −5	*	*	*	*	*	*	*	*	*	*	*	*
4					+3.6 −2	*	*	*	*	*	*	*	*	*	*	*
5						0 −16	*	+2 −3.6	+2 −3.6	*	0 −16	+16 0	+2 −3.6	*	+16 0	*
6							*	*	*	*	*	*	0 −16	*	*	*
7								+5 −5	*	*	*	*	*	*	*	*
8									*	*	*	*	*	*	*	*
9										+5 −5	*	*	*	*	*	*
10											*	*	*	*	*	*
11												*	*	*	*	*
12													0 −16	*	*	*
13														+5 −5	*	*
14															*	*
15																+16 0
16																

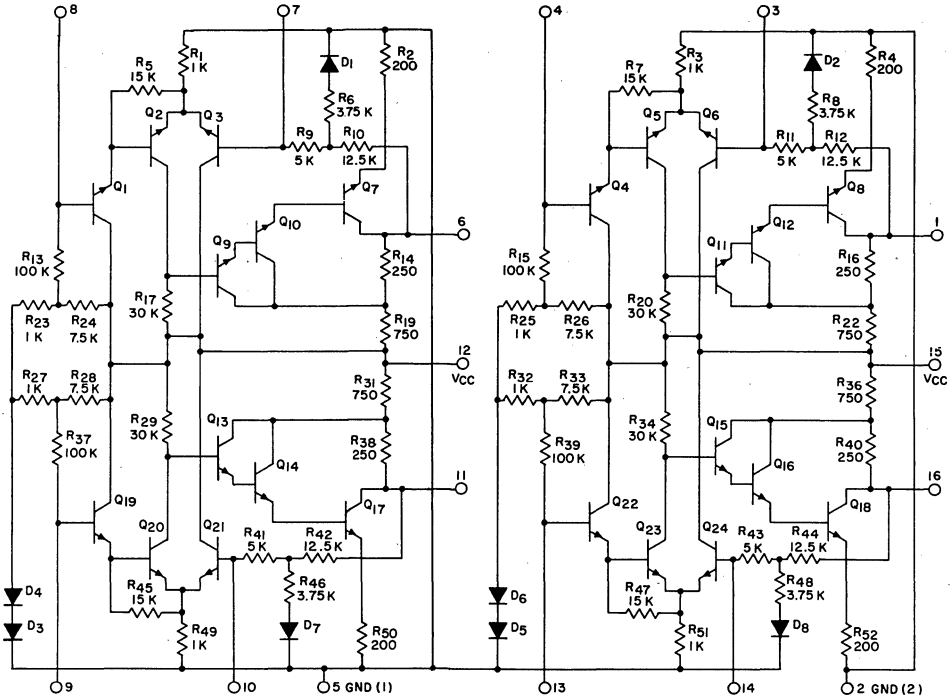
* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	TEST CIR-CUIT	LIMITS CA3052			UNITS	TYPICAL CHARAC-TERISTICS CURVES
			FIG.	MIN.	TYP.	MAX.		FIG.
STATIC								
Current drain per amplifier pair	I_{12} or I_{15}	V_{CC} = +12 V	3	9.5	13.5	17.5	mA	4, 5
DC Voltage at Output Terminals	V_1, V_6, V_{11}, V_{16}	V_{CC} = +12 V	3	6.1	6.9	8.1	V	–
DC Voltage at Feedback Terminals	V_3, V_7, V_{10}, V_{14}	V_{CC} = +12 V	3	1.7	2.0	2.3	V	–
DC Voltage at Input Terminals	V_4, V_8, V_9, V_{13}	V_{CC} = +12 V	3	2.2	2.5	2.8	V	–
DYNAMIC each amplifier with no AC feedback unless otherwise noted–terminals 3, 7, 10, & 14 bypassed to ground								
Open-Loop Gain	A_{OL}	V_{CC} = +12 V E_{IN} = 2 mV f = 10 kHz	6	53	58	–	dB	7, 8
Open-Loop Output Voltage Swing	$V_{O(rms)}$	V_{CC} = +12 V f = 1 kHz THD = 5%	6	2.0	2.4	–	V	–
Open-Loop -3 dB Bandwidth	BW	V_{CC} = +12 V E_{IN} = 2 mV	6	–	300	–	kHz	9
Open-Loop Total Harmonic Distortion	THD	V_{CC} = +12 V, f = 1 kHz E_{OUT} = 2 V rms	6	–	0.65	–	%	10
Input Resistance	R_I	V_{CC} = +12 V, f = 1 kHz	–	–	90	–	kΩ	–
Input Capacitance	C_I	V_{CC} = +12 V, f = 1 MHz	–	–	9	–	pF	–
Output Resistance	R_O	V_{CC} = +12 V, f = 1 kHz	–	–	1	–	kΩ	–
Feedback Capacitance (Output to non-inverting Input)	C_{FB}	V_{CC} = +12 V f = 1 MHz	–	–	<0.1	–	pF	–
Equivalent Input Noise Voltage (Amplifiers 1 & 4), "C" Filter at Output*	E_{NI}‡	V_{CC} = +10 V R_S = 5 kΩ A = 45 dB	12	–	1.7	6.4	μV	–
Equivalent Input Noise Voltage (Amplifiers 2 & 3) RIAA Compensated*	E_{N2}‡	V_{CC} = +10 V R_S = 5 kΩ A = 64 dB (1 kHz)	11	–	4	15.0	μV	–
Inter-Amplifier Audio Separation "Cross Talk"		V_{CC} = +12 V f = 1 kHz 0 dB = 0.78 V	13	–	<-45	–	dB	–
Inter-Amplifier Capacitance (Any amplifier output to any *other* amplifier input)	C	V_{CC} = +12 V f = 1 MHz	–	–	< 0.02	–	pF	–

*Per IHF Standard Methods of Measurement for Audio Amplifiers IHF–A–201, 1966

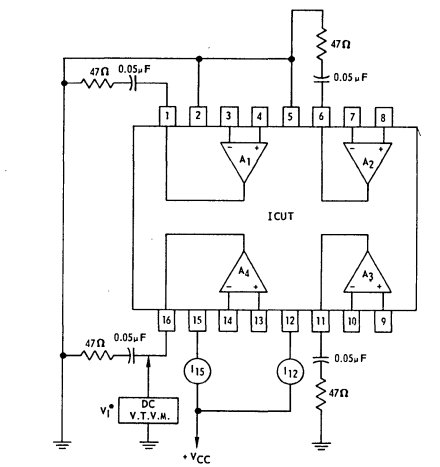
‡ ac feedback included in test circuit



NOTE: ALL RESISTOR VALUES ARE IN OHMS

92CM-15412

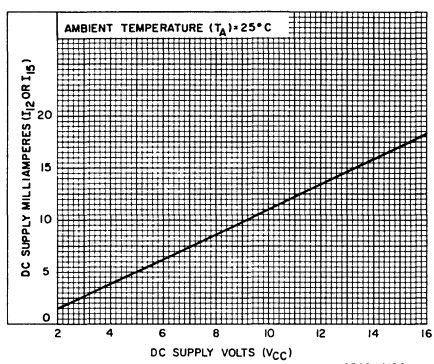
Fig. 2 - Schematic diagram for CA3052.



* CONNECT TO APPROPRIATE TERMINAL TO READ VOLTAGE

92CS-15473

Fig. 3 - Test circuit for measurement of collector supply voltage and currents.



92SS-4120

Fig. 4 - Typical DC supply current vs supply voltage.

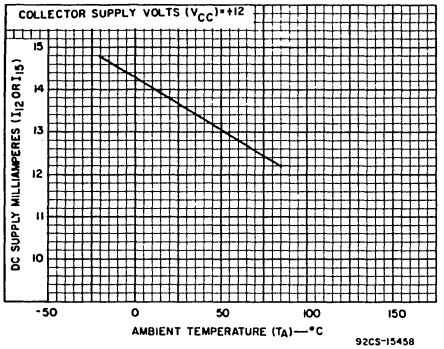


Fig. 5 - Typical DC supply current vs ambient temperature.

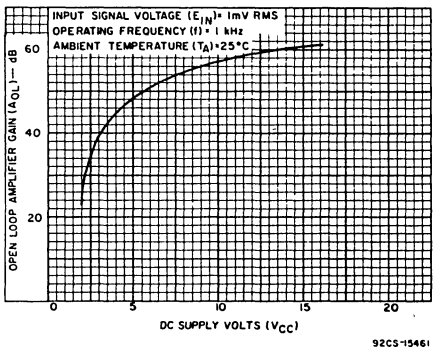
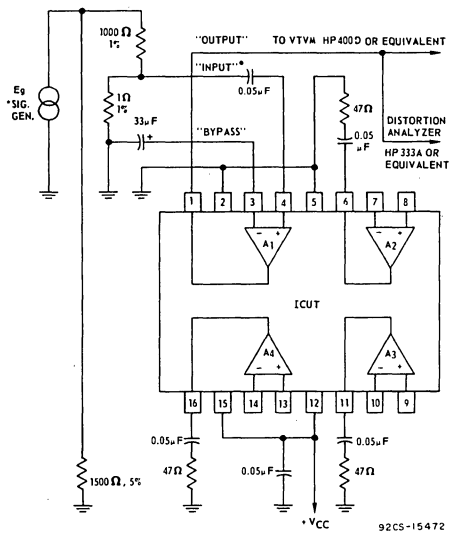


Fig. 7 - Typical amplifier gain vs DC supply voltage.



* Sig. Gen. should be a low distortion type (0.2% THD or less) HP206A or equivalent.

● Adjustment of E_g to 2 volts will make E_S = 2 mV.

Test Circuit shows Amplifier #1 under test, to test Amplifiers 2, 3, or 4; Connect terminals as shown in Table.

AMPLIFIER	TERMINALS		
	OUTPUT	INPUT	BYPASS
1	1	4	3
2	6	8	7
3	11	9	10
4	16	13	14

Fig. 6 - Test circuit for measurement of distortion, open-loop gain, and bandwidth characteristics.

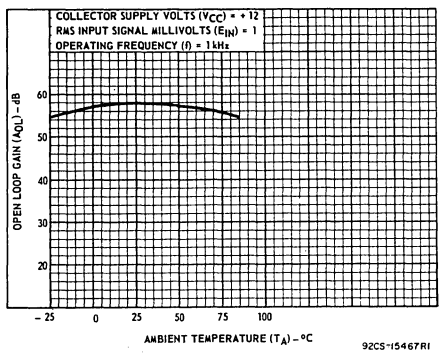


Fig. 8 - Typical open-loop gain vs ambient temperature.

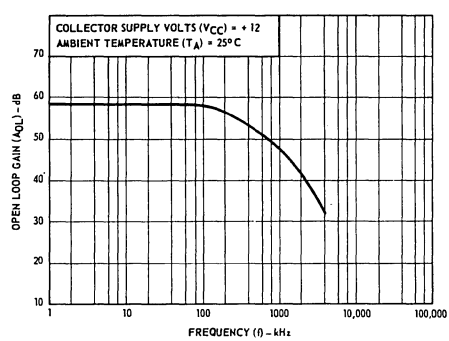


Fig. 9 - Typical open-loop gain vs frequency.

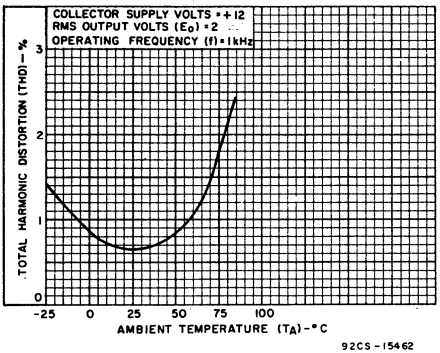
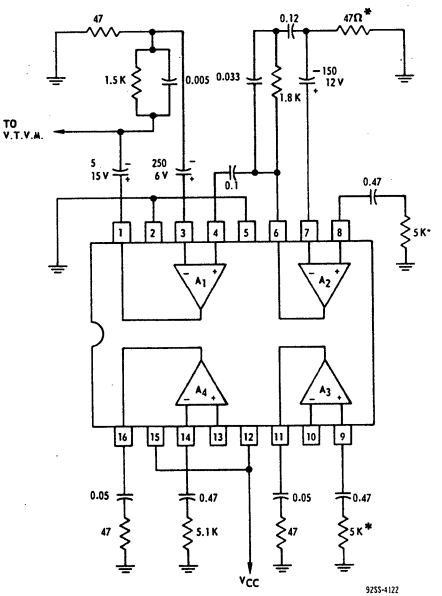
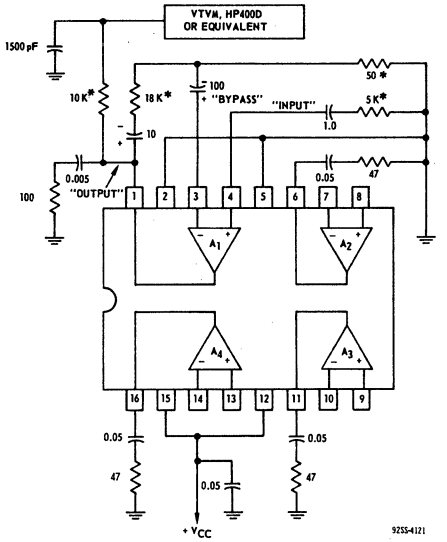


Fig. 10. - Typical total harmonic distortion vs ambient temperature.



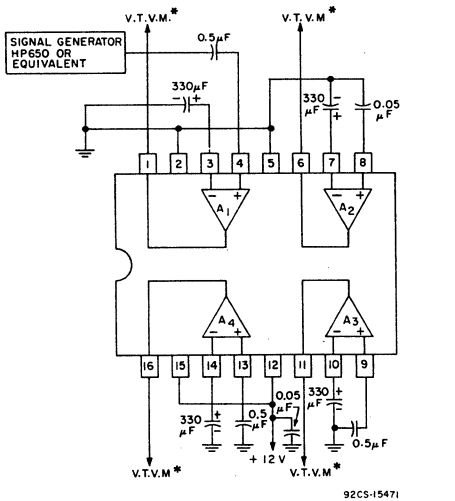
*Resistors are low noise precision (1%) Metal Film type.

Fig. 11 - Test circuit for equivalent input noise voltage measurement, RIAA compensated.



*Resistors are low noise precision, (1%) Metal Film type. Resistor values are in ohms; capacitance values are in microfarads, unless otherwise specified.

Fig. 12 - Test circuit for measurement of equivalent input noise voltage of amplifiers 1 and 4.

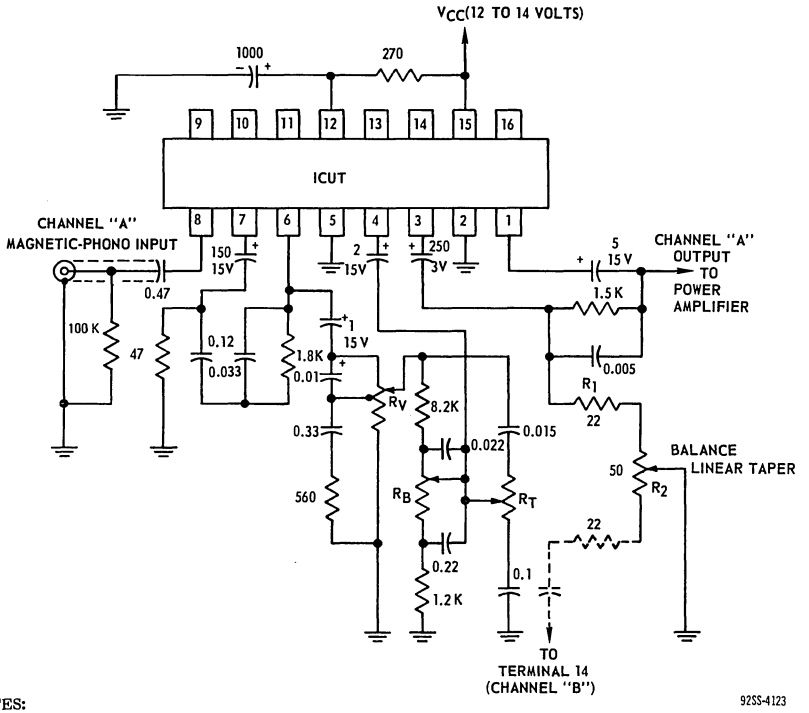


*V.T.V.M. - Hewlett-Packard Model 400D or equivalent.

Procedure:

1. Adjust Signal Generator for 0 dB output at reference terminal.
2. Read voltage at other output terminals (Figure shows terminal #1 used as reference).

Fig. 13 - Test circuit for measurement of inter-amplifier audio separation "cross talk" characteristic.



NOTES:

1) Resistor values are in ohms, capacitance values are in microfarads, unless otherwise specified.
2) R_1 and R_2 resistor values are selected for a sensitivity of 3 mV input at 1 kHz.
3) R_V, volume control potentiometer, 15000 ohms tap at 6000 ohms with logarithmic taper *
4) R_B, bass control potentiometer, 25000 ohms.
5) R_T, treble control potentiometer, 25000 ohms.

Fig. 14 - Typical magnetic phono pre-amplifier using CA3052.

*This control, (part No. 11782-JM, type Q-T4-2G) may be obtained by contacting CTS Asheville Inc., Mills Gap Rd., Skyland, N. C. 28872. Guide for potentiometer manufacturers refer to Buyers'.

Typical Performance Data/Channel For Stereo Preamplifier

Magnetic-Phono Input

Voltage Gain at f = 1 kHz . . . . . 47 dB

Noise and Hum:*
- Full volume . . . . . -60 dB below 40 W
- Zero volume . . . . . -80 dB below 40 W

Boost and Cut:
- Bass at f = 100 Hz . . . . . ± 10 dB
- Treble at f = 10 kHz . . . . . ± 10 dB

Channel Separation at f = 1 kHz . . . . . > 40 dB

Input Equalization, RIAA . . . . . ± 2 dB

*Measurement made with preamplifier connected to 40-watt Quasi-Complementary Symmetry audio amplifier circuit. For circuit details see RCA publication, Form No. 2L1111.

To construct channel B circuit, duplicate channel A component circuit values to the appropriate channel B terminal as shown in table.

Channel B Terminal No.	Channel A Terminal No.	Circuit Description
9	8	input
10	7	feedback
11	6	interstage output
13	4	interstage input
14	3	feedback
16	1	output

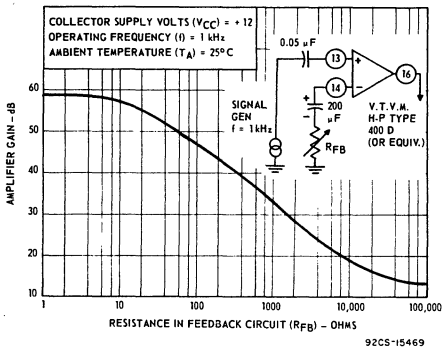


Fig. 15 - Typical amplifier gain vs feedback resistance

OPERATING CONSIDERATIONS

Economical Gain Control

The CA3052 is designed to permit flexibility in the methods by which amplifier gain can be controlled. Fig. 15 shows a curve of the gain of an amplifier when when the internal resistive feedback of the device is used in conjunction with an external resistor. Although measured gain of various amplifiers will not be uniform, because of tolerances of internal resistances, this method is very economical and easy to apply.

Stability

The CA3052, as in other devices having high gain-band-width product, requires some attention to circuit layout, design, and construction to achieve stability.

Should the CA3052 be left unterminated, socket capacitance alone will provide sufficient feedback to cause high frequency oscillations; therefore, all test circuits in this data bulletin include loading networks that provide stability under all conditions.

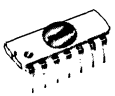
RCA
Solid State Division

Linear Integrated Circuits

Monolithic Silicon

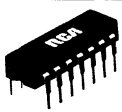
CA3058,CA3059,CA3079

CA3058



14-Lead Dual-In-Line Ceramic Package H-1553

CA3059
CA3079



14-Lead Dual-In-Line Plastic Package H-1517

Zero-Voltage Switches

For 50/60 and 400 Hz Thyristor Control Applications

Features	CA3058	CA3059	CA3079
■ 24V, 120V, 208/230V, 277V at 50 60, or 400 Hz operation	√	√	√
■ Differential Input	√	√	√
■ Low Balance Input Current (max.) -μA	1	1	2
■ Built-in Protection Circuit for opened or shorted sensor (Term. 14)	√	√	
■ Sensor Range (R_X) - kΩ	2 to 100	2 to 100	2 to 50
■ DC Mode (Term 12)	√	√	
■ External Trigger (Term. 6)	√	√	
■ External Inhibit (Term. 1)	√	√	
■ DC Supply Volts (max.)	14	14	10
■ Operating Temperature Range - °C	-55 to 125	-40 to 85	-40 to 85

RCA CA3058, CA3059, and CA3079 zero-voltage switches are monolithic silicon integrated circuits designed to control a thyristor in a variety of AC power switching applications for AC input voltages of 24 V, 120 V, 208/230 V, and 277 V at 50/60 and 400 Hz. Each of the zero-voltage switches incorporates 4 functional blocks (See Fig. 2) as follows:

1. Limiter-Power Supply - - Permits operation directly from an AC line.
2. Differential On/Off Sensing Amplifier - - Tests the condition of external sensors or command signals. Hysteresis or proportional-control capability may easily be implemented in this section.
3. Zero-Crossing Detector - - Synchronizes the output pulses of the circuit at the time when the AC cycle is at zero voltage point; thereby eliminating radio-frequency interference (RFI) when used with resistive loads.
4. Triac Gating Circuit - - Provides high-current pulses to the gate of the power controlling thyristor.

In addition, the CA3058 and CA3059 provide the following important auxiliary functions (See Fig. 2):

1. A built-in protection circuit that may be actuated to remove drive from the triac if the sensor opens or shorts.
2. Thyristor firing may be inhibited through the action of an internal diode gate connected to Terminal 1.
3. High-power dc comparator operation is provided by overriding the action of the zero-crossing detector. This is accomplished by connecting Terminal 12 to Terminal 7. Gate current to the thyristor is continuous when Terminal 13 is positive with respect to Terminal 9.

For an explanation of these functions see Operating Considerations, page 8. For detailed application information, see companion Application Note ICAN-6182, "Features and Applications of RCA Integrated-Circuit Zero-Voltage Switches (CA3058, CA3059, and CA3079)".

Applications

- Relay control ■ Heater control ■ Photosensitive control
- Valve control ■ Lamp control ■ Power one-shot control
- Synchronous switching of flashing lights
- On-off motor switching
- Differential comparator with self-contained power supply for industrial applications

MAXIMUM RATINGS, *Absolute Maximum Values, at* $T_A = 25^oC$

DC Supply Voltage (between Terms. 2 and 7):		
CA3058, CA3059	14	V
CA3079	10	V
DC Supply Voltage (between Terms. 2 and 8):		
CA3058, CA3059	14	V
CA3079	10	V
Peak Supply Current (Terms. 5 and 7)	±50	mA
Output Pulse Current (Term. 4)	150	mA
Power Dissipation:		
Up to T_A = 75°C - - CA3058	700	mW
Up to T_A = 55°C - - CA3059, CA3079	700	mW
Above T_A = 75°C - - CA3058	Derate Linearly 8	mW/°C
Above T_A = 55°C - CA3059, CA3079	Derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating	-55 to +125	°C
Storage	-65 to +150	°C
Lead Temperature (During Soldering) At distance 1/16 ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.	+ 265	°C

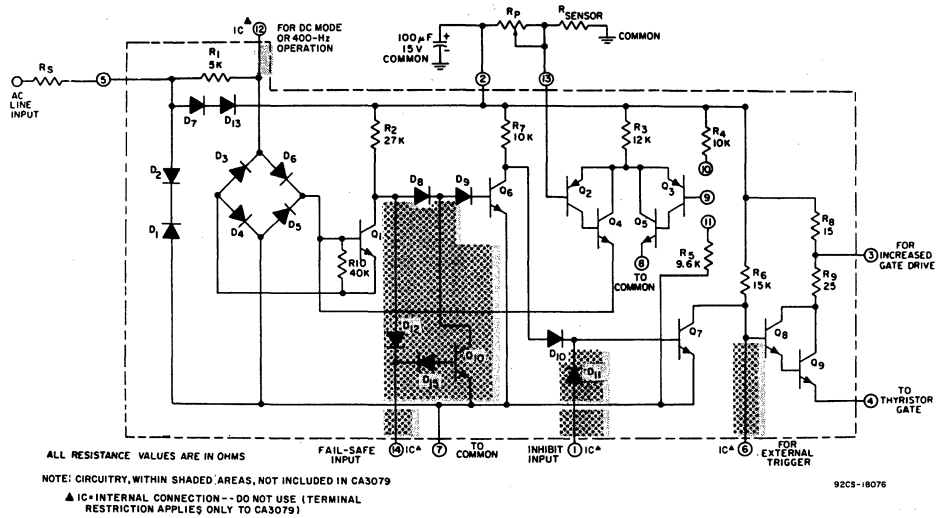


Fig. 1—Schematic diagram of zero-voltage switches CA3058, CA3059 and CA3079. For functional block diagram see Fig. 2.

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

TERMINAL NO.	1 Note 3	2	3	4	5 Note 1	6 Note 3	7	8	9	10	11	12 Note 3	13	14 Note 2,3
1 Note 3		*	*	*	*	15 0	10 −2	*	*	*	*	*	*	*
2			0 −15	0 −15	2 −14	0 −14	0▲ −14	0▲ −14	0 −14	0 −14	0 −14	* .	0 −14	0 −14
3				0 −15	*	*	*	*	*	*	*	*	*	*
4					*	2 −10	*	*	*	*	*	*	*	*
5 Note 1						*	7 −7	*	*	*	*	*	*	*
6 Note 3							14 0	*	*	*	*	*	*	*
7								*	14 0	*	20 0	2.5 −2.5	14 0	6 −6
8									10 0	*	*	*	*	*
9										*	*	*	*	*
10											*	*	*	*
11												*	*	*
12 Note 3													*	*
13														*
14 Note 3														

This chart gives the range of voltages which can be applied to the terminals listed horizontally with respect to the terminals listed vertically. For example, the voltage range of horizontal Terminal 6 to vertical Terminal 4 is 2 to −10 volts.

Note 1 – Resistance should be inserted between Term. 5 and external supply or line voltage for limiting current into Term. 5 to less than 50 mA.

Note 2 – Resistance should be inserted between Term. 14 and external supply for limiting current into Term. 14 to less than 2 mA.

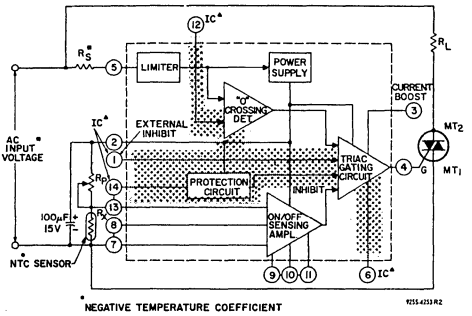
NOTE 3: For the CA3079 indicated terminal is internally connected and therefore, should not be used.

▲For CA3079 (0 to-10V)

*Voltages are not normally applied between these terminals; however, voltages appearing between these terminals are safe, if the specified voltage limits between all other terminals are not exceeded.

MAXIMUM CURRENT RATINGS

I_{IN} mA	I_{OUT} mA
10	0.1
150	10
*	*
0.1	150
50	10
*	*
*	*
0.1	2
*	*
*	*
*	*
50	50
*	*
2	2



AC Input Voltage (50/60 or 400 Hz) V AC	Input Series Resistor (R_S) k Ω	Dissipation Rating for R_S W
24	2	0.5
120	10	2
208/230	20	4
277	25	5

NOTE:
Circuitry, within shaded areas, not included in CA3079

■ See chart above

▲ IC = Internal Connection - - DO NOT USE (Terminal Restriction applies only to CA3079).

Fig.2—Functional block diagrams of the zero-voltage switches CA3058, CA3059 and CA3079. For schematic diagram see Fig. 1.

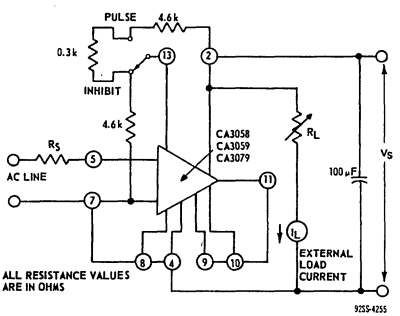


Fig.3a—DC supply voltage test circuit for CA3058, CA3059 and CA3079.

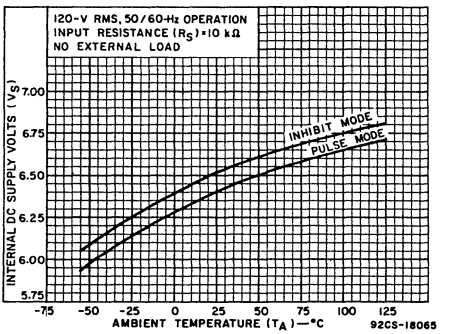


Fig.3b—DC supply voltage vs. T_A for CA3058, CA3059 and CA3079.

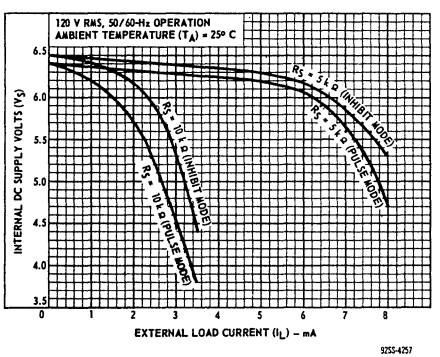


Fig.3c—DC supply voltage vs. external load current for CA3058, CA3059 and CA3079.

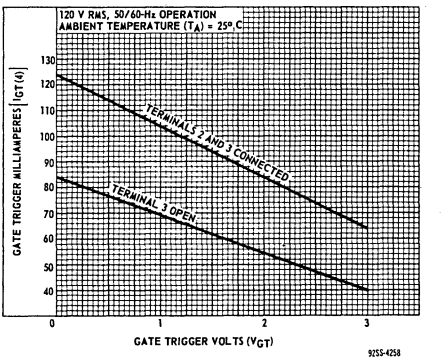


Fig.4—Gate trigger current vs. gate trigger voltage for CA3058, CA3059 and CA3079.

ELECTRICAL CHARACTERISTICS (For all types, unless indicated otherwise)
All voltages are measured with respect to Terminal 7.

CHARACTERISTIC	SYMBOL	TEST CONDITIONS			LIMITS			UNITS
		CIRCUIT Fig. No.	T_A = 25°C (Unless Indicated Otherwise)	Typical Characteristics Curves Fig. No.	Min.	Typ.	Max.	
For Operating at 120V rms, 50-60 Hz (AC Line Voltage)•								
DC Supply Voltage:								
Inhibit Mode								
At 50/60 Hz	V_S	3a	R_S = 10 kΩ, I_L = 0	3b	6.1	6.5	7	V
At 400 Hz			R_S = 10 kΩ, I_L = 0	–	–	6.8	–	V
At 50/60 Hz			R_S = 5 kΩ, I_L = 2 mA	3c	–	6.4	–	V
Pulse Mode								
At 50/60 Hz			R_S = 10 kΩ, I_L = 0	3b	6	6.4	7	V
At 400 Hz			R_S = 10 kΩ, I_L = 0	–	–	6.7	–	V
At 50/60 Hz			R_S = 5 kΩ, I_L = 2 mA	3c	–	6.3	–	V
At 50/60 Hz (CA3058)			R_S = 10 kΩ, I_L = 0 T_A=-55 to 125°C	–	5.5	–	7.5	V
Gate Trigger Current	I_{GT}(4)	5a	Terms 3 and 2 connected, V_{GT}=1V	4	–	105	–	mA
Peak Output Current (Pulsed):								
With Internal Power Supply	I_{OM}(4)	5a	Term. 3 open, Gate Trigger Voltage (V_{GT}) = 0	5b	50	84	–	mA
			Terms.3 and 2 connected, Gate Trigger Voltage (V_{GT}) = 0	5b	90	124	–	mA
With External Power Supply	I_{OM}(4)	6a	Term. 3 open, V^+ = 12V, V_{GT} = 0	6b, c	–	170	–	mA
			Terms 3 and 2 connected V^+ = 12V, V_{GT} = 0	6b, c	–	240	–	mA
Inhibit Input Ratio:		7a	Voltage Ratio of Term. 9 to 2	7b	0.465	0.485	0.520	–
All Types	V_9/V_2							
CA3058			T_A = -55 to 125°C		0.450	–	0.520	
Total Gate Pulse Duration:*								
For positive dv/dt								
50-60 Hz	t_P	8a	C_{EXT} = 0	8b	70	100	140	μs
400 Hz	t_P		C_{EXT} = 0, R_{EXT} = ∞	8d	–	12	–	μs
For negative dv/dt								
50-60 Hz	t_N	8a	C_{EXT} = 0	8b	70	100	140	μs
400 Hz	t_N		C_{EXT} = 0, R_{EXT} = ∞	8d	–	10	–	μs
Pulse Duration After Zero Crossing (50-60Hz):								
For positive dv/dt	t_{P1}	8a	C_{EXT} = 0 R_{EXT} = ∞	8c	–	50	–	μs
For negative dv/dt	t_{N1}	8a		8c	–	60	–	μs
Output Leakage Current Inhibit Mode:								
All Types	I_4	–		9	–	0.001	10	μA
CA3058			T_A = -55 to 125°C		–	–	20	μA
Input Bias Current:								
CA3058, CA3059,	I_{IB}	10			–	220	1000	nA
CA3079						220	2000	nA
Common-Mode Input Voltage Range	V_{CMR}		Terms. 9 and 13 connected		–	1.5 to 5	–	V
Sensitivity ≠ (Pulse Mode)	ΔV_{13}	5a	Term. 12 open	12	–	6	–	mV

≠Required voltage change at Term.13 to either turn OFF the triac when ON or turn ON the triac when OFF.

*Pulse duration in 50 Hz applications is approximately 15% longer than shown in Fig. 8b

•The values given in the Electrical Characteristics Chart at 120V also apply for operation at input voltages of 24V, 208/230V, and 277V , except for Pulse Duration. However, the series resistor (R_S) must have the indicated value, shown in the chart in Fig. 2, for the specified input voltage.

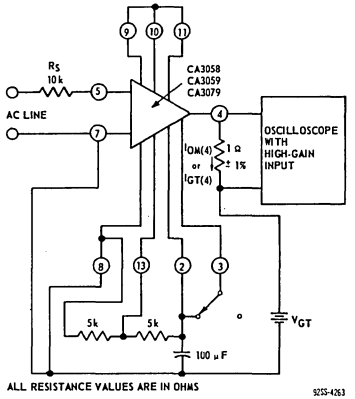


Fig.5a—Peak output (pulsed) and gate trigger current with internal power supply test circuit for CA3058, CA3059 and CA3079.

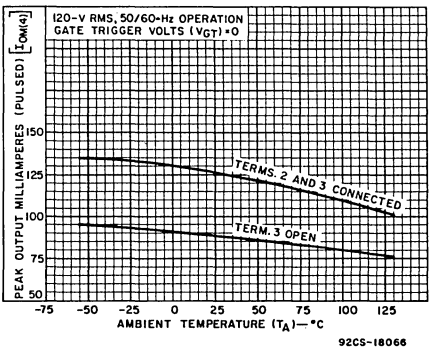


Fig.5b—I_{OM} vs. T_A for CA3058, CA3059 and CA3079.

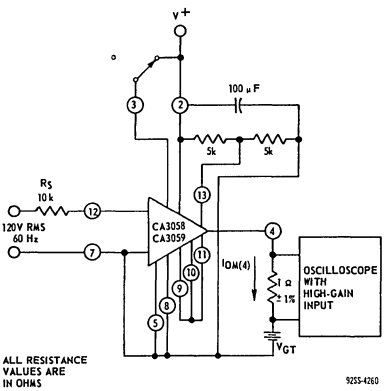


Fig.6a—Peak output current (pulsed) with external power supply test circuit for CA3058 and CA3059.

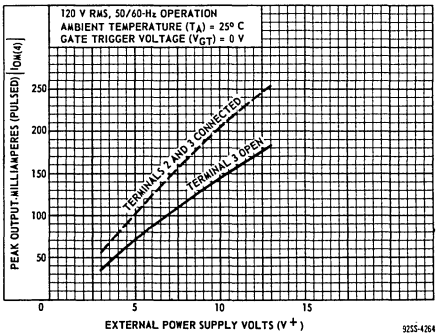


Fig.6b—I_{OM} vs. external power supply voltage for CA3058 and CA3059.

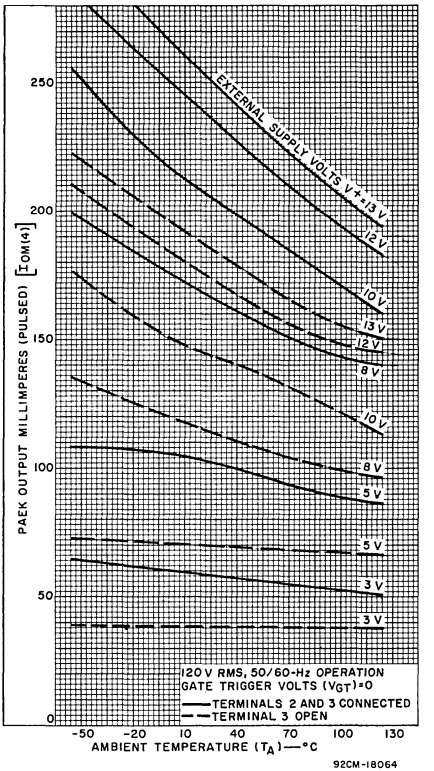


Fig.6c—I_{OM} with external power supply vs. T_A for CA3058 and CA3059.

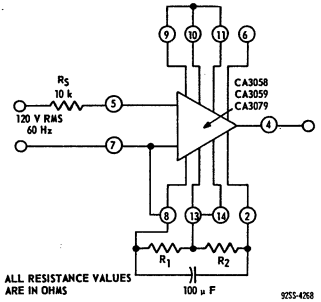


Fig.7a—Input inhibit ratio test circuit for CA3058, CA3059 and CA3079.

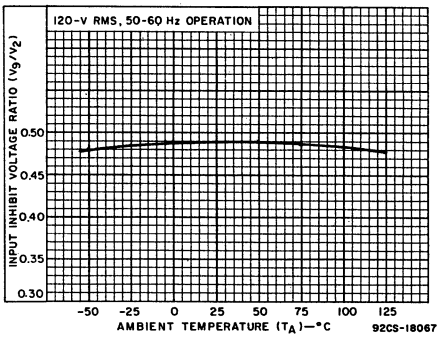


Fig.7b—Input inhibit voltage ratio vs. T_A for CA3058, CA3059 and CA3079.

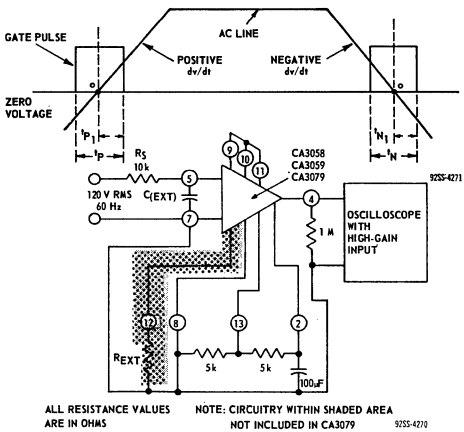


Fig.8a—Gate pulse duration test circuit with associated waveform for CA3058, CA3059 and CA3079.

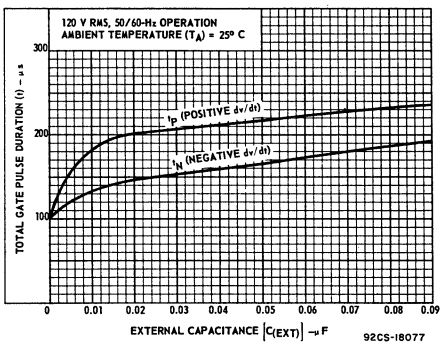


Fig.8b—Total gate pulse duration vs. external capacitance for CA3058, CA3059 and CA3079.

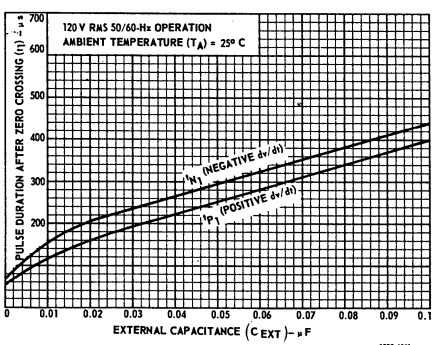


Fig.8c—Pulse duration after zero crossing vs. external capacitance for CA3058, CA3059 and CA3079.

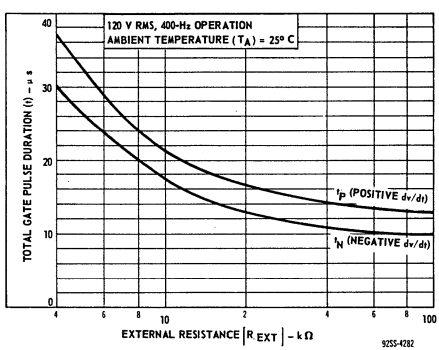


Fig.8d—Total gate pulse duration vs. external resistance for CA3058 and CA3059.

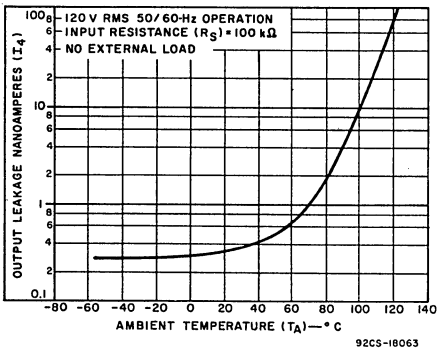


Fig.9—Output leakage current (inhibit mode) vs. T_A for CA3058, CA3059 and CA3079.

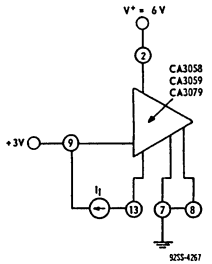
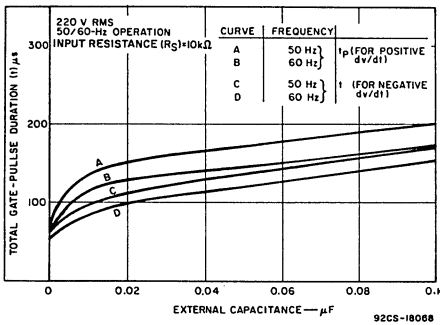
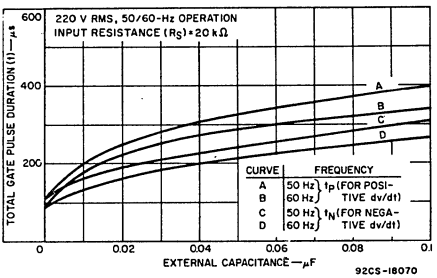
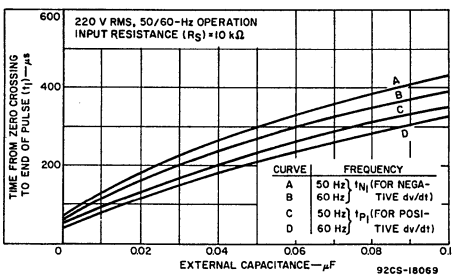


Fig.10—Input bias current test circuit for CA3058, CA3059 and CA3079.







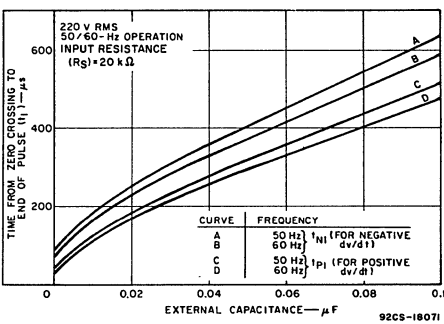


Fig.11—Relative pulse width and location of zero-voltage crossing for 220-volt operation for CA3058, CA3059 and CA3079.

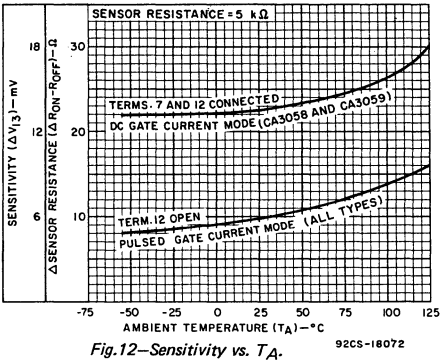


Fig.12—Sensitivity vs. T_A.

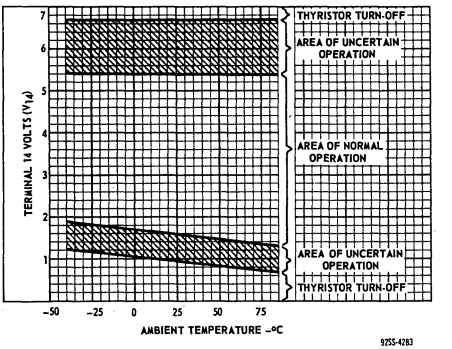


Fig.13—Operating regions for built-in protection circuit for CA3058 and CA3059.

OPERATING CONSIDERATIONS

Power Supply Considerations for CA3058, CA3059 and CA3079

The CA3058, CA3059 and CA3079 are intended for operation as self-powered circuits with the power supplied from an AC line through a dropping resistor. The internal supply is designed to allow for some current to be drawn by the auxiliary power circuits. Typical power supply characteristics are given in Figs. 3b and 3c.

Power Supply Considerations for CA3058 and CA3059

The output current available from the internal supply may not be adequate for higher power applications. In such applications an external power supply with a higher voltage should be used with a resulting increase in the output level. (See Fig. 5 for the peak output current characteristics). When an external power supply is used, Terminal 5 should be connected to Terminal 7 and the synchronizing voltage applied to Terminal 12 as illustrated in Fig. 5a.

Operation of Built-in Protection for the CA3058, CA3059

A special feature of the CA3058 and CA3059 is the inclusion of a protection circuit which, when connected, removes power from the load if the sensor either shorts or opens. The protection circuit is activated by connecting Terminal 14 to Terminal 13 as shown in Fig. 2. To assure proper operation of the protection circuit the following conditions should be observed:

1. Use the internal supply and limit the external load current to 2mA with a 5k Ω dropping resistor.
2. Set the value of Rp and sensor resistance (R_X) between 2k Ω and 100k Ω.
3. The ratio of R_X to Rp, typically, should be greater than 0.33 and less than 3. If either of these ratios is not met with an unmodified sensor over the entire anticipated temperature range, then either a series of shunt resistor must be added to avoid undesired activation of the circuit.

If operation of the protection circuit is desired under conditions other than those specified above, then apply the data given in Fig. 13.

External Inhibit Function for the CA3058 and CA3059

A priority inhibit command may be applied to Terminal 1. The presence of at least + 1.2V at 10 μA will remove drive from the thyristor. This required level is compatible with DTL or T^2L logic. A logical 1 activates the inhibit function.

DC Gate Current Mode for the CA3058 and CA3059

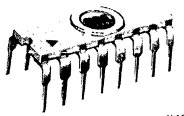
Connecting Terminals 7 and 12 disables the zero-crossing detector and permits the flow of gate current on demand from the differential sensing amplifier. This mode of operation is useful when comparator operation is desired or when inductive loads are switched. Care must be exercised to avoid overloading the internal power supply when operating in this mode. A sensitive gate thyristor should be used with a resistor placed between Terminal 4 and the gate in order to limit the gate current.

Companion Application Notes, ICAN-6168 and ICAN-6268 provide detailed descriptions of the circuit operation and include many useful control applications for the zero-voltage switches.

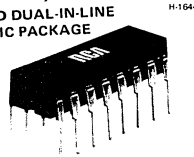
RCA
Solid State Division

Linear Integrated Circuits

CA3060AD CA3060BD CA3060D CA3060E



16-LEAD DUAL-IN-LINE CERAMIC PACKAGE H-1644



16-LEAD DUAL-IN-LINE PLASTIC PACKAGE H-1622

Operational Transconductance Amplifier Arrays

APPLICATIONS

- For low power conventional operational amplifier applications
- Active filters
- Comparators
- Gyrators
- Mixers
- Modulators
- Multiplexers
- Multipliers
- Strobing and gating functions
- Sample and hold functions

FEATURES

- Low power consumption — as low as 100 μW per amplifier
- Independent biasing for each amplifier
- High forward transconductance
- Programmable range of input characteristics
- Low input bias and input offset current
- High input and output impedance
- No effect on device under output short-circuit conditions
- Zener diode bias regulator

RCA-CA3060AD, CA3060BD, CA3060D, and CA3060E, monolithic integrated circuits, are arrays of three independent Operational Transconductance Amplifiers. This type of amplifier is a new circuit concept that has the generic characteristics of an operational voltage amplifier with the exception that the forward gain characteristic is best described by transconductance rather than voltage gain (open-loop voltage gain is the product of the transconductance and the load resistance, $g_m R_L$). When operated into a suitable load resistor and with provisions for feedback, these amplifiers are well suited for a wide variety of operational-amplifier and related applications. In addition, the extremely high output impedance makes these types particularly well suited for service in active filters.

The three amplifiers in the CA3060 family are identical push-pull Class A types which can be independently biased to achieve a wide range of characteristics for specific applications. The electrical characteristics of each amplifier are a function of the amplifier bias current (I_{ABC}). This feature offers the system designer maximum flexibility with regard to output current capability, power consumption, slew rate, input resistance, input bias current, and input offset current. The linear variation of the parameters with respect to bias and the ability to maintain a constant dc level between input and output of each amplifier also makes the CA3060 suitable for a variety of non-linear applications such as mixers, multipliers, and modulators.

In addition; the types in the CA3060 family incorporate a unique Zener diode regulator system that permits current regulation below supply voltages normally associated with such systems.

Generic applications of the OTA are described in ICAN-6668, Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers.

The CA3060AD, CA3060BD, and CA3060D are supplied in a hermetic 16-lead dual-in-line ceramic package which can be operated over the full military temperature range, -55°C to +125°C. The CA3060E is supplied in a 16-lead dual-in-line plastic package and is operational from -40°C to +85°C.

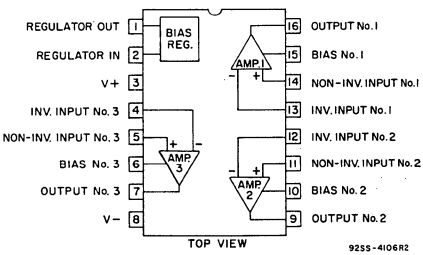


Fig. 1—Functional block diagram for each type in the CA3060 family.

MAXIMUM RATINGS, *Absolute Maximum Values at T_A = 25°C*

DC Supply Voltage (between V^+ and V^- terminals):
CA3060AD, CA3060BD, CA3060E ..... 36V (±18V)
CA3060D ..... 14 V (±7V)

Differential Input Voltage (each amplifier):
CA3060AD, CA3060BD, CA3060E ..... ±5V
CA3060D ..... ±5V

DC Input Voltage ..... V^+ to V^-

Input Signal Current (each amplifier of each type): ..... ±1 mA

Amplifier Bias Current (each amplifier of each type) ..... 2 mA

Bias Regulator Input Current ..... -5 mA

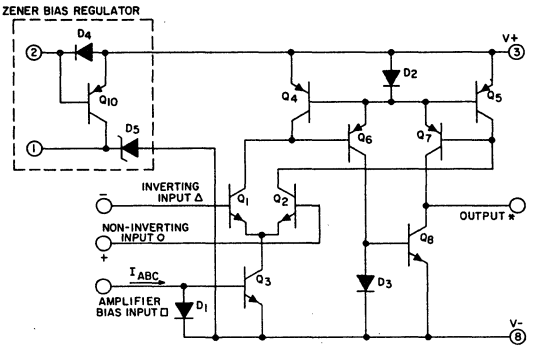
Output Short-Circuit Duration* ..... No limitation

Device Dissipation:
Total Package of each type up to T_A = 75°C ..... 490 mW
Above T_A = 75°C ..... Derate linearly 6.67 mW/°C

Temperature Range:
Operating –
CA3060AD, CA3060BD, CA3060D ..... -55 to +125°C
CA3060E ..... -40 to +85°C
Storage –
CA3060AD, CA3060BD, CA3060D,
CA3060E ..... -65 to +150°C

Lead Temperature (During Soldering):
At distance 1/16 ±1/32 in. (1.59 ±0.79 mm)
from case for 10s max ..... +300°C

*Short circuit may be applied to ground or to either supply.



Δ INVERTING INPUT OF AMPLIFIERS 1, 2, AND 3 IS ON TERMINAL Nos. 13, 12 AND 4, RESPECTIVELY
O NON-INVERTING INPUT OF AMPLIFIERS 1, 2, AND 3 IS TERMINAL Nos. 14, 11, AND 5, RESPECTIVELY
* OUTPUT OF AMPLIFIERS 1, 2, AND 3 IS ON TERMINAL Nos. 16, 9, AND 7, RESPECTIVELY
□ AMPLIFIER BIAS CURRENT OF AMPLIFIERS 1, 2, AND 3 IS ON TERMINAL Nos. 15, 10, AND 6, RESPECTIVELY

NOTE: A complete schematic diagram of the OTA is shown on Page 6. 92CS-15860RI

Fig.2—Simplified schematic diagram showing bias regulator and one operational transconductance amplifier for each type of the CA3060 family.

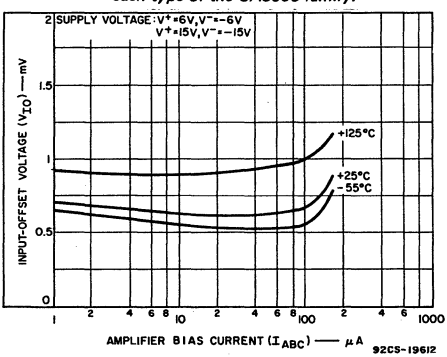


Fig.3—Input offset voltage vs. amplifier bias current.

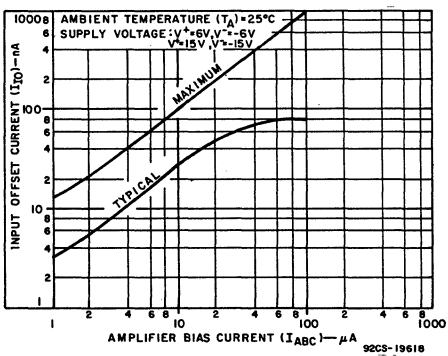


Fig.4—Input offset current vs. amplifier bias current.

ELECTRICAL CHARACTERISTICS (CA3060D)

For each amplifier at T_A = 25°C, V^+ = 6 V, V^- = -6 V

CHARACTERISTIC	SYMBOL	TYPICAL CHARACTERISTICS CURVES Fig.	LIMITS — Amplifier Bias Current I_{ABC} = 1 μA MIN.	TYP.	MAX.	I_{ABC} = 10 μA MIN.	TYP.	MAX.	I_{ABC} = 100 μA MIN.	TYP.	MAX.	UNITS
STATIC CHARACTERISTICS												
Input Offset Voltage	V_{IO}	3	–	1	5	–	1	5	–	1	5	mV
Input Offset Current	I_{IO}	4	–	3	14	–	30	100	–	250	1000	nA
Input Bias Current	I_{IB}	5a, b	–	33	70	–	300	550	–	2500	5000	nA
Peak Output Current	I_{OM}	6a, b	1.3	2.3	–	15	26	–	150	240	–	μA
Peak Output Voltage: Positive	V_{OM^+}	7	4.6	5	–	4.5	4.8	–	4.5	4.7	–	V
Negative	V_{OM^-}		5.8	5.95	–	5.8	5.95	–	5.7	5.9	–	
Amplifier Supply Current (each amplifier)	I_A	8a, b	–	8.5	14	–	85	120	–	850	1200	μA
Power Consumption (each amplifier)	P	–	–	0.10	0.17	–	1	1.45	–	10	14.5	mW
Input Offset-Voltage Sensitivity■: Positive	$\Delta V_{IO}/\Delta V^+$	–	–	1.5	120	–	2	120	–	2	120	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$		–	20	120	–	20	120	–	30	120	
Amplifier Bias Voltage*	V_{ABC}	9	–	0.54	–	–	0.60	–	–	0.66	–	V
DYNAMIC CHARACTERISTICS (at 1 kHz unless specified otherwise)												
Forward Transconductance (large signal)	g_{21}	10a, b	0.3	1.55	–	3	18	–	30	102	–	mmho
Common-Mode Rejection Ratio	CMRR	–	70	110	–	70	110	–	70	90	–	dB
Common-Mode Input-Voltage Range	V_{ICR}	–	4.4 to -5.1 min. 4.7 to -5.3 typ.			4.3 to -5 min. 4.6 to -5.2 typ.			4.3 to -5 min. 4.6 to -5.2 typ.			V
Slew Rate (Test ckt., Fig. 13	SR		–	0.1	–	–	1	–	–	8	–	V/μs
Open-Loop (g_{21}) Bandwidth	BW_{OL}	11	–	20	–	–	45	–	–	110	–	kHz
Input Impedance Components: Resistance	R_I	12	800	1600	–	90	170	–	10	20	–	kΩ
Capacitance at 1 MHz	C_I	–	–	2.7	–	–	2.7	–	–	2.7	–	pF
Output Impedance Components: Resistance	R_O	14	–	200	–	–	20	–	–	2	–	MΩ
Capacitance at 1 MHz	C_O	–	–	4.5	–	–	4.5	–	–	4.5	–	pF

ZENER BIAS REGULATOR CHARACTERISTICS (at T_A = 25°C, I_Z = 0.1 mA)				MIN.	TYP.	MAX.	
Voltage	V_Z	15	Temp. Coeff. = 3 mV/°C	6.2	6.7	7.9	V
Impedance	Z_Z	–			200	300	Ω

* Temperature-Coefficient; -2.2 mV/°C (at V_{ABC} = 0.54 V, I_{ABC} = 1 μA; -2.1 mV/°C (at V_{ABC} = 0.060 V, I_{ABC} = 10 μA); -1.9 mV/°C (at V_{ABC} = 0.66 V, I_{ABC} = 100 μA)

■ Conditions for Input Offset Voltage and Supply Sensitivity:

(a) Bias current derived from the regulator with an appropriate resistor connected from terminal No. 1 to the bias terminal on the amplifier under test —

V^+ is reduced to 5 volts for V^+ sensitivity

V^- is reduced to -5 volts for V^- sensitivity

(b) V^+ sensitivity in μV/V = $\frac{\text{Voffset - Voffset for +5 V and -6 V supplies}}{\text{1 volt}}$

V^- sensitivity in μV/V = $\frac{\text{Voffset - Voffset for -5 V and +6 V supplies}}{\text{1 volt}}$

ELECTRICAL CHARACTERISTICS (CA3060AD, CA3060BD, CA3060E)

For each amplifier at T_A = 25°C, V^+ = 15 V, V^- = -15 V

CHARACTERISTIC	SYMBOL	TYPICAL CHARACTERISTICS CURVE Fig.	LIMITS — Amplifier Bias Current I_{ABC} = 1 μA (CA3060BD) MIN.	TYP.	MAX.	I_{ABC} = 10 μA (CA3060BD) MIN.	TYP.	MAX.	I_{ABC} = 100 μA (CA3060AD, CA3060BD, CA3060E) MIN.	TYP.	MAX.	UNITS
STATIC CHARACTERISTICS												
Input Offset Voltage	V_{IO}	3	–	1	5	–	1	5	–	1	5	mV
Input Offset Current	I_{IO}	4	–	3	14	–	30	100	–	250	1000	nA
Input Bias Current	I_{IB}	5a,b	–	33	70	–	300	550	–	2500	5000	nA
Peak Output Current	I_{OM}	6a,b	1.3	2.3		15	26	–	150	240	–	μA
Peak Output Voltage: Positive	V_{OM^+}	7	12	13.6	–	12	13.6	–	12	13.6	–	V
Peak Output Voltage: Negative	V_{OM^-}	7	12	14.7	–	12	14.7	–	12	14.7	–	V
Amplifier Supply Current (each amplifier)	I_A	8a,b	–	8.5	14	–	85	120	–	850	1200	μA
Power Consumption (each amplifier)	P	–	–	0.26	0.42	–	2.6	3.6	–	26	36	mW
Input Offset-Voltage Sensitivity■: Positive	$\Delta V_{IO}/\Delta V^+$	–	–	1.5	150	–	2	150	–	2	150	μV/V
Input Offset-Voltage Sensitivity■: Negative	$\Delta V_{IO}/\Delta V^-$	–	–	20	150	–	20	150	–	30	150	μV/V
Amplifier Bias Voltage*	V_{ABC}	9	–	0.54	–	–	0.60	–	–	0.66	–	V
DYNAMIC CHARACTERISTICS (at 1 kHz unless specified otherwise)												
Forward Transconductance (large signal)	g_{21}	10a,b	0.3	1.55	–	3	18	–	30	102	–	mmho
Common-Mode Rejection Ratio	CMRR	–	70	110	–	70	110	–	70	90	–	dB
Common-Mode Input Voltage Range	V_{ICR}	–	+12 to -12 min. +13 to -14 typ.			+12 to -12 min. +13 to -14 typ.			+12 to -12 min. +13 to -14 typ.			V
Slew Rate (Test ckt., Fig. 13)	SR	–	–	0.1	–	–	1	–	–	8	–	V/μs
Open-Loop (g_{21}) Bandwidth	BW_{OL}	11	–	20	–	–	45	–	–	110	–	kHz
Input Impedance Components: Resistance	R_I	12	800	1600	–	90	170	–	10	20	–	kΩ
Input Impedance Components: Capacitance at 1 MHz	C_I	–	–	2.7	–	–	2.7	–	–	2.7	–	pF
Output Impedance Components: Resistance	R_O	14	–	200	–	–	20	–	–	2	–	MΩ
Output Impedance Components: Capacitance at 1 MHz	C_O	–	–	4.5	–	–	4.5	–	–	4.5	–	pF

ZENER BIAS REGULATOR CHARACTERISTICS (at T_A=25°C, I_Z = 0.1 mA)

				MIN.	TYP.	MAX.	
Voltage	V_Z	15	Temp. Coeff. = 3 mV/°C	6.2	6.7	7.9	V
Impedance	Z_Z	–			200	300	Ω

* Temperature-Coefficient; -2.2 mV/°C (at V_{ABC} = 0.54 V, I_{ABC} = 1 μA; -2.1 mV/°C (at V_{ABC} = 0.060 V, I_{ABC} = 10 μA); -1.9 mV/°C (at V_{ABC} = 0.66 V, I_{ABC} = 100 μA)

■ Conditions for Input Offset Voltage and Supply Sensitivity:

(a) Bias current derived from the regulator with an appropriate resistor connected from terminal No. 1 to the bias terminal on the amplifier under test ---

V^+ is reduced to 13 volts for V^+ sensitivity

V^- is reduced to −13 volts for V^- sensitivity

(b) V^+ sensitivity in μV/V = $\frac{\text{Voffset} - \text{Voffset for +13 V and −15 V suppl}}{\text{1 volt}}$

V^- sensitivity in μV/V = $\frac{\text{Voffset} - \text{Voffset for −13 V and +15 V suppl}}{\text{1 volt}}$

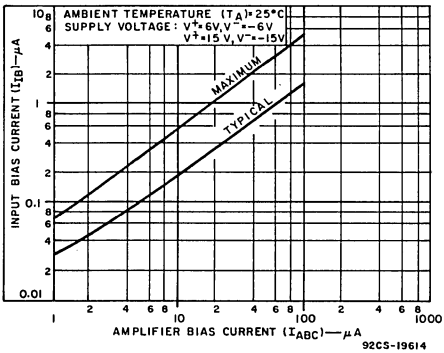


Fig.5a—Input bias current vs. amplifier bias current

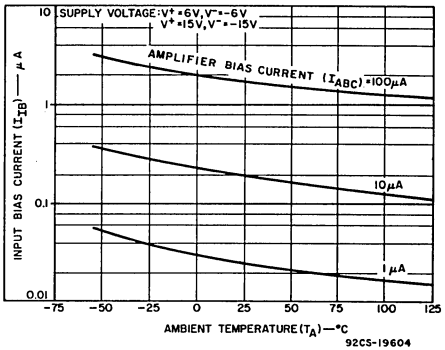


Fig.5b—Input bias current vs. ambient temperature.

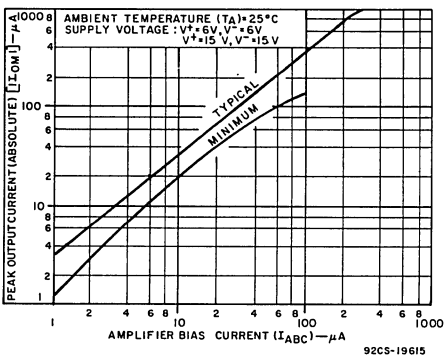


Fig.6a—Peak output current vs. amplifier bias current.

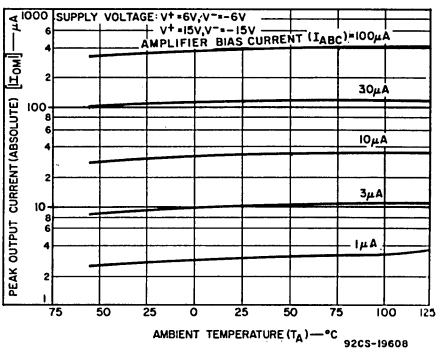


Fig.6b—Peak output current vs. ambient temperature.

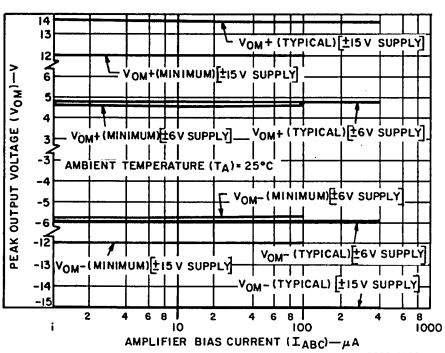


Fig.7—Peak output voltage vs. amplifier bias current.

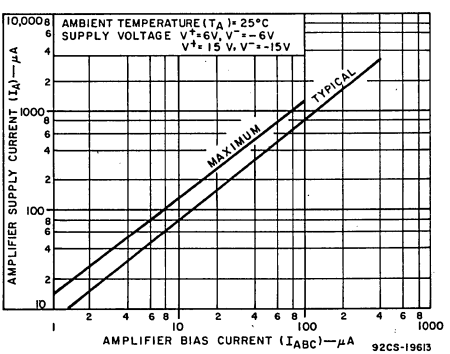


Fig.8a—Amplifier supply current (each amplifier) vs. amplifier bias current.

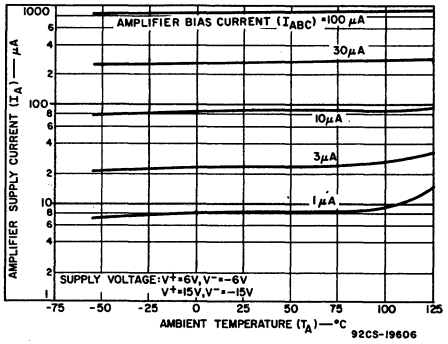


Fig.8b—Amplifier supply current (each amplifier) vs. ambient temperature.

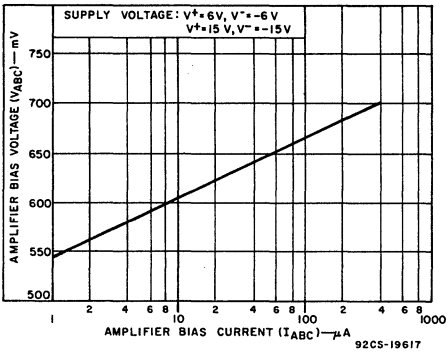


Fig.9—Amplifier bias voltage vs. amplifier bias current.

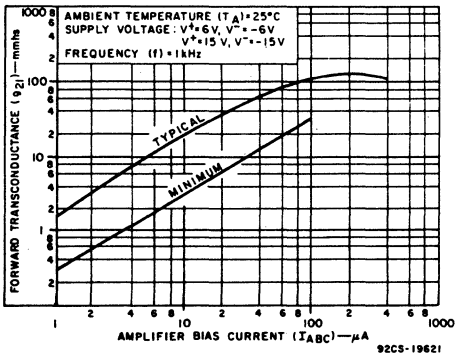


Fig.10a—Forward transconductance vs. amplifier bias current.

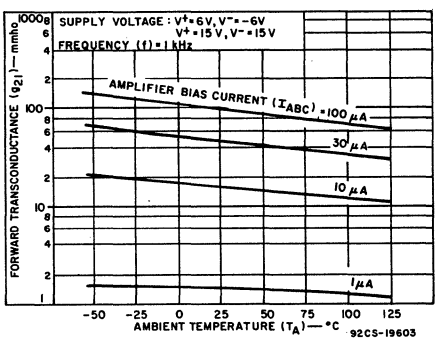


Fig.10b—Forward transconductance vs. ambient temperature.

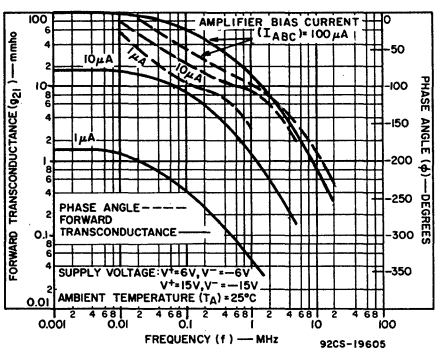


Fig.11—Forward transconductance vs. frequency.

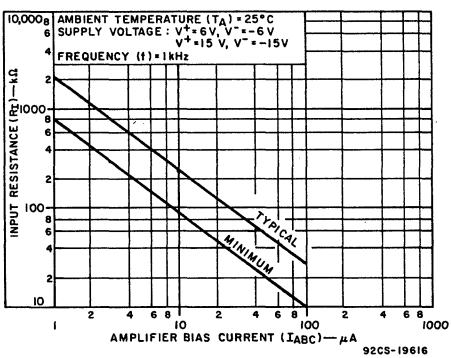
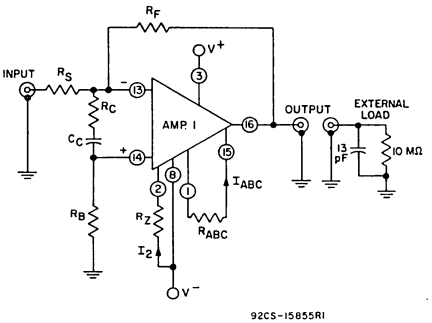


Fig.12—Input resistance vs. amplifier bias current.



V_Z is measured between terminals 1 and 8.

V_{ABC} is measured between terminals 15 and 8.

$$R_Z = \frac{[(V^+)-(V^-)-0.7]}{I_2}, \quad R_{ABC} = \frac{V_Z - V_{ABC}}{I_{ABC}}$$

Supply Voltage: for both ±6 V and ±15 V.

TYPICAL SLEW RATE TEST CIRCUIT PARAMETERS

I_{ABC}	SLEW RATE	I_2	R_{ABC}	R_S	R_F	R_B	R_C	C_C
μA	V/μs	μA	ohms					μF
100	8	200	62 k	100k	100k	51k	100	0.02
10	1	200	620k	1M	1M	510k	1k	0.005
1	0.1	2	6.2M	10M	10M	5.1M	∞	0

Fig. 13—Slew rate test circuit for amplifier No. 1 of CA3060.

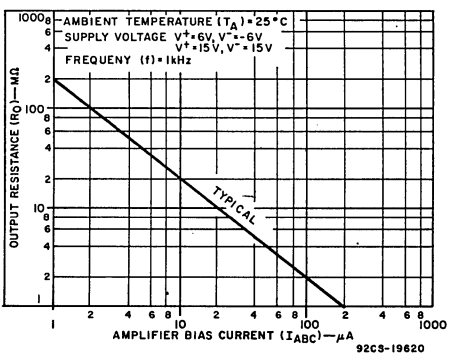


Fig. 14—Output resistance vs. amplifier bias current.

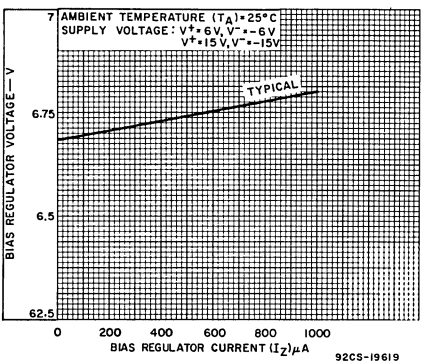


Fig. 15—Bias regulator voltage vs. bias regulator current.

OPERATING CONSIDERATIONS

The CA3060 consists of three operational amplifiers similar in form and application to conventional operational amplifiers but sufficiently different from the standard operational amplifier (op-amp) to justify some explanation of their characteristics. The amplifiers incorporated in the CA3060 are best described by the term Operational Transconductance Amplifier (OTA). The characteristics of an ideal OTA are similar to those of an ideal op-amp except that the OTA has an extremely high output impedance. Because of this inherent characteristic the output signal is best defined in terms of current which is proportional to the difference between the voltages of the two input terminals. Thus, the transfer characteristic is best described in terms of transconductance rather than voltage gain. Other than the difference given above, the characteristics tabulated on pages 3 and 4 of this data bulletin are similar to those of any typical op-amp.

The OTA circuitry incorporated in the CA3060 (See Fig. 16) provides the equipment designer with a wider variety of circuit arrangements than does the standard op-amp; because as the curves in the data bulletin indicate, the user may select the optimum circuit conditions for a specific application simply by varying the bias conditions of each amplifier. If low power consumption, low bias, and low offset current, or high input impedance are primary design requirements, then low current operating conditions may be selected. On the other hand, if operation into a moderate load impedance is the primary consideration, then higher levels of bias may be used.

Bias Considerations for Op-Amp Applications

The operational transconductance amplifiers allow the circuit designer to select and control the operating conditions of the circuit merely by the adjustment of the input bias current I_{ABC}. This enables the designer to have complete control over transconductance, peak output current and total power consumption independent of supply voltage.

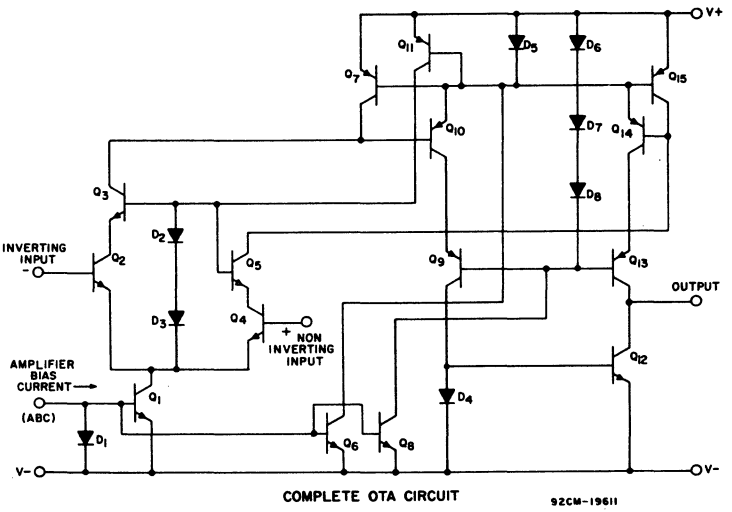


Fig.16 – Complete schematic diagram showing one of the three operational transconductance amplifiers.

In addition, the high output impedance makes these amplifiers ideal for applications where current summing is involved.

The design of a typical operational amplifier circuit (See Fig. 17) would proceed as follows:

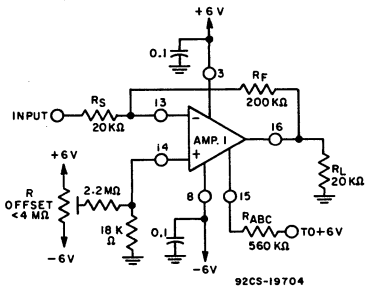


Fig.17–20-dB amplifier using the CA3060.

Circuit Requirements
Closed loop voltage gain = 10 (20 dB)
Offset voltage adjustable to zero
Current drain as low as possible
Supply voltage = ±6 V
Maximum input voltage = ±50 mV
Input resistance = 20 kΩ
Load resistance = 20 kΩ
Device: CA3060

Calculation

1. Required transconductance g_{21}.
   Assume that the open loop gain A_{OL} must be at least ten times the closed loop gain. Therefore, the forward transconductance required is given by

$$g_{21} = A_{OL}/R_L$$

$$= 100/18\ k\Omega$$

$$\cong 5.5\ mmho$$

(R_L = 20 kΩ in parallel with 200 kΩ

≅ 18 kΩ)

2. Selection of suitable amplifier bias current.
   The amplifier bias current is selected from the minimum value curve of transconductance (Fig. 10a) to assure that the amplifier will provide sufficient gain. For the required g_{21} of 5.5 mmho an amplifier bias current I_{ABC} of 20 μA is suitable.

3. Determination of Output Swing Capability.
   For a loop gain of 10 the output swing is ±0.5 V and the peak load current 25 μA. However, the amplifier must also supply the necessary current through the feedback resistor and for R_S = 20 kΩ than R_F = 200 kΩ if A_{OL} = 10. Therefore, the feedback loading = 0.5/200 kΩ = 2.5 μA.

   The total amplifier current output requirements are, therefore, ±27.5 μA. Referring to the data given in Fig. 6a we see that for an amplifier bias current of 20 μA the amplifier output current is ±40 μA. This is obviously adequate and it is not necessary to change the amplifier bias current I_{ABC}.

4. Calculation of bias resistance.
   For minimum supply current drain the amplifier bias current I_{ABC} should be fed directly from the supplies and not from the bias regulator. The value of the resistor R_{ABC} may be directly calculated using Ohm's law.

$$R_{ABC} = \frac{V_{SUP} - V_{ABC}}{I_{ABC}}$$

$$R_{ABC} = \frac{12 - 0.63}{20 \times 10^{-6}}$$

$$= 568.5\ k\Omega \text{ or } \cong 560\ k\Omega$$

5. Calculation of offset adjustment circuit.
   In order to reduce the loading effect of the offset adjustment circuit on the power supply, the offset control should be arranged to provide the necessary offset current. The source resistance of the non-inverting input is made equal to the source resistance of the inverting input.

i.e. $$\frac{20 \times 200 \times 10^{6}\ \text{ohms}}{220 \times 10^{3}} \cong 18\ k\Omega$$

Because the maximum offset voltage is 5 mV and an additional increment due to the offset current (Fig. 4) flowing through the source resistance

(i.e. $200 \times 10^{-9} \times 18 \times 10^{3}$ volts), therefore,

the Offset Voltage Range = 5 mV + 3.6 mV = ±8.6 mV

The current necessary to provide this offset is

$$\frac{8.6 \times 10^{-3}}{18 \times 10^{3}} \text{ or } 0.48\ \mu A$$

With a supply voltage of ±6 V, this current can be provided by a 10 MΩ resistor. However, the stability of such a resistor is often questionable and a more realistic value of 2.2 MΩ was used in the final circuit.

OTHER CONSIDERATIONS

Capacitance Effects

The CA3060 is designed to operate at such low power levels that high impedance circuits must be employed. In designing such circuits, particularly feedback amplifiers, stray circuit capacitance must always be considered because of its adverse effect on frequency response and stability. For example a 10-kΩ load with a stray capacitance of 15 pF has a time constant of 1 MHz. Fig. 18 illustrates how a 10-kΩ 15-pF load modifies the frequency characteristic.

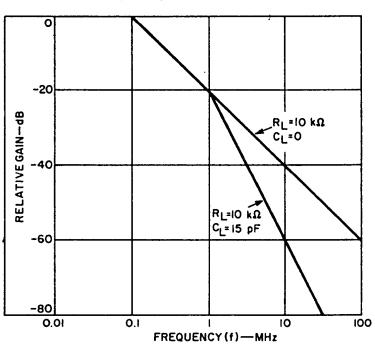


Fig.18—Effect of capacitive loading on frequency response.

Capacitive loading also has an effect on slew rate; because the peak output current is established by the amplifier bias current, I_{ABC} (see Fig. 6a), the maximum slew rate is limited to the maximum rate at which the capacitance can be charged by the I_{OM}. Therefore,

$$SR = dV/dt = I_{OM}/C_L$$

where C_L is the total load capacitance including strays. This relationship is shown graphically in Fig. 19. When measuring slew rate for this data bulletin, care was taken to keep the total capacitive loading to 13 pF.

Phase Compensation

In many applications phase compensation will not be required for the amplifiers of the CA3060. When needed, compensation may easily be accomplished by a simple RC network at the input of the amplifier as shown in Fig. 13. The values given in Fig. 13 provide stable operation for the critical unity gain condition, assuming that capacitive loading on the output is 13 pF or less. Input phase compensation is recommended in order to maintain the highest possible slew rate.

In applications such as integrators, two OTAs may be cascaded to improve current gain. Compensation is best accomplished in this case with a shunt capacitor at the output of the first amplifier. The high gain following compensation assures a high slew rate.

APPLICATIONS

Having determined the operating points of the CA3060 amplifiers, they can now function in the same manner as conventional op-amps, and thus, are well suited for most op-amp applications, including inverting and non-inverting amplifiers, integrators, differentiators, summing amplifiers etc.

TRI-LEVEL COMPARATOR

Tri-level comparator circuits are an ideal application for the CA3060 since it contains the requisite three amplifiers. A tri-level comparator has three adjustable limits. If either the upper or lower limit is exceeded, the appropriate output is activated until the input signal returns to a selected intermediate limit. Tri-level comparators are particularly suited to many industrial control applications.

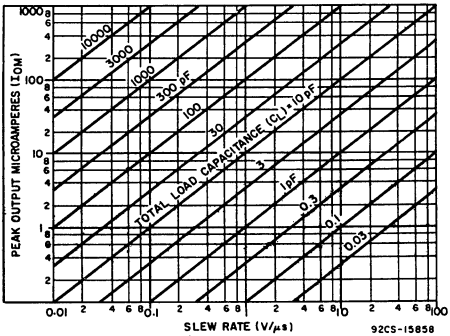


Fig.19—Effect of load capacitance on slew rate.

Circuit Description

Fig. 20 shows the block diagram of a tri-level comparator using the CA3060. Two of the three amplifiers are used to compare the input signal with the upper-limit and lower-limit reference voltages. The third amplifier is used to compare the input signal with a selected value of intermediate-limit reference voltage. By appropriate selection or resistance ratios this intermediate-limit may be set to any voltage between the upper-limit and lower-limit values. The output of the upper-limit and lower-limit comparator sets the corresponding upper or lower-limit flip-flop. The activated flip-flop retains its state until the third comparator (intermediate-limit) in the CA3060 initiates a reset function, thereby indicating that the signal voltage has returned to the intermediate-limit selected. The flip-flops employ two CA3086 transistor-array IC's, with circuitry to provide separate "SET" and "POSITIVE OUTPUT" terminals.

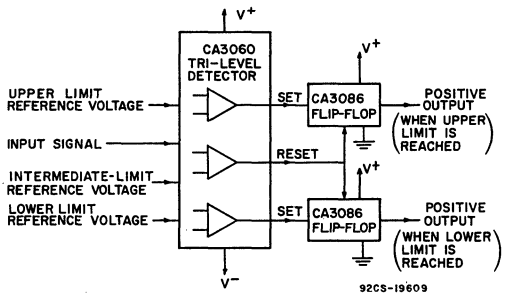


Fig.20—Functional block diagram of a tri-level comparator.

The circuit diagram of a tri-level comparator appears in Fig. 21. Power is provided for the CA3060 via terminals 3 and 8 by ±6-volt supplies and the built-in regulator provides amplifier-bias-current (I_{ABC}) to the three amplifiers via terminal 1. Lower-limit and upper-limit reference voltages are selected by appropriate adjustment of potentiometers R1 and R2, respectively. When resistors R3 and R4 are equal in value (as shown), the intermediate-limit reference voltage is automatically established at a value midway between the lower-limit and upper-limit values. Appropriate variation of resistors R3 and R4 permits selection of other values of intermediate-limit voltages. Input signal (E_S) is applied to the three comparators via terminals 5, 12, and 14. The "SET" output lines trigger the appropriate flip-flop whenever the input signal reaches a limit value. When the input signal returns to an intermediate-value, the common flip-flop "RESET" line is energized. The loads in the circuits, shown in Fig. 21 are 5-V, 25-mA lamps.

Active Filters — Using the CA3060 as a Gyrator

The high output impedance of the OTAs makes the CA3060 ideally suited for use as a gyrator in active filter applications. Fig. 22 shows two OTAs of the CA3060 connected as a gyrator in an active filter circuit. The OTAs in this circuit can make a 3-μF capacitor function as a floating 10-kilohenry inductor across Terminals A and B. The measured Q of 13 (at a frequency of 1 Hz) of this inductor compares favorably with a calculated Q of 16. The 20-kilohm to 2-megohm attenuators in this circuit extend the dynamic range of the OTA by a factor of 100. The 100-kilohm potentiometer, across V+ and V-, tunes the inductor by varying the g_{21} of the OTAs, thereby changing the gyration resistance.

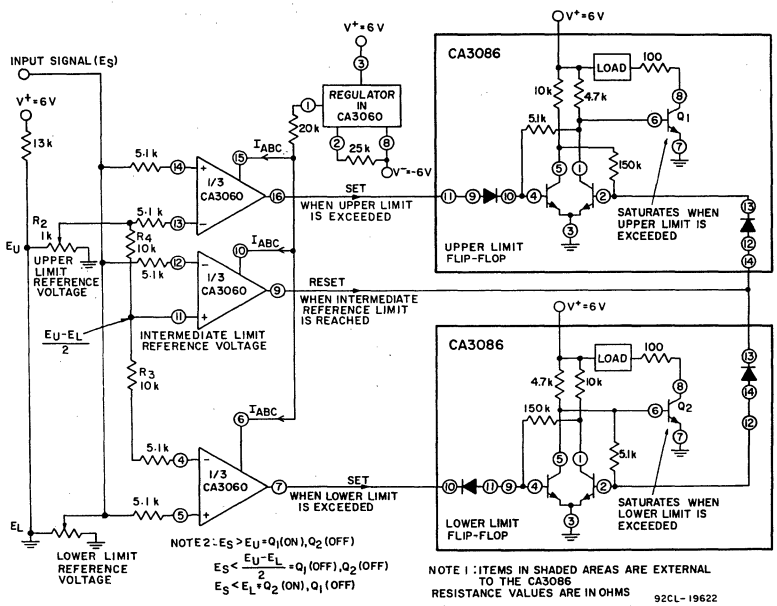


Fig.21—Tri-level comparator circuit.

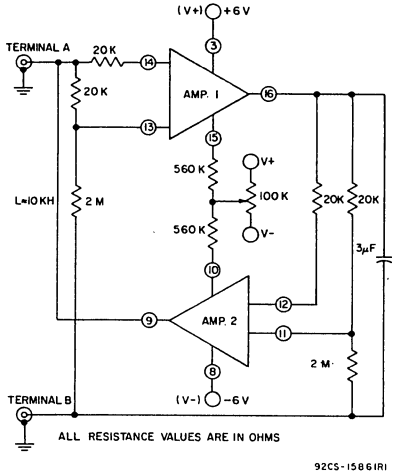


Fig.22—Two operational transconductance amplifiers of the CA3060 connected as a gyrator in an active filter circuit.

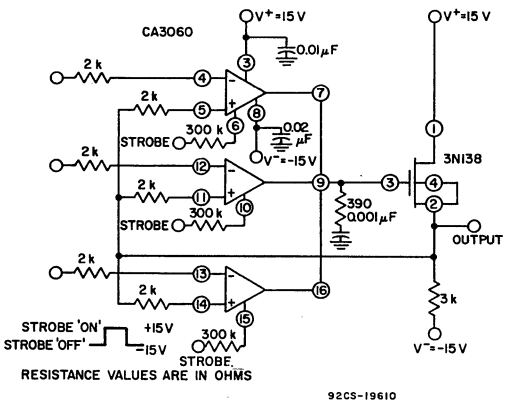


Fig.23—Three-channel multiplexer.

THREE CHANNEL MULTIPLEXER

Fig. 23 shows a schematic of a three channel multiplexer using a single CA3060 and a 3N138 MOS/FET as a buffer and power amplifier.

When the CA3060 is connected as a high-input impedance voltage follower, and strobe "ON," each amplifier is activated and the output swings to the level of the input of that amplifier. The cascade arrangement of each CA3060 amplifier with the MOS/FET provides an open loop voltage gain in excess of 100 dB, thus assuring excellent accuracy in the voltage follower mode with 100% feedback.

Operation at ±6 volts is also possible with several minor changes. First, the resistance in series with amplifier bias current (I_{ABC}) terminal of each amplifier should be decreased to maintain 100 μA of strobe—"ON" current at this lower supply voltage. Second, the drain resistance for the MOS/FET should be decreased to maintain the same value of source current. The low cost dual-gate protected MOS/FET, RCA-40841, may be used when operating at the low supply voltage.

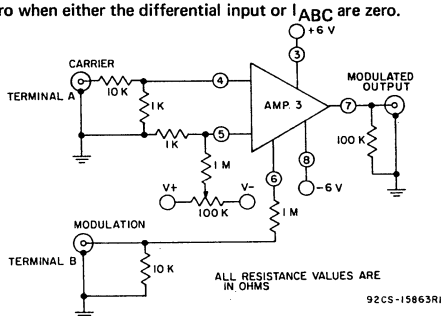
The phase compensation network consists of a single 390Ω resistor and a 1000-pF capacitor, located at the interface of the CA3060 output and the MOS/FET gate. The bandwidth of the system is 1.5 MHz and the slew rate is 0.3 volts/μsec. The system slew rate is directly proportional to the value of the phase compensation capacitor. Thus, with higher gain settings where lower values of phase compensation capacitors are possible, the slew rate is proportionally increased.

NON LINEAR APPLICATIONS

AM Modulator (Two-Quadrant Multiplier)

Fig. 24 shows Amplifier No. 3 of the CA3060 used in an AM modulator or 2-quadrant multiplier circuit. When modulation is applied to the amplifier bias input, Terminal B, and the carrier frequency to the differential input, Terminal A, the waveform, shown in Fig. 24, is obtained. Fig. 24 is a result of adjusting the input offset control to balance the circuit so that no modulation can occur at the output without a carrier input. The linearity of the modulator is indicated by the solid trace of the superimposed modulating frequency. The maximum depth of modulation is determined by the ratio of the peak input modulating voltage to V⁻.

The two-quadrant multiplier characteristic of this modulator is easily seen if modulation and carrier are reversed as shown in Fig. 24. The polarity of the output must follow that of the differential input; therefore, the output is positive only during, the positive half cycle of the modulation and negative only in the second half cycle. Note, that both the input and output signals are referenced to ground. The output signal is zero when either the differential input or I_{ABC} are zero.



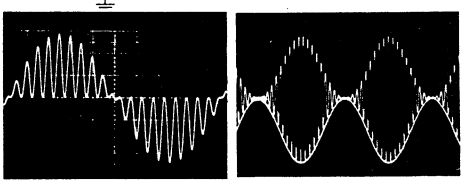


Fig.24—Two-quadrant multiplier circuit using the CA3060 with associated waveforms.

Four-Quadrant Multiplier

The CA3060 is also useful as a four-quadrant multiplier. A block diagram of such a multiplier, utilizing Amplifier Nos. 1, 2, and 3, is shown in Fig. 25 and a typical circuit is shown in Fig. 26. The multiplier consists of a single CA3060 and, as in the two-quadrant multiplier, exhibits no level shift between input and output. In Fig. 25, Amplifier No. 1 is connected as an inverting amplifier for the X-input signal. The output current of Amplifier No. 1 is calculated as follows:

$$I_O(1) = [-V_X]\ [g_{21}(1)] \qquad \text{(Eq.3)}$$

Ampl. No. 2 is a non-inverting amplifier so that

$$I_O(2) = [+V_X]\ [g_{21}(2)] \qquad \text{(Eq. 4)}$$

Because the amplifier output impedances are high, the load current is the sum of the two output currents, for an output voltage

$$V_O = V_X R_L\ [g_{21}(2) - g_{21}(1)] \qquad \text{(Eq. 5)}$$

The transconductance is approximately proportional to the amplifier bias current; therefore, by varying the bias current the g_{21} is also controlled. Amplifier No. 2 bias current is proportional to the Y-input signal and is expressed as

$$I_{ABC(2)} \approx \frac{(V\text{-}) + V_Y}{R_1} \qquad \text{(Eq. 6)}$$

Hence,

$$g_{21}(2) \approx k\ [(V\text{-}) + V_Y]. \qquad \text{(Eq. 7)}$$

Bias for Amplifier No. 1 is derived from the output of Amplifier No. 3 which is connected as a unity-gain inverting amplifier. $I_{ABC(1)}$, therefore, varies inversely with V_Y. And by the same reasoning as above

$$g_{21}(1) \approx k\ [(V\text{-}) - V_Y]. \qquad \text{(Eq. 8)}$$

Combining equation 5, 7, and 8 yields:

$$V_O \approx V_X \cdot k \cdot R_L \left\{[(V\text{-}) + V_Y] - [(V\text{-}) - V_Y]\right\} \text{ or}$$

$$V_O \approx 2\,k\,R_L\,V_X\,V_Y$$

Fig. 26 shows the actual circuit including all the adjustments associated with differential input and an adjustment for equalizing the gains of Amplifiers No. 1 and No. 2. Adjustment of the circuit is quite simple. With both the X and Y voltages at zero, connect Terminal 10 to Terminal 8. This procedure disables Amplifier No. 2 and permits adjusting the offset voltage of Amplifier No. 1 to zero by means of the 100-kΩ potentiometer. Next, remove the short between Terminals 10 and 8 and connect Terminal 15 to Terminal 8. This step disables Amplifier No. 1 and permits Amplifier No. 2 to be zeroed with the other potentiometer. With AC signals on both the X and Y input, R3 and R11 are adjusted for symmetrical output signals. Fig. 27 shows the output waveform with the multiplier adjusted. The voltage waveform in Fig. 27a shows suppressed carrier modulation of 1-kHz carrier with a triangular wave.

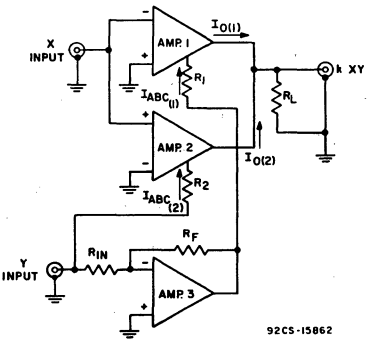


Fig.25—Four-quadrant multiplier using the CA3060.

Figures 27b and 27c, respectively, show the squaring of a triangular wave and a sine wave. Notice that in both cases the outputs are always positive and return to zero after each cycle.

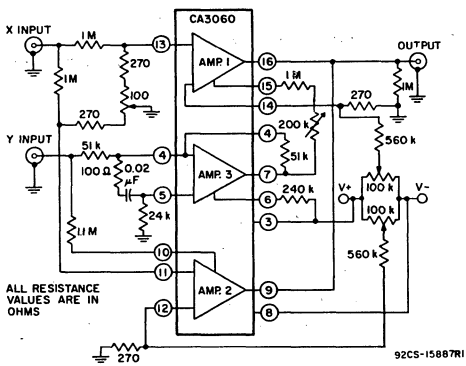


Fig.26—Typical four-quadrant multiplier circuit.

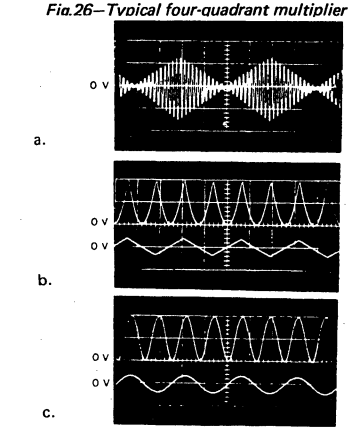


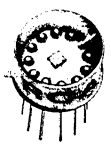
Fig.27—Voltage waveforms of four-quadrant multiplier circuit.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3062



Approx. 2½ times actual size

Modified 12-lead TO-5 style package

H-1625

Photo Detector and Power Amplifier

For Photoelectric Control Applications

Features

- **100 mA output-current capability — can drive a relay or thyristor directly**
- **5 to 15 volt dc supply voltage**
- **Compact — complete system in a TO-5 style package**

The CA3062* is an integrated circuit consisting of a photosensitive section, an amplifier, and a pair of high-current output transistors on a single monolithic chip.

The photosensitive section consists of Darlington pairs and affords high sensitivity. The power amplifier has a differential configuration which provides complementing outputs in response to a light input — normally "ON" and normally "OFF". The separate photodetector, amplifier, and high-current switch provide flexibility of circuit arrangement. This feature plus the high current capability of the output section, can now provide the user with a complete system particularly useful in photoelectric control applications utilizing IR emitters and visible-light sources.

*Formerly developmental type TA5371B.

Applications

- **Counters**
- **Sorting**
- **Level controls**
- **Inspection**
- **Intrusion alarms**
- **Position sensor**
- **Edge monitoring**
- **Isolators**
- **See ICAN-6538, "Applications of the RCA-CA3062 IC Photodetector and Power Amplifier in Switching Circuits"**

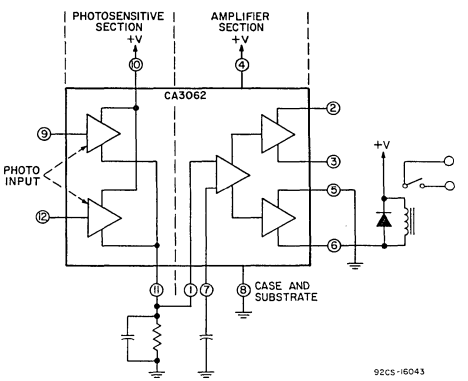


Fig. 1 - Light operated relay using CA3062.

ABSOLUTE-MAXIMUM RATINGS

DISSIPATION:
- Up to T_A = 55°C . . . . . 700 mW
- Above T_A = 55°C . . . . . Derate linearly 5.6 mW/°C
- At Case Temperature (T_C) ≤ 55°C . . . . . 1.5 W
- Above T_C = 55°C . . . . . Derate linearly 16 mW/°C

TEMPERATURE RANGE:
- Operating . . . . . -55°C to +125°C
- Storage . . . . . -65°C to +150°C

LEAD TEMPERATURE (During soldering):
- At distance ≥ 1/32 in (3.17 mm) from seating plane for 10 s max . . . . . +300°C

Maximum Voltage Ratings

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 and horizontal terminal 3 is +15 to 0 volts.

TERMINAL No.	10	11	12	1	2	3	4	5	6	7	8
9	0 -9	+2 -5	*	*	*	*	*	*	*	*	*
10		+9 0	+9 0	*	*	*	*	*	*	*	+15 0
11			+5 -2	*	*	*	*	*	*	*	*
12				*	*	*	*	*	*	*	*
1					*	*	*	*	*	+5 -5	+3 -3
2						+15 0	*	*	*	*	+15 0
3							*	*	*	*	+5 0
4								*	*	*	+9 0
5									0 -15	*	+5 0
6										*	+15 0
7											+3 -3
8	Reference Substrate and Case										

Maximum Current Ratings

TERMINAL No.	I_{IN} mA	I_{OUT} mA
9	1	0.1
10	5	0.1
11	0.1	5
12	1	0.1
1	1	0.1
2	100	0.1
3	0.1	100
4	10	1
5	0.1	100
6	100	0.1
7	1	0.1
8	1	10

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	MEASUREMENT TERMINAL Nos.	TEST CIRCUIT FIG.	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS
STATIC CHARACTERISTICS								
Photo Darlington Section: Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	E = 0 lumens/ft^2, I_C = 1 mA	10-11	–	10	–	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 0.1 mA, E = 0	9-11, 12-11	–	10	–	–	V
Dark Current	I_{DARK}	V_{CE} = 7.5 V, E = 0	10	3	–	0.1	30	μA
Photo Current	I_P	V_{CE} = 7.5 V, E = 8 lumens/ft^2	10	3	–	60	–	μA
Wavelength of Max. Sensitivity	λ max.			3	–	725	–	Note 2 nm
Relative Angular Sensitivity				–	–	–	–	–
Area of Each Photo Transistor				–	1.3 x 10^{-4} cm^2			
Amplifier Section Output Transistor: Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO\ 6}$ $V_{(BR)CEO\ 7}$	I_C = 1 mA	2-3, 6-5	–	15	–	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO\ 6}$ $V_{(BR)EBO\ 7}$	I_E = 1 mA	3-8, 6-8	–	5	–	–	V
DC Supply Current	I_{SUPPLY}	V_4 = 7.5 V	4	–	–	5.5	10	mA
Sensitivity: Illumination, For Normal "OFF" Output	E_{ON}	Set light input for I_6 = 70 mA	6	7, 15, 17	–	8	70	Notes 1, 3 lumens per ft^2
For Normal "ON" Output	E_{OFF}	Set light input for I_2 = 5 mA	2	7, 15, 17	–	10	–	Notes 1, 3 lumens per ft^2
DYNAMIC CHARACTERISTICS								
Overall Response Time: Turn-On Time	t_{on}	E = 700 μW/cm^2 at λ = 930 nm	——	12	–	38	–	μs
Rise Time	t_r			12	–	125	–	μs
Turn-Off Time	t_{off}			12	–	43	–	μs
Fall Time	t_f			12	–	20	–	μs

NOTES

(1) Tungsten filament light source at a color temperature of 2854K.
(2) One (1) nanometer = 10 Angstrom units.
(3) A radiant flux density of 7.5 μW/cm^2 at 725 nm produces the same photocurrent as 1 lumen/ft^2 from a tungsten filament lamp at a color temperature of 2854K.

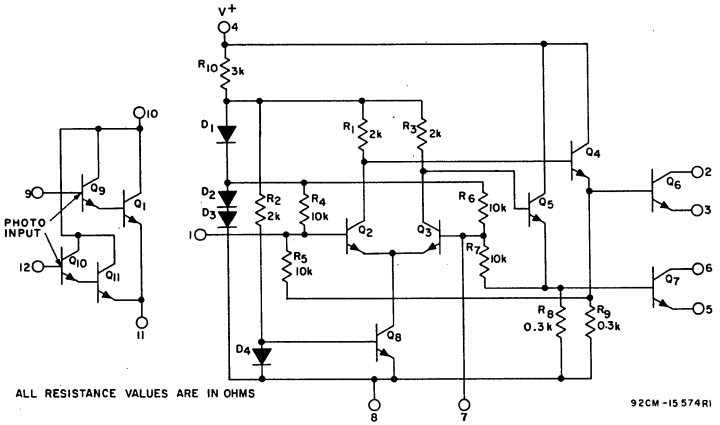


Fig. 2 - Schematic diagram of CA3062.

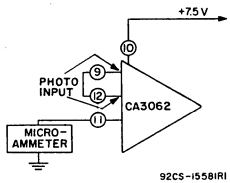


Fig. 3 - Test circuit for photocurrent and typical spectral response of photosensitive Darlington unit.

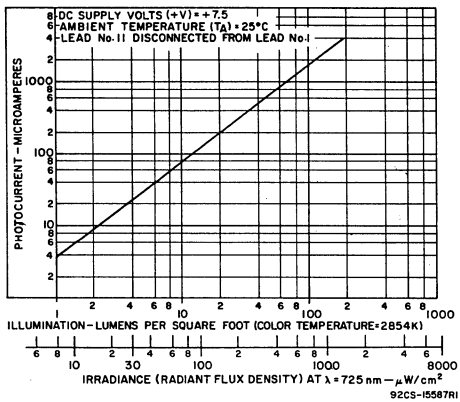


Fig. 4 - Photocurrent as a function of radiant flux.

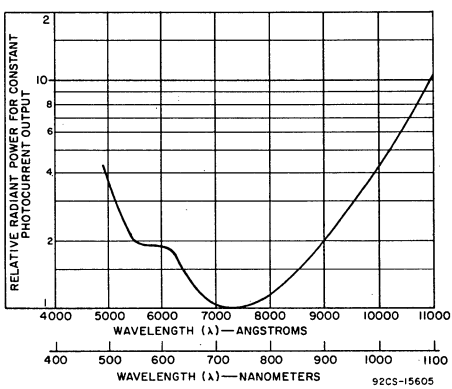


Fig. 5 - Typical spectral response of photosensitive Darlington unit.

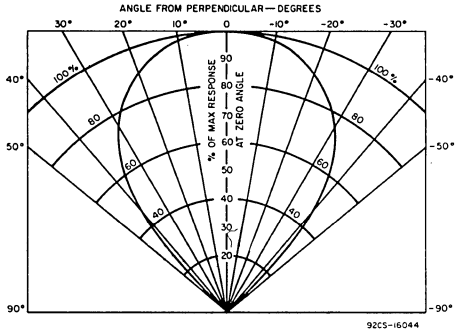


Fig. 6 - Relative angular sensitivity.

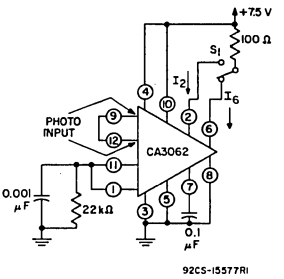


Fig. 7 - Test circuit for sensitivity and dc current measurement.

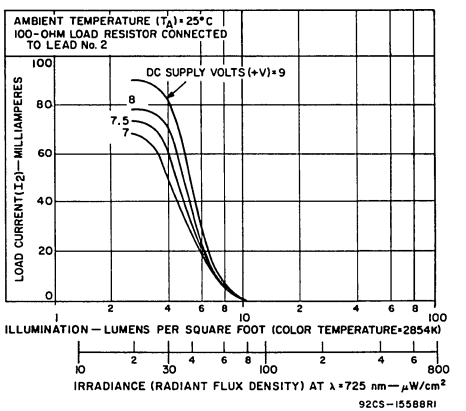


Fig. 8 - Load current (I_{2}) vs. illumination as a function of supply volts.

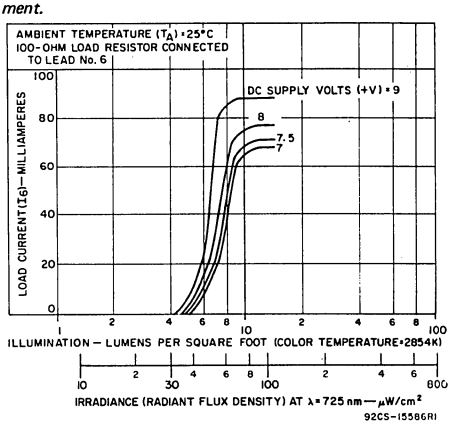


Fig. 9 - Load current (I_{6}) vs. illumination as a function of supply volts.

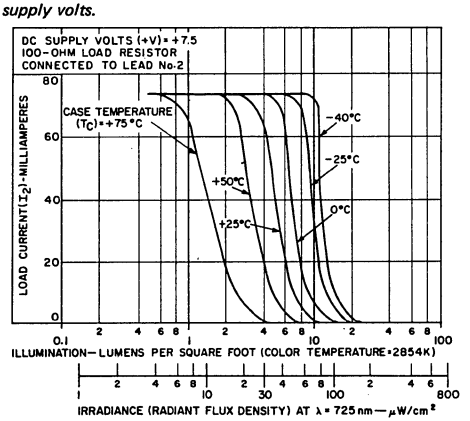


Fig. 10 - Load current (I_{2}) vs. illumination as a function of case temperature.

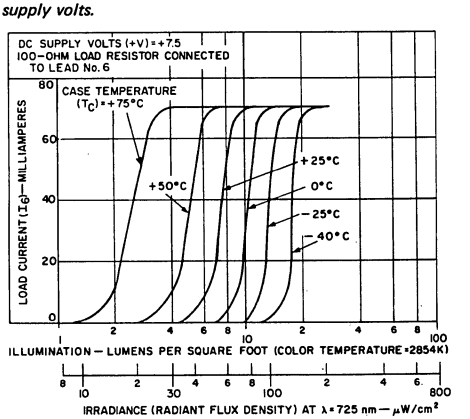


Fig. 11 - Load current (I_{6}) vs. illumination as a function of case temperature.

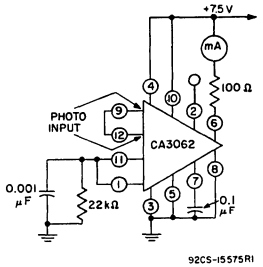


Fig. 12 - Response time test circuit.

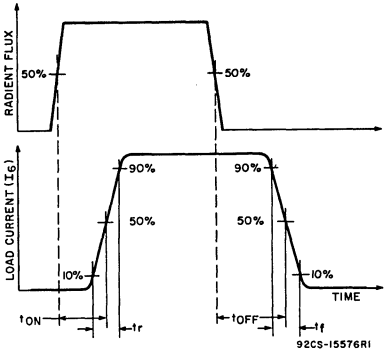


Fig. 13 - Waveforms for measurement of response time.

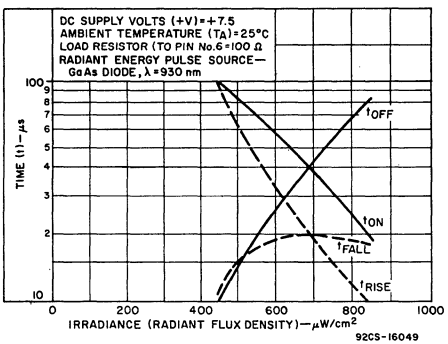


Fig. 14 - Response time as a function of radiant flux density.

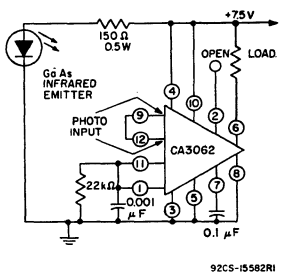


Fig. 15 - Circuit diagram for "ON-OFF" photoelectric control applications.

OPERATING CONSIDERATIONS

Switching Service

The CA3062 is primarily intended to provide "ON-OFF" output in response to a light signal. Optimum performance of this device is achieved when the output transistors are operated at values of load current sufficient to saturate the device in the "ON" state. Operation of the CA3062 at values of load current between the condition of no load current and saturation will cause substantial power to be dissipated in the silicon chip. This condition of operation is therefore not recommended because the heat rise in the silicon chip induced by the increased power dissipation causes the load current to shift in the same direction as though additional illumination were applied to the CA3062, a condition which will substantially alter the switching characteristics of the device.

The signal voltages at the input terminals (terminal No. 1 and No. 7) must not exceed 3 volts, because any increase in the signal voltage beyond the value specified will cause both output transistors to be turned "ON". In the circuit shown in Fig. 7, this condition will occur for values of illumination greater than 60 lumens/ft ^{2}. This adverse operating condition can be avoided by either limiting the maximum illumination or by clamping the input so that the voltage does not exceed 3 volts.

Linear Service

The CA3062 can be connected as shown in Fig. 16 to give a linear output. The value of the load resistor should be greater than 1000 ohms in order to limit the power dissipation and thus minimize the heating effects. Because of the many possible variations in circuit configurations, the CA3062 has not been characterized for linear service applications. A guide-line circuit for this class of service is shown in Fig. 16.

Specific inquiries for use of the CA3062 in this type of service should be addressed to your local RCA Field Technical Representative.

Precautions

Because of the high amplification of the CA3062, care should be taken, when wiring, to keep all lead lengths as short as possible. A recommended breadboard layout is shown in Fig. 17.

If the CA3062 is operated with an inductive load impedance, such as a relay, it is recommended that a diode be connected across the load to absorb the energy of the pulse voltages generated during switching.

Many of the graphs are shown with two sets of abscissa values for light energy input, one expressed in illumination values (lumens/sq. ft.) and the other in irradiance values (μW/sq. cm.)

Correlation between these two sets of abscissa values is accomplished by having the light source operating at the maximum sensitivity wavelength of the CA3062. See Notes on page three.

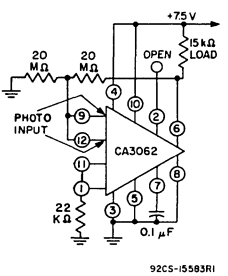


Fig. 16 - Circuit diagram for linear output photoelectric applications.

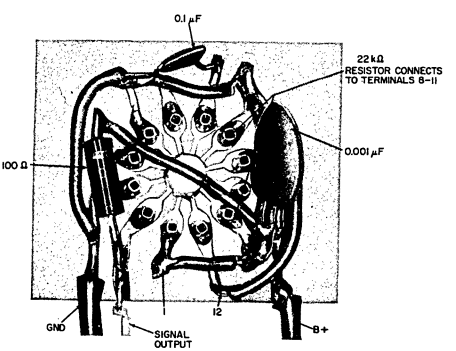


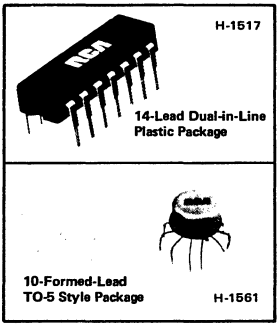
Fig. 17 - Breadboard layout of test circuit, shown in Fig. 7 for the CA3062.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3064
CA3064E



TV Automatic Fine Tuning Circuit

Features:

- Cascode type high-gain amplifier (18 mV input for rated output)
- Internal voltage regulator
- Differential detector
- For use with either color or monochrome
- Differential amplifier
- Bipolar outputs
- Wide operating-temperature range; −55 to +125°C

RCA-CA3064 and CA3064E represent the third generation of integrated circuits designed primarily for AFC (Automatic-Frequency-Control) applications. They provide all of the signal-processing components needed (with the exception of the tuned-phase-detector transformer) to derive the AFT correction signals from the output of the video-if amplifier. The CA3064 is supplied in the 10-formed-lead TO-5 style package, and the CA3064E in the 14-lead dual-in-line plastic package. Both types operate over the temperature range of −55 to +125°C.

The CA3064 and CA3064E are functionally similar to the CA3044 and CA3044V1 but embody a higher-gain input amplifier which provides a 20-dB improvement in sensitivity. The increased sensitivity extends the application of a proven AFT system to the low-level if-amplifier stages in TV receivers.

Because the CA3064 and CA3064E are functionally similar to the CA3044 and CA3044V1, refer to Application Note ICAN-5831, "Application of the RCA CA3044 and CA3044V1 Integrated Circuits in Automatic Fine-Tuning Systems" for general application information.

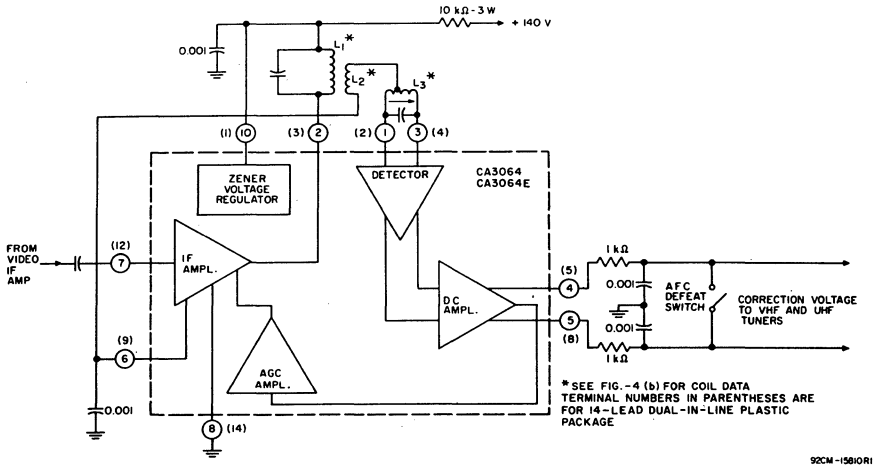


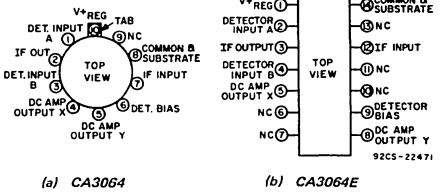
Fig. 1 – Block diagram of typical operating circuit utilizing the CA3064 and CA3064E.

MAXIMUM RATINGS, *Absolute-Maximum Values:*

DEVICE DISSIPATION:
- Up to T_A = 25°C . . . . . 700 mW
- Above T_A = 25°C . . . . . derate linearly 5.6 mW/°C

AMBIENT TEMPERATURE RANGE:
- Operating . . . . . −55 to +125°C
- Storage . . . . . −65 to +150°C

LEAD TEMPERATURE (During Soldering):
- At distance 1/16" ±1/32" (1.59 mm ±0.79 mm) from case for 10 s max.. . . . . 265°C



(a) CA3064 *(b) CA3064E*

Fig.2 – Terminal assignment diagrams.

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical terminal 2 (3) and horizontal terminal 6 (9) is +20 to 0 volts. Terminal nos. in parentheses are for the 14-lead dual-in-line plastic package.

TERMINAL No.	9(6,7, 10,11, 13)	10 (1)	1 (2)	2 (3)	3 (4)	4 (5)	5 (8)	6 (9)	7 (12)	8 (14)
9(6,7,♦ 10,11, 13)	← NO INTERNAL CONNECTION →									
10 (1)			+12 0	+10 -10	+12 0	+12 0	+12 0	+10 0	+20 0	▲
1 (2)				*	+10 - 10	*	*	+5 - 5	*	+5 - 6
2 (3)					*	*	*	+20 0	*	+20 0
3 (4)						*	*	+5 - 6	*	+5 - 6
4 (5)							*	*	*	+12 0
5 (8)								*	*	+12 0
6 (9)									+5 - 2	+2 0
7 (12)										+2 - 10
8 (14)										REF.SUBSTRATE & CASE■

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
9(6,7, 10,11, 13)	-	-
10 (1)	50	50
1 (2)	1	0.1
2 (3)	20	20
3 (4)	1	0.1
4 (5)	5	5
5 (8)	5	5
6 (9)	5	5
7 (12)	1	1
8 (14)	50	50

▲ Terminal number 10 (1) may be connected to any positive voltage source greater than the internal zener regulating voltage through a suitable dropping resistor – provided the dissipation rating is not exceeded.

■ This terminal should be connected to the most negative potential of the complete circuit.

* Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if the specified limits between all other terminals are not exceeded.

♦ It is recommended that unused terminals 6,7,10,11, and 13 on the 14-lead dual-in-line-plastic package and terminal 9 on the TO-5 package be grounded to act as shields.

ELECTRICAL CHARACTERISTICS at T_A = 25°C, Unless Otherwise Specified

CHARACTERISTICS	SYMBOLS	TEST CIRCUITS FIG.	TEST CONDITIONS		LIMITS CA3064, CA3064E MIN.	TYP.	MAX.	UNITS	CHARACTERISTIC CURVES FIG.
STATIC CHARACTERISTICS									
Device Dissipation	P_D	4	V^+ = 30V, R_S = 1.5kΩ	T_A					
				−25°C	-	135	150	mW	-
				+25°C	130	140	150		-
				+85°C	-	145	150		-
Current Drain at 10.5 Volts	I_T	4	$V_{10(1)}$ = 10.5 V		4	6.5	9.5	mA	-
Zener Regulated Voltage – DC Supply Voltage at terminal 10(1)*	$V_{10(1)}$	4	V^+ = 30 V, R_S = 1.5 kΩ		10.9	11.8	12.8	V	-
Quiescent Operating Current into Terminal 2(3)	$I_{2(3)}$	4			1	2	4	mA	-
Quiescent Operating Voltage at Terminal 4(5)	$V_{4(5)}$	-			5	6.9	8	V	-
Quiescent Operating Voltage at Terminal 5(8)	$V_{5(8)}$	-			5	6.9	8	V	-
Output Offset Voltage between Terminals 4 and 5(5 and 8)	V_{4-5} (5-8)	-			-1	0	1	V	-
DYNAMIC CHARACTERISTICS (AS RF AMPLIFIER IN TO-5 STYLE PACKAGE)									
Input Voltage Sensitivity	V_I sensitivity	5	V^+ = +30 V, V_I = 18 mV		Correction Voltage Output as shown in table below.				
Input Admittance	y_{11}	-	f = 45.75 MHz, V^+ = 30 V, R_S = 1.5 kΩ		-	0.41 + j1.0	-	mmho	-
Reverse Transfer Admittance	y_{12}	-			-	0 +j3.4	-	μmho	-
Forward Transfer Admittance	y_{21}	-			-	24.5 - j29	-	mmho	-
Output Admittance	y_{22}	-			-	0.04 + j0.9	-	mmho	-
OUTPUT vs FREQUENCY DEVIATION - AFC									
Correction-Control Voltage at Terminal 4(5)	$V_{corr.\ 4(5)}$	5	V^+ = +30 V, V_I = 18 mV RMS, f_0 = MHz as indicated	% of V_{10} (1)			% of V_{10} (1)		
			45.750 - 0.030	85	-	-	V		6,7
			45.750 + 0.030	-	-	25	V		
			45.750 - 0.900	80	-	-	V		7
			45.750 + 0.900	-	-	35	V		
			45.750 - 1.500		-	80	V		
			45.750 + 1.500	35	-		V		
Correction-Control Voltage at Terminal 5(8)	$V_{corr.\ 5(8)}$	5	45.750 - 0.030	-	-	25	V		6,7
			45.750 + 0.030	85	-	-	V		
			45.750 - 0.900	-	-	35	V		7
			45.750 + 0.900	80	-	-	V		
			45.750 - 1.500	35	-	-	V		
			45.750 + 1.500	-	-	80	V		

* Terminal numbers in parentheses are for 14-lead dual-in-line plastic package.

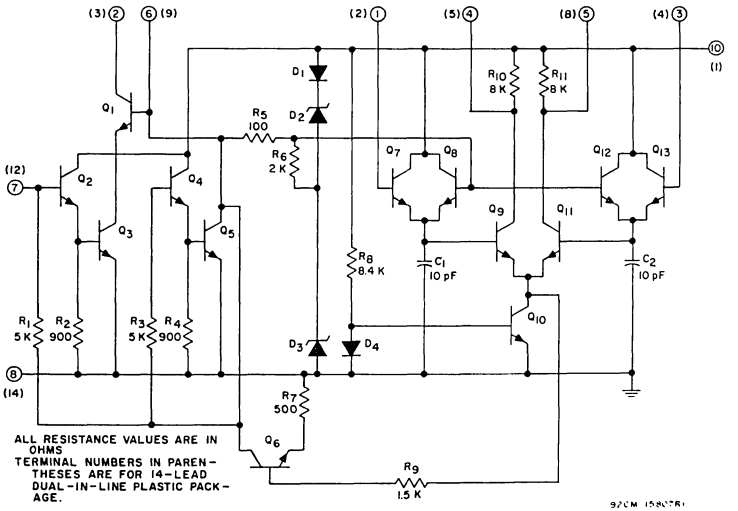


Fig.3 – Schematic diagram for CA3064 and CA3064E.

Circuit Description

The CA3064 and CA3064E integrated circuits can be considered as five functional blocks; an if amplifier-limiter, a balanced detector, a differential dc amplifier, an internally used AGC amplifier, and a zener voltage regulator. The 45-MHz amplifier limiter combination consists of emitter-follower input stage Q2 followed by a cascode-type amplifier Q1, Q3. The emitter-follower input stage Q2 is internally biased, therefore, capacitor coupling must be provided to the input at pin 7 (12) . The external load is connected to pin 2 (3) and should present a load impedance of about 1800 ohms at 45.75 MHz. The detector inputs at pins 1 (2) and 3 (4) from the external transformer are biased through the tertiary winding connected to pin 6 (9), which must be bypassed. The balanced detector is a high-efficiency type consisting of Q7/C1 and Q13/C2, which are internally biased by matching transistors Q8 and Q12. The dc amplifier consists of the differential-amplifier Q9, Q10, Q11, and D4.

The amplifier detector system provides the sharply defined pull-in characteristics shown in figures 5 and 6. The AGC amplifier Q6 senses the detected signals at the collector of A10 and adjusts the gain to compensate for signal changes such as airplane flutter conditions. Diodes D1, D2, and D3 provide the internal voltage regulation.

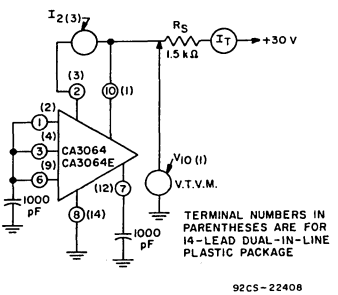
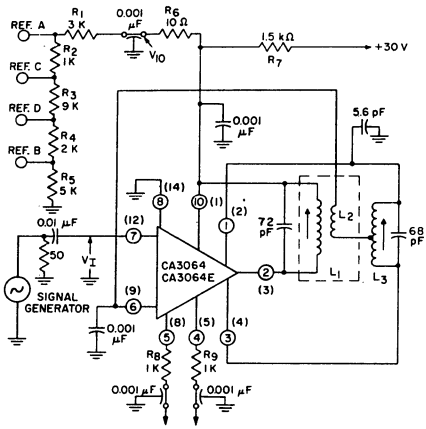


Fig.4 – Test setup: Measurement of total device dissipation, zener regulating voltage, quiescent operating current at terminal 2 (3).



ALL RESISTORS ARE 1% TOLERANCE AND ARE IN OHMS.
TERMINAL NUMBERS IN PARENTHESES ARE FOR 14-LEAD DUAL-IN-LINE PLASTIC PACKAGE

92CS-15813R1

L_1 IS ALIGNED FOR SYMMETRICAL BANDWIDTH ON EITHER SIDE OF 45.750 MHz
L_2 TERTIARY WINDING WOUND ON L_1 COIL FORM
L_3 IS ALIGNED FOR ZERO DIFFERENTIAL OUTPUT BETWEEN TERMINALS 4 AND 5 AT f_o = 45.750 MHz
* FOR COIL CONSTRUCTION DATA, SEE FIG.4(b).

REFERENCE VOLTAGE PERCENTAGES	
Ref. A	85% of $V_{10(1)}$
Ref. B	25% of $V_{10(1)}$
Ref. C	80% of $V_{10(1)}$
Ref. D	35% of $V_{10(1)}$

Coil	RCA Distributor Part No.
(L_1, L_2)	122 213
L_3	122 203

Fig.5 (a) — Correction voltage test circuit for CA3064 and CA3064E.

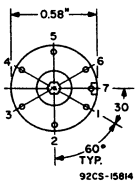
COIL DATA FOR DISCRIMINATOR WINDINGS

L_1 — Discriminator Primary: 3-1/6 turns; #20 Enamel-covered wire — close-wound, at bottom of coil form. Inductance of L_1 = 0.165 μH; Q_O = 120 at f_O = 45.75 MHz. Start winding at terminal #6; finish at Terminal #1. See Notes below.

L_2 — Tertiary Windings: 2-1/6 turns; #20 Enamel-covered wire — close wound over bottom end of L_1. Start winding at Terminal #3; finish at Terminal #4. See Notes below.

L_3 — Discriminator Secondary: 3-1/2 turns; center-tapped, space wound at bottom of coil form. Inductance of L_3 = 0.180 μH; Q_O = 150 at f_O = 45.75 MHz. Start winding at Terminal #2; finish at Terminal #5; connect center tap to Terminal #7. See Notes below.

Notes:
1. Coil Forms; Cylindrical; —0.30" Dia. max.
2. Tuning Core: 0.250" Dia. x 0.37" Length.
   : Material: Carbinal J or equivalent
3. Coil Form Base: See drawing below.
4. End of coil nearest terminal board to be designated the winding start end.



92CS-15814

Fig.5 (b) Coil form base terminal diagram.

The CA3064 and CA3064E are specifically intended for use in the AFT system of color television receivers. These devices are tested so that the control voltages generated by the circuit meet the critical requirements of the system. Figure 5 (a) is the schematic diagram of the test circuit.

Figures 5, 6, and 7 show the control voltages generated at terminals 4(5) and 5(8) of the Integrated Circuit as a function of the frequency deviation from the nominal center frequency. Figure 6 shows the region within 30 kHz of the center frequency while Figure 7 covers the entire bandwidth of the system. The horizontal reference lines on the figures are generated by a voltage divider connected between the power supply voltage on terminal 10(1) and ground. The dynamic control voltages are compared with these references according to the Output vs Frequency Deviation Table. For example: when the frequency deviation is —30 kHz the control voltage at terminal 4(5) is greater than the reference A voltage; the control voltage at terminal 5(8) is less than the reference B voltage.

The shape of the correction voltage characteristics is dependent to a large degree upon transformer characteristics and the parts layout. In order to closely duplicate the curves shown, the printed circuit boards shown in Figures 8 and 10 and the parts layouts shown in Figures 9 and 11 should be followed as closely as possible.

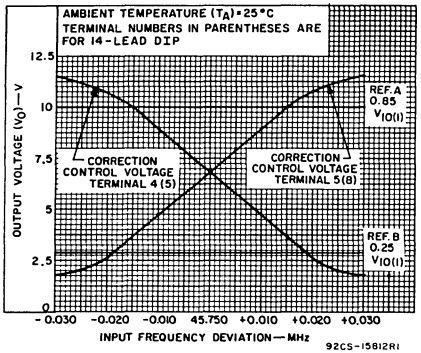


Fig.6 – Typical narrow-band dynamic control voltage characteristics.

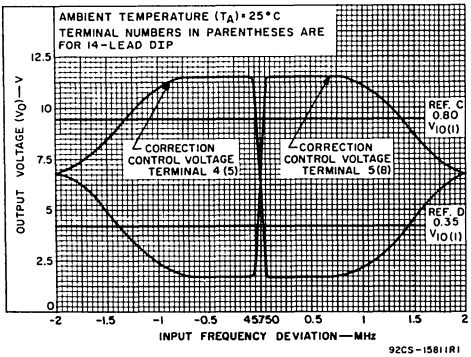


Fig.7 – Typical wide-band dynamic control voltage characteristics.

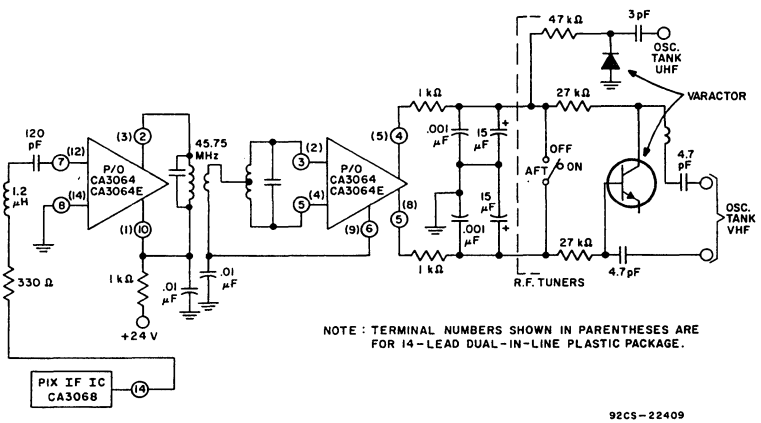
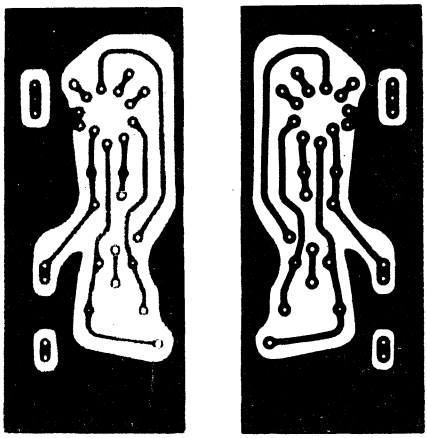


Fig.8 – Typical application of CA3064 and CA3064E AFT IC.



(a) Top view (b) Bottom view

Fig.9 – Printed circuit board for test circuit, (for TO-5 style package).

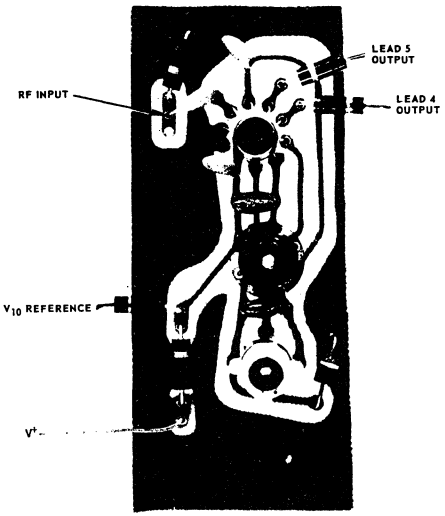
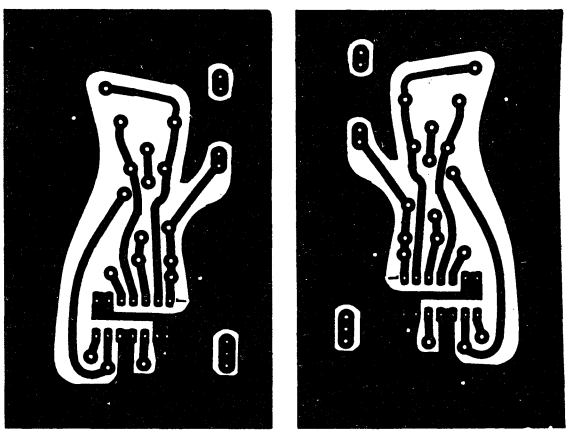


Fig.10 – Top view of wired test board (for TO-5 style package).



(a) Top view (b) Bottom view

Fig.11 – Printed circuit board for test circuit, (for 14-lead dual-in-line plastic package).

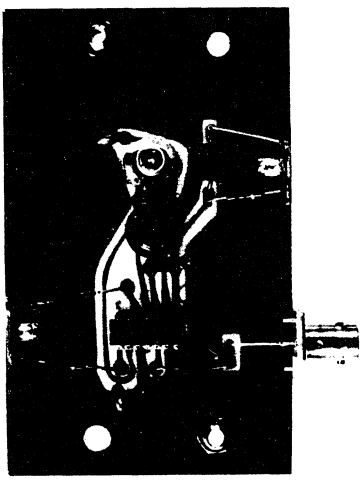


Fig.12 – Top view of wired test board (for 14-lead dual-in-line plastic package)

RCA
Solid State
Division

Linear Integrated Circuits

CA3065

IF AMPLIFIER-LIMITER, FM DETECTOR, ELECTRONIC ATTENUATOR, AUDIO DRIVER



For Television Sound-System Applications

The RCA CA3065 Television Sound System is a monolithic integrated circuit which combines a multistage IF amplifier limiter, an FM detector, an electronic attenuator, a zener diode regulated power supply, and an audio amplifier-driver that is designed to directly drive an npn power transistor or high-transconductance tube. Because the circuit is so inclusive, a minimum number of external components is required. A block diagram of the integrated circuit television sound system is shown in Fig. 1.

The CA3065 with its advanced circuit design provides a high-performance multistage subsystem for the sound system of a television receiver. A particular feature of the CA3065 is the electronic attenuator which performs the conventional volume control function. Volume control is accomplished when the bias levels in the attenuator are changed by means of a variable resistor connected between Terminal 6 and ground (attenuation in excess of 60 dB is attained). Because no audio signal is present in this control, hum or noise pickup can be bypassed. In most cases, only a single unshielded wire is required between the IF board and the variable resistor (volume control).

The CA3065 utilizes a 14-lead dual-in-line plastic package with leads specially formed to facilitate automatic insertion of the device into suitably punched printed-circuit boards.

FEATURES:

- Electronic attenuator - replaces conventional volume control
- Differential peak detector - requires one single tuned coil
- Internal Zener diode regulated supply
- Inherent high stability
- Excellent AM rejection - 50 dB typ. at 4.5 MHz
- Low harmonic distortion
- High sensitivity - 200 μV limiting (knee) at 4.5 MHz
- Audio drive capability - 6 mA p-p
- Undistorted audio output voltage - 7 V p-p

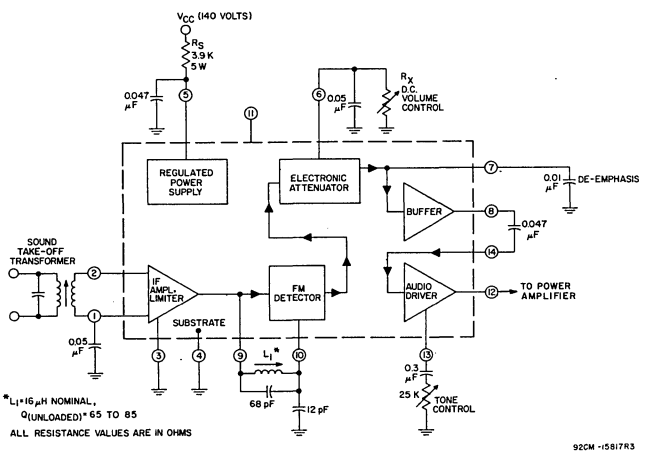


Fig. 1 - Block diagram of CA3065 in a typical circuit application.

MAXIMUM RATINGS, *Absolute Maximum Values*, at T_A = 25°C

Input Signal Voltage (between Terminals 1 and 2) . . .	±3	V
Power Supply Current (Terminal 5)	50	mA
Power Dissipation:		
Up to T_A = 25°C	850	mW
Above T_A = 25°C	Derate linearly 6.67	mW/°C
Ambient Temperature Range:		
Operating	−40 to +85	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C

MAXIMUM VOLTAGE RATINGS at T_A = 25°C

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range of the vertical terminal 9 with respect to terminal 3 is 0 to +4 volts.

TERMINAL No.	4	5	6	7	8	9	10	11	12	13	14	1	2	3
4		SUBSTRATE CONNECTION – ALWAYS CONNECT TO TERMINAL 3												
5			+13 0	+13 0	+13 0	*	*	INTERNAL CONNECTION DO NOT USE	+13 0	+13 0	*	*	*	NOTE 1
6				*	*	*	*		*	*	*	*	*	+13 −5
7					+1 −4	*	*		*	*	*	*	*	+13 0
8						*	*		*	*	*	*	*	*
9							*		*	*	*	*	*	+4 0
10									*	*	*	*	*	+4 −5
11									INTERNAL CONNECTION DO NOT USE					
12										+4 −1	*	*	*	*
13											*	*	*	*
14												*	*	+3 −5
1													+5 −5	+5 −5
2														+4 −5
3														

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
4	SUBSTRATE: CONNECT TO TERMINAL 3	
5	50	1
6	1	1
7	1	1
8	0.5	6
9	1	1
10	1	0.1
11	INT. CONN. DO NOT USE	
12	0.5	6
13	1	2
14	1	0.1
1	1	0.1
2	1	0.1
3	0.1	50

Note 1: Terminal No. 5 may be connected to any positive voltage through a suitable resistor provided that the current and dissipation ratings of the CA3065 are not exceeded.

*Voltages are not normally applied between these terminals. Voltages appearing between these terminals will be safe if specified limits between all other terminals are not exceeded.

ELECTRICAL CHARACTERISTICS *at T_A = 25° C, V_{CC} = +140 V applied to Terminal 5 through R_S = 3.9 kΩ, and DC Volume Control (R_x) = 0 unless otherwise indicated.*

CHARACTERISTIC	SYMBOL	SPECIAL TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
Static Characteristics						
Zener Regulating Voltage Terminal No. 5	V_5		10.3	11.2	12.2	V
Current into Terminal 5	I_5	Connect Terminal 5 to +9 V	10	16	24	mA
Total Device Dissipation	P_T		343	370	400	mW
Terminal Voltages: 1	V_1		–	2	–	V
6	V_6		–	4.8	–	
7	V_7		–	6.1	–	
9	V_9		–	3.7	–	
12	V_{12}		4	5.1	5.8	
Dynamic Characteristics						
IF AMPLIFIER Input Limiting Voltage (at −3 dB point)	$V_{i(lim)}$	f_o = 4.5 MHz, f_m = 400 Hz, Deviation = ±25 kHz,	–	200	400	μV
AM Rejection	AMR	Amplitude Modulation = 30% f = 4.5 MHz	40	50	–	dB
Transconductance Magnitude	$\|G_m\|$(IF)	f = 4.5 MHz IF Input Terminals: 2, 1 IF Output Terminals: 9, 3	–	500	–	mmho
Phase Angle	θ(IF)		–	46	–	degrees
Feedback Capacitance	C_{fb}	f = 1 MHz; Terminals 2 and 9	–	<0.02	–	pF
Input Impedance Components: Parallel Input Resistance	R_i(IF)	Measured between Terminal Nos. 1 and 2	–	17	–	kΩ
Parallel Input Capacitance	C_i(IF)	f = 4.5 MHz	–	4	–	pF
Output Impedance Components: Parallel Output Resistance	R_o(IF)	Measured between Terminal No. 9 and gnd	–	3.25	–	kΩ
Parallel Output Capacitance	C_o(IF)	f = 4.5 MHz	–	7.5	–	pF
DETECTOR Recovered AF Voltage	V_o(af)	f = 4.5 MHz; V_I = 100 mV Δf = ±25 kHz fm = 400 Hz	0.5	0.75	–	V(rms)
Total Harmonic Distortion	THD		–	0.9	2	%
Output Resistance: Terminal 7	R_o		–	7.5	–	kΩ
Terminal 8			–	300	–	Ω
ATTENUATOR		See Fig. 7				
Max. Attenuation	–	R_X = ∞	60	80	–	dB
Max. "Play-through" Voltage*	–	R_X = ∞	–	0.075	1	mV
AUDIO AMPLIFIER Voltage Gain	A(af)	V_I = 0.1 V(rms), f = 400 Hz	17.5	20	–	dB
Total Harmonic Distortion	THD	V_O = 2 V(rms), f = 400 Hz	–	1.5	–	%
Undistorted Output Voltage	–	THD = 5%, f = 400 Hz	2	2.5	–	V(rms)
Input Resistance	R_i(af)	f = 400 Hz	–	70	–	kΩ
Output Resistance	R_o(af)	f = 400 Hz	–	270	–	Ω

*"Playthrough" voltage is the unwanted signal, measured at Terminal 8, when the volume control is set for minimum output.

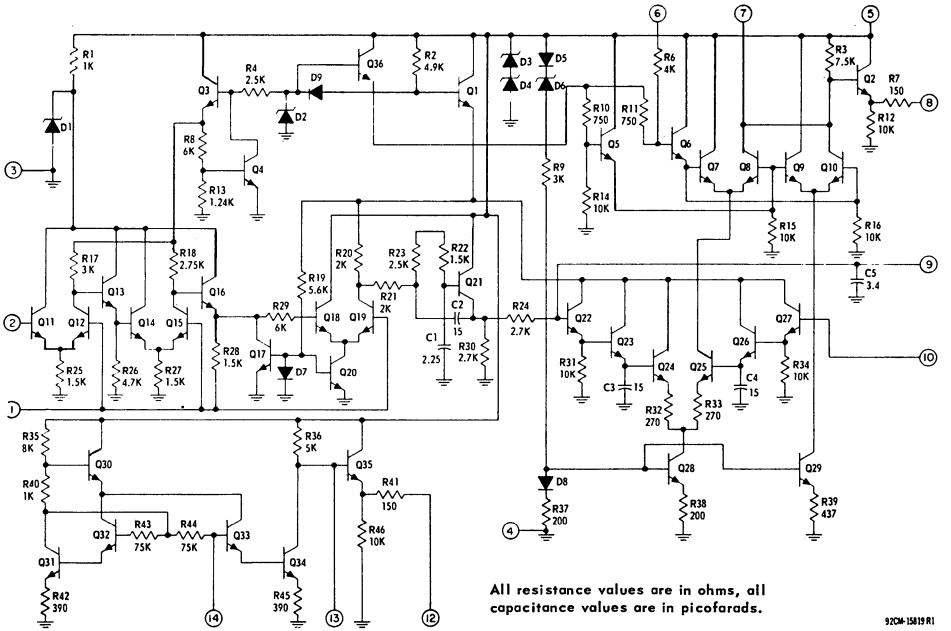


Fig. 2 - Schematic diagram of CA3065

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturers in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ±30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

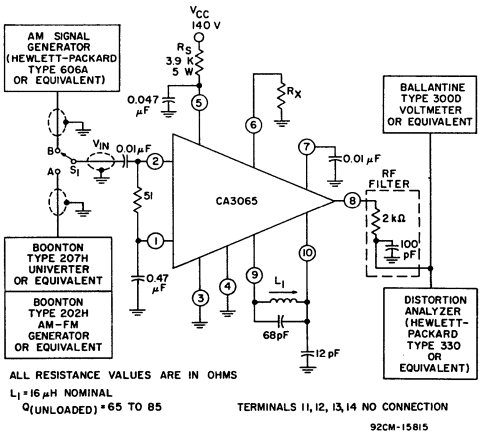


Fig. 3 - Input limiting voltage, AM rejection, recovered audio, total harmonic distortion, maximum attenuation, maximum "play-through" test circuit.

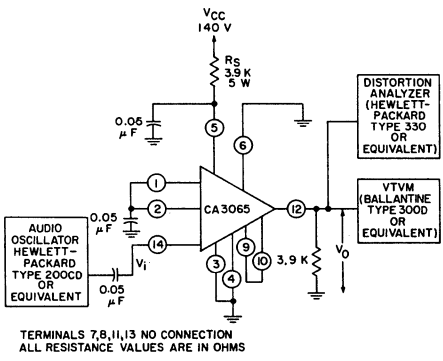
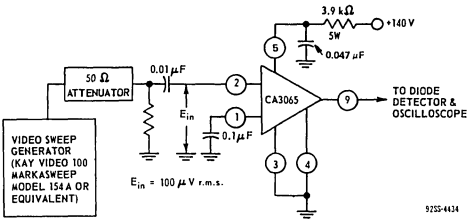
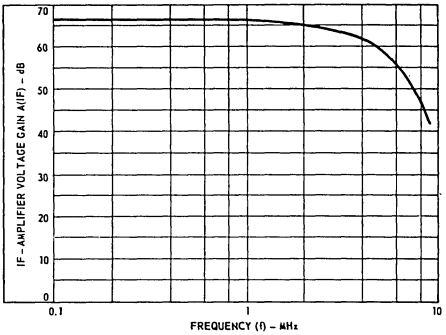


Fig. 4 - Audio voltage gain (undistorted output) test circuit.

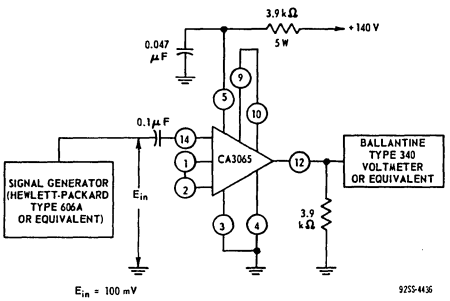


(a) Test circuit

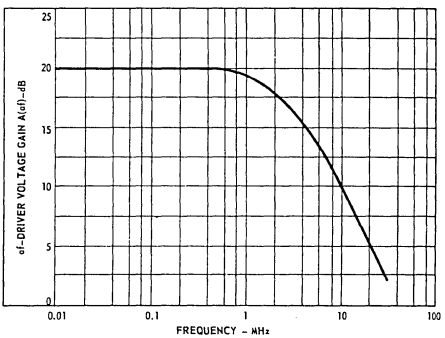


(b) Response curve

Fig. 5 - Frequency response of IF-amplifier section of CA3065



(a) Test circuit



(b) Response curve

Fig. 6 - Frequency response of af-amplifier section of CA3065

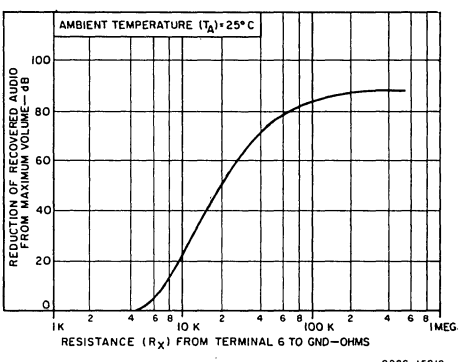


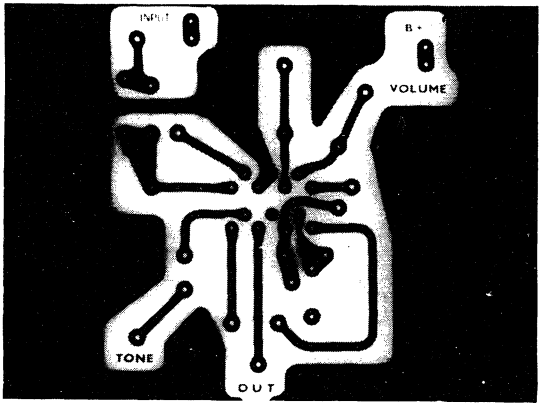
Fig. 7 - Gain reduction vs. resistance (terminal 6 to gnd)

OPERATING CONSIDERATIONS

The CA3065 may be used to drive a video output transistor or a high-transconductance output tube.

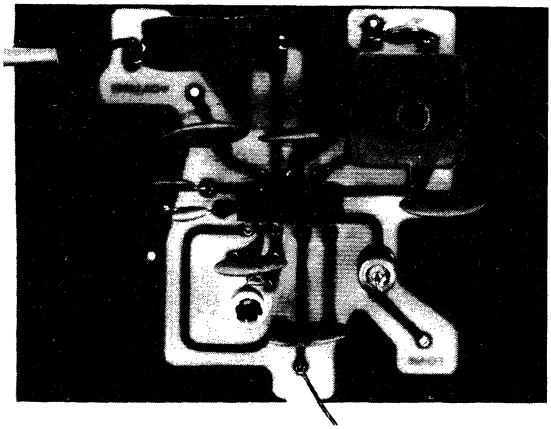
As in all TV receivers, precaution should be taken to prevent destruction of the CA3065 in the event of cascade arcs originating in the picture tube or in the output tube. In the case of arcing in the output tube a resistor of 150k in series with terminal No. 12 and the grid of the tube is usually sufficient protection.

To prevent damage from picture tube arcs, a careful analysis of board layout and coupling modes (electrostatic or magnetic) may be necessary to suggest alternate layouts or appropriate locations for the placement of spark gaps to absorb the high energy discharge.



*(a) Printed circuit board – bottom view**

92SS-4438



92SS-4439

*(b) Parts layout – top view**

Fig. 8 - Recommended parts layout for TV receiver sound strip using CA3065.

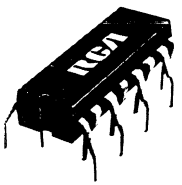
* A 200 mil square grid was used in the layout of passive components on the printed circuit board. The Quad-in-line formed leads conform to a standard grid spacing of 100 mil centers.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3066
CA3067



16-lead Quad-in-line plastic package

CA3066 – CA3067

H-1706

Television Chroma System

The RCA CA3066 and CA3067 are monolithic silicon integrated circuits that constitute a complete chroma system for color television receivers. The CA3066 provides subcarrier regeneration and total chroma signal processing prior to demodulation; the CA3067 performs the demodulation and tint control functions. Each device utilizes a 16-lead quad-in-line plastic package.

System Features

CA3066
CHROMA SIGNAL PROCESSOR

- Complete Color Sync Circuit
- Blanked Chroma Amplifier
- Chroma Band-Pass Amplifier
- Low Output Impedance Chroma Driver
- ACC Detector-Amplifier
- Killer Detector-Amplifier
- DC Chroma Gain Control
- Zener Diode for Regulated Voltage Reference
- Short-Circuit Protection on All Terminals

CA3067
CHROMA DEMODULATOR

- Balanced Chroma Demodulators
- Color Difference Matrix
- DC Tint Control
- Three Low Output Impedance Drivers for Direct Coupling
- Reference Subcarrier Limiter
- Zener Diode for Regulated Voltage Reference
- Internal RF Filtering

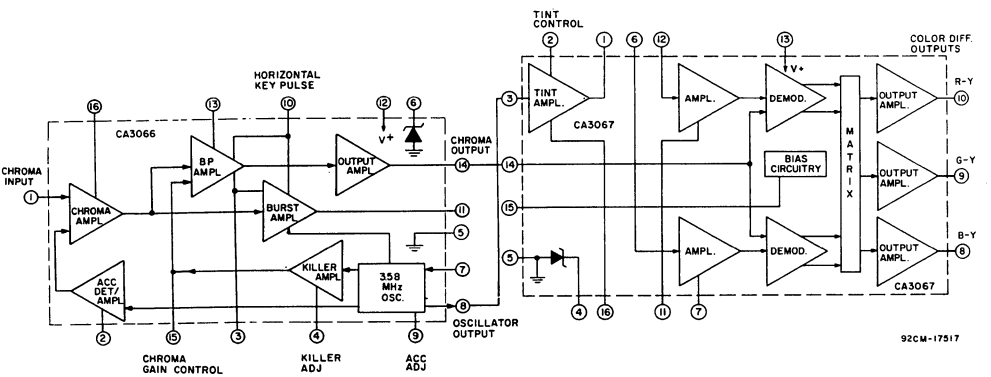


Fig. 1 - TV chroma system functional block diagram.

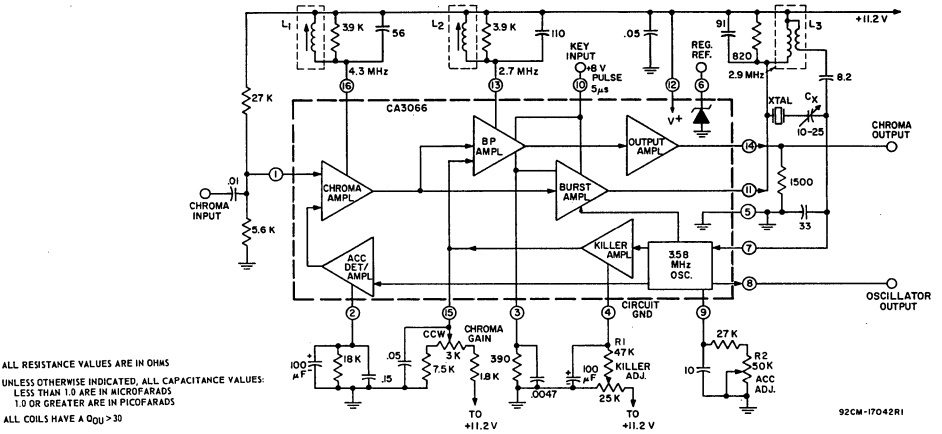


Fig. 2 - Functional diagram of CA3066.

The CA3066 contains substantially all the color processing circuitry exclusive of the tint control and demodulating circuits. The chroma amplifier sections of the CA3066 consist of the chroma and bandpass amplifiers. The chroma amplifier receives the chroma input signal at terminal No. 1. This amplifier is gain controlled by the automatic chroma control (ACC) detector-amplifier. The chroma signal is internally coupled from the output of the chroma amplifier to the input of the chroma bandpass amplifier and burst separator amplifier. The horizontal keying pulse (+8V) is used to gate the burst portion of the chroma signal from the input of the bandpass amplifier to the input of the burst separator amplifier. The bandpass amplifier is gain controlled by the dc chroma gain control and can also be controlled by the killer detector-amplifier. The bandpass amplifier output is internally coupled to the chroma output amplifier stage of the CA3066. The coils of the chroma amplifier and the bandpass amplifier are stagger-tuned to provide a combined typical bandpass of 3.08 to 4.08 MHz. The burst separator amplifier injects the burst signal into the 3.58 MHz oscillator. The oscillator amplitude is dependent on the terminal No. 9 impedance to ground and is also responsive to the burst signal amplitude at terminal No. 11. The ACC detector and killer detector sense the burst level or absence of burst, respectively, by monitoring the oscillators response to the burst injection level. The thresholds for the ACC and killer are independently adjusted by resistors R2 and R1 at terminals No. 9 and No. 4, respectively. The chroma output is at terminal No. 14 and the oscillator output is at terminal No. 8. Terminal No. 6 is a zener diode for use as a regulated voltage reference at 11.9 volts. When the zener reference element is not used, the power supply voltage should be maintained at 11.2 ± 0.5 volts.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$

Supply Voltages and Currents (see charts below)

Device Dissipation:

Up to $T_A = 70^oC$ . . . . . 600 mW

Above $T_A = 70^oC$ . . . . derate linearly 7.7 mW/°C

Ambient Temperature Range:

Operating . . . . . −40 to +85 °C

Storage . . . . . −65 to +150 °C

Lead Temperature (During soldering for 10s max. at not less than 1/32" from package) . . . +265 °C

Voltage with respect to Terminal No. 5.

Terminal No.	$V_{min.}$ (volts)	$V_{max.}$ (volts)
6	See Note N1	
7	–	–
8	–	–
9	—	–
10	−5.0	N2
11	0.0	18.0
12	0.0	12.0
13	0.0	15.0
14	–	–
15	0.0	N2
16	0.0	15.0
1	−5.0	5.0
2	–	–
3	–	–
4	–	–

Current

Terminal No.	I_I mA	I_O mA
6	20	0.1
7	5	0.1
8	1	2
9	0.1	2
10	1	0.1
11	10	1
12	50	1
13	10	1
14	0.1	6
15	3	1
16	6	1
1	1	0.1
2	0.1	2
3	0.1	20
4	1	1

N1 Terminal No. 6 is connected to a zener reference element, that, if used, should be biased by a positive voltage through a resistor that limits the current to a value which is less than the maximum current rating of terminal No. 6.

N2 The upper voltage limit cannot exceed the power supply input voltage at terminal 12.

ELECTRICAL CHARACTERISTICS at T_A = 25°C and V^+ = 11.2 V

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST FIG. AND CURVES
Static Characteristics							
Voltages:							
ACC Reference	V_2		–	0.5	–		
Burst-Chroma Ampl. Bias Current Term.	V_3		–	2.9	–		
Killer Reference	V_4		–	1.0	–		
Zener Reg. Reference	V_6		10.6	11.9	12.6	V	4
Oscillator Input	V_7		–	1.4	–		
Oscillator Output	V_8		–	2.35	–		
Balance (ACC Control)	V_9		–	1.65	–		
Chroma Output	V_{14}		–	4.6	–		
Currents:							
Total Supply	I_5		14	24	33		
Burst Separator Output	I_{11}	S_1 Closed	–	6.5	–	mA	
Band-Pass Ampl. Output	I_{13}		–	4.8	–		
Chroma Ampl. Output	I_{16}		–	1.27	–		
Dynamic Characteristics							
Oscillator Output	v_8	v_1 = 0 v_{p-p} v_1 = 1.25 v_{p-p}	0.8 –	1.2 2.5	– 3.5	v_{p-p}	6
Chroma Output:							
100%	v_{14}	v_1 = 1.25 v_{p-p}	0.5	1.0	–	v_{p-p}	6, 5
Killed		v_1 = 0.025 v_{p-p}	–	–	12		
ACC Detector Output	v_2	v_1 = 1.25 v_{p-p}	–	0.9	–	V	6
Small-Signal Input Resistance (Term. No.1)	r_i		–	50	–	kΩ	–
Small-Signal Input Capacitance (Term. No.1)	c_i		–	2.4	–	pF	
Small-Signal Output Impedance (Term. No.14)	r_o		–	250	–	Ω	

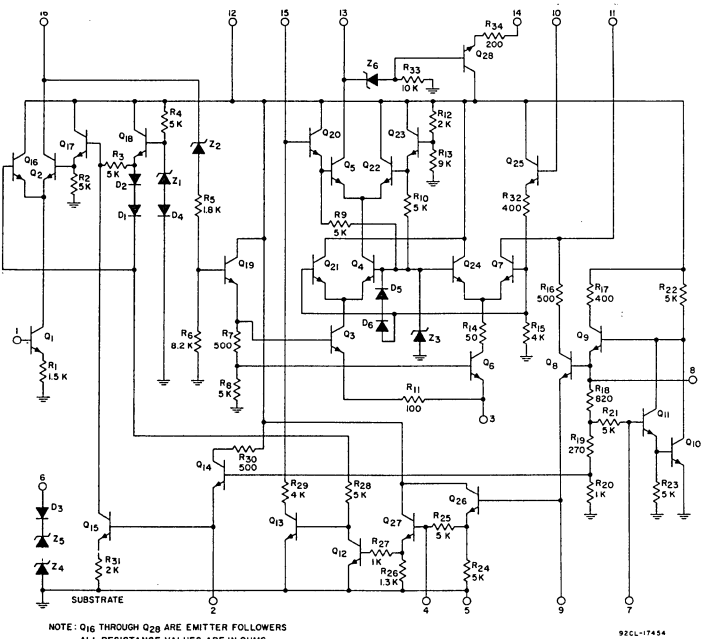


Fig. 3 - CA3066 schematic diagram.

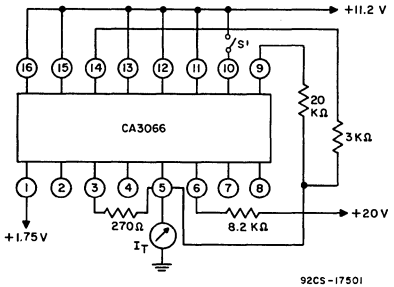


Fig. 4 - Static characteristics test circuit.

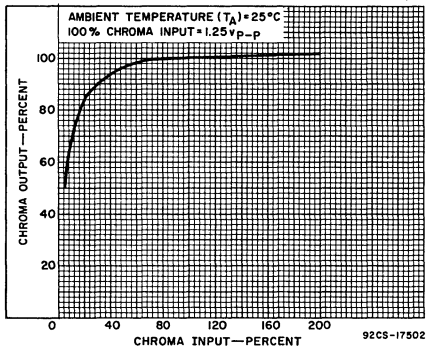


Fig. 5 - Typical ACC characteristic of chroma output vs chroma input.

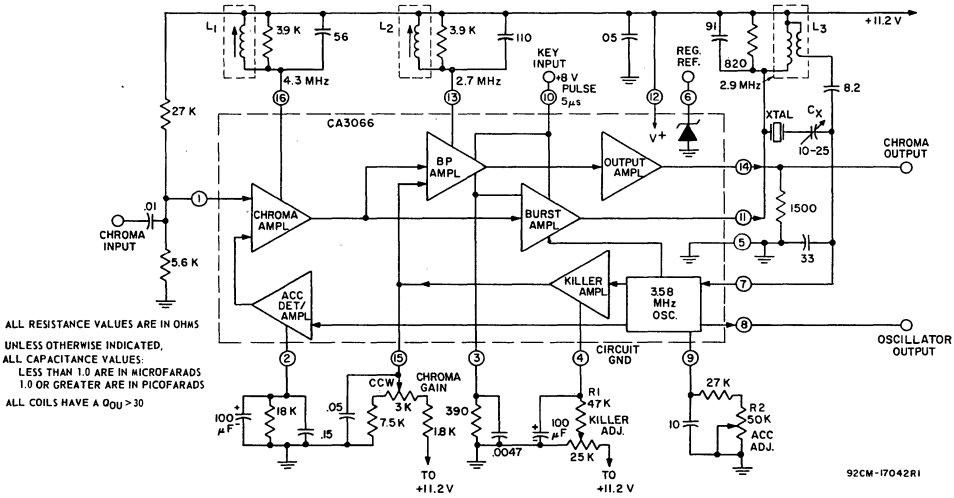


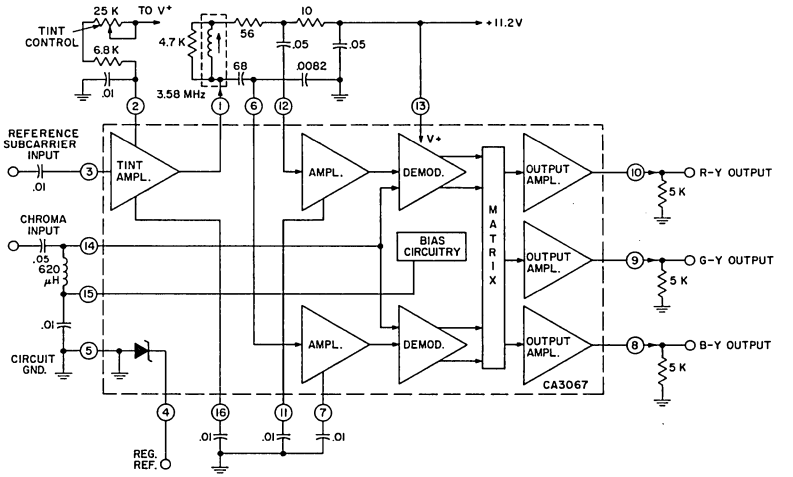
Fig. 6 - Dynamic characteristics test circuit.

DYNAMIC CHARACTERISTICS TEST PROCEDURE

Steps 1, 2, and 3 are performed with no Chroma input ($v_1 = 0$)

1. Adjust ACC potentiometer for $V_2 = +0.65V$.
2. Adjust Killer potentiometer for $V_4 = +1.2V$.
3. Adjust capacitor C_X (crystal trimmer) so that frequency of oscillator is 3.579545 MHz.
4. Unless otherwise noted, the chroma gain control is at maximum gain (fully clockwise).
5. The chroma input test signal is a 52.5 μs "line" at subcarrier frequency, and 10 cycles of burst at 46.5% of the "line" amplitude. The chroma input (v_1) is in peak-to-peak volts of "line" amplitude.
6. The chroma output (v_{14}) is the same as the chroma input (v_1) except that the burst is removed and keying overshoot occurs in the retrace period. The chroma output is in peak-to-peak volts of "line" amplitude.
7. The oscillator output (v_8) is the CW output at terminal No. 8 and is in peak-to-peak volts. Some modulation of oscillation dampening between burst injection is visible.

CA3067 Chroma Demodulator



ALL RESISTANCE VALUES ARE IN OHMS

UNLESS OTHERWISE INDICATED, ALL CAPACITANCE VALUES LESS THAN 1.0 ARE IN MICROFARADS 1.0 OR GREATER ARE IN PICOFARADS

Fig. 7 - Functional diagram of CA3067.

The CA3067 contains the separate functional systems of a dc tint control and a demodulator. The phase shift of the tint amplifier system is accomplished by functional control of the fixed phase signal from the CA3066 oscillator output. This regenerated reference subcarrier is applied to terminal No. 3 and driven differentially into phase shift circuits. The tint adjustment controls the vector addition of phase shifted signals after which a limiting amplifier removes any remaining amplitude modulation. The output of the tint amplifier at terminal No. 1 is phase separated for the required reference subcarrier phase at terminal No. 6 and No. 12 (terminal No. 12 lags terminal No. 6 by approximately 76°). These terminals are inputs to the demodulator drive amplifiers. The demodulators consist of two sets of balanced detectors which receive their reference subcarrier from the demodulator drive amplifiers. The chroma signal input from the CA3066 is applied to terminal No. 14. The chroma signal differentially drives the demodulators. The demodulation components are matrixed and dc-shifted in voltage to give R–Y, G–Y, and B–Y color difference components with close dc balance and proper amplitude ratios. The output amplifiers of the CA3067 are specially designed to meet the low-impedance driving source requirements of the high-level color output amplifiers. A special feature of the CA3067 is R–C filtering of high frequency demodulation components. Terminal No. 4 is a zener diode for use as a regulated voltage reference at 11.9V. When the zener reference element is not used, the power supply should be maintained at +11.2 ±0.5 volts.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$

Supply Voltages and Currents (see charts below)

Device Dissipation:

Up to $T_A = 70^oC$ . . . . . 600 mW

Above $T_A = 70^oC$ . . . . derate linearly 7.7 mW/^oC

Ambient Temperature Range:

Operating . . . . . −40 to +85 ^oC

Storage . . . . . −65 to +150 ^oC

Lead Temperature (During soldering for 10s max. at not less than 1/32" from package) . . . +265 ^oC

N1 Terminal No. 4 is connected to a zener reference element, that, if used, should be biased by a positive voltage through a resistor that limits the current to a value which is less than the maximum current rating of terminal No. 4.

N2 The upper voltage limit cannot exceed the power supply input voltage at terminal 13.

N3 Terminal No. 16 should be bypassed for normal operation.

Voltage with respect to Terminal No. 5

Terminal No.	$V_{min.}$ (volts)	$V_{max.}$ (volts)
6	0	N2
7	0	N2
8	0	N2
9	0	N2
10	0	N2
11	0	N2
12	0	N2
13	0	12
14	−3	N2
15	0	N2
16	N3	N3
1	0	15
2	0	N2
3	0	5
4	N1	

Current

Terminal No.	I_i (mA)	I_o (mA)
6	3	3
7	3	3
8	20	20
9	20	20
10	20	20
11	3	3
12	3	3
13	50	1
14	1	0.1
15	6	2
16	N3	N3
1	3	3
2	3	0.1
3	3	3
4	20	0.1

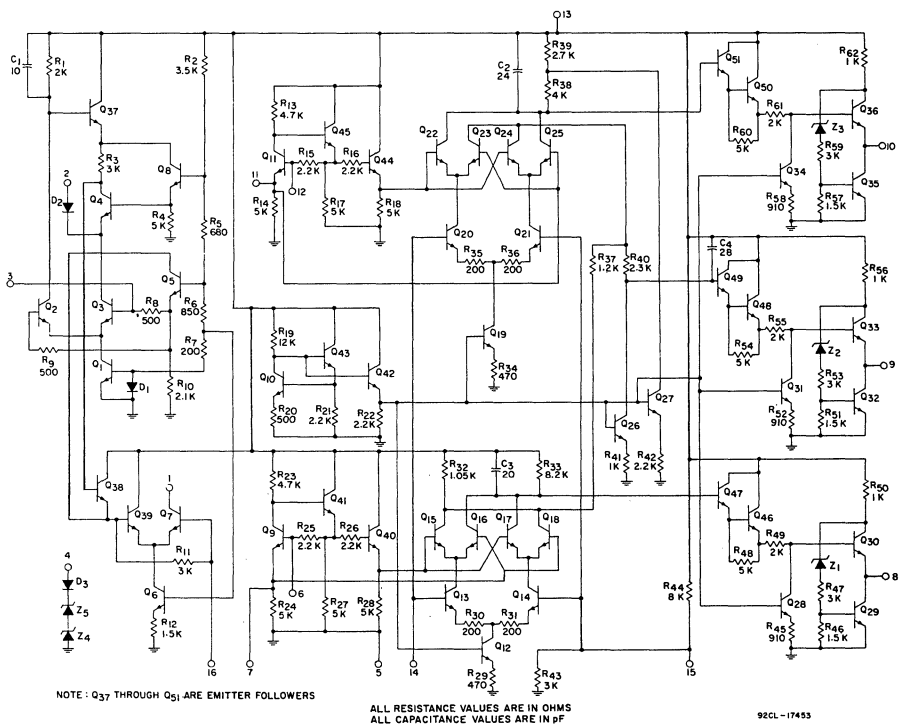


Fig. 8 - CA3067 schematic diagram.

ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$ and $V^+ = 11.2$ V

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST FIG. AND CURVES
Static Characteristics							
Voltages:							
Tint Control Input	V_2	$I_2 = 0.25$ mA	–	3.5	–	V	9
Reference Subcarrier	V_3		–	2.1	–		
Zener Regulator Ref.	V_4		10.6	11.9	12.6		
B–Y, R–Y Oscillator Ref. Inputs	V_6, V_{12}		–	5.7	–		
Balance (B–Y, R–Y)	V_7, V_{11}		–	5.0	–		
B–Y, G–Y, R–Y Outputs	$V_{8, 9, 10}$		4.2	5.0	5.8		9, 11, 12
Difference Outputs*	$\Delta V_8, \Delta V_9, \Delta V_{10}$		–0.3	–	0.3		9
Chroma Inputs	V_{14}, V_{15}		–	3.0	–		
Tint Ampl. Balance	V_{16}		–	4.7	–		
Currents:						mA	
Tint Ampl. Output (min.)	I_1(min.)	$V_{16} = 8$ V	0.16	0.37	–		
Total Supply	$I_1 + I_{13}$		15	24	33		
Dynamic Characteristics							
Tint Amplifier Output							10
Sensitivity	V_1	$V_3 = 7$ mV (RMS)	160	250	–	mV (RMS)	
Limiting Knee		$V_3 = 35$mV (RMS)	–	300	–		
Limiting		$V_3 = 350$mV (RMS)	–	–	380		
Tint Ampl. Phase Ref.▲	ϕ_6	$V_3 = 70$mV (RMS)	185	220	235	deg.	
Tint Ampl. Phase Shift‡	$\Delta\phi_6$	$V_3 = 70$mV (RMS)	90	105	–	deg.	
Demodulated Chroma Output:							
R–Y	V_{10}	$V_3 = 70$mV (RMS), $V_{14} = 35$mV (RMS)	150	250	–	V(RMS)	
Ratio of G–Y to R–Y	V_9/V_{10}		0.28	0.36	0.44		
Ratio of B–Y to R–Y	V_8/V_{10}		1.0	1.2	1.4		
Color Difference Output BW at 3.3 dB	$BW_{Diff.}$		450	550	–	kHz	
Color Difference Outputs (max. input signals):							
R–Y	v_{10}	$V_3 = 70$mV (RMS), $V_{14} = 212$mV (RMS)	–	3.0	–	V_{p-p}	
G–Y	v_9		–	1.1	–		
B–Y	v_8		–	3.6	–		
Small Signal Input Resistance						Ω	
Terminal No. 3	r_i		–	550	–		
Terminal Nos. 6 & 12			–	22	–		
Small Signal Output Resistance							
Terminal Nos. 8, 9, & 10	r_o		–	5	–		

*$\Delta V_8 = V_8 - \left(\frac{V_8 + V_9 + V_{10}}{3}\right)$, $\Delta V_9 = V_9 - \left(\frac{V_8 + V_9 + V_{10}}{3}\right)$, $\Delta V_{10} = V_{10} - \left(\frac{V_8 + V_9 + V_{10}}{3}\right)$

▲ Terminal No. 3 is phase reference

‡ read phase shift as tint control is varied

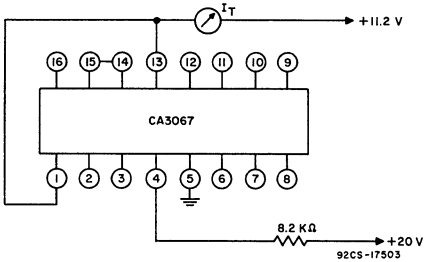


Fig. 9 - Static characteristics test circuit.

DYNAMIC CHARACTERISTICS TEST PROCEDURE

1. The reference subcarrier input (v_3) is a 3.58 MHz CW signal from a 50Ω source.
2. The chroma input (v_{14}) is a 3.53 MHz CW signal from a 50Ω source.
3. Phase and amplitude at terminal Nos. 1, 3, 6 and 12 are measured with a vector voltmeter (HP8405A or equivalent).
4. Signals at terminal Nos. 8, 9, and 10 are measured with an ac voltmeter (HP400E or equivalent) or an oscilloscope.
5. Unless otherwise noted the Tint control is at maximum resistance.

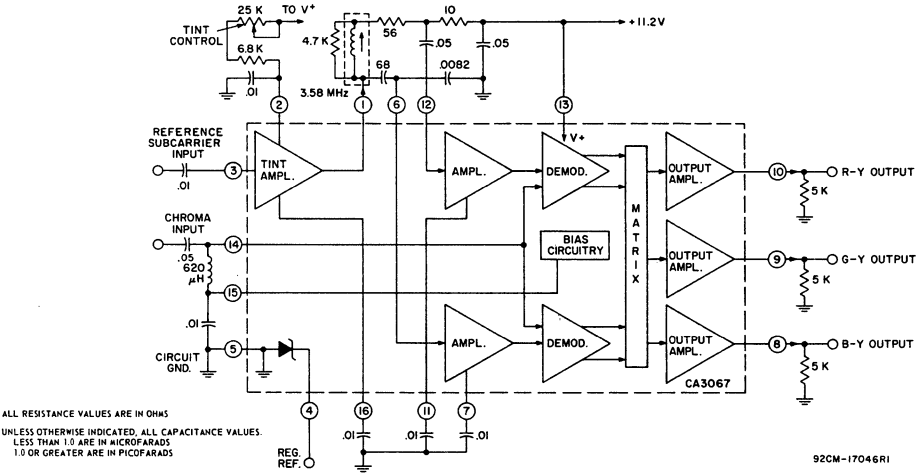


Fig. 10 - Dynamic characteristics test circuit.

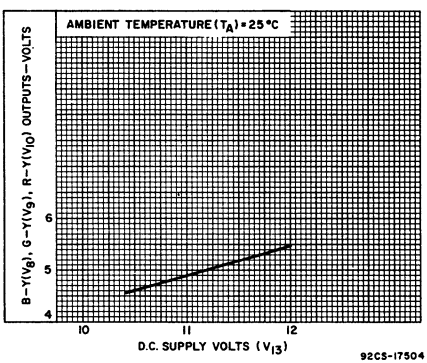


Fig. 11 - DC voltage at color-difference outputs vs supply voltage.

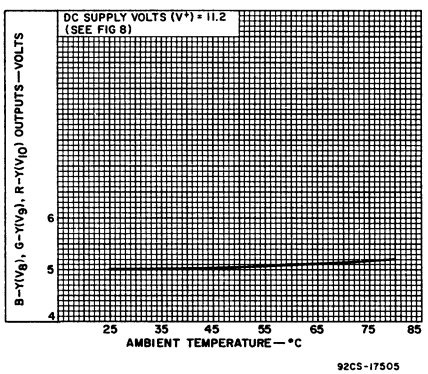


Fig. 12 - Temperature drift of DC voltage at color-difference outputs.

Application Information

TYPICAL CHROMA SYSTEM UTILIZING THE CA3066 AND THE CA3067

CA3066

A typical circuit using the CA3066 is shown in Fig. 13. This circuit is designed for a peak-to-peak chroma input level (v_1) of 1.25 volts, a horizontal keying pulse amplitude (V_{10}) of +8 peak volts, and a regulated supply votlage (V_{12}) of +11.2 volts. The chroma signal should be derived from the 1st or 2nd video amplifier and the luminance should be filtered out before the signal is applied to the CA3066 chroma input at terminal No. 1. For proper switching, the horizontal keying pulse (V_{10}) should be at least +7.5 peak volts but must not exceed the dc supply voltage level (V_{12}) which should be maintained at the recommended value of +11.2V. The dc supply can be externally regulated or the regulation circuit shown in Fig. 13 may be used. An RCA 2N3053 (or equivalent) transistor in an emitter follower configuration is used as a basic regulator in the circuits shown in Figs. 13 or 17. The zener diodes (connected to terminal No. 6 in the CA3066 or terminal No. 4 in the CA3067) are intended as reference-voltage sources for this circuit and may be used separately.

If either the CA3066 or CA3067 can be separately removed from the operating circuit, paralleling the zeners (to establish a regulator reference) is recommended to avoid excessive voltage on the remaining unit. For best voltage tracking and bias stability the zener diode reference element of the CA3066 should be used for the CA3066 supply voltage regulator circuit. The setup adjustments for the circuit of Fig. 13 are the killer (R_1), automatic chroma control (R_2), and oscillator frequency (c_x). The chroma gain control is a dc adjustment that controls the color drive level to the demodulator circuit and is normally a front panel adjustment. The killer and ACC adjustments are initial setup controls to optimize performance. The killer control (R_1) setting adjusts the threshold level at which the chroma bandpass amplifier will be cutoff. This threshold level is normally set at +1.2 V at terminal No. 4. The ACC adjustment (R_2) controls the oscillator loop gain and sets the ACC threshold level at which the chroma output signal ceases to increase linearly with increases in the chroma-input-signal level. When R_2 is properly adjusted, the voltage at terminal No. 2 is +0.6 to +0.7 volts (normally set at +0.65 volts).

The L_1 coil in Fig. 13 has two slugs, one for setting the frequency and another which serves as a Q "spoiler." In this way it is possible to control the tilt of the chroma bandpass frequency response and to compensate for overall-system phase errors. Coils L_1 and L_2 are single-tuned; the transformer T_1 is fix-tuned. The secondary of T_1 provides the reverse phase signal to neutralize the 3.58 MHz crystal and, with the series 12 pF capacitor, provides the correct compensation to terminal No. 7. An adjustable trimmer capacitor in series with the crystal is set for a free-running frequency of 3.579545 MHz ±10 Hz and will, for the typical circuit shown, stay within a nominal drift variation of 30 Hz during warm up.

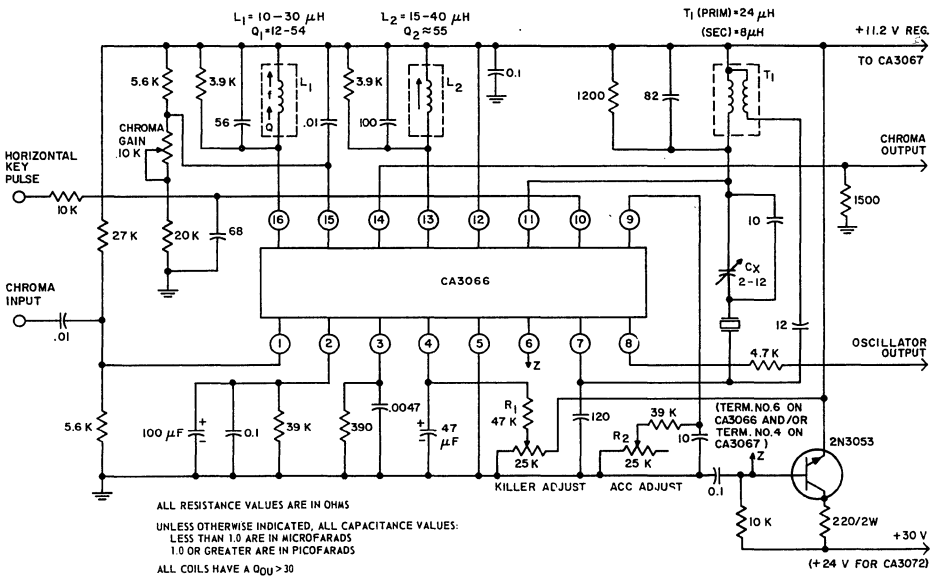


Fig. 13 - CA3066 chroma amplifier-oscillator circuit.

92CM-17506

System performance curves for the CA3066 are shown in Fig. 14. The chroma and oscillator outputs and the killer and ACC reference voltage are plotted as a function of the input chroma signal. Because the killer threshold is a function of the killer reference voltage, a typical curve for the threshold variation is shown in Fig. 15. This curve was generated for various settings of R_1 (killer reference points) with no signal applied to terminal No. 1. At each setting a signal was applied and reduced in magnitude until the bandpass amplifier was cutoff by the killer amplifier. Oscilloscope photographs of the terminal voltage signals and frequency response curves are shown in Fig. 16.

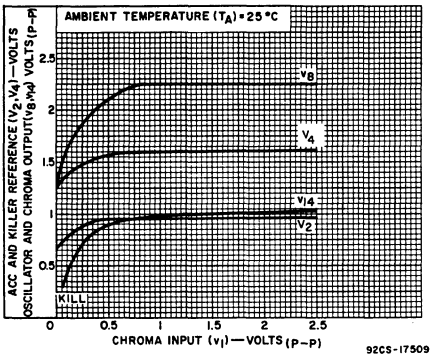


Fig. 14 - Typical chroma system parameters vs NTSC chroma input signal for CA3066.

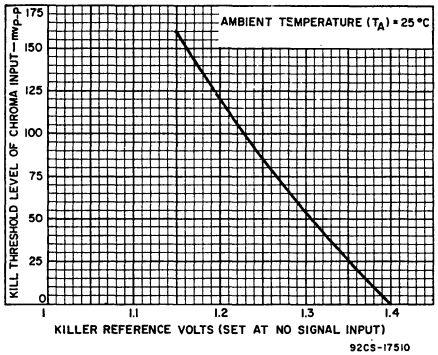
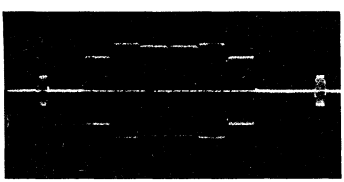
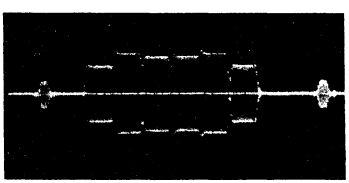


Fig. 15 - Typical killer threshold of chroma input vs killer reference voltage (V_4) using NTSC signal.

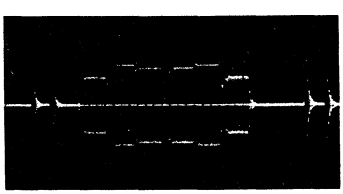
Fig. 16 a thru 16 k



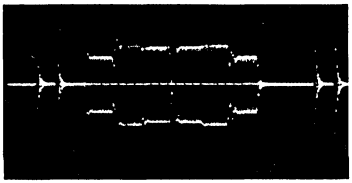
(a) Terminal No. 1.
One horizontal line
1.25 v_{p-p} of NTSC signal at chroma input (v_1 = 1.25 v_{p-p}).



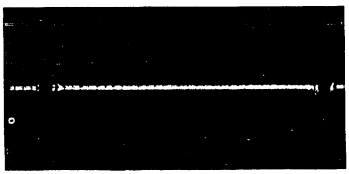
(b) Terminal No. 16.
One horizontal line
0.2 v_{p-p} of chroma amplifier output (v_1 = 1.25 v_{p-p}).



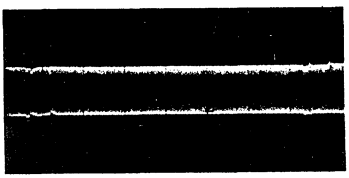
(c) Terminal No. 13.
One horizontal line
1.0 v_{p-p} bandpass amplifier output (v_1 = 1.25 v_{p-p}).



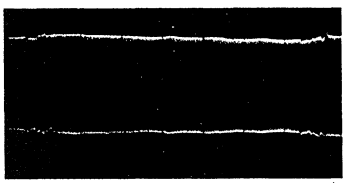
(d) Terminal No. 14.
One horizontal line
1.0 $v_{p\text{-}p}$ of chroma output (v_1 = 1.25 $v_{p\text{-}p}$).



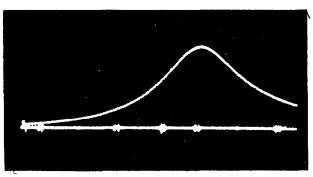
(e) Terminal No. 11.
One horizontal line
2.3 $v_{p\text{-}p}$ of separated burst (v_1 = 1.25 $v_{p\text{-}p}$).



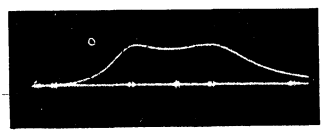
(f) Terminal No. 8.
One horizontal line
1.2 $v_{p\text{-}p}$ of oscillator output with no input signal (v_1 = 0).



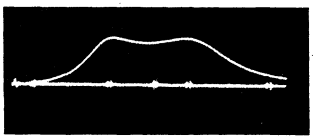
(g) Terminal No. 8.
One horizontal line
2.5 $v_{p\text{-}p}$ of oscillator output (v_1 = 1.25 $v_{p\text{-}p}$).



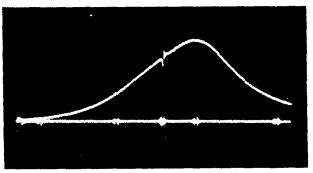
(h) Terminal No. 16.
Frequency response sweep 0.5 MHz/horizontal division peak response at 4.08 MHz (Terminal No. 4 connected through 24 kΩ to +11.2V).



(i) Terminal No. 13.
Frequency response sweep 0.5 MHz/horizontal division (terminal No. 4 connected through 24 kΩ to +11.2V).



(j) Terminal No. 14.
Frequency response sweep 0.5 MHz/horizontal division (terminal No. 4 connected through 24 kΩ to +11.2V).



(k) Terminal No. 11.
Frequency response sweep 0.5 MHz/horizontal division
Terminal No. 4 connected through 24 kΩ to +11.2V
Terminal No. 7 connected through 4.7 kΩ to +11.2V
Terminal No. 10 connected through 10 kΩ to +11.2V

CA3067

The Tint Amplifier-Demodulator, CA3067 is shown in Fig. 17. The oscillator output from Terminal No. 8 of the CA3066 is buffer-connected through a 4.7 KΩ resistor to the reference subcarrier input, Terminal No. 3. The chroma output from the CA3066, available on Terminal No. 14, is connected through a series tuned circuit consisting of a 150 pF capacitor, a 560Ω resistor, and a 47 μH coil to terminal Nos. 14 and 15. Terminal Nos. 14 and 15 are biased through an interconnected choke network to provide a balanced bias to the chroma demodulator drivers Q13 and Q14. If desired, the phase polarity of the output of the CA3067 circuit can be reversed by reversing the input connections at terminal Nos. 14 and 15. The regulated 11.2 V dc supply voltage for the CA3067 is obtained from Terminal No. 12 of the CA3066.

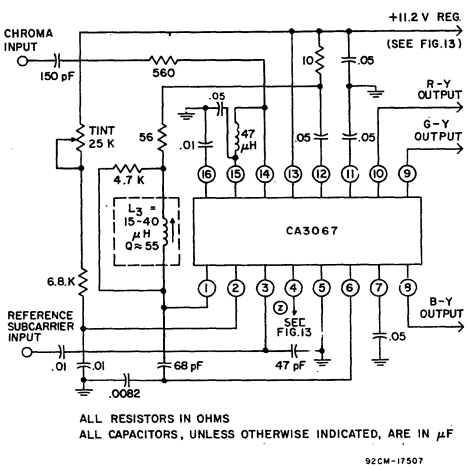


Fig. 17 - CA3067 tint control-chroma demodulator circuit.

In Table I the amplitude and phase values are given with the 0° phase reference at terminal No. 3 and the tint amplifier adjusted to a B–Y signal reference which can be recognized by the waveform on terminal No. 8. Typical terminal voltage values are given for the CA3066 and CA3067 in Table II.

TABLE I – Typical Voltage and Phase Relationships for the CA3067 Tint-Control Amplifier.

TERMINAL NO.	AC VOLTAGE-mv	PHASE ANGLE
3	70	0°
1	200	– 93°
6	1.5	– 67°
12	2.5	–143°

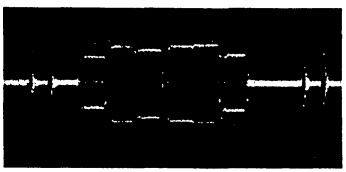
Reference Condition: **Tint control centered on B–Y phase at terminal No. 8.**

TABLE II – Typical DC Terminal Voltages with no Input Signals for CA3066 and CA3067.

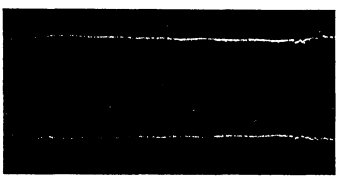
TERMINAL NO.	DC VOLTS	
	CA3066	CA3067
1	1.75	11.2
2	0.68	3.5
3	2.8	2.1
4	1.25	11.9
5	0	0
6	11.9	5.7
7	1.4	5.0
8	2.2	5.0
9	1.9	5.0
10	0	5.0
11	11.2	5.0
12	11.2	5.7
13	11.2	11.2
14	4.6	3.0
15	4.4	3.0
16	11.2	4.8

The demodulation angles are determined by the phase of the reference subcarrier signals at terminal Nos. 6 and 12. These signals are amplified and applied to the demodulators such that their respective demodulated signals are present at terminal Nos. 8 and 10. The phase shift network from terminal No. 1 resolves the signal into two components that are phase separated by 76°. Relative to the terminal No. 6 phase, which is directly represented by the B–Y phase, the terminal No. 12 phase is shifted 180° and the demodulation angle at terminal No. 10 is 180° minus 76° or typically 104°. While the output signals at terminals Nos. 8, 9, and 10 are given as B–Y, G–Y, and R–Y respectively, it is obvious that the phase angles as recognized by the waveforms in the oscilloscope photographs of Fig. 18 are not precisely the NTSC standard representation of color difference signals. The latest developments in color TV picture tubes, such as the 18VANP22, require some phase shift for color correction.

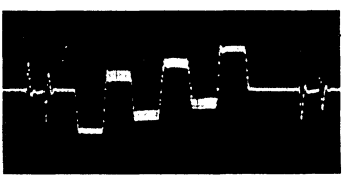
Fig. 18 a thru 18 e.



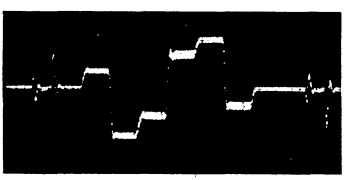
(a) Terminal No. 14.
One horizontal line
0.2 $v_{p\text{-}p}$ chroma input to demodulator.



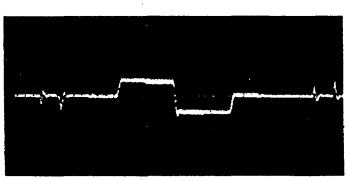
(b) Terminal No. 3.
One horizontal line
0.25 $v_{p\text{-}p}$ oscillator injection input to tint control amplifier.



(c) Terminal No. 8.
One horizontal line
1.0 $v_{p\text{-}p}$ at B–Y output.



(d) Terminal No. 10.
One horizontal line
1.2 $v_{p\text{-}p}$ at R–Y ouyput.



(e) Terminal No. 9.
One horizontal line
0.4 $v_{p\text{-}p}$ at G–Y output.

The tint amplifier of the CA3067 is unique in that all phase shift requirements are satisfied by dc bias control to terminal No. 2. Resistor R1 and capacitor C1 of Fig. 8 provide the basic requirements for a phase shifting of the tint-controlled signal. The reference subcarrier signal at terminal No. 3 is separated 180° by the differential amplifier Q2 and Q3. The output of Q2 is shifted in phase by the R1, C1 time constant. The output of Q3 is directed to a recombination adder junction at the collector of Q4. The tint control determines the Q4 output signal by directing more or less signal to ac ground through diode, D2. The tint-controlled signal is then passed through an amplifier-limiter circuit to terminal No. 1.

The output amplifiers of the CA3067 are very-low-impedance followers that allow for direct coupling to high-level amplifiers. As shown in Figs. 11 and 12, the difference outputs vary linearly with voltage and temperature. Typically, the red and blue difference outputs have a 3-volt peak-to-peak maximum voltage-swing capability with a 5 $k\Omega$ load.

CA3072 Alternate Demodulator Circuit

The circuit shown in Fig. 19 represents an alternate tint amplifier-chroma demodulator. This circuit provides greater color-difference output levels than the CA3067. When the CA3072-2N3933 demodulator and tint amplifier circuit is used in conjunction with the CA3066, +24 volts should be used to provide the proper V^+ for the CA3072. Both the 2N3053 and 2N3933 are typical of the type of transistors that may be used with the CA3066, CA3067, and CA3072 integrated circuits. For complete data information on the RCA types 2N3053, 2N3933, and CA3072, refer to their respective Technical Bulletins.

Construction Information

Fig. 20 is a photograph and template of a circuit board layout for the CA3066 and CA3067 combination. Particular information for most of the components is given in Figs. 13 and 17. Special attention must be given to bypassing at terminal Nos. 2 and 15 in the CA3066. Terminal No. 2 requires a high-Q capacitor (0.1 μF) in parallel with the 100 μF electrolytic bypass. Terminal No. 15 requires bypassing to the power supply lines for best results. To assure complete cutoff at the minimum chroma-gain-control setting, the power supply side of L2 must be well bypassed to ground, and preferably to a common ground point that also includes the 1500-ohm resistor at terminal No. 14 and the CA3067 terminal No. 15 bypass.

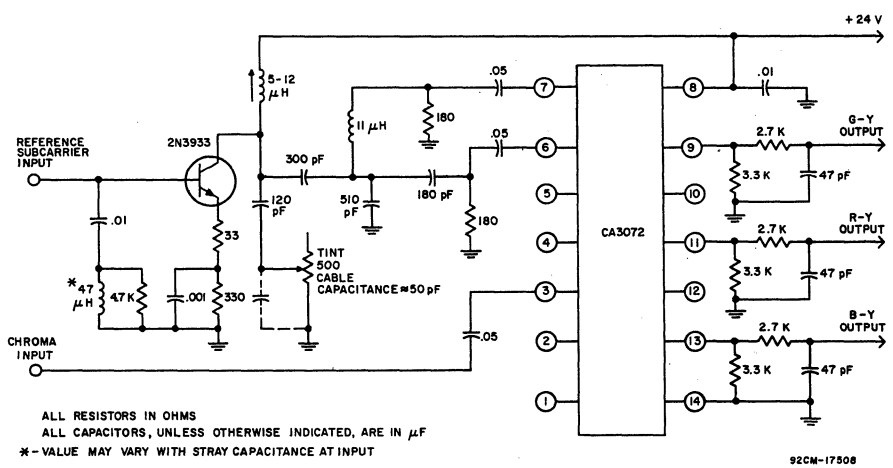


Fig. 19 - CA3072 chroma demodulator with 2N3933 tint control amplifier circuit.

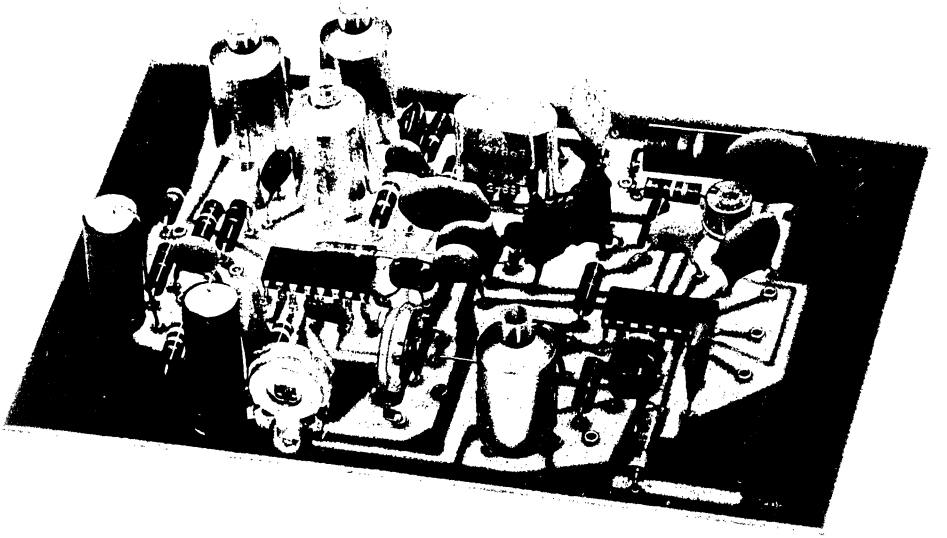
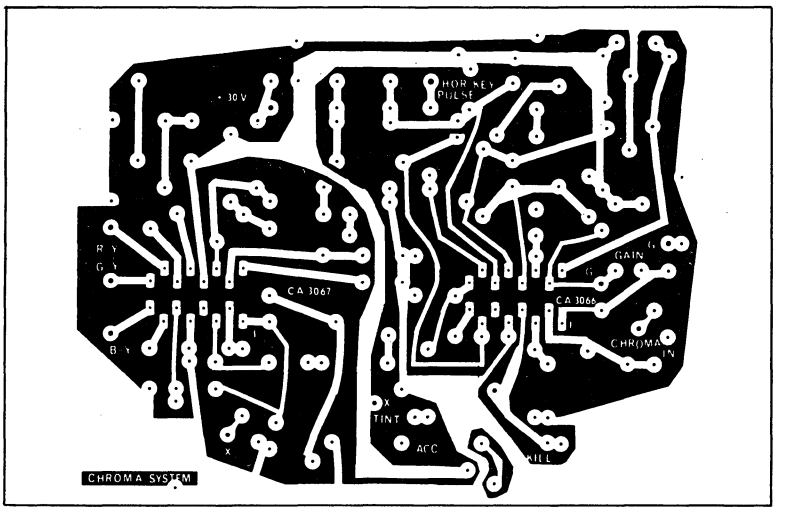


Fig. 20 a - Circuit board layout.



FULL SIZE

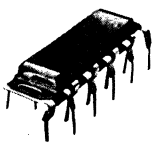
Fig. 20 b - Template for circuit board layout

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3068



20-lead quad-in-line
plastic package

H-1704

Television Video IF System

FEATURES:

- High-gain wide-band IF amplifier: 75 dB typ. at 45 MHz
- Gain reduction with excellent stability: 50 dB typ. at 45 MHz
- Video detector with linear characteristics
- Video amplifier: 12 dB gain
- Impulse noise limiter
- Keyed AGC with noise immunity circuits
- Delayed AGC for tuner
- Buffered AFT output
- Separate sound IF intercarrier amplification
- Sound carrier detector
- 4.5 MHz sound carrier amplifier
- Isolated zener reference diode for regulated voltage supply
- See ICAN-6303, "A Single IC for the Complete PIX-IF System in TV Receivers"

RCA-CA3068* is a monolithic integrated circuit that incorporates an entire video TV-IF subsystem on a single chip. Innovations in integrated circuit design, in addition to the many active devices and closely matched components utilized in the circuit, make the CA3068 ideally suited for use in color and black-and-white TV receivers.

The primary functions performed by the IF subsystem are video IF amplification, linear detection, video output amplification, AGC from a keyed supply, AGC delay for tuner, sound carrier detection, sound carrier amplification, and a buffered AFT output. The advanced circuit design of the CA3068 also includes secondary functions for improved noise immunity and minimal airplane flutter. An isolated zener reference diode, incorporated in the IC, provides a convenient and economical means for controlling the regulated voltage supply. The inherent wide bandwidth capability (10-70 MHz) and high overall gain (87 dB) make the CA3068 suitable for other AM IF applications whose frequencies range within this bandwidth.

The CA3068 utilizes a unique 20-lead quad-in-line plastic package. This package also includes a wrap-around shield that serves to minimize interlead capacitances.

* Formerly Developmental No. TA5914

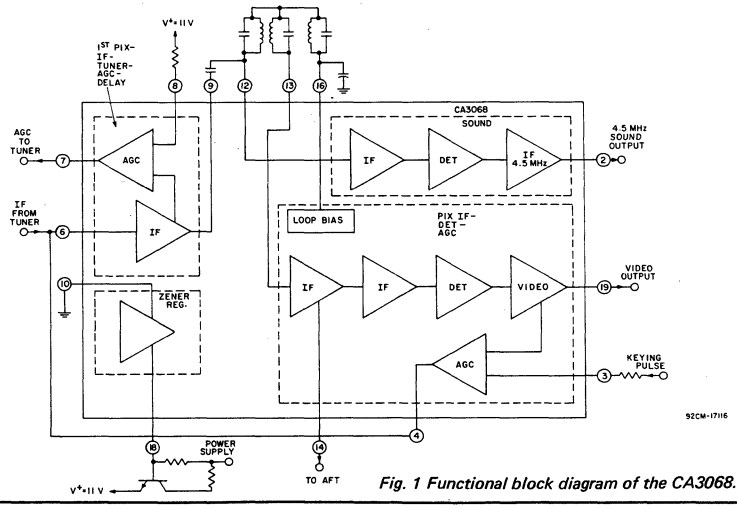


Fig. 1 Functional block diagram of the CA3068.

MAXIMUM RATINGS, *Absolute Maximum Values, at $T_A = 25°C$*

DC Supply Voltage:		
Between Terminals 15 and 5*	11.3	V
Terminal 7 (Collector to ground)	20	V
Terminal 9 (Collector to ground)	20	V
DC Current (into Terminal 18)	2	mA
Device Dissipation:		
Up to $T_A = 60°C$	600	mW
Above $T_A = 60°C$	derate linearly 6.7 mW/°C	
Ambient Temperature Range:		
Operating	−40 to +85	°C
Storage	− 65 to +150	°C
Lead Temperature (During soldering):		
At distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C

* This rating does not apply when using the internal zener reference in conjunction with the pass transistor.

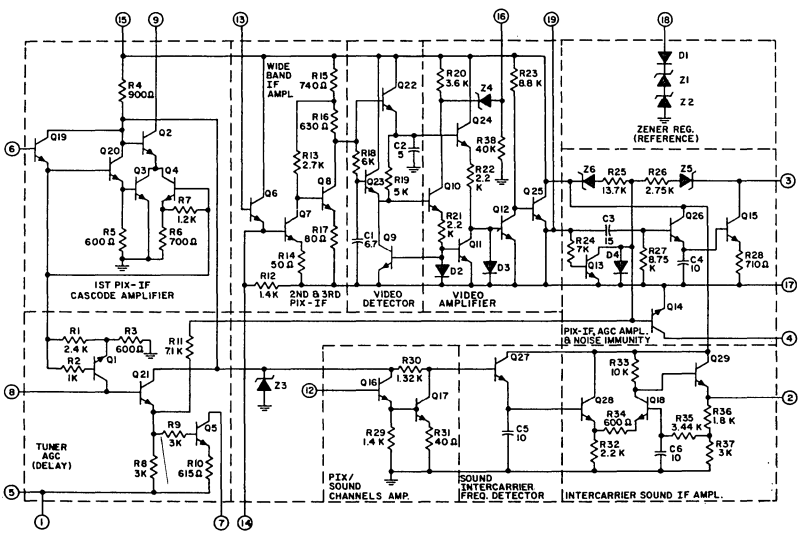


Fig. 2 - CA3068 — Simplified schematic diagram.

ELECTRICAL CHARACTERISTICS at $T_A = 25°C$

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	CIRCUIT Fig. No.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Static (DC) Characteristics							
Quiescent Circuit Current	I_{15}	–	3	15	–	45	mA
DC Voltages:							
Terminal 2 (Sound)	V_2	–	5	–	6	–	V
Terminal 3 (Keying Input)	V_3	–	3	6.4	–	10	V
Terminal 7 (1) (AGC)	V_7	–	3	16	–	21	V
Terminal 7 (2) (AGC)	V_7	–	4	–	1	–	V
Terminal 8 (AGC Delay)	V_8	–	4	–	4	–	V
Terminal 9 (Cascode Collector)	V_9	–	3	–	8.5	–	V
Terminal 16 (Bias)	V_{16}	–	3	1.1	–	2.3	V
Terminal 18 (Zener)	V_{18}	$V_5 = V_{17} = 0$ V, $I_{18} = 1$ mA	–	10.6	11.9	13.2	V
Terminal 19 (White Level)	V_{19}	–	5	6	–	10	V
Dynamic Characteristics							
Video Sensitivity	e_I	f_O = 45.75 MHz, Mod. (AM) = 85% at 400 Hz; Adjust e_I for 4 V_{p-p} at Term. 19	6	40	100	200	μV
Sync. Tip Level Voltage	V_{19}	f_O = 45.75 MHz, e_I(CW) = 10 mV	6	0.4	0.8	1.6	V
Automatic Fine Tuning (AFT) Drive Level Voltage	V_{14}		6	–	15	–	mV
Delay Bias Voltage:	V_7						
At e_I = 10 mV		f_O = 45.75 MHz, e_I(CW) = 20 mV;	6	16	–	–	V
At e_I = 30 mV		Adjust R_1 for V_7 = 14 V		0.5	–	2	V
3.58 MHz Chroma Output Voltage	V_{19}	f_O = 45.75 MHz, e_I(step mod.) = 10 mV; f_1 = 42.17 MHz, e_I(step mod.) = 3.33 mV	6	0.5	0.8	—	V
4.5-MHz Sound Output Voltage	V_2	f_O = 45.75 MHz, e_I(step mod.) = 10 mV; f_2 = 41.25 MHz, e_I(step mod.) = 2.5 mV	6	50	200	—	mV
Parallel Input Impedance:		f_O = 45.75 MHz Impedance and Admittance measured at bias conditions as developed by circuit shown in Fig. 7					
Resistance at Term. 6	R_{I-6}		7	4	–	–	kΩ
Capacitance at Term. 6	C_{I-6}			–	2	–	pF
Resistance at Term. 12	R_{I-12}		7	–	4.5	–	kΩ
Capacitance at Term. 12	C_{I-12}			–	4	–	pF
Resistance at Term. 13	R_{I-13}		7	–	5	–	kΩ
Capacitance at Term. 13	C_{I-13}			–	4	–	pF
Parallel Output Impedance:							
Resistance at Term. 9	R_{O-9}		7	30	–	–	kΩ
Capacitance at Term. 9	C_{O-9}			–	3	–	pF
Cascode Transfer Characteristics: Magnitude of Forward Transadmittance	$\lvert Y_f \rvert$		7	–	50	–	mmho
Reverse Transfer Capacitance	C_r		7	–	0.001	–	pF

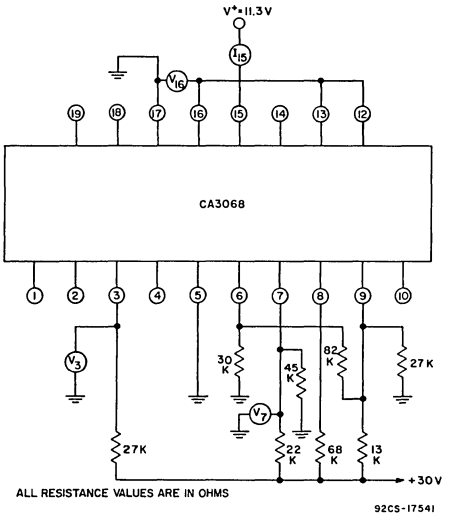


Fig. 3 – Test circuit for measurement of quiescent current (I_{15}), keying terminal voltage (V_3), bias voltage (V_{16}), AGC terminal voltage 1 (V_7), and cascode collector voltage (V_9)

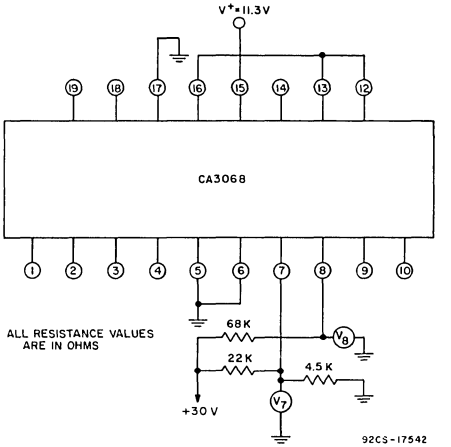


Fig. 4 – Test circuit for measurement of AGC terminal voltage 2 (V_7) and terminal 8 voltage (V_8).

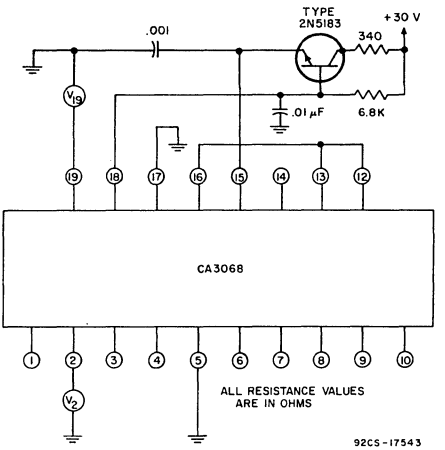
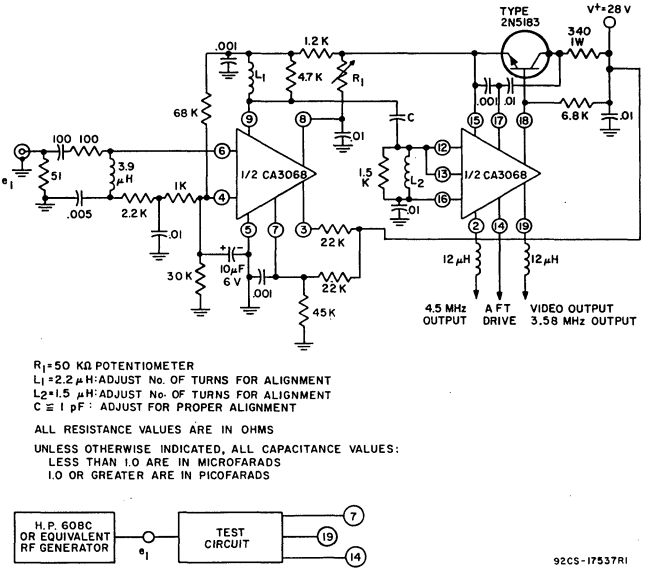
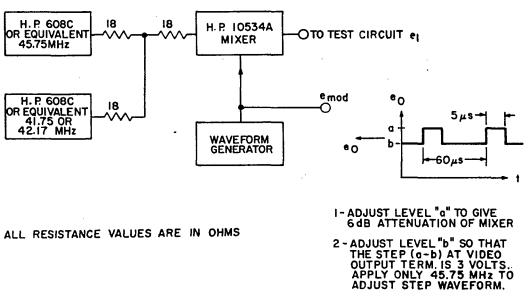


Fig. 5 – Test circuit for measurement of white level (V_{19}) and terminal 2 voltage (V_2).



(a) Test setup for measurement of video sensitivity, sync. tip level, delay bias, AFT drive voltage.



(b) Test setup for measurement of sound and chroma outputs.

Fig. 6 — Typical dynamic test circuit diagrams.

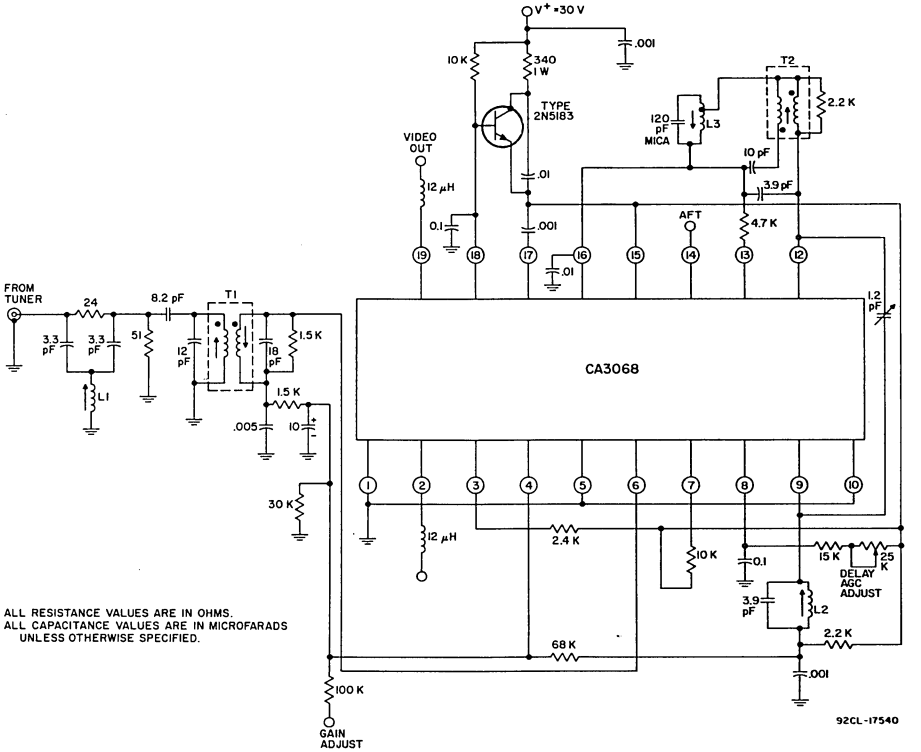


Fig. 7a – Color TV-IF amplifier test circuit.

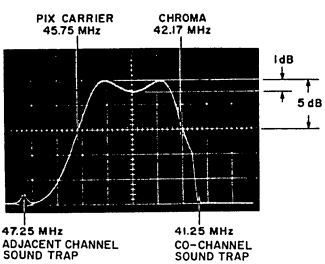


Fig. 7b – Color TV-IF amplifier with associated waveform and test circuit.

Alignment of the IF Amplifier

1. Apply a 2 to 4 mV signal from a sweep generator, Telonic SV13 or equivalent to the input of the IF amplifier.
2. Apply a negative DC supply voltage, to the Gain Adjust Terminal.
3. Set the gain supply voltage to provide a peak-to-peak output of 6 volts.
4. The overall response curve should conform to the waveform shown in Fig. 7b.

A TYPICAL COLOR–TV VIDEO SYSTEM

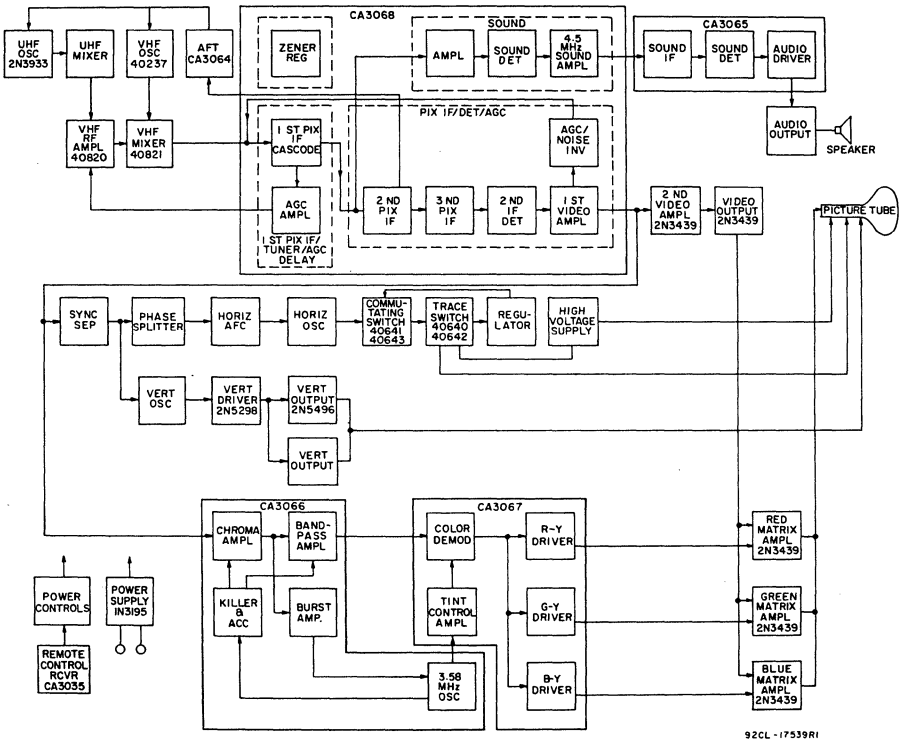


Fig. 8 – Block diagram of a typical color TV receiver utilizing the CA3068.

Application Information

A block diagram of a typical color TV application of the CA3068 is shown in Fig. 8. The input from the TV tuner is applied to the IF cascode amplifier of the IC. The cascode amplifier has a gain reduction of 50 dB typ. and a gain of 35 dB typ. The cascode output is coupled to succeeding stages via the IC lead interconnections. Associated with the cascode amplifier is an AGC delay network that provides gain control for the RF amplifier. This arrangement enables the circuit designer to introduce the desired bandpass-shaping circuitry between the cascode input stages and the remaining IF stages. These IF stages provide an additional gain of 40 dB typ. The output, taken from the emitter of the second IF stage, also provides a buffered AFT signal that is designed to drive the RCA-CA3064 TV Automatic Fine-Tuning IC.

The IF detector circuit provides an extremely linear output signal that is DC coupled to the first video amplifier. The first video amplifier has a voltage gain of 12 dB typ. The detector and video amplifier circuits provide a signal which has in addition to its linear output an extremely sharp limiting characteristic. The maximum video output level is approximately 7 volts peak-to-peak. The sharp limiting action of this circuit clips any signal (e.g. impulse noise) that exceeds this 7-volt value.

The video amplifier also provides a signal which drives a keyed AGC signal. The unique keyed AGC circuits utilize active devices that virtually eliminate noise from interfering with the action of the AGC. A separate sound section provides amplification at intercarrier frequencies, sound carrier detection, and sound carrier amplification. This sound section is designed to drive the RCA-CA3065 TV Sound System IC.

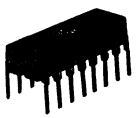
A color IF circuit with associated performance data is shown in Fig. 7. For a more detailed description of the CA3068 and related performance and IF printed circuit construction information, refer to the RCA Application Note ICAN-6544.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3070, CA3071 CA3072



CA3070
16-Lead Dual-In-Line Plastic Package
H-1622



CA3071
CA3072
14-Lead Dual-In-Line Plastic Package
H-1517

Television Chroma System

SYSTEM FEATURES

CA3070

- Voltage Controlled Oscillator
- Keyed APC & ACC Detectors
- DC Hue Control
- Shunt Regulator

CA3071

- ACC Controlled Chroma Amplifier
- DC Chroma Gain Control
- Color Killer
- Amplifier Short-Circuit Protection

CA3072

- Synchronous Detector with Color Difference Matrix
- Emitter-Follower Output Amplifiers with Short-Circuit Protection

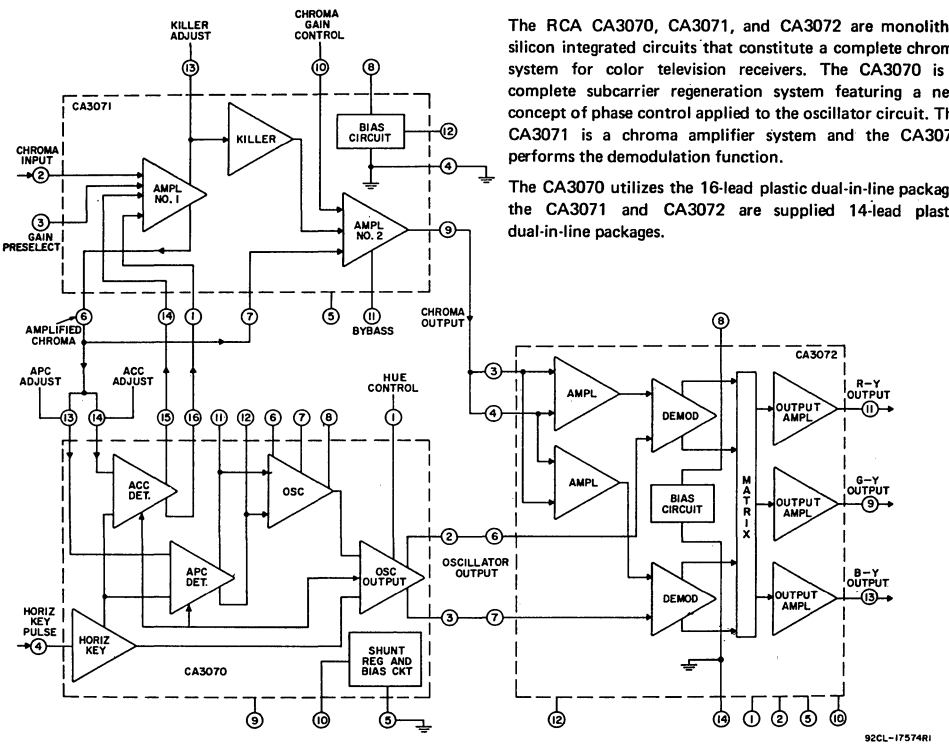


Fig. 1 – Simplified block diagram of TV chroma system.

The RCA CA3070, CA3071, and CA3072 are monolithic silicon integrated circuits that constitute a complete chroma system for color television receivers. The CA3070 is a complete subcarrier regeneration system featuring a new concept of phase control applied to the oscillator circuit. The CA3071 is a chroma amplifier system and the CA3072 performs the demodulation function.

The CA3070 utilizes the 16-lead plastic dual-in-line package; the CA3071 and CA3072 are supplied 14-lead plastic dual-in-line packages.

CA3070 Chroma Signal Processor

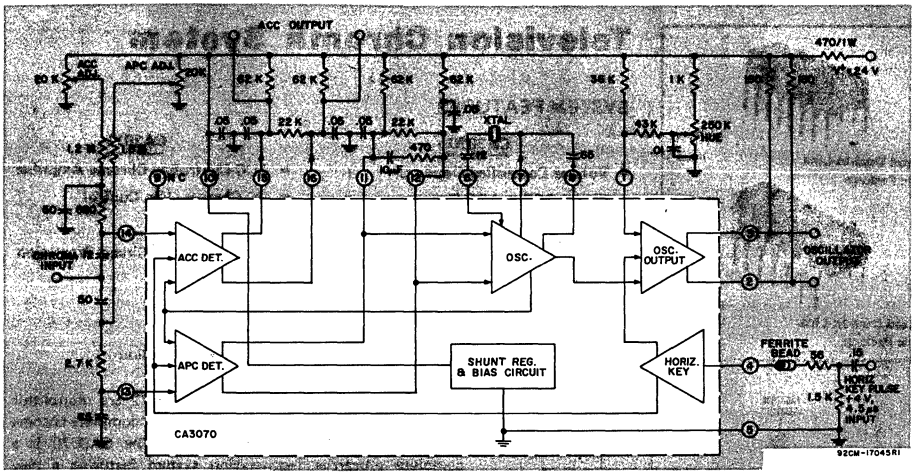


Fig. 2 – Functional diagram of RCA-CA3070.

The CA3070 is a complete subcarrier regeneration system with automatic phase control applied to the oscillator. An amplified chroma signal from the CA3071 is applied to terminals No. 13 and No. 14, which are the automatic phase control (APC) and the automatic chroma control (ACC) inputs. APC and ACC detection is keyed by the horizontal pulse which also inhibits the oscillator output amplifier during the burst interval.

The ACC system uses a synchronous detector to develop a correction voltage at the differential output terminal Nos. 15 & 16. This control signal is applied to the input terminal Nos. 1 & 14 of the CA3071. The APC system also uses a synchronous detector. The APC error voltage is internally coupled to the 3.58 MHz oscillator at balance; the phase of the signal at terminal No. 13 is in quadrature with the oscillator.

To accomplish phasing requirements, an RC phase shift network is used between the chroma input and terminal Nos. 13 and 14. The feedback loop of the oscillator is from terminal Nos. 7 and 8 back to No. 6. The same oscillator signal is available at terminal Nos. 7 and 8, but the dc output of the APC detector controls the relative signal levels at terminal Nos. 7 or 8. Because the output at terminal No. 8 is shifted in phase compared to the output at terminal No. 7, which is applied directly to the crystal circuit, control of the relative amplitudes at terminal Nos. 7 and 8 alters the phase in the feedback loop, thereby changing the frequency of the crystal oscillator. Balance adjustments of dc offsets are provided to establish an initial no-signal offset control in the ACC output, and a no-signal, on-frequency adjustment through the APC detector-amplifier circuit which controls the oscillator frequency. The oscillator output stage is differentially controlled at terminal Nos. 2 and 3 by the hue control input to terminal No. 1. The hue phase shift is accomplished by the external R, L, and C components that couple the oscillator output to the demodulator input terminals. The CA3070 includes a shunt regulator to establish a 12-volt dc supply.

MAXIMUM RATINGS, *Absolute Maximum-Values at $T_A = 25°C$*

DC Supply Voltage and Current .....See Charts Below
Device Dissipation:
Up to $T_A = +70°C$ ..... 530 mW
Above $T_A = +70°C$ ... Derate Linearly at 6.7 mW/°C
Ambient Temperature Range:
Operating ..... –40 to +85 °C
Storage ..... –65 to +150 °C
Lead Temperature (During Soldering):
At distance 1/32 in. (3.17 mm) from seating plane for 10 s max. ..... +265 °C

▲ With respect to terminal No.5 and with terminal No. 10 connected through 470Ω to +24 V.
N1 Regulated voltage at terminal No. 10.
N2 Controlled by max. input current.
N3 Limited by dissipation.

Maximum Voltage and Current Ratings at $T_A = +25°C$

Voltage▲			Current		
Terminal No.	Min. Volts	Max. Volts	Terminal No.	I_I mA	I_O mA
1	0	*	1	20	1
2	0	+16	2	–	–
3	0	+16	3	–	–
4	–5	N2	4	20	1
6	–	–	10	N3	1
7	–	–	11	–	–
8	–	–	12	–	–
10	0	N3	13	20	1
11	0	N1	14	20	1
12	0	N1			
13	0	N1			
14	0	N1			
15	0	+16			
16	0	+16			

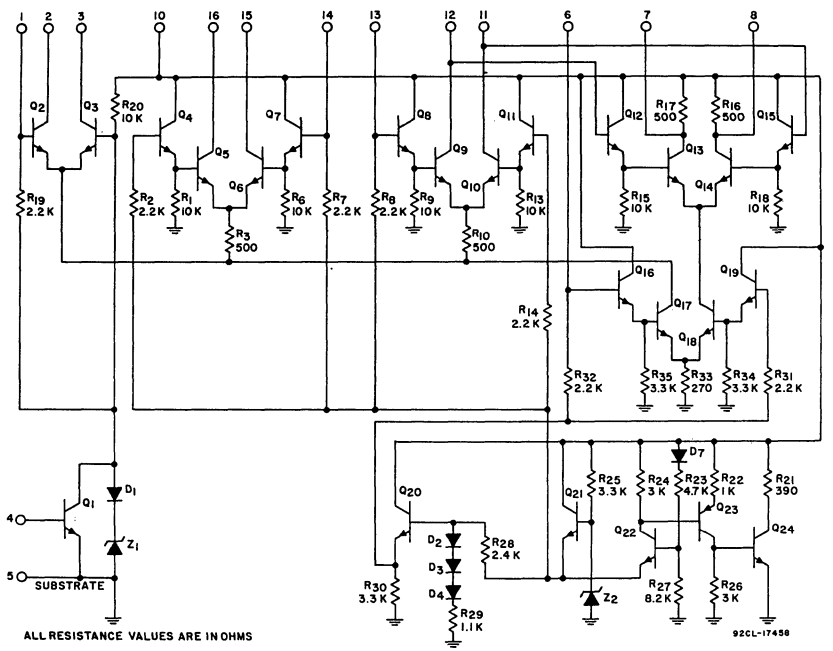


Fig. 3 – Schematic diagram CA3070.

ELECTRICAL CHARACTERISTICS, at T_A = 25°C and V^+ = +24 V unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS CA3070			UNITS	TEST CIRCUITS
			MIN.	TYP.	MAX.		FIG.
Static Characteristics							
Voltage:							
Hue Control	V_1	Switch in position 2	6.9	7.7	8.6	V	4c
Oscillator Input	V_6		–	2.8	–		4a
APC Input	V_{13}		–	6.5	–		
Regulator	V_{10}	V^+ = 21 V	11	12.3	13.5		
Regulator Change	V_{10}	V^+ = 27 V	−0.2	–	+0.2		
Horizontal Key Input	V_4	I_4 = −10 μA	5	—	—		
Currents:							
Oscillator Output	I_2		–	5.8	–	mA	4c
APC Output	I_{11}, I_{12}		–	1.45	—		4b
ACC Output	I_{15}, I_{16}		–	1.45	–		
Dynamic Characteristics							
Oscillator Outputs:							
Terminal No. 2	V_2	S_1 in position 1	0.75	1.0	–	V_{p-p}	5
Terminal No. 3	V_3	S_1 in position 2	0.75	1.0	–		
ACC Detected Output	V_{16}–V_{15}	S_1 in position 1	115	150	–	mV	5
Oscillator Pull-In Range	–		–	±400	–	Hz	5

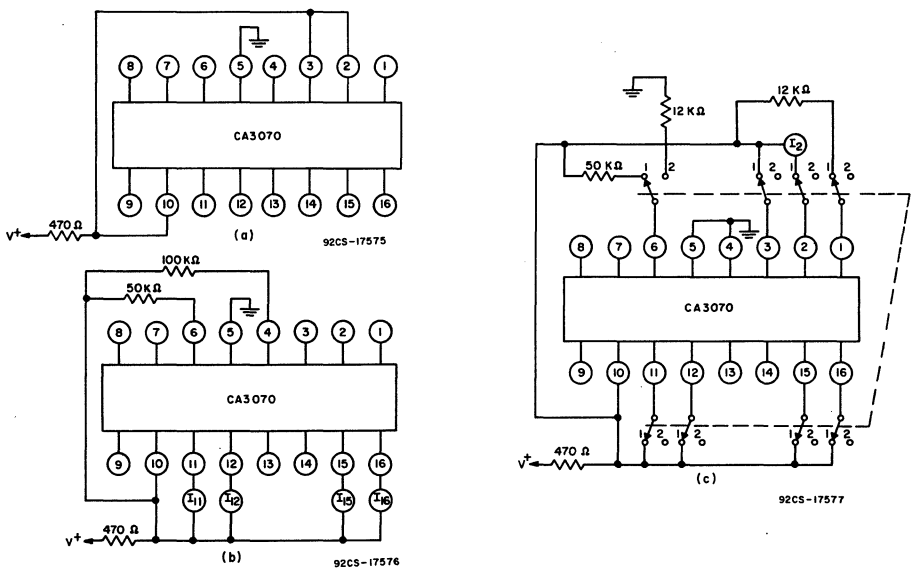
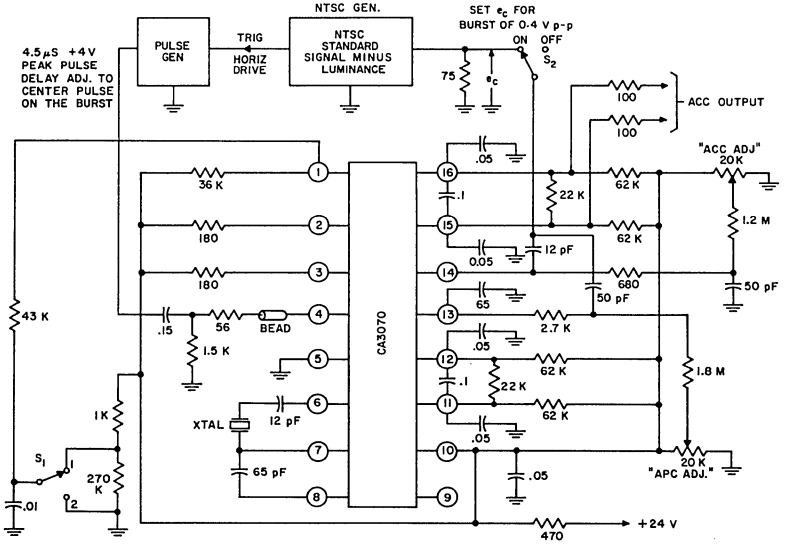


Fig. 4 — Static characteristics test circuits.



NOTES:

1. ALL RESISTANCES IN OHMS.
2. UNLESS OTHERWISE SPECIFIED ALL CAPACITANCES ARE IN MICROFARADS.
3. v_2 & v_3 MEAS'D WITH LOW-CAPACITY SCOPE PROBE ≤ 20 pF.

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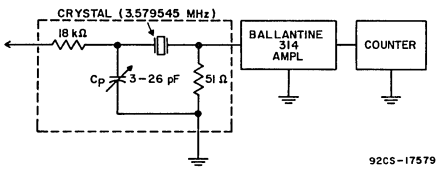
Fig. 5 – CA3070 Dynamic test circuit.

Dynamic Test Initial Adjustments

1. APC ADJUST: With S2 in "OFF" position adjust the "APC ADJ" potentiometer to set oscillator frequency at 3.579545MHz ±25 Hz. With S1 in position 1 measure frequency at terminal No. 2 output, using crystal probe shown in Fig. 6.
2. ACC ADJUST: With S2 in "OFF" position adjust "ACC ADJ" potentiometer to give an ACC output reading of 0 ±2 mV.

Procedure to Pull-in Range Measurement

1. Set S1 in position 1 and connect the crystal probe to terminal No. 2.
2. Turn S2 to "OFF" and set "APC ADJ." arm to ground.
3. Turn S2 to "ON" and gradually adjust "APC ADJ" until oscillator "locks" as witnessed by a sharp increase in ACC output voltage between terminal Nos. 15 and 16.
4. Turn S2 to "OFF" and adjust capacitor Cp of crystal probe for maximum deflection on Ballantine Meter.
5. Switch Ballantine meter to "Amplifier" position and read oscillator frequency on counter.
6. Repeat steps 2 – 5 with "APC ADJ" arm set to terminal No. 10 instead of to ground.



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Fig. 6 – Crystal probe for frequency measurements.

CA3071 Chroma Amplifier

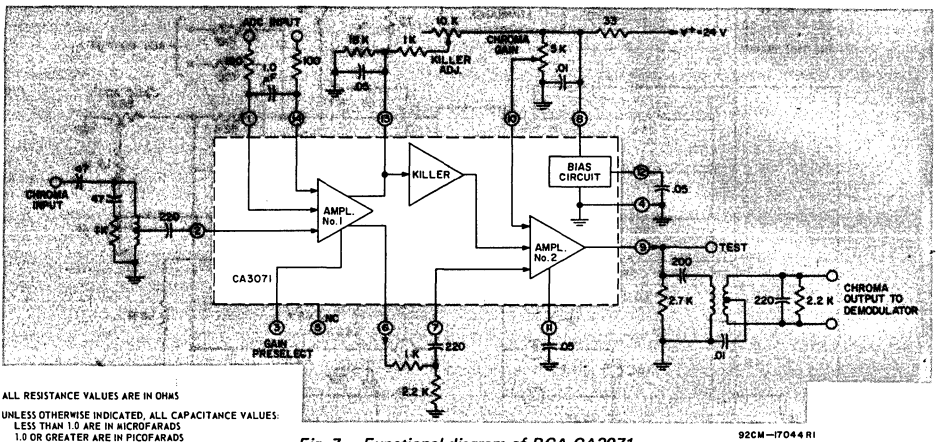


Fig. 7 – Functional diagram of RCA-CA3071.

The CA3071 is a combined two-stage chroma amplifier and functional control circuit. The input signal is received from the video amplifier and applied to terminal No. 2 of the input amplifier stage. The first amplifier stage is part of the ACC system and is controlled by differential adjustment from the ACC input terminal Nos. 1 and 14. The output of the 1st amplifier is directed to terminal No. 6 from where the signal may be applied to the ACC detection system of the CA3070 or an equivalent circuit. The output at terminal No. 6 is also applied to terminal No. 7 which is the input to the 2nd amplifier stage. Another output of the 1st amplifier at terminal No. 13 is directed to the killer adjustment circuit.

The dc voltage level at terminal No. 13 rises as the ACC differential voltage decreases with a reduction in the burst amplitude. At a pre-set condition determined by the killer adjustment resistor the killer circuit is activated and causes the 2nd chroma amplifier stage to be cut off. The 2nd chroma amplifier stage is also gain controlled by the adjustment of dc voltage at terminal No. 10. The output of the 2nd chroma amplifier stage is available at terminal No. 9. The typical output termination circuit that is shown, provides differential chroma drive signal to the demodulator circuit. Both amplifier outputs utilize emitter-followers with short-circuit protection.

MAXIMUM RATINGS, *Absolute Maximum-Values at* $T_A = 25°C$

DC Supply Voltage (Terminal 8 to Terminal 4) ..... 30 VDC

Device Dissipation:
- Up to $T_A = +70°C$ ..... 530 mW
- Above $T_A = +70°C$ .... Derate Linearly at 6.7 mW/°C

Ambient Temperature Range:
- Operating ..... –40 to +85 °C
- Storage ..... –65 to +150 °C

Lead Temperature (During Soldering):
- At distance 1/32 in (3.17 mm) from seating plane for 10 s max. ..... +265 °C

Maximum Voltage and Current Ratings @ $T_A = +25°C$

Terminal No.	Current I_I mA	Current I_O mA	Terminal No.	Voltage* MIN VOLTS	Voltage* MAX VOLTS
1	5	1.0	1	–5	+15
2	5	1.0	2	–5	+5
3	10	10	3	0	+2
6	1.0	20	6	0	+24
7	5	1.0	7	–5	+5
9	1.0	20	8	0	+30
12	1.0	5	9	0	+24
14	5	1.0	10	0	+24
			11	0	+24
			12	0	+20
			13	0	+20
			14	–5	+15

* With reference to terminal No. 4 and with +24 V on terminal No. 8 except for the rating given for terminal No. 8.

ELECTRICAL CHARACTERISTICS, at T_A = 25° C

CHARACTERISTICS	SYMBOLS (Measure)	SPECIAL TEST CONDITIONS	LIMITS CA3071 MIN.	LIMITS CA3071 TYP.	LIMITS CA3071 MAX.	UNITS	CURVES & TEST CIRCUITS FIG.
Static Characteristics							
Voltages:							
Bias Reference Terminal	V_{12}	S_1 Open, S_2 Open	–	17.3	–	V	8
Ampl. No. 1 Chroma Input	V_2	S_1 Open, S_2 Open	–	1.75	–		
Ampl. No. 1 Chroma Output Balanced	V_6	S_1 Open, S_2 Open	–	20	–		
Unbalanced	V_6	S_1 Open, S_2 Closed	–	13.5	–		
Ampl. No. 2 Chroma Input	V_7	S_1 Open, S_2 Open	–	1.5	–		
Ampl. No. 2 Chroma Output	V_9	S_1 Closed, S_2 Open	–	20.6	–		
Supply Current	I_T	S_1 Open, S_2 Open	17	24.5	31	mA	
Dynamic Characteristics							
Amplifier No. 1 Voltage Gain	A_{V1}	E_g = 30 mV_{RMS} Measure v_6	14	–	–	dB	9
Amplifier No. 2 Voltage Gain	A_{V2}	V_g = 1.0 V (RMS) Measure v_7	–	14	–	dB	
Max. Chroma Output Voltage	v_9		–	2	–	V_{RMS}	13
10% Chroma Gain Control Reference Voltage	$V_8 - V_{10}$	E_g = 50 mV_{RMS}, adjust Chroma Gain Control to Change v_9 to 10% of Maximum Chroma Output	2.1	3.8	6.8	V	9
Output Voltage, Killer Off	v_9	S_1 in Position 2 E_g = 50 mV_{RMS}, adjust "Killer Adjust" for an abrupt decrease in V_9	--	–	12	mV RMS	
Output Voltage, Chroma Off	v_9	E_g = 50 mV_{RMS}, adjust Chroma control to min. Chroma Output	–	–	12	mV RMS	
Bandwidth: Amplifier No. 1	BW		–	12	–	MHz	11, 12
Amplifier No. 2			–	30	–		
Ampl. No. 1 Input Impedance	r_{i1}		–	2	–	kΩ	9
	c_{i1}		–	4	–	pF	
Ampl. No. 1 Output Impedance	r_{o1}		–	85	–	Ω	
Ampl. No. 2 Input Impedance	r_{i2}		–	2.1	–	kΩ	
	c_{i2}		–	3.5	–	pF	
Ampl. No. 2 Output Impedance	r_{o2}		–	85	–	Ω	

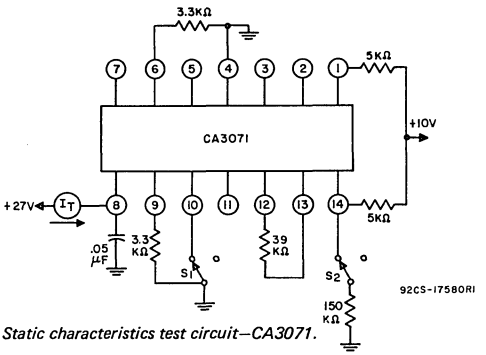


Fig. 8 – Static characteristics test circuit–CA3071.

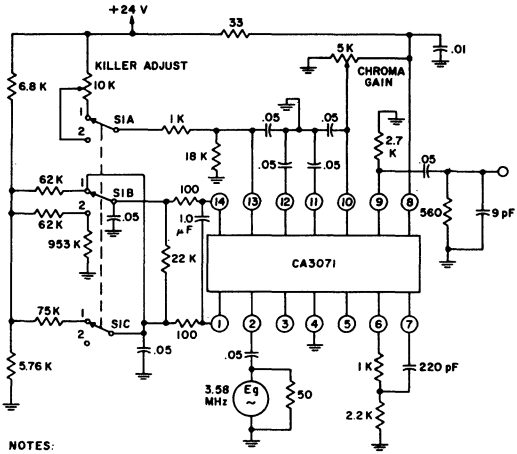


Fig. 9 — Dynamic characteristics circuit-CA3071.

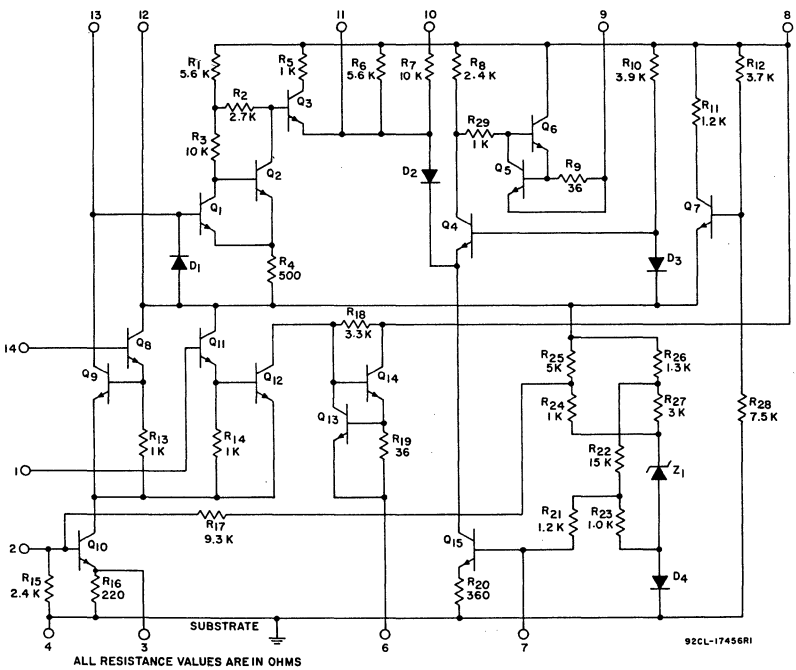


Fig.10—Schematic diagram for CA3071.

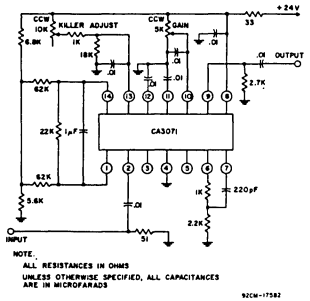


Fig. 11 – CA3071 Wideband amplifier circuit.

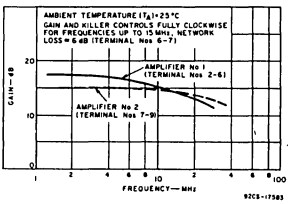


Fig. 12 – Frequency response for wideband amplifier CA3071.

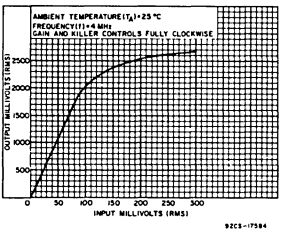


Fig. 13 – Typical CA3071 wideband amplifier linearity

CA3072 Chroma Demodulator

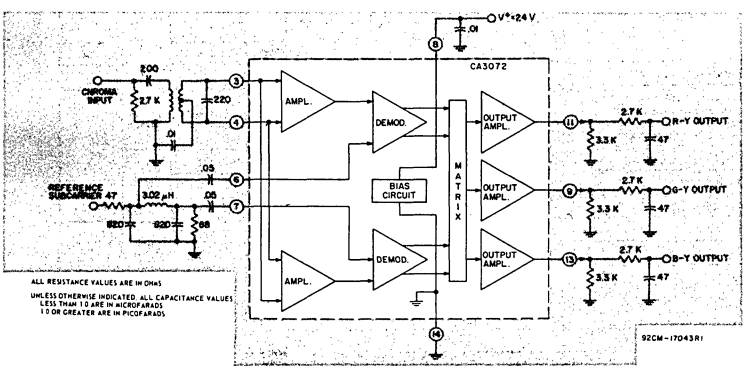


Fig. 14 – Functional diagram of RCA-CA3072.

The CA3072 has two sets of synchronous detectors with matrix circuits to achieve the R-Y, G-Y, and B-Y color difference output signals. The chroma input signal is applied to terminal Nos. 3 and 4 while the oscillator injection signal is applied to terminal Nos. 6 and 7. The color difference signals, after matrix, have a fixed relationship of amplitude and phase nominally equal dc voltage levels. The outputs of the CA3072 are suitable for driving high level color difference or R, G, B output amplifiers. Emitter-follower output stages used to drive the high level color amplifiers have short-circuit protection.

ELECTRICAL CHARACTERISTICS, at T_A = 25° C and V^+ = +24 V unless otherwise specified

CHARACTERISTICS	SYMBOLS	SPECIAL TEST CONDITIONS	LIMITS CA3072			UNITS	TEST CIRCUITS
			MIN.	TYP.	MAX.		FIG.
Static Characteristics							
Supply Current With Output Loads	I_T	S_1 Closed	16.5	–	26.5	mA	15
With No Output Loads		S_1 Open	–	9			
G-Y, R-Y, B-Y Outputs	V_9, V_{11}, V_{13}	S_1 Closed	13.2	14.7	15.8	V	
Chroma Inputs	V_3, V_4	S_1 Open	–	3.3	–		
Reference Subcarrier	V_6, V_7	S_1 Open	–	6.2	–		
Dynamic Characteristics							
Demodulator Unbalance	v_9, v_{11}, v_{13}	$V_3 = V_4 = 0$	–	–	0.8	V_{p-p}	16
Maximum Color Difference Output Voltage	v_{13}	$V_3 = V_4 = 0.6$ V_{p-p}	8.0	–	–	V_{p-p}	
	v_{11}		5.5	–	–		
	v_9		1.2	–	–		
Chroma Input Sensitivity	v_3	Adjust e_c for 5.0 v_{p-p} @ term No. 13 (B-Y)	–	0.2	0.35		
Relative R-Y Output	v_{11}		3.5	–	4.2		
Relative G-Y Output	v_9		0.75	–	1.25		
V_{DC} Difference Between any two Output Terminals	$\lvert V_9 \rvert - \lvert V_{11} \rvert$	$e_c = 0$	–	–	0.6	V	
	$\lvert V_9 \rvert - \lvert V_{13} \rvert$						
	$\lvert V_{11} \rvert - \lvert V_{13} \rvert$						
Input Impedance Reference Subcarrier Inputs	r_i 6, 7		–	1.7	–	kΩ	
	c_i 6, 7		–	6	–	pF	
Input Impedance at Chroma Inputs	r_i 3, 4		–	0.95	–	kΩ	
	c_i 3, 4		–	6	–	pF	
Output Resistance	r_o 9, r_o 11, r_o 13		–	180	–	Ω	

MAXIMUM RATINGS, *Absolute Maximum-Values at T_A = 25° C*

DC Supply Voltage (Terminal 8 to Terminal 14) ..... 27 V
Reference Input Voltage ..... 5 V_{p-p}
Chroma Input Voltage ..... 5 V_{p-p}
Device Dissipation:
Up to T_A = +70°C ..... 530 mW
Above T_A = +70°C ..... Derate Linearly at 6.7 mW/°C
Ambient Temperature Range:
Operating ..... -40 to +85°C
Storage ..... -65 to +150°C
Lead Temperature (During Soldering):
At distance 1/32 in (3.17 mm) from seating plane for 10 s max ..... +265°C

*With reference to terminal No. 14 and with the voltage between terminal No. 8 and terminal No. 14 at +24 V except as given in rating for terminal No. 8.

Maximum Voltage and Current Ratings at T_A = +25°C

Voltage*

Terminal No.	MIN VOLTS	MAX VOLTS
3	0	+5
4	0	+5
6	0	+12
7	0	+12
8	0	+27
9	0	+20
11	0	+20
13	0	+20

Current

Terminal No.	I_I mA	I_O mA
3	–	–
4	–	–
6	–	–
7	–	–
8	–	–
9	1.0	20
11	1.0	20
13	1.0	20

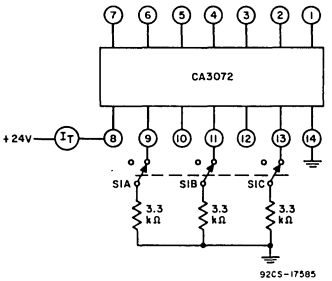


Fig. 15 – Static characteristics test circuit–CA3072.

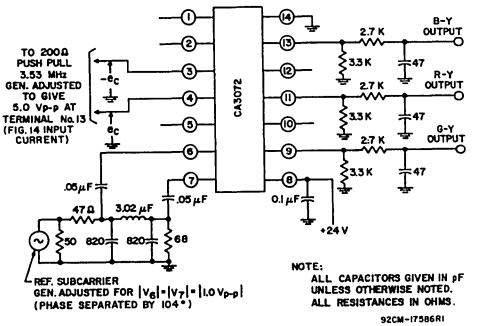


Fig. 16 – Dynamic characteristics test circuit for CA3072.

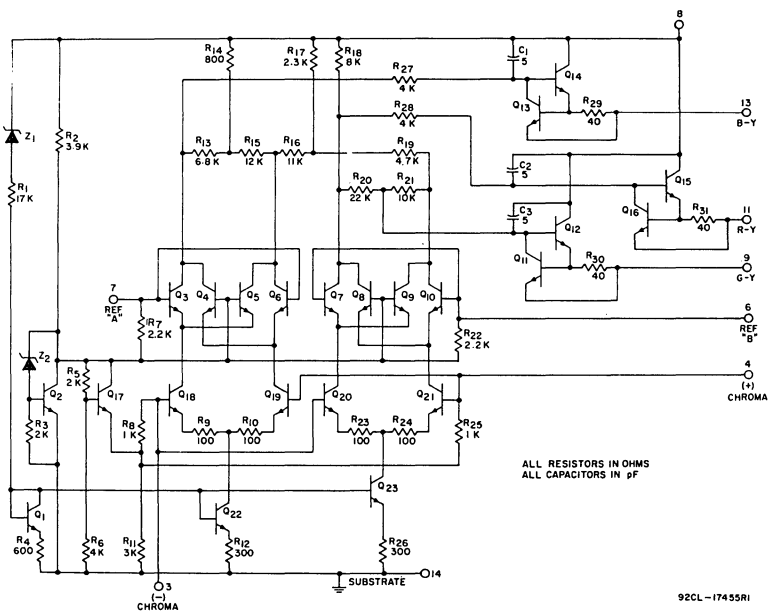


Fig.17–Schematic diagram for CA3072.

Application Information

TYPICAL APPLICATION CIRCUIT FOR THE CHROMA SYSTEM

The circuit of Fig. 18 is a complete signal processing system for color TV. The RCA types CA3070, CA3071 and CA3072 monolithic integrated circuits are respectively used as the subcarrier regenerator, chroma amplifier, and chroma demodulator.

The input to the system is the chroma signal which may be taken from the first or second video stage and is coupled into the CA3071 chroma amplifier through a bandpass filter. The outputs from the system are the color difference signals which are intended to drive high level amplifiers. Luminance mixing may be external to the picture tube or, the difference signals may be amplified and applied to the picture tube grid or cathode, where they are internally mixed with the luminance signal.

Other input requirements to the system are the power supply voltage of +24 volts and the horizontal keying pulse. The power supply voltage should be maintained within ±3 volts of the recommended value of +24 volts. The total current for the system is approximately 70 milliamperes. The horizontal keying pulse input to the subcarrier regenerator is approximately +4 volts peak and centered on the burst as seen at terminal Nos. 13 and 14 of the CA3070. The pulse width should be maintained as close as possible to the recommended value of 4.5 microseconds.

CA3070 Circuit Operation

The CA3070 circuit as shown in Fig. 3, consists of an oscillator, automatic phase control (APC) detector, automatic chroma control (ACC) detector, gated oscillator output amplifier and a shunt regulator. The shunt regulator provides the necessary bias stability for the 3.579545 MHz oscillator, as well as the bias to all functions of the CA3070 circuit. The regulation voltage is nominally +12 volts as measured at terminal No. 10.

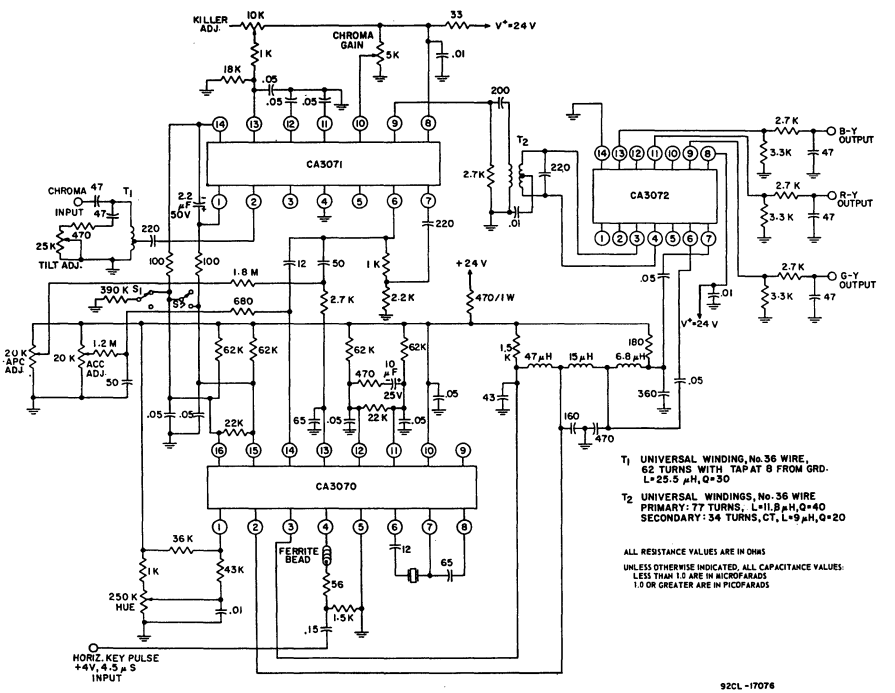


Fig. 18 – Typical chroma system for color-TV receivers utilizing RCA-CA3070, CA3071, and CA3072.

The APC and ACC detectors are synchronous detectors which are keyed by the horizontal input pulse. This form of detection eliminates the need for a burst separator as an individual amplifier stage. When a positive pulse is present at terminal No. 4, the oscillator output is cutoff and the oscillator drive signal is diverted to the APC and ACC detectors. Referring to Fig. 3, the APC detector (Q_9 & Q_{10}) and the ACC detector (Q_5 & Q_6) are emitter driven from the oscillator transistor (Q_{17}), when the oscillator output amplifier transistors (Q_2 & Q_3) are cutoff. The chroma signal is applied to terminal Nos. 13 and 14. There is oscillator current drive to the APC and ACC detectors during the keying interval; burst separation is effectively accomplished by the gating action of the detectors. A further advantage of the keying action is the high gain made possible as a result of the low average current flow of the APC and ACC detectors. High resistor values of 62 kilohms at the detector output terminals provide proper detector bias consistent with the duty factor of the keying pulse. For a wider keying pulse, it is necessary that smaller values of detector load resistors be used.

In the absence of the keying pulse (line period), the resistor, R_{20}, biases the oscillator's output amplifier transistors (Q_2 & Q_3) on by keeping their emitters at a higher potential than the base bias voltages of Q_5, Q_6, Q_9, and Q_{10}. The 3.58 MHz signal is now present at terminal Nos. 2 & 3. Photographs of oscilloscope traces for one line period at the terminal Nos. 1, 2, and 3 are shown in Fig. 19 The effect of the keying pulse is shown in Fig. 19a, and the cutoff of the oscillator output amplifier is shown in Fig. 19b and 19c.

The oscillator section of the CA3070 consists of the loop formed by Q_{18} and the emitter driven differential pair, Q_{13} & Q_{14}. The signal output from terminal Nos. 7 & 8 is coupled through the series tuned crystal circuit back through terminal No. 6 to Q_{16} & Q_{17}. The collector of Q_{17} drives the oscillator output amplifier and the APC & ACC detectors. Q_{17} is emitter coupled to transistor Q_{18}. The oscillator frequency and phase control is accomplished by the differential drive from the APC detector to transistors Q_{12} & Q_{15} which control the balance of Q_{13} & Q_{14}. The resulting phase of the feedback loop is determined by the relative amplitudes of the oscillator output signal at terminal Nos. 7 and 8. The 65 pF capacitor between terminal No. 7 and 8 provides the phase shifting component as the balance of Q_{13} and Q_{14} is varied. In this way the APC detector controls the crystal frequency at which the phase shift is cancelled in the feedback loop.

The controls for the CA3070 subcarrier regenerator circuit are the APC balance, the ACC balance, and the hue control. The hue control is a dc balance adjustment of the oscillator output amplifier transistors Q_2 & Q_3. A phase delay network between the output terminals Nos. 2 & 3 determines the range of the hue control, which for the value shown in Fig. 18, is approximately 90°.

The ACC adjustment sets the initial balance of the ACC drive to the input of the CA3071 in Fig. 18 (terminal Nos. 1 and 14 of the CA3071). The APC is a frequency adjustment of the oscillator through the balance control of the APC detector.

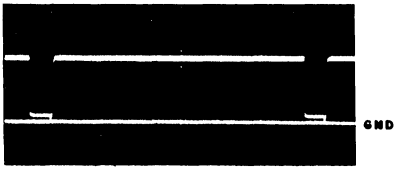


Fig. 19(a) - CA3070 terminal No. 1 7.5 V oscillator "gate off" pulse.

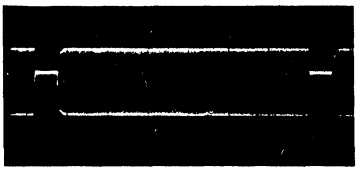


Fig. 19(b) - CA3070 terminal No. 2, 3.5 $V_{p\text{-}p}$ oscillator output; one horizontal line, (gated off during burst).

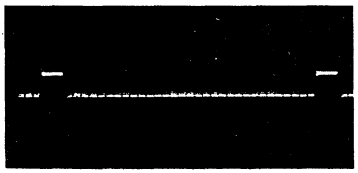


Fig. 19(c) - CA3070 terminal No. 3, 2.0 $V_{p\text{-}p}$ oscillator output; one horizontal line, (gated off during burst).

As a setup adjustment, for both the ACC and APC, switch S1 is opened and S2 is closed. The chroma input to the system is removed and the dc voltage at terminal No. 6 of the CA3071 is noted. The switch S2 is then opened and the ACC adjusted to set the voltage at terminal No. 6 to that previously noted. Alternatively, the differential dc voltage at terminal Nos. 15 & 16 of the CA3070 may be set to 0 mV (±2 mV) when S1 and S2 are open, and the CA3071 is removed from the circuit.

With the chroma signal still removed, the APC adjustment sets the frequency of the oscillator to 3.579545 MHz. Due to the gated off interval, a counter will not accurately record the frequency at the oscillator output amplifier terminals. Two simple and accurate methods are as follows: (1) a buffered crystal filter circuit, connected to the oscillator output amplifier terminals will continue to ring and fill the gated off window providing the proper interface to a counter; (2) the other method involves monitoring the demodulated output at the color difference output terminals

of the CA3072. A zero beat signal, at the color difference outputs may be seen on an oscilloscope.

When these adjustments are made, similar oscilloscope traces should be seen as shown in Fig. 20.



Fig. 20(a) - CA3070 terminal No. 6, oscillator waveform 1.1 V_{p-p} 3.58 MHz.

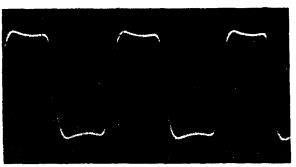


Fig. 20(b) - CA3070 terminal No. 7, oscillator waveform 1.4 V_{p-p} 3.58 MHz.

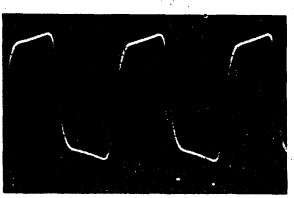


Fig. 20(c) - CA3070 terminal No. 8, oscillator waveform 1.6 V_{p-p} 3.58 MHz.

CA3071 CIRCUIT OPERATION

The CA3071 is the basic amplifier and control circuit of the chroma system. It contains the gain control functions of the ACC loop, the color killer, and the dc chroma gain control. The CA3071 is a wide band amplifier having two stages of voltage gain. Curves of frequency-response and linearity are shown in Figs. 12 & 13 for the wideband circuits shown in Fig. 11. This is the same basic amplifier as the one in the system shown in Fig. 18 except for the omission of the tuned circuits and the ACC loop connection. The amplifiers have bandwidths of greater than 10 MHz. and are usable well beyond 30 MHz. The signal swing of the wide band amplifier is in excess of 5 V_{p-p}., even with the typical load coupling as shown in Fig. 18. Fig. 21 (a, b and c) show the oscilloscope traces for an NTSC signal at the chroma input. The overall frequency-response curves are shown in Fig. 22.

CA3071 operation is as follows (Refer to Figs. 10 & 18). The input chroma signal is applied to terminal No. 2. This signal is amplified in a cascode differential circuit from Q_{10} to Q_{12}



Fig. 21(a) - CA3071 chroma input 1.25 V_{p-p}; one horizontal line of NTSC input signal.

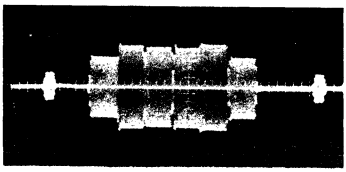


Fig. 21(b) - CA3071 terminal No. 6, amplifier No. 1 chroma output 2.3 V_{p-p}; one horizontal line for 1.25 V_{p-p} chroma input

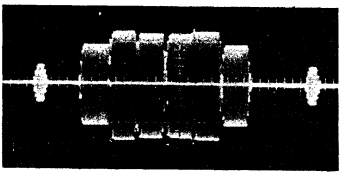


Fig. 21(c) - CA3071 terminal No. 9, amplifier No. 2 chroma output 5.5 V_{p-p}; one horizontal line for 1.25 V_{p-p} chroma input

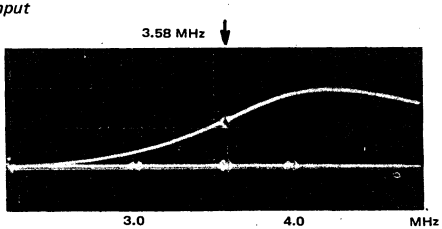


Fig. 22(a) - Frequency response sweep curve between terminal Nos. 2 & 6 for CA3071. f = 250 KHz/div.

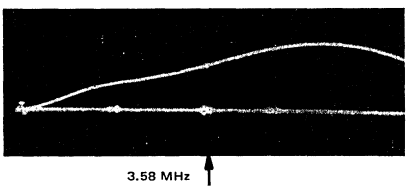


Fig. 22(b) - Frequency response sweep curve between terminal No. 2 of CA3071 and terminal No. 3 of CA3072. f = 250 KHz/div.

and the output is an emitter follower, Q_{14} (Terminal No. 6.) The signal is divided in the Q_9 & Q_{12} differential amplifier, depending on the applied ACC error signal amplitude at terminal Nos. 1 & 14. The ACC error signal is derived from terminal Nos. 15 & 16 of the CA3070 and after filtering, is applied to terminal Nos. 1 & 14 of the CA3071.

At low signal drive, the 390 kilohm resistor at switch S1 (normally closed) unbalances the differential amplifier for high signal gain through Q_{12}. As the burst level at the chroma input increases, the ACC drive changes differentially in a positive direction at terminal No. 14 and a negative direction at terminal No. 14 and a negative direction at terminal No. 1. At strong signal levels the gain is reduced by diverting the balance of ac current in the differential amplifier from Q_{12} to Q_9, which is shunted to ac ground at terminal Nos. 12 and 13. The ACC loop is completed through the chroma signal at terminal No. 6 of the CA3071 to terminal No. 14 (input) of the CA3070. A typical ACC characteristic is shown in Fig. 23.

The chroma signal is buffer connected from terminal No. 6 to terminal No. 7 of the CA3071 and is amplified in the 2nd

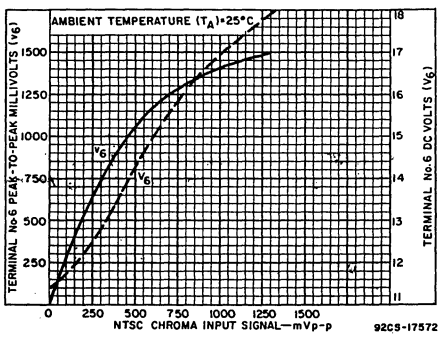


Fig. 23 - Typical ACC characteristics for chroma system of Fig. 18

stage of voltage gain. Both the color killer adjustment and the dc chroma gain control are applied to the 2nd stage to control the chroma output at terminal No. 9. The color killer section of the CA3071 is a Schmitt trigger & amplifier circuit consisting of transistors Q_1, Q_2 and Q_3. Under maximum chroma output conditions, the diode D_2 is reversed biased, and the signal path is through Q_{15}, Q_4 and Q_5 to terminal No. 9. When the color killer circuit is actuated, or the chroma gain control is adjusted to a higher positive voltage at terminal No. 10, the anode voltage of diode D_2 is increased to draw current from the signal path at the emitter of Q_4. This decreases the chroma gain as the potential at terminal No. 10 is increased. When the potential at terminal No. 10 is the same as terminal No. 8, the chroma output at terminal 9 is cutoff.

The color killer circuit provides an abrupt voltage swing at the anode of D_2 to cutoff the chroma output when the Schmitt trigger circuit is forward biased at terminal No. 13. In the circuit of Fig. 18, the color killer adjustment is a resistance divider circuit which establishes the threshold of burst level at which the killer operates the chroma amplifier.

CA3072 CIRCUIT OPERATION

The CA3072 is a chroma demodulator having full color difference signal demodulation capability. The chroma signal is applied to terminal Nos. 3 & 4 and the reference subcarrier signal is applied to terminals Nos. 6 & 7 of the CA3072. The output color difference signals are B-Y at terminal No. 13, R-Y at terminal No. 11, and G-Y at terminal No. 9. The typical level of differential chroma drive required at terminal Nos. 3 & 4 is 400 mV_{p-p}. The amplitude of chroma at terminal No. 6 & 7 is approximately 1.0 volt at 104° relative phase difference which results in a B-Y output amplitude of $5V_{p-p}$. The voltages of the R-Y & G-Y outputs are at 3.8 and 1.0 V_{p-p} respectively, when there is $5V_{p-p}$ output at B-Y. These comparative signals are based upon a complete phase rotation of the chroma relative to the subcarrier signal reference. The relative demodulation phase and amplitude ratios of the Fig. 18 circuit are shown in the oscilloscope trace photographs of Fig. 24. Using the hue control setting for B-Y phase at the B-Y output, the G-Y color-difference signal is approximately −104° and the R-Y color-difference signal is approximately +106°. Since the amplitude ratios are a function of the applied signal phase relationship, the NTSC color difference output signals are shown here primarily for phase reference conditions.

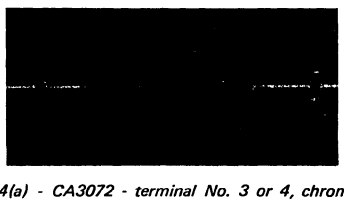


Fig. 24(a) - CA3072 - terminal No. 3 or 4, chroma input signal, 220 mV_{p-p}, one horizontal line

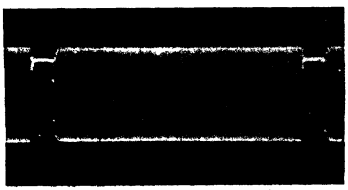


Fig. 24(b) - CA3072 - terminal No. 6 or 7, reference subcarrier 1.2 V_{p-p}, one horizontal line

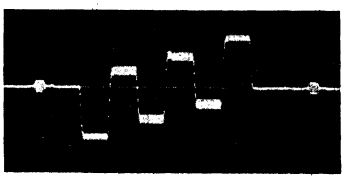


Fig. 24(c) - CA3072 terminal No. 13, 4.8 v_{p-p} B-Y output, one horizontal line



Fig. 24(d) - CA3072 - terminal No. 9, 1.2 v_{p-p} G-Y output, one horizontal line

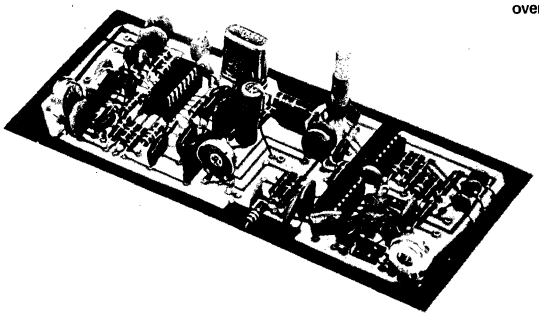


Fig. 25 (a) - Circuit layout and template (printed circuit board) for TV chroma system CA3070, CA3071, and CA3072.

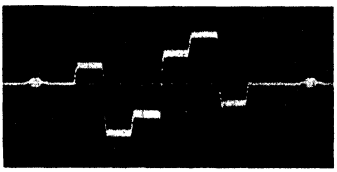


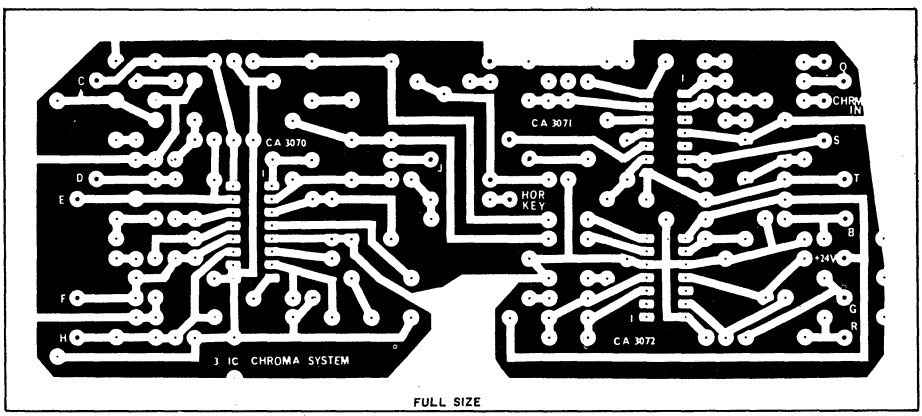
Fig. 24(e) - CA3072 - terminal No. 11, 5.2 v_{p-p} R-Y output, one horizontal line

CHROMA SYSTEM CONSTRUCTION

Fig. 25 shows the complete CA3070, CA3071 and CA3072 chroma system in the Fig. 18 circuit. Table I lists the dc terminal voltages for the system. The chroma gain and hue controls, as well as the switches S1 and S2 are removed. The template circuit board layout is also shown for duplication purposes. It should be noted that a few component values are modified in Fig. 18 from the dynamic circuit values of the data sheet. These are necessary for system matching and overall filter requirements.

TABLE 1 TYPICAL CHROMA SYSTEM TERMINAL DC VOLTAGES (NO SIGNAL INPUT)

TERMINAL No.	DC VOLTS		
	CA3070	CA3071	CA3072
1	7.6	7.3	–
2	11.5	1.7	–
3	11.5	–	3.3
4	−1.7	0	3.3
5	0	–	–
6	2.8	11.4	5.9
7	11.2	1.4	5.9
8	11.2	23.0	24.0
9	–	VARIABLE	14.7
10	12.0	VARIABLE	–
11	7.8	VARIABLE	14.7
12	7.8	15.0	–
13	6.7	VARIABLE	14.7
14	6.7	7.1	0
15	7.3	–	–
16	7.1	–	–



FULL SIZE

(b) — Printed circuit board template.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3075



14-lead dual-in-line plastic package, leads in quad-formed arrangement.

H-1596

FM IF Amplifier - Limiter, Detector, and Audio Preamplifier

For FM IF Amplifier Applications Up To 20 MHz In Communications Receivers And High-Fidelity Receivers

Features:

- Good sensitivity: Input limiting voltage (knee) = 250 μV typ. at 10.7 MHz
- Excellent AM rejection: 55 dB typ. at 10.7 MHz
- Internal Zener diode regulation for the IF amplifier section
- Low harmonic distortion
- Differential peak detection: Permits simplified single-coil tuning
- Audio preamplifier voltage gain: 21 dB typ.
- Minimum number of external parts required

RCA CA3075 is an integrated circuit which provides, in a single monolithic chip, an FM IF subsystem for Communications and High-Fidelity Receivers. This device, shown in the schematic diagram (Fig. 2), consists of a multistage IF amplifier-limiter section with a Zener regulated power supply, an FM detector stage, and an AF preamplifier section. A typical application of the CA3075, in FM receiver circuits, is shown in the block diagram (Fig. 1).

The three-stage, emitter-follower-coupled IF amplifier section provides a 60-dB typ. voltage gain at an operating frequency of 10.7 MHz and features, because of its transistor constant-current sink, an output stage with exceptionally good limiting characteristics.

The FM detector section, which utilizes a differential-peak-detection circuit, requires only a single coil in the associated outboard detector circuit; hence, tuning the detector circuit is a simple procedure.

The audio preamplifier circuit provides a 21-dB voltage gain with low impedance output for driving subsequent audio amplifier stages.

The CA3075 utilizes a 14-lead dual-in-line plastic package with leads in a special quad-formed arrangement.

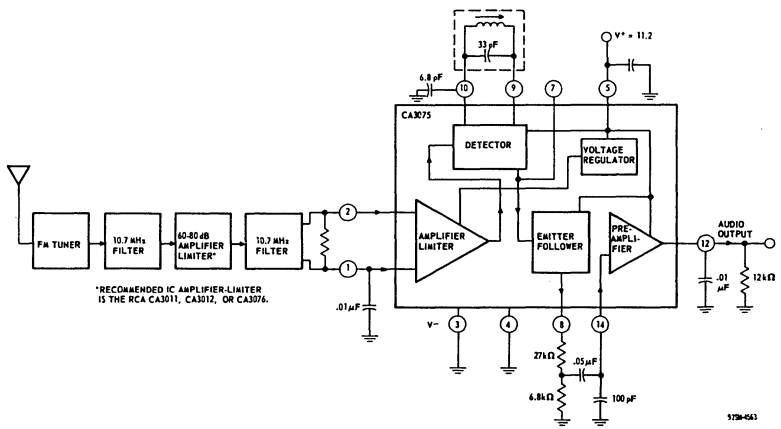


Fig. 1 - Block diagram of typical FM receiver utilizing the CA3075

MAXIMUM RATINGS, *Absolute-Maximum Values at* T_A = 25°C

DC Supply Voltage [between Terminals 5 (V^+) and 3 (V^-)]		12.5	V
DC Current (into Terminal 5)		30	mA
Device Dissipation:			
Up to T_A = 50°C		760	mW
Above T_A = 50°C	derate linearly	7.6	mW/°C
Ambient Temperature Range:			
Operating		−55 to +125	°C
Storage		−65 to +150	°C
Lead Temperature (During soldering for 10 s max.)		+265	°C

ELECTRICAL CHARACTERISTICS at T_A = 25° C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST CIRCUIT FIG. NO.
Static Characteristics							
DC Voltage:							
At Terminal 7	V_7	V^+ = 11.2 V	–	6.1	–	V	6
At Terminal 8	V_8		–	5.4	–	V	
At Terminal 12	V_{12}		–	5.2	–	V	
DC Current (into Terminal 5):							
At V^+ = 8.5 V			8.5	15	–	mA	
At V^+ = 11.2 V	I_5	–	–	17.5	–	mA	6
At V^+ = 12.5 V			–	19	29	mA	
Dynamic Characteristics at V^+ = 11.2							
IF AMPLIFIER Input Limiting Voltage (knee, – 3 dB point)	V_I(lim)	f_0 = 10.7 MHz f(Modulation) = 400 Hz Deviation = ±75 kHz	–	250	600	μV	3
AM Rejection	AMR	f_0 = 10.7 MHz f(Modulation) = 400 Hz FM: Deviation = ±75 kHz AM: Modulation = 30%	–	55	–	dB	5
Input Impedance Components:							
Parallel Resistance	R_I	f_0 = 10.7 MHz	–	4.5	–	kΩ	–
Parallel Capacitance	C_I	V_{IN} = 10 mV RMS	–	4.5	–	pF	
DETECTOR Recovered AF Voltage (at Terminal 12)	V_O(AF)	f_0 = 10.7 MHz f(Modulation) = 400 Hz	–	1.5	–	V	3
Total Harmonic Distortion	THD	Deviation = ± 75 kHz	–	1	2	%	
AUDIO PREAMPLIFIER Voltage Gain	A(AF)	V_{IN} = 100 mV, f_0 = 400 Hz	–	21	–	dB	4
Total Harmonic Distortion	THD	V_{OUT} = 2 V, f_0 = 400 Hz	–	1.5	5	%	4

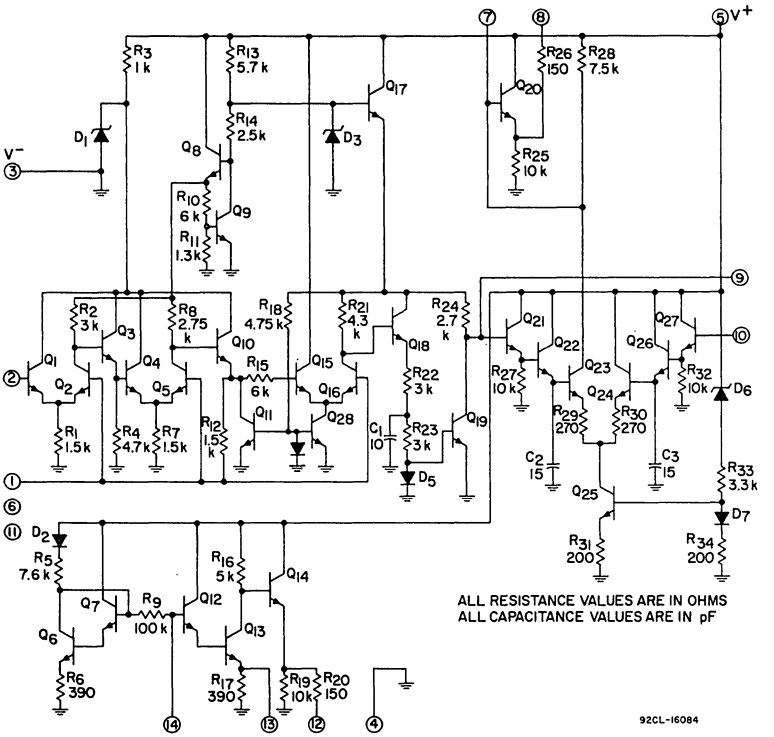


Fig. 2-Schematic diagram of CA3075

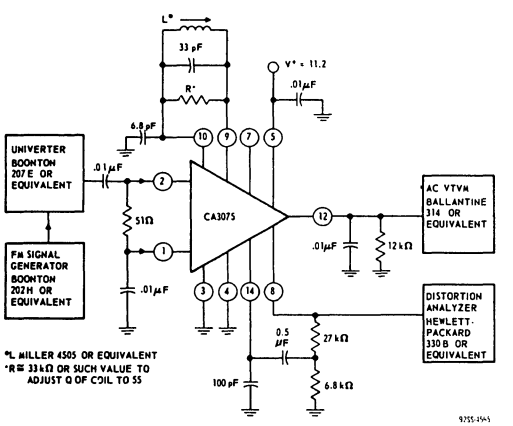


Fig.3 – Test Circuit for input limiting voltage, recovered AF voltage, and total harmonic distortion

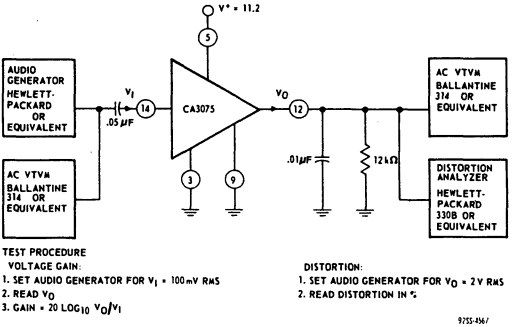


Fig.4 – Test circuit for audio preamplifier voltage gain and total harmonic distortion

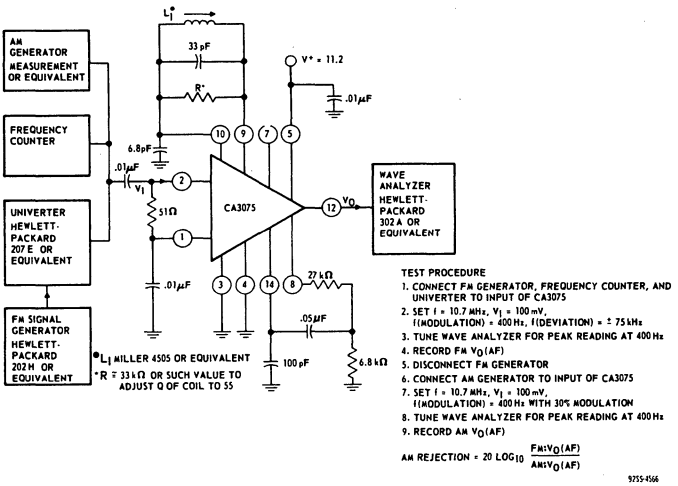


Fig.5 – Test circuit for AM rejection

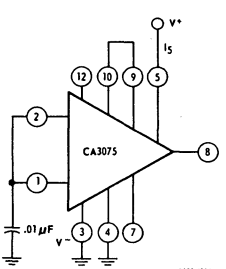


Fig. 6-Test circuit for static characteristics

RCA Solid State Division

Linear Integrated Circuits

CA3076



8-Lead TO-5

High-Gain Wide-Band IF Amplifier-Limiter

For FM IF Amplifier Applications in Communications Receivers

Features:

- exceptionally good sensitivity: input limiting voltage (knee) = 50 μV typ. at 10.7 MHz
- high gain: 80 dB with 2-kilohm load
- internal voltage supply regulator
- wide frequency capability: > 20 MHz

RCA CA3076, monolithic integrated circuit, is a high-gain wide-band amplifier-limiter for use in the IF sections of Communications and High-Fidelity FM Receivers. The CA3076, shown in the schematic diagram (Fig. 2), consists of a four stage IF amplifier-limiter section with a voltage regulator section. A typical application of the CA3076 in FM receiver circuits is shown in the block diagram (Fig. 1).

The four-stage emitter-follower-coupled IF amplifier section provides an 80-dB voltage gain with a 2-kilohm load at a frequency of 10.7 MHz. The output stage has exceptionally good limiting characteristics because of its transistor constant-current sink. The voltage regulator section provides zener-regulated, decoupled voltages for the IF amplifier.

The CA3076 utilizes an hermetically-sealed 8-lead TO-5 package.

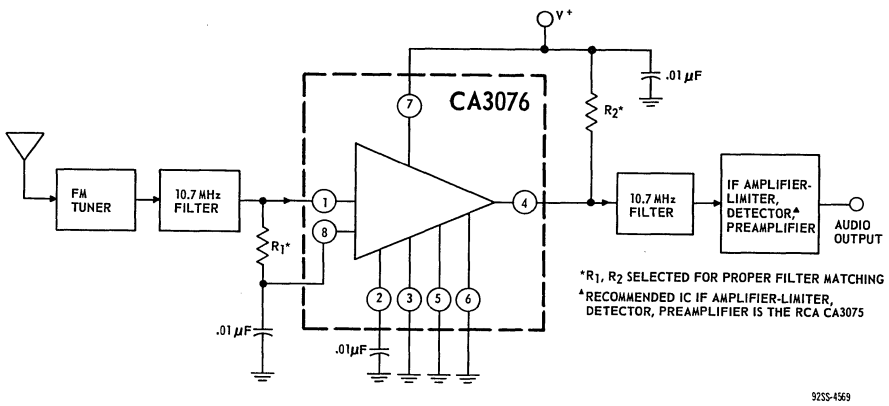


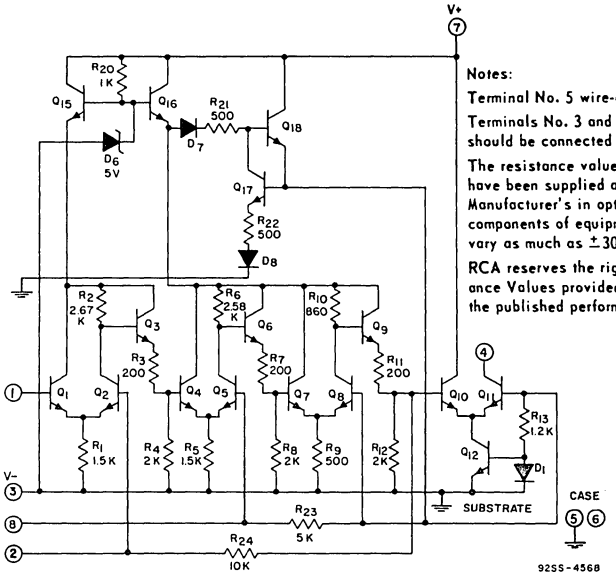
Fig. 1 - Block diagram of typical FM receiver utilizing the CA3076.

MAXIMUM RATINGS, *Absolute Maximum-Values at* T_A = 25°C

DC Supply Voltage [between Terminals 7 (V^+) and 3 (V^-)]	15	V
DC Current (into Terminal 7)	35	mA
Device Dissipation:		
Up to T_A = 50°C	500	mW
Above T_A = 50°C	derate linearly 5 mW/°C	
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/32 in (3.17 mm) from seating plane for 10 s max	+265	°C

ELECTRICAL CHARACTERISTICS at T_A = 25°C

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	LIMITS MIN.	LIMITS TYP.	LIMITS MAX.	UNITS	TEST CIRCUIT FIG. NO.
***Static Characteristics* – V^+ = 8.5 V**							
DC Current (into Term. 7)	I_7	–	10	15	24	mA	3
Quiescent Operating Current (into Term. 4)	I_4	–	–	0.65	–	mA	3
***Dynamic Characteristics* – V^+ = 8.5 V, f_O = 10.7 MHz**							
Input Limiting Voltage (knee, −3 dB point)	V_I (lim.)	–	–	50	200	μV	–
Output Voltage	V_O	V_I = 20 μV	4	12	–	mV	5
Output Noise Voltage	V_N	V_I = 0	–	1	–	mV	5
Forward Transfer Admittance:							
Magnitude	$\lvert Y_{21} \rvert$	V_I =	–	6	–	mho	4
Phase	θ_{21}	10 μV	–	80	–	degrees	
Reverse Transfer Admittance:							
Magnitude	$\lvert Y_{12} \rvert$	–	–	0.1	–	μmho	–
Phase	θ_{12}		–	−90	–	degrees	
Input-Impedance Components:							
Parallel Resistance	R_I	–	–	7.5	–	kΩ	–
Parallel Capacitance	C_I		–	4	–	pF	
Output-Impedance Components:							
Parallel Resistance	R_O	–	50	–	–	kΩ	–
Parallel Capacitance	C_O		–	1.7	–	pF	



Notes:

Terminal No. 5 wire-connected to the case.

Terminals No. 3 and 6 which are connected to the substrate should be connected to the most negative point in the circuit.

The resistance values included on the schematic diagram have been supplied as a convenience to assist Equipment Manufacturer's in optimizing the selection of "outboard" components of equipment designs. The values shown may vary as much as ±30%.

RCA reserves the right to make any changes in the Resistance Values provided such changes do not adversely affect the published performance characteristics of the device.

Fig. 2 - Schematic diagram of CA3076.

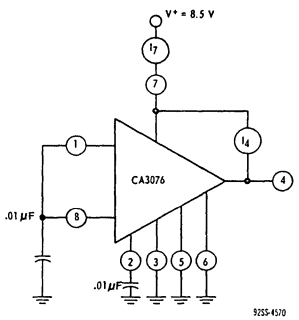


Fig. 3 - Test circuit for DC current (Terminal 7) and operating current (Terminal 4).

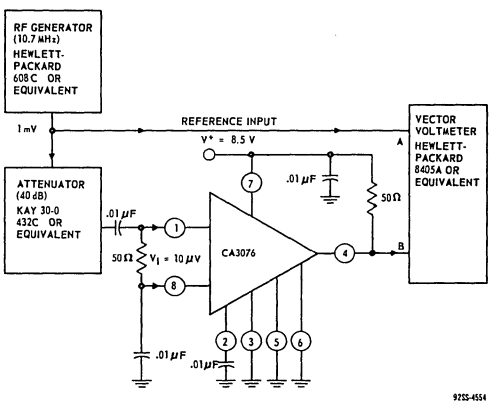


Fig. 4 - Forward transfer admittance (Y_{21}) test circuit

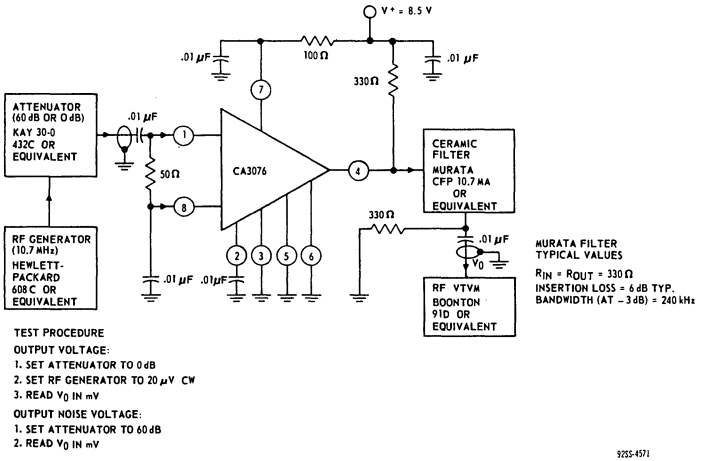


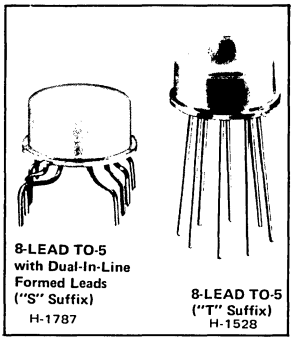
Fig. 5 - 10.7 MHz voltage gain and noise test circuit

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3078S CA3078T
CA3078AS CA3078AT



8-LEAD TO-5
with Dual-In-Line
Formed Leads
("S" Suffix)
H-1787

8-LEAD TO-5
("T" Suffix)
H-1528

Micropower Operational Amplifier

Features:

- Low standby power: as low as 700 nW
- Wide supply voltage range: ±0.75 to ±15 V
- High peak output current: 6.5 mA min.
- Adjustable quiescent current
- Output short-circuit protection

Applications:

- Portable electronics
- Medical electronics
- Instrumentation
- Telemetry

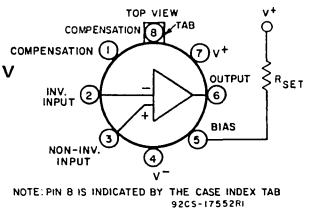


Fig.1 – Functional diagram of the CA3078T and CA3078AT.

The RCA CA3078T* and CA3078AT▲ are high-gain monolithic operational amplifiers which can deliver milli-amperes of current yet only consume microwatts of standby power. Their operating points are externally adjustable and frequency compensation may be accomplished with one external capacitor. The CA3078T and CA3078AT provide the designer with the opportunity to tailor the frequency response and improve the slew rate without sacrificing power. Operation with a single 1.5-volt battery is a practical reality with these devices.

* Formerly developmental type TA5807

▲ Formerly developmental type TA5807X

The CA3078AT is a premium device having a supply voltage range of $V^{\pm}$ = 0.75V to $V^{\pm}$ = 15V and an operating temperature range of -55°C to +125°C. The CA3078T has the same lower supply voltage limit but the upper limit is V^{+} = +6V and V^{-}=-6V. The operating temperature range is from 0°C to +70°C.

The CA3078 and CA3078A are supplied in either the standard 8-lead TO-5 package ("T" suffix), or in the 8-lead dual-in-line formed-lead "DIL-CAN" package ("S" suffix).

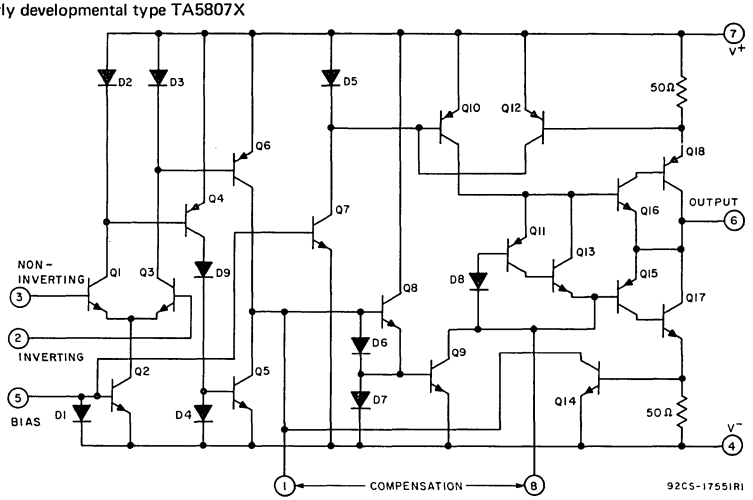


Fig.2-Schematic diagram of the CA3078T and CA3078AT.

ELECTRICAL CHARACTERISTICS
For Equipment Design

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			CA3078AT LIMITS R_{SET} = 5.1 MΩ, I_Q = 20 μA					UNITS
		V^+ & V^-	R_S KΩ	R_L KΩ	T_A = 25°C MIN	T_A = 25°C TYP	T_A = 25°C MAX	T_A = −55 to 125°C MIN	T_A = −55 to 125°C MAX	
Input Offset Voltage	V_{IO}	6	≤10	–	–	0.70	3.5	–	4.5	mV
Input Offset Current	I_{IO}		–	–	–	0.50	2.5	–	5.0	nA
Input Bias Current	I_{IB}		–	–	–	7	12	–	50	nA
Open-Loop Diff. Voltage Gain	A_{OL}		–	≥10	92	100	–	90	–	dB
Total Quiescent Current	I_Q		–	–	–	20	25	–	45	μA
Device Dissipation	P_D		–	–	–	240	300	–	540	μW
Maximum Output Voltage	V_{OM}		–	≥10	±5.1	±5.3	–	±5	–	V
Common-Mode Input Voltage Range	V_{ICR}		≤10	–	–	-5,5 to +5.8	–	-5 to +5	–	V
Common-Mode Rejection Ratio	CMRR		≤10	–	80	115	–	–	–	dB
Maximum Output Current	I_{OM}^+ or I_{OM}^-		–	–	–	12	–	6.5	30	mA
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V^+$		≤10	–	76	105	–	–	–	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$			–	76	105	–	–	–	
					R_{SET} = 13 MΩ, I_Q = 20 μA					
Input Offset Voltage	V_{IO}	15	≤10	–	–	1.4	3.5	–	4.5	mV
Open-Loop Diff. Voltage Gain	A_{OL}		–	≥10	92	100	–	88	–	dB
Total Quiescent Current	I_Q		–	–	–	20	30	–	50	μA
Device Dissipation	P_D		–	–	–	600	750	–	1350	μW
Maximum Output Voltage	V_{OM}		–	≥10	±13.7	±14.1		±13.5	–	V
Common-Mode Rejection Ratio	CMRR		≤10	–	80	106	–	–	–	dB
Input Bias Current	I_{IB}		–	–	–	7	14	–	55	nA
Input Offset Current	I_{IO}		–	–	–	0.50	2.7	–	5.5	nA

MAXIMUM RATINGS, *Absolute Maximum Values at T_A = 25°C*

	CA3078AT	CA3078T
DC Supply Voltage (between V^+ and V^- terminal	36V	14V
Differential Input Voltage	±6V	±6V
DC Input Voltage	V^+ to V^-	V^+ to V^-
Input Signal Current	0.1 mA	0.1 mA
Output Short-Circuit Duration*	No Limitation	No Limitation
Device Dissipation	50 mW (up to 125°C)	500 mW (up to 70°C)
Temperature Range:		
Operating	-55 to +125°C	0 to +70°C
Storage	-65 to +150°C	-65 to +150°C
Lead Temperature (During Soldering):		
At distance 1/16 ±1/32 in. (1.59 ±0.79 mm) from case for 10s max.	+300°C	+300°C

*Short circuit may be applied to ground or to either supply.

ELECTRICAL CHARACTERISTICS
For Equipment Design

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			CA3078T LIMITS R_{SET} = 1 MΩ, I_Q = 100 μA					UNITS
		V^+ & V^-	R_S KΩ	R_L KΩ	T_A = 25°C MIN	T_A = 25°C TYP	T_A = 25°C MAX	T_A = 0 to 70°C MIN	T_A = 0 to 70°C MAX	
Input Offset Voltage	V_{IO}	6	≤10	–	–	1.3	4.5	–	5	mV
Input Offset Current	I_{IO}		–	–	–	6	32	–	40	nA
Input Bias Current	I_{IB}		–	–	–	60	170	–	200	nA
Open-Loop Diff. Voltage Gain	A_{OL}		–	≥10	88	92	–	86	–	dB
Total Quiescent Current	I_Q		–	–	–	100	130	–	150	μA
Device Dissipation	P_D		–	–	–	1200	1560	–	1800	μW
Maximum Output Voltage	V_{OM}		–	≥10	±5.1	±5.3	–	±5	–	V
Common-Mode Input Voltage Range	V_{ICR}		≤10	–	–	-5.5 to +5.8	–	-5 to +5	–	V
Common-Mode Rejection Ratio	CMRR		≤10	–	80	110	–	–	–	dB
Maximum Output Current	I_{OM}^+ or I_{OM}^-		–	–	–	12	–	6.5	30	mA
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V^+$		≤10	–	76	93	–	–	–	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$			–	76	93	–	–	–	
Input Offset Voltage	V_{IO}	15	≤10	–	–	–	–	–	–	mV
Open-Loop Diff. Voltage Gain	A_{OL}		–	≥10	–	–	–	–	–	dB
Total Quiescent Current	I_Q		–	–	–	–	–	–	–	μA
Device Dissipation	P_D		–	–	–	–	–	–	–	μW
Maximum Output Voltage	V_{OM}		–	≥10	–	–	–	–	–	V
Common-Mode Rejection Ratio	CMRR		≤10	–	–	–	–	–	–	dB
Input Bias Current	I_{IB}		–	–	–	–	–	–	–	nA
Input Offset Current	I_{IO}		–	–	–	–	–	–	–	nA

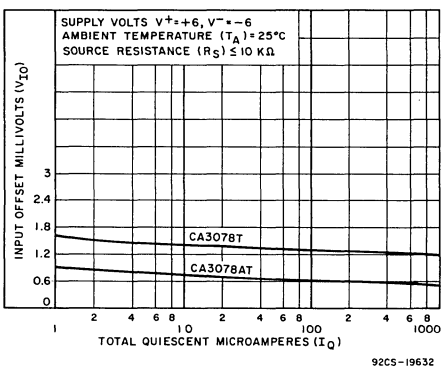


Fig.3 – Input offset voltage vs. total quiescent current.

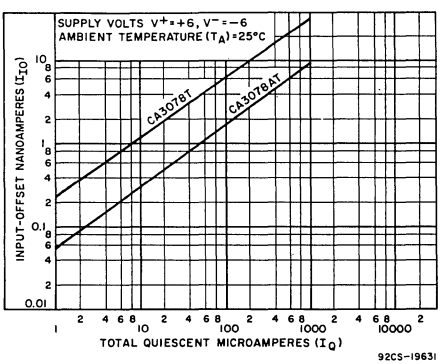


Fig.4 – Input offset current vs. total quiescent current.

ELECTRICAL CHARACTERISTICS, at $T_A = 25^oC$
Typical Values Intended Only for Design Guidance

CHARACTERISTICS SYMBOLS	TYPICAL VALUES				UNITS
	CA3078AT		CA3078T		
	$V^+ = +1.3V$, $V^- = -1.3V$, $R_{SET} = 2\ M\Omega$, $I_Q = 10\ \mu A$	$V^+ = +0.75V$, $V^- = -0.75V$, $R_{SET} = 10\ M\Omega$, $I_Q = 1\ \mu A$	$V^+ = +1.3V$, $V^- = -1.3V$, $R_{SET} = 2\ M\Omega$, $I_Q = 10\ \mu A$	$V^+ = 0.75V$, $V^- = -0.75V$, $R_{SET} = 10\ M\Omega$, $I_Q = 1\ \mu A$	
V_{IO}	0.7	0.9	1.3	1.5	mV
I_{IO}	0.3	0.054	1.7	0.5	nA
I_{IB}	3.7	0.45	9	1.3	nA
A_{OL}	84	65	80	60	dB
I_Q	10	1	10	1	μA
P_D	26	1.5	26	1.5	μW
V_{OPP}	1.4	0.3	1.4	0.3	V
V_{ICR}	−0.8 to +1.1	−0.2 to +0.5	−0.8 to +1.1	−0.2 to +0.5	V
CMRR	100	90	100	90	dB
$I_{OM}^{\pm}$	12	0.5	12	0.5	mA
$\Delta V_{IO}/\Delta V^{\pm}$	20	50	20	50	$\mu V/V$

Typical Values Intended Only for Design Guidance at $T_A = 25^oC$ and $V^+ = +6\ V$, $V^- = 6\ V$

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS	CA3078AT		CA3078T	UNITS
			$R_{SET} = 5.1\ M\Omega$, $I_Q = 20\ \mu A$	$R_{SET} = 1\ M\Omega$, $I_Q = 100\ \mu A$	$R_{SET} = 1\ M\Omega$, $I_Q = 100\ \mu A$	
Input Offset Voltage Drift	$\Delta V_{IO}/\Delta T_A$	$R_S \leq 10\ K\Omega$	5	6	6	$\mu V/^oC$
Input Offset Current Drift	$\Delta V_{IO}/\Delta T_A$	$R_S \leq 10\ K\Omega$	6.3	70	70	$pA/^oC$
Open-Loop Bandwidth	BW_{OL}	3dB pt.	0.3	2	2	kHz
Slew Rate: Unity Gain	SR	See Figs. 20, 21	0.027	0.04	0.04	$V/\mu s$
Comparator			0.5	1.5	1.5	
Transient Response		10% to 90% Rise Time	3	2.5	2.5	μs
Input Resistance	R_I		7.4	1.7	0.87	$M\Omega$
Output Resistance	R_O		1	0.8	0.8	$K\Omega$
Equiv. Input Noise Voltage	$e_N(10Hz)$	$R_S = 0$	40	—	25	$nV/\sqrt{Hz}$
Equiv. Input Noise Current	$i_N(10Hz)$	$R_S = 1\ M\Omega$	0.25	—	1	$pA/\sqrt{Hz}$

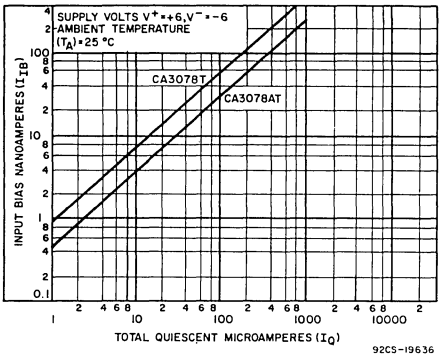


Fig.5 – Input bias current vs. total quiescent current.

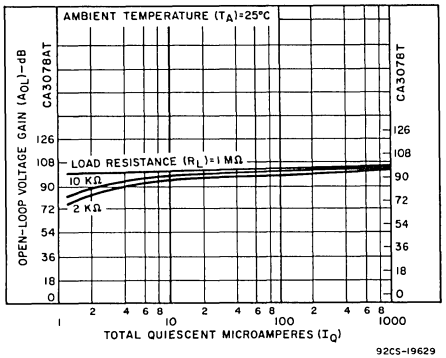


Fig.6 – Open-loop voltage gain vs. total quiescent current.

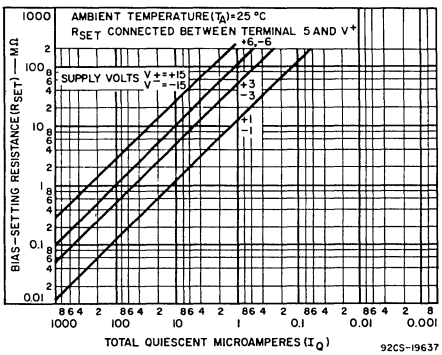


Fig.7 – Bias-setting resistance vs. total quiescent current.

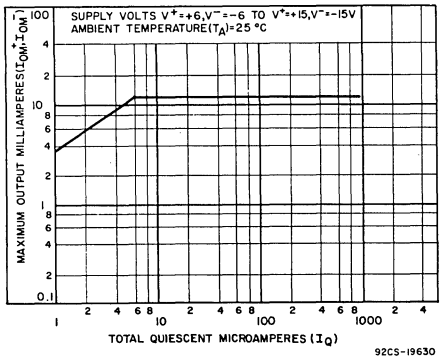


Fig.8 – Maximum output current vs. total quiescent current.

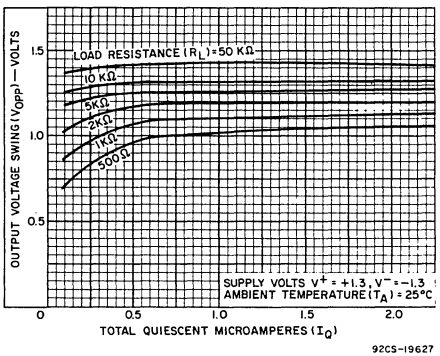


Fig.9 – Output voltage swing vs. total quiescent current.

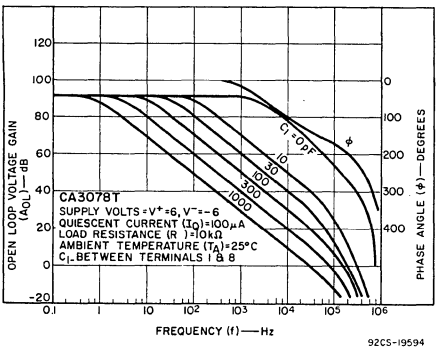


Fig.10 – Open-loop voltage gain vs. frequency for I_Q = 100 μA – CA3078T.

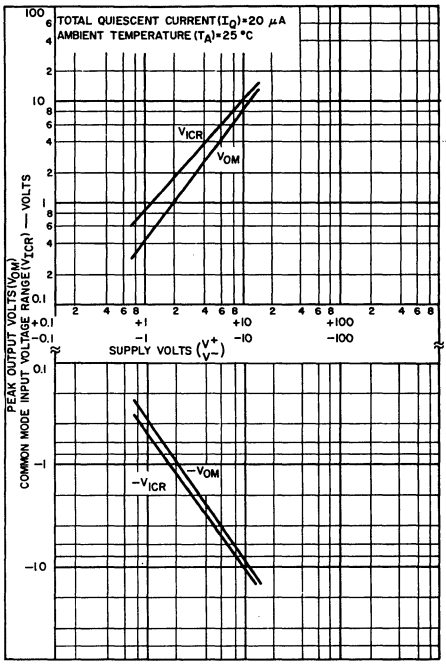


Fig.11 — Output and common-mode voltage vs. supply voltage.

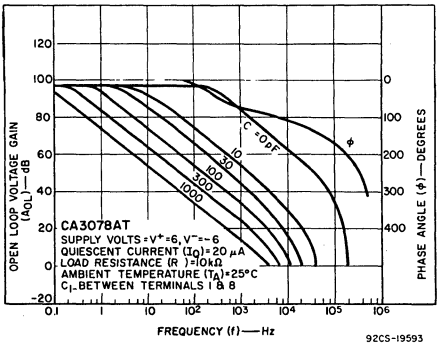


Fig.12 — Open-loop voltage gain vs. frequency for I_Q = 20 μA — CA3078AT.

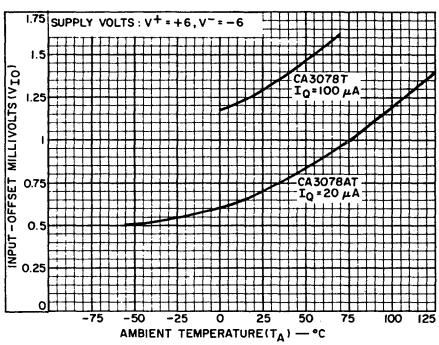


Fig.13 — Input offset voltage vs. temperature.

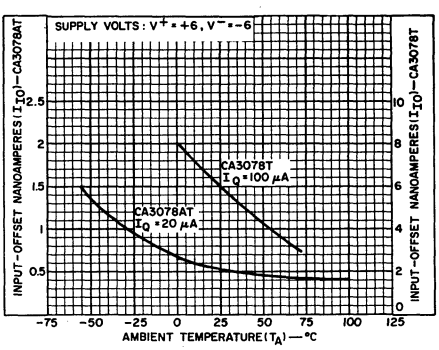


Fig.14 — Input offset current vs. temperature.

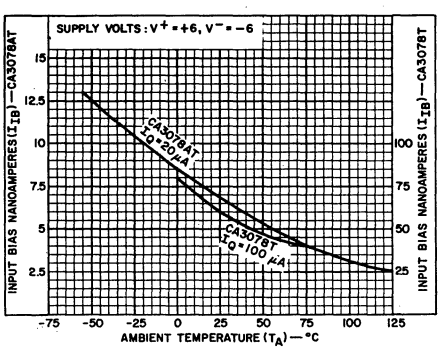


Fig.15 — Input bias current vs. temperature.

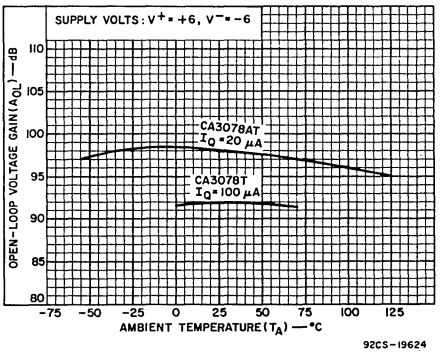


Fig.16 — Open-loop voltage gain vs. temperature.

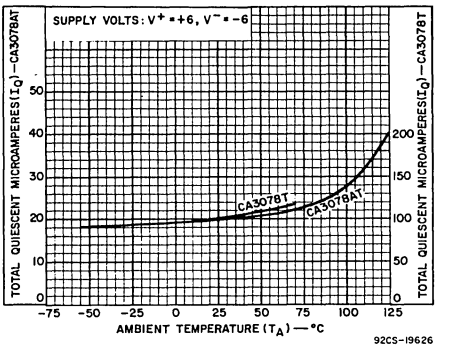


Fig.17 — Total quiescent current vs. temperature.

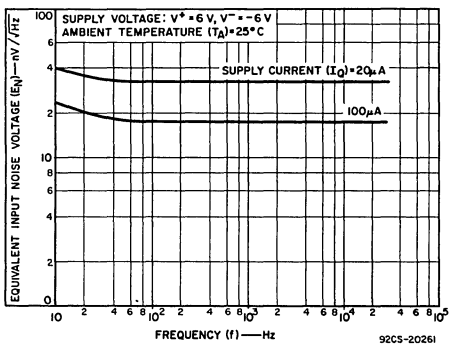


Fig.18 — Equivalent input noise voltage vs. frequency.

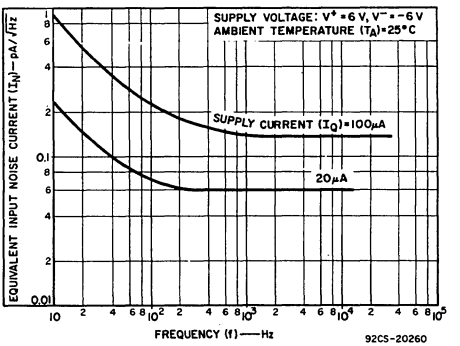


Fig.19 — Equivalent input noise current vs. frequency.

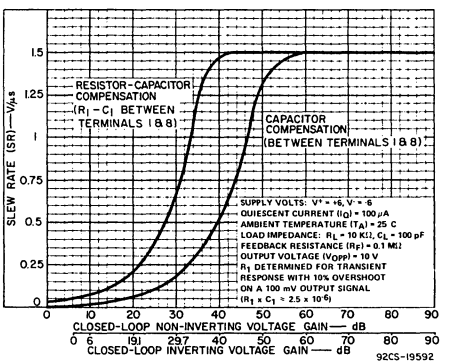


Fig.20 — Slew rate vs. closed-loop gain for I_Q = 100 μA — CA3078T.

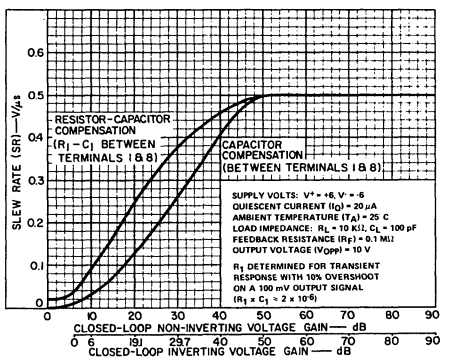


Fig.21 — Slew rate vs. closed-loop gain for I_Q = 20 μA — CA3078AT.

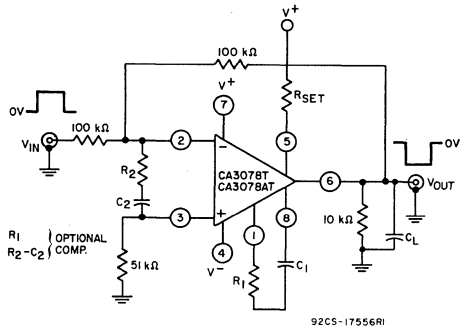


Fig.22 – Transient response and slew-rate, unity gain (inverting) test circuit.

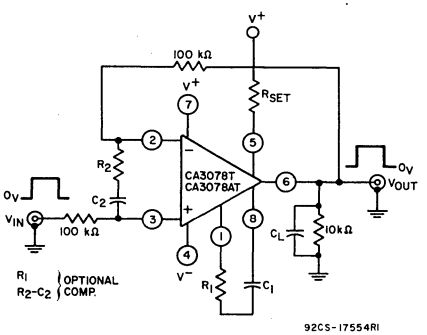


Fig.23 – Slew, rate, unity gain (non-inverting) test circuit.

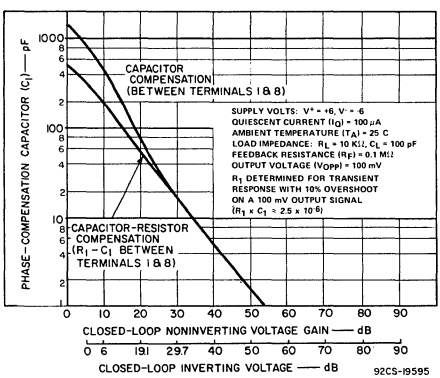


Fig.24 – Phase compensation capacitance vs. closed-loop gain – CA3078T.

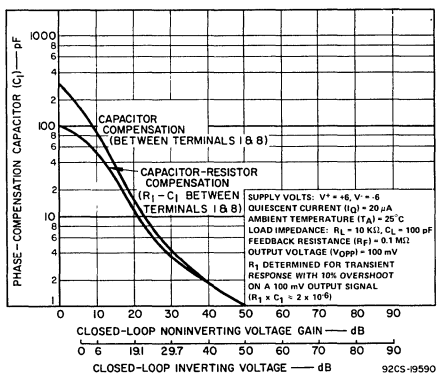


Fig.25 – Phase compensation capacitance vs. closed-loop gain – CA3078AT.

Table I - Unity-gain slew rate vs. compensation – CA3078T and CA3078AT

SUPPLY VOLTS: $V^+ = 6$, $V^- = -6$
OUTPUT VOLTAGE (V_O) = ±5V
LOAD RESISTANCE (R_L) = 10 kΩ

TRANSIENT RESPONSE: 10% OVERSHOOT FOR AN OUTPUT VOLTAGE of 100 mV
AMBIENT TEMPERATURE (T_A) = 25°C

COMPENSATION TECHNIQUE	UNITY GAIN (INVERTING) Fig. 22					UNITY GAIN (NON-INVERTING) Fig. 23				
	R1	C1	R2	C2	SLEW RATE	R1	C1	R2	C2	SLEW RATE
CA3078T – I_Q = 100 μA	kΩ	pF	kΩ	μF	V/μs	kΩ	pF	kΩ	μF	V/μs
Single Capacitor	0	750	∞	0	0.0085	0	1500	∞	0	0.0095
Resistor & Capacitor	3.5	350	∞	0	0.04	5.3	500	∞	0	0.024
Input	∞	0	0.25	0.306	0.67	∞	0	0.311	0.45	0.67
CA3078AT – I_Q = 20 μA										
Single Capacitor	0	300	∞	0	0.0095	0	800	∞	0	0.003
Resistor & Capacitor	14	100	∞	0	0.027	34	125	∞	0	0.02
Input	∞	0	0.644	0.156	0.29	∞	0	0.77	0.4	0.4

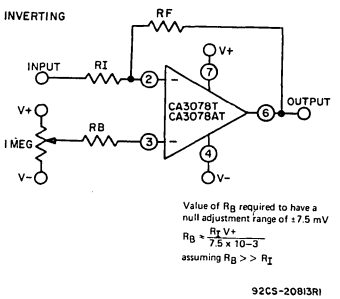
OPERATING CONSIDERATIONS

Compensation Techniques

The CA3078AT and CA3078T can be phase-compensated with one or two external components depending upon the closed-loop gain, power consumption, and speed desired. The recommended compensation is a resistor in series with a capacitor connected from terminal 1 to terminal 8. Values of the resistor and capacitor required for compensation as a function of closed loop gain are shown in Figs. 24 and 25. These curves represent the compensation necessary at quiescent currents of 20 μA and 100 μA, respectively, for a transient response with 10% overshoot. Figs. 21 and 22 show the slew rates that can be obtained with the two different compensation techniques. Higher speeds can be achieved with input compensation, but this increases noise output. Compensation can also be accomplished with a single capacitor connected from terminal 1 to terminal 8, with speed being sacrificed for simplicity. Table 1 gives an indication of slew rates that can be obtained with various compensation techniques at quiescent currents of 20 μA and 100 μA.

Single Supply Operation

The CA3078AT and CA3078T can operate from a single supply with a minimum total supply voltage of 1.5 volts. Figs. 27 and 28 show the CA3078AT or CA3078T in inverting and non-inverting 20-dB amplifier configurations utilizing a 1.5-volt type "AA" cell for a supply. The total power consumption for either circuit is approximately 675 nanowatts. The output voltage swing in this configuration is 300 mV p-p with a 20 kΩ load.



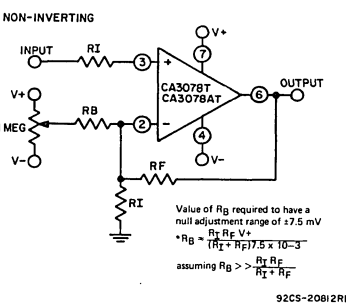


Fig.26 – Offset voltage null circuits.

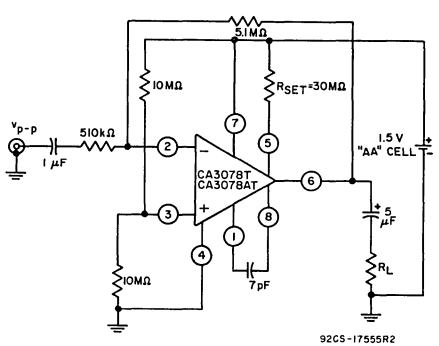


Fig.27 – Inverting 20-dB amplifier circuit.

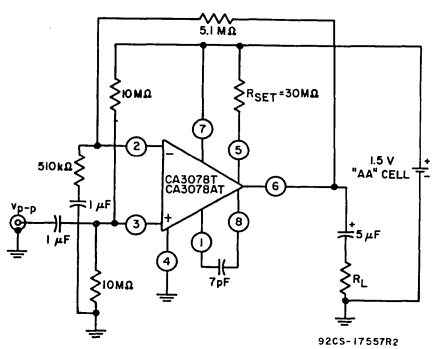


Fig.28 – Non-inverting 20-dB amplifier circuit.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3080, CA3080A Types



CA3080S
CA3080AS
8-LEAD TO-5
with Dual-In-Line
Formed Leads
"DIL-CAN"
H-1787



CA3080
CA3080A
8-LEAD TO-5
H-1528



CA3080E
8-LEAD Dual-
In-Line Plastic
Package "MINI-DIP"
H-1817

Operational Transconductance Amplifiers (OTA's)

Gateable-Gain Blocks

Features:

- Slew rate (unity gain, compensated): 50 V/μs
- Adjustable power consumption: 10μW to 30 mW
- Flexible supply voltage range: ±2 V to ±15 V
- Fully adjustable gain: 0 to $g_m R_L$ limit
- Tight g_m spread: CA3080 (2:1), CA3080A (1.6:1)
- Extended g_m linearity: 3 decades

Applications:

- Sample and hold
- Multiplex
- Voltage follower
- Multiplier
- Comparator

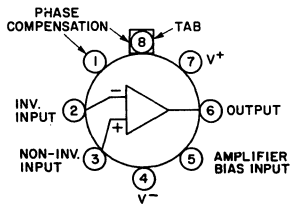
The RCA-CA3080 and CA3080A are Gateable-Gain Blocks which utilize the unique Operational Transconductance Amplifier (OTA) concept described in Application Note ICAN-6668, "Applications of the CA3080 and CA3080A High-Performance Operational Transconductance Amplifiers".

The CA3080 and CA3080A have differential input and a single-ended, push-pull, class A output. In addition, these types have an amplifier bias input which may be used either for gating or for linear gain control. These types also have a high output impedance and their transconductance (g_m) is directly proportional to the amplifier bias current (I_{ABC}).

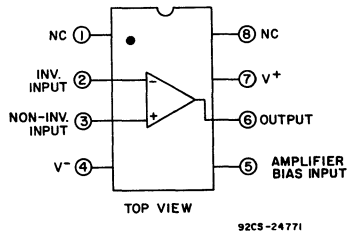
The CA3080 and CA3080A are notable for their excellent slew rate (50 V/μs), which makes them especially useful for multiplex and fast unity-gain voltage followers. These types are especially applicable for multiplex applications because power is consumed only when the devices are in the "ON" channel state.

The CA3080A is rated for operation over the full military-temperature range (−55 to +125°C) and its characteristics are specifically controlled for applications such as sample-hold, gain-control, multiplex, etc. Operational transconductance amplifiers are also useful in programmable power-switch applications, e.g., as described in Application Note ICAN-6048, "Some Applications of a Programmable Power Switch/Amplifier" (CA3094, CA3094A, CA3094B).

These types are supplied in the 8-lead TO-5 style package (CA3080, CA3080A), and in the 8-lead TO-5 style package with dual-in-line formed leads ("DIL-CAN", CA3080S, CA3080AS). The CA3080E is also supplied in the 8-lead dual-in-line plastic ("MINI-DIP") package (CA3080E), and in chip form (CA3080H).



TO-5 Style Package



Plastic Package (CA3080E)

Fig. 1—Functional diagrams.

ELECTRICAL CHARACTERISTICS
For Equipment Design **CA3080**

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS			LIMITS			UNITS
		Circuit Fig.	$V^+ = 15$ V, $V^- = -15$ V, $I_{ABC} = 500\ \mu A$, $T_A = 25^oC$ (unless indicated otherwise)	Typical Characteristics Curves Fig.	Min.	Typ.	Max.	
Input Offset Voltage	V_{IO}	–		3	–	0.4	5	mV
			T_A = 0 to 70°C		–	–	6	
Input Offset Current	I_{IO}	–		4	–	0.12	0.6	μA
Input Bias Current	I_I	–		5	–	2	5	μA
			T_A = 0 to 70°C		–	–	7	
Forward Transconductance (large signal)	g_m	–		14	6700	9600	13000	μmho
			T_A = 0 to 70°C		5400	–	–	
Peak Output Current	$\|I_{OM}\|$	–	$R_L = 0$	6	350	500	650	μA
			$R_L = 0$, T_A = 0 to 70°C		300	–	–	
Peak Output Voltage: Positive	V^+_{OM}	–	$R_L = \infty$	7	12	13.5	–	V
Negative	V^-_{OM}				−12	−14.4	–	
Amplifier Supply Current	I_A	–		8	0.8	1	1.2	mA
Device Dissipation	P_D	–		9	24	30	36	mW
Input Offset Voltage Sensitivity: Positive	$\Delta V_{IO}/\Delta V^+$	–		–	–	–	150	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$	–		–	–	–	150	
Common-Mode Rejection Ratio	CMRR	–		–	80	110	–	dB
Common-Mode Input-Voltage Range	V_{ICR}	–		7	12 to −12	13.6 to −14.6	–	V
Input Resistance	R_I	–		15	10	26	–	kΩ

ELECTRICAL CHARACTERISTICS
Typical Values Intended Only for Design Guidance **CA3080**

Input Offset Voltage	V_{IO}	–	$I_{ABC} = 5\ \mu A$	3	0.3	mV
Input Offset Voltage Change	$\|\Delta V_{IO}\|$	–	Change in V_{IO} between $I_{ABC} = 500\ \mu A$ and $I_{ABC} = 5\ \mu A$	–	0.2	mV
Peak Output Current	I_{OM}	–	$I_{ABC} = 5\ \mu A$	6	5	μA
Peak Output Voltage: Positive	V^+_{OM}	–	$I_{ABC} = 5\ \mu A$	7	13.8	V
Negative	V^-_{OM}				−14.5	
Magnitude of Leakage Current		10	$I_{ABC} = 0$, $V_{TP} = 0$	11	0.08	nA
			$I_{ABC} = 0$, $V_{TP} = 36$ V		0.3	
Differential Input Current		12	$I_{ABC} = 0$, $V_{DIFF} = 4$ V	13	0.008	nA
Amplifier Bias Voltage	V_{ABC}	–		16	0.71	V
Slew Rate: Maximum (uncompensated)	SR			–	75	V/μs
Unity Gain (compensated)		23			50	
Open-Loop Bandwidth	BW_{OL}	–		–	2	MHz
Input Capacitance	C_I	–	f = 1 MHz	17	3.6	pF
Output Capacitance	C_O	–	f = 1 MHz	17	5.6	pF
Output Resistance	R_O	–		18	15	MΩ
Input-to-Output Capacitance	$C_{I\text{-}O}$	19	f = 1 MHz	20	0.024	pF

ELECTRICAL CHARACTERISTICS

For Equipment Design — **CA3080A**

CHARACTERISTICS	SYMBOLS	Circuit Fig.	TEST CONDITIONS (V^+ = 15 V, V^- = −15 V, I_{ABC} = 500 μA, T_A = 25°C (unless indicated otherwise))	Typical Characteristics Curves Fig.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Input Offset Voltage	V_{IO}	–	I_{ABC} = 5 μA	3	–	0.3	2	mV
					–	0.4	2	
			T_A = −55 to +125°C		–	–	5	
Input Offset Voltage Change	$\lvert\Delta V_{IO}\rvert$	–	Change in V_{IO} between I_{ABC} = 500 μA and I_{ABC} = 5 μA	3	–	0.1	3	mV
Input Offset Current	I_{IO}	–		4	–	0.12	0.6	μA
Input Bias Current	I_I	–		5	–	2	5	μA
			T_A = −55 to +125		–	–	8	
Forward Transconductance (large signal)	g_m	–		14	7700	9600	12000	μmho
			T_A = −55 to + 125°C		4000	–	–	
Peak Output Current	$\lvert I_{OM}\rvert$		I_{ABC} = 5 μA, R_L = 0	6	3	5	7	μA
			R_L = 0		350	500	650	
			R_L = 0, T_A = −55 to +125°C		300	–	–	
Peak Output Voltage:								
Positive	V^+_{OM}	–	I_{ABC} = 5 μA	7	12	13.8	–	V
Negative	V^-_{OM}		R_L = ∞		−12	−14.5	–	
Positive	V^+_{OM}	–	R_L = ∞		12	13.5	–	
Negative	V^-_{OM}				−12	−14.4	–	
Amplifier Supply Current	I_A	–		8	0.8	1	1.2	mA
Device Dissipation	P_D	–		9	24	30	36	mW
Input Offset Voltage Sensitivity:								
Positive	$\Delta V_{IO}/\Delta V^+$	–		–	–	–	150	μV/V
Negative	$\Delta V_{IO}/\Delta V^-$				–	–	150	
Magnitude of Leakage Current		10	I_{ABC} = 0, V_{TP} = 0	11	–	0.08	5	nA
			I_{ABC} = 0, V_{TP} = 36 V		–	0.3	5	
Differential Input Current		12	I_{ABC} = 0, V_{DIFF} = 4 V	13	–	0.008	5	nA
Common-Mode Rejection Ratio	CMRR	–		–	80	110	–	dB
Common-Mode Input-Voltage Range	V_{ICR}	–		7	12 to −12	13.6 to −14.6	–	V
Input Resistance	R_I	–		15	10	26	–	kΩ

ELECTRICAL CHARACTERISTICS

Typical Values Intended Only For Design Guidance — **CA3080A**

Amplifier Bias Voltage	V_{ABC}	–		16	0.71	V
Slew Rate:						
Maximum (uncompensated)	SR			–	75	V/μs
Unity Gain (compensated)		23			50	
Open-Loop Bandwidth	BW_{OL}	–	–	–	2	MHz
Input Capacitance	C_I	–	f = 1 MHz	17	3.6	pF
Output Capacitance	C_O	–	f = 1 MHz	17	5.6	pF
Output Resistance	R_O	–		18	15	MΩ
Input-to-Output Capacitance	$C_{I\text{-}O}$	19	f = 1 MHz	20	0.024	pF

MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25^oC$

DC SUPPLY VOLTAGE (between V^+ and V^- terminals)	36 V
DIFFERENTIAL INPUT VOLTAGE	±5 V
DC INPUT VOLTAGE	V^+ to V^-
INPUT SIGNAL CURRENT	1 mA
AMPLIFIER BIAS CURRENT	2 mA
OUTPUT SHORT-CIRCUIT DURATION*	Indefinite
DEVICE DISSIPATION	125 mW
TEMPERATURE RANGE:	
Operating	
CA3080	0 to + 70 °C
CA3080A	−55 to + 125 °C
Storage	−65 to + 150 °C
LEAD TEMPERATURE (During Soldering):	
At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 s max..	+265 °C

* Short circuit may be applied to ground or to either supply.

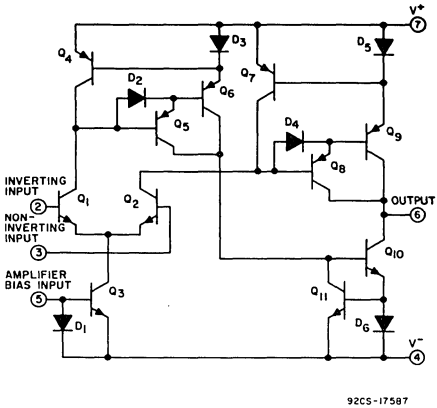


Fig.2 – Schematic diagram for CA3080 and CA3080A.

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS FOR THE CA3080 AND CA3080A

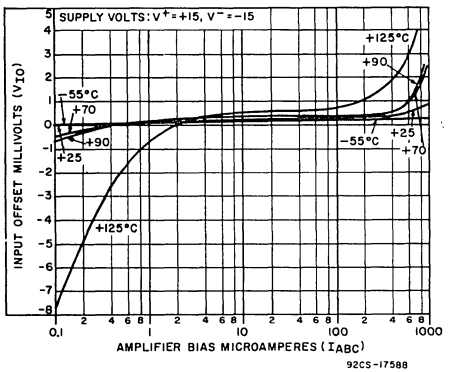


Fig.3 – Input offset voltage vs. amplifier bias current.

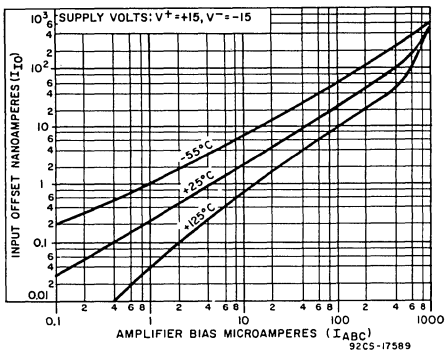


Fig.4 – Input offset current vs. amplifier bias current.

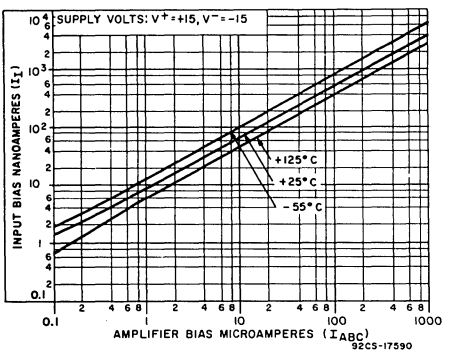


Fig.5 – Input bias current vs. amplifier bias current.

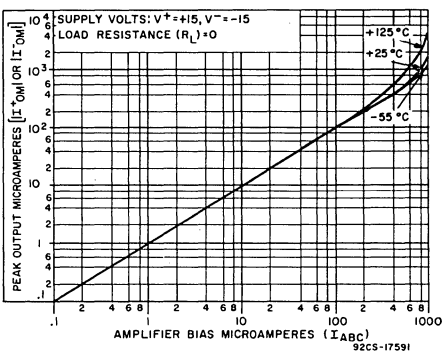


Fig.6 – Peak output current vs. amplifier bias current.

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS (Cont'd)

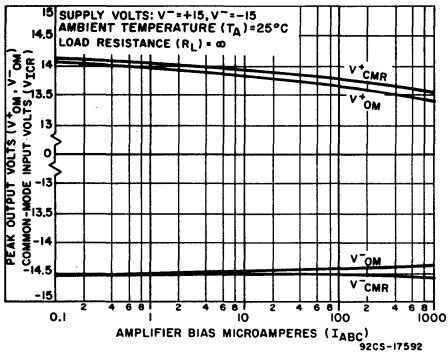


Fig.7 – Peak output voltage vs. amplifier bias current.

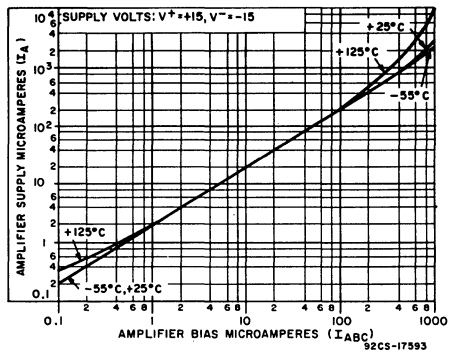


Fig.8 – Amplifier supply current vs. amplifier bias current.

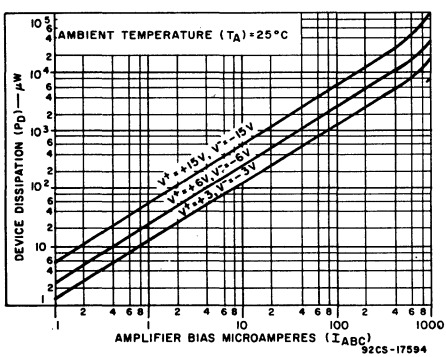


Fig.9 – Total power dissipation vs. amplifier bias current.

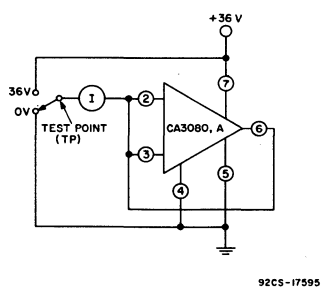


Fig.10 – Leakage current test circuit.

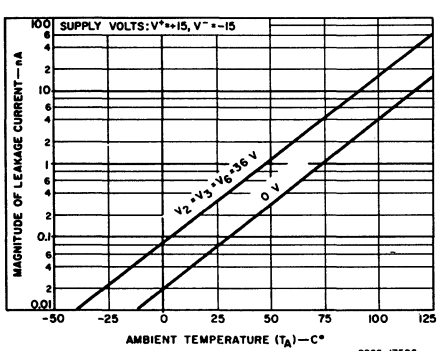


Fig.11 – Leakage current vs. temperature.

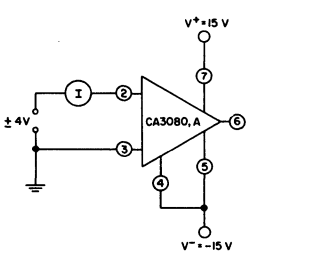


Fig.12 – Differential input current test circuit.

TYPICAL CHARACTERISTIC CURVES AND TEST CIRCUITS (Cont'd)

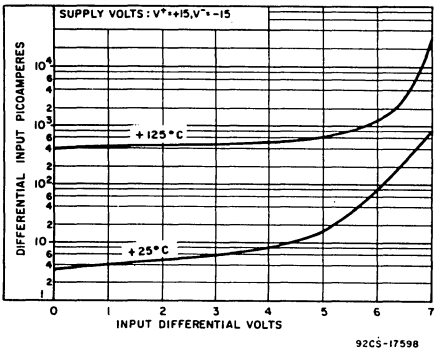


Fig. 13 — Input current vs. input differential voltage.

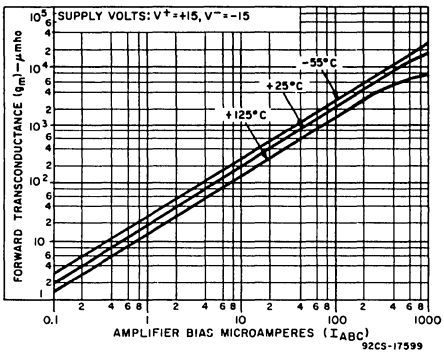


Fig. 14 — Transconductance vs. amplifier bias current.

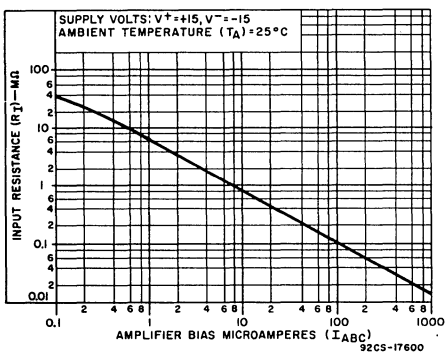


Fig. 15 — Input resistance vs. amplifier bias current.

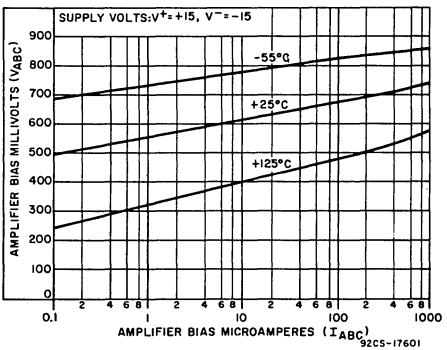


Fig. 16 — Amplifier bias voltage vs. amplifier bias current.

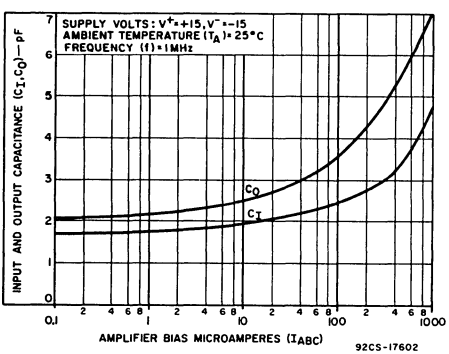


Fig. 17 — Input and output capacitance vs. amplifier bias current.

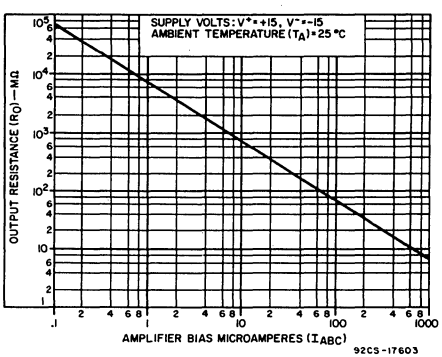


Fig. 18 — Output resistance vs. amplifier bias current.

TYPICAL CHARACTERISTICS CURVES AND TEST CIRCUITS (Cont'd)

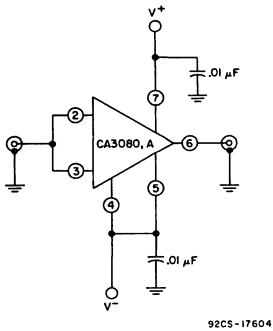


Fig.19 — Input-to-output capacitance test circuit.

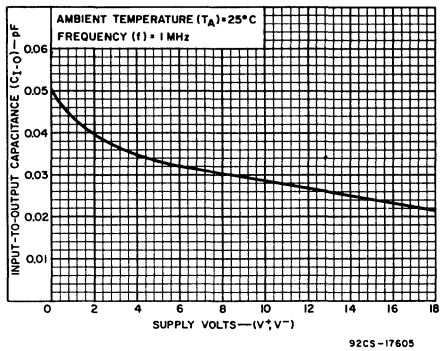
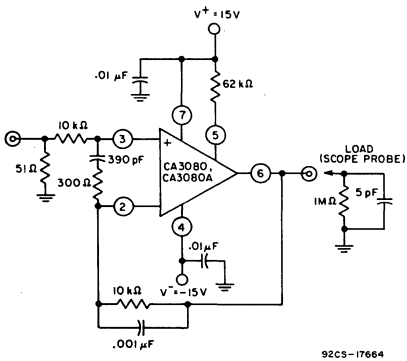


Fig.20 — Input-to-output capacitance vs. supply voltage.



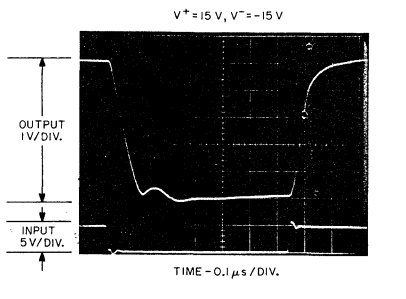
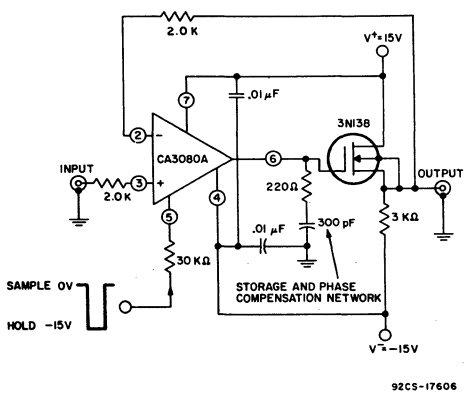


Fig.21 — Schematic diagram of the CA3080 and CA3080A in a unity-gain voltage follower configuration and associated waveform.



SLEW RATE(IN SAMPLE MODE)= 1.3 V/μs
ACQUISITION TIME* = 3 μs

* TIME REQUIRED FOR OUTPUT TO SETTLE WITHIN ±3mV OF A 4-VOLT STEP

Fig.22 — Schematic diagram of the CA3080A in a sample-hold configuration.

RCA Solid State Division

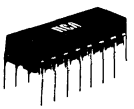
Linear Integrated Circuits

Monolithic Silicon

CA3081, CA3081F
CA3082, CA3082F



16-Lead Dual-In-Line
Frit-Seal (Hermetic)
Package H-1807



16-Lead Dual-In-Line
Plastic Package H-1622

General-Purpose High-Current N-P-N Transistor Arrays

CA3081–Common-Emitter Array CA3082–Common-Collector Array

Directly Drive 7-Segment Incandescent Displays and Light-Emitting-Diode (LED) Displays

Features

- 7 transistors permit a wide range of applications in either a common-emitter (CA3081) or common-collector (CA3082) configuration
- High I_C: 100 mA max.
- Low $V_{CE\ sat}$ (at 50 mA): 0.4 V typ.

Applications

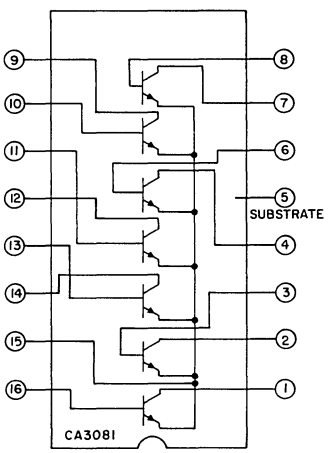
- Drivers for:
  - Incandescent display devices (e.g. RCA NUMITRON DR2000 Series and lamps)
  - LED (e.g. RCA-SG1002 GaAs High-Efficiency Emitting Diode)
  - Relay control
  - Thyristor firing

RCA-CA3081* and CA3082* consist of seven high-current (to 100 mA) silicon n-p-n transistors on a common monolithic substrate. The CA3081 is connected in a common-emitter configuration and the CA3082 is connected in a common-collector configuration.

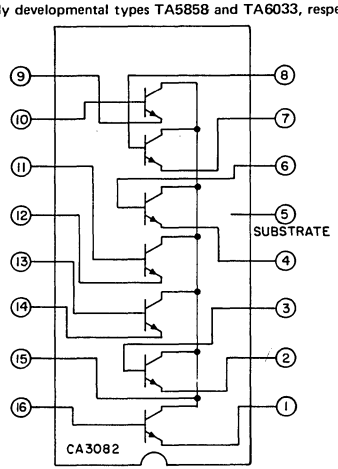
The CA3081 and CA3082 are capable of directly driving seven-segment displays, such as the RCA NUMITRON devices (DR2000 and DR2010), and light-emitting diode (LED) displays. These types are also well-suited for a variety of other driver applications, including relay control and thyristor firing.

The CA3081 and CA3082 are supplied in a 16-lead dual-in-line plastic package, and the CA3081F and CA3082F in a 16-lead dual-in-line frit-seal ceramic package, which includes a separate substrate connection for maximum flexibility in circuit design.

* Formerly developmental types TA5858 and TA6033, respectively.



(a)
COMMON-EMITTER CONFIGURATION
92CS-17958



(b)
COMMON-COLLECTOR CONFIGURATION
92CS-17957

Fig. 1–Functional diagrams of types CA3081 and CA3082.

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Power Dissipation:

Any one transistor	500	mW
Total package	750	mW
Above 55°C	Derate linearly 6.67	mW/°C

Ambient Temperature Range:

Operating	−55 to +125	°C
Storage	−65 to + 150	°C

Lead Temperature (During Soldering):

At distance 1/16" ±1/32" (1.59 mm ±0.79 mm)

from case for 10 seconds max.	265	°C

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage (V_{CEO})	16	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})■	20	V
Emitter-to-Base Voltage (V_{EBO})	5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	20	mA

■ The collector of each transistor of the CA3081 and CA3082 is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (5) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTIC	SYMBOL	TEST CONDITIONS	Typ. Char. Curve Fig. No.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
Collector-to-Base Breakdown Voltage	$V_{(BR)CES}$	I_C = 500 μA, I_E = 0	–	20	60	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 500 μA, I_E = 0, I_B = 0	–	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	–	16	24	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_C = 500 μA	–	5	6.9	–	V
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 0.5 V, I_C = 30 mA	–	30	68	–	
		V_{CE} = 0.8 V, I_C = 50 mA	–	40	70	–	
Base-to-Emitter Saturation Voltage	$V_{BE\ sat}$	I_C = 30 mA, I_B = 1 mA	3	–	0.87	1.0	V
Collector-to-Emitter Saturation Voltage: CA3081, CA3082	$V_{CE\ sat}$	I_C = 30 mA, I_B = 1 mA	–	–	0.27	0.5	V
CA3081		I_C = 50 mA, I_B = 5 mA	4	–	0.4	0.7	
CA3082		I_C = 50 mA, I_B = 5 mA	4	–	0.4	0.8	
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10 V, I_B = 0	–	–	–	10	μA
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10 V, I_E = 0	–	–	–	1	μA

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR OF TYPES CA3081 AND CA3082

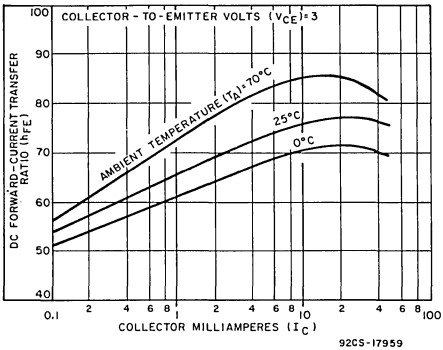


Fig.2—h_{FE} vs. I_C

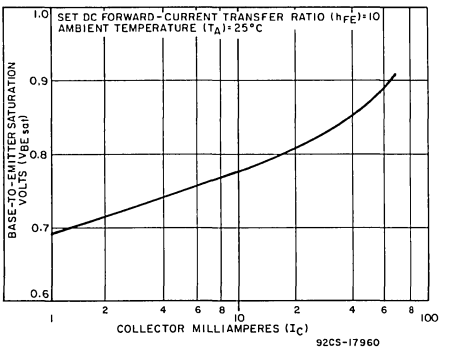


Fig.3—V_{BEsat} vs. I_C

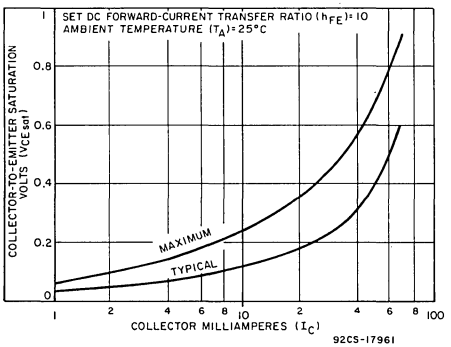


Fig.4—V_{CEsat} vs. I_C at T_A = 25°C.

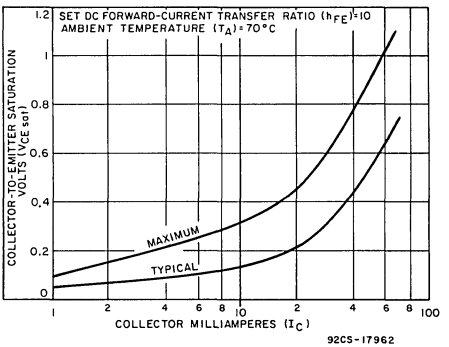


Fig.5—V_{CEsat} vs. I_C at T_A = 70°C.

TYPICAL READ-OUT DRIVER APPLICATIONS

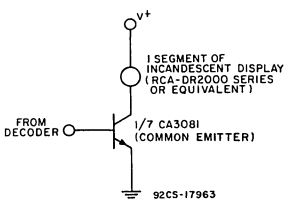
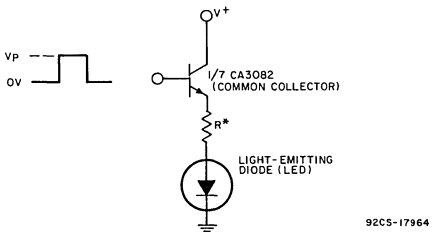


Fig.6—Schematic diagram showing one transistor of the CA3081 driving one segment of an incandescent display.



*THE RESISTANCE FOR R IS DETERMINED BY THE RELATIONSHIP

$$R = \frac{V_P - V_{BE} - V_F(LED)}{I\,(LED)}$$

WHERE: V_P = INPUT PULSE VOLTAGE

V_F = FORWARD VOLTAGE DROP ACROSS THE DIODE

R = 0 FOR $V_P = V_{BE} + V_F(LED)$

Fig.7—Schematic diagram showing one transistor of the CA3082 driving a light-emitting diode (LED).

RCA Solid State Division

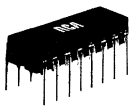
Linear Integrated Circuits

Monolithic Silicon

CA3083, CA3083F



16-Lead Dual-In-Line Frit-Seal (Hermetic) Package H-1807



16-Lead Dual-In-Line Plastic Package H-1622

General-Purpose High-Current N-P-N Transistor Array

Applications

- Signal processing and switching systems operating from DC to VHF
- Lamp and relay driver
- Differential amplifier
- Temperature-compensated amplifier
- Thyristor firing
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Circuit Transistor Array" for suggested applications

RCA-CA3083 is a versatile array of five high-current (to 100mA) n-p-n transistors on a common monolithic substrate. In addition, two of these transistors (Q1 and Q2) are matched at low currents (i.e. 1mA) for applications in which offset parameters are of special importance.

Independent connections for each transistor plus a separate terminal for the substrate permit maximum flexibility in circuit design. The CA3083 is supplied in a 16-lead dual-in-line plastic package, and the CA3083F in a 16-lead dual-in-line frit-seal ceramic package.

Features

- High I_C: 100mA max.
- Low V_{CEsat} (at 50mA): 0.7V max.
- Matched pair (Q1 and Q2)–
  V_{IO} (V_{BE} matched): ± 5 mV max.
  I_{IO} (at 1mA): 2.5 μA max.
- 5 independent transistors plus separate substrate connection

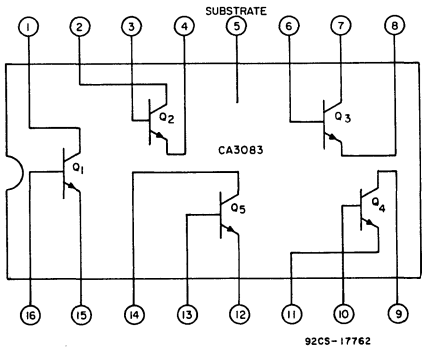


Fig.1–Functional diagram of the CA3083.

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

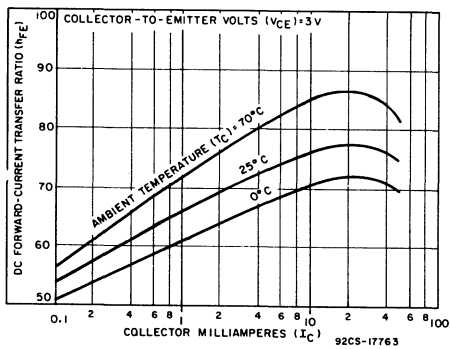


Fig.2 – h_{FE} vs I_C

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Power Dissipation:

Any one transistor	500	mW
Total package	750	mW
Above 55°C	Derate linearly 6.67	mW/°C

Ambient Temperature Range:

Operating	−55 to +125	°C
Storage	−65 to +150	°C

Lead Temperature (During Soldering):

At distance 1/16" ±1/32" (1.59 mm ±0.79 mm) from case for 10 seconds max.	265	°C

The following ratings apply for each transistor in the device:

Collector-to-Emitter Voltage (V_{CEO})	15	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})■	20	V
Emitter-to-Base Voltage (V_{EBO})	5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	20	mA

■ The collector of each transistor of the CA3083 is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (5) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	Typ. Char. Curve Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
For Each Transistor:							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0	–	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0	–	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0	–	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0	–	5	6.9	–	V
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	–	10	μA
Collector-Cutoff-Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	–	–	1	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 10mA	2	40	76	–	
		V_{CE} = 3V, I_C = 50mA	2	40	75	–	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA	3	0.65	0.74	0.85	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_C = 50mA, I_B = 5mA	4	–	0.40	0.70	V
For Transistors Q1 and Q2 (As a Differential Amplifier):							
Absolute Input Offset Voltage	$\lvert V_{IO} \rvert$	V_{CE} = 3V, I_C = 1mA	7	–	1.2	5	mV
Absolute Input Offset Current	$\lvert I_{IO} \rvert$	V_{CE} = 3V, I_C = 1mA	8	–	0.7	2.5	μA

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

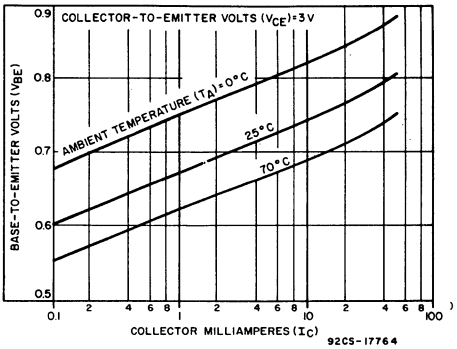


Fig.3 – V_{BE} vs I_C

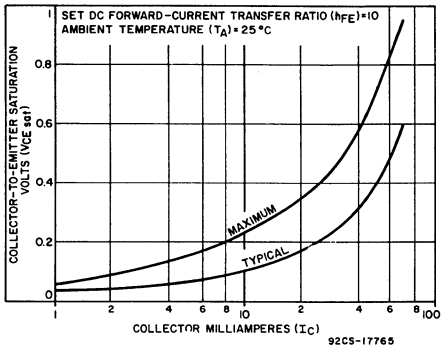


Fig.4 – V_{CEsat} vs I_C at 25°C

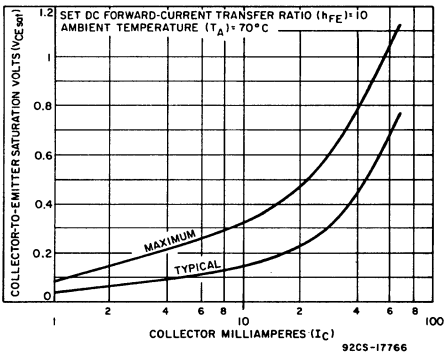


Fig.5 – V_{CEsat} vs I_C at 70°C

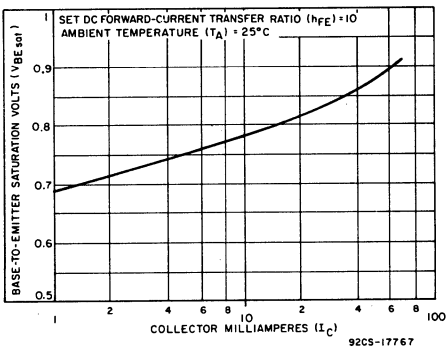


Fig.6 – V_{BEsat} vs I_C

TYPICAL STATIC CHARACTERISTICS FOR DIFFERENTIAL AMPLIFIER

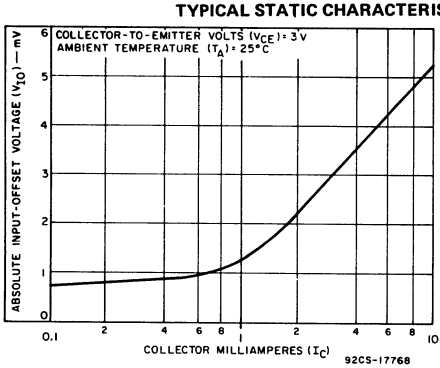


Fig.7 – V_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier).

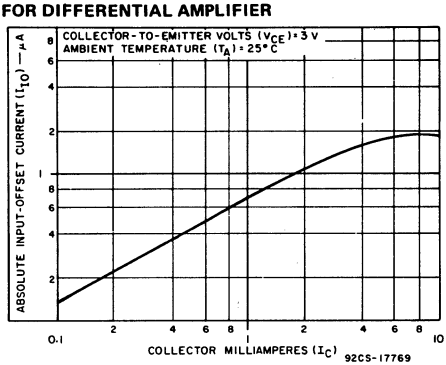


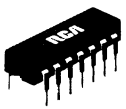
Fig.8 – I_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier).

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3084



H-1517

14-Lead Dual-In-Line
Plastic Package

General-Purpose P-N-P Transistor Array

FEATURES

- Matched transistor pair (Q1 and Q2)
  V_{IO} (V_{BE} matched): ± 6mV max.
  I_{IO} (at 100 μA): ± 0.6 μA
- Wide operating current range
- Low noise figure - - 3.2 dB typ. at 1 kHz

RCA-CA3084* is a general-purpose silicon p-n-p transistor array incorporating two independent transistors, a Darlington circuit, and a current-mirror pair with a shared diode.

The two independent transistors in the array may be used in a variety of circuit applications. The Darlington pair may be employed as the equivalent of a single high-beta transistor. The current-mirror pair is well suited for constant-current applications and can also be used as the active loads in a differential amplifier which uses n-p-n transistors.

The total array is especially useful for a wide range of applications in systems having low-power and low-frequency requirements. Although the transistors may be used as discrete units in conventional circuits, they offer the advantages inherent in integrated-circuit construction, that is, to provide close electrical and thermal matching.

The CA3084 utilizes the 14-lead dual-in-line plastic package.

*Formerly developmental type TA5799A.

APPLICATIONS

- General use in signal processing systems having low-power and low-frequency requirements
- Differential amplifiers
- Temperature compensated amplifiers
- Active loads for differential amplifiers using n-p-n transistors
- Complementary uses with RCA n-p-n transistor arrays

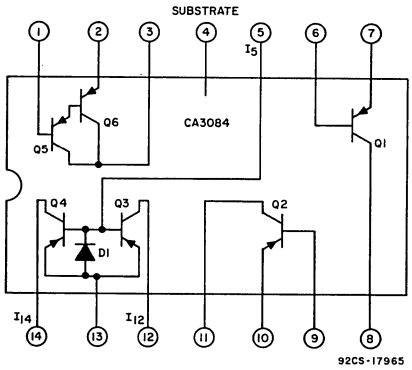


Fig.1 – Functional diagram of the CA3084.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	Typ. Characteristics Curve Fig. No.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
For Each Transistor:							
Collector-Cutoff Current	I_{CBO}	V_{CB} = −10V, I_E = 0	2	–	−0.055	−100	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = −10V, I_B = 0	3	–	−0.12	−100	nA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_{CE} = −100μA, I_B = 0	–	−40	−70	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_{CB} = −100μA, I_E = 0	–	−40	−80	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_{EB} = −100μA, I_C = 0	–	−40	−100	–	V
Emitter-to-Substrate Breakdown Voltage	$V_{(BR)EIO}$	I_{EI} = 100μA	–	−40	−100	–	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_E = 1mA, I_B = 100μA	4	–	−0.125	−0.25	V
Base-to-Emitter Voltage	V_{BE}	I_E = 100μA, V_{CE} = −10V	5	−0.50	−0.59	−0.68	V
DC Forward-Current Transfer Ratio	h_{FE}		7	15	40	–	
For Transistors Q1 and Q2 (As a Differential Amplifier):							
Magnitude of Input Offset Voltage	$\lvert V_{IO} \rvert$	I_E = 100μA, V_{CE} = −10V	8	–	0.422	6	mV
Input Offset Current	I_{IO}		–	−0.6	0	0.6	μA
For Transistors Q3 and Q4 (Current-Mirror Configuration):							
Collector Current (Normalized)	I_C/I_5	V_{CE} = −5V, V_{CIO} = −5V, Term. 13 = Gnd. I_5 = −100μA,	10	0.85	1.00	1.15	
Magnitude of Collector Current Ratio	$\lvert I_{C(Q3)}/I_{C(Q4)} \rvert$		11	0.90	1.00	1.10	
For Transistors Q5 and Q6 (Darlington Configuration):							
Collector-Cutoff Current	I_{CEO}	V_{CE} = −10V, I_B = 0	–	–	–	−1.0	μA
Base-to-Emitter Voltage	V_{BE}	I_E = 100μA, V_{CE} = −10V	13	0.92	1.07	1.20	V
DC Forward-Current Transfer Ratio	h_{FE}		15	100	1230	–	

ELECTRICAL CHARACTERISTICS at T_A = 25°C
Typical Values Intended Only For Design Guidance

Magnitude of Temperature Coefficient:							
V_{BE} (for each transistor)	$\lvert \Delta V_{BE}/\Delta T \rvert$	I_E = 100μA, V_{CE} = −10V	6		−1.78		mV/°C
V_{IO} (as a differential amplifier)	$\lvert \Delta V_{IO}/\Delta T \rvert$		9		0.54		μV/°C
V_{BE} (Darlington configuration)	$\lvert \Delta V_{BE}/\Delta T \rvert$		14		−3.7		mV/°C
For Each Transistor:							
Input Resistance	R_I	f = 1kHz, V_{CE} = −10V, I_C = −100μA	19		9		kΩ
Output Resistance	R_O		20	–	600	–	kΩ
Forward Transconductance	g_m		22	–	3	–	mmho
Collector-to-Base Capacitance	C_{CBO}	I_{CB} = 0	23	–	3.3	–	pF
Collector-to-Emitter Capacitance	C_{CEO}	I_{CE} = 0	23	–	2.5	–	pF
Base-to-Substrate Capacitance	C_{BIO}	I_{CIO} = 0	23	–	4.5	–	pF

MAXIMUM RATINGS, Absolute-Maximum Values at T_A = 25°C

Dissipation:		
Any one transistor	200	mW*
Total package	750	mW
Above T_A = 55°C	derate linearly6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (During Soldering): At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C
The following ratings apply for each transistor in the device:		
Collector-to-Emitter Voltage (V_{CEO})	−40	V
Collector-to-Base Voltage (V_{CBO})	−40	V
Base-to-Substrate Voltage (V_{BIO})*	−40	V
Emitter-to-Base Voltage (V_{EBO})	−40	V
Collector Current (I_C)	−10	mA

*The base of each transistor of the CA3084 is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any base voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal (4) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

STATIC CHARACTERISTICS FOR EACH TRANSISTOR

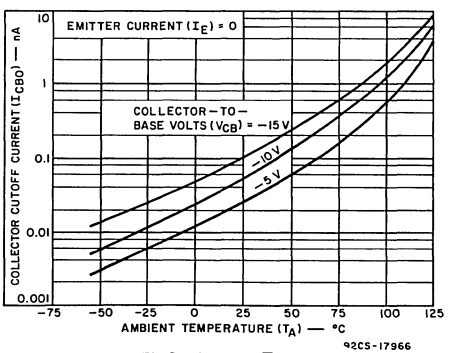


Fig.2– I_{CBO} vs T_A.

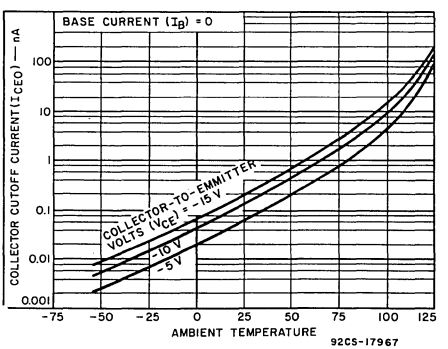


Fig.3– I_{CEO} vs T_A.

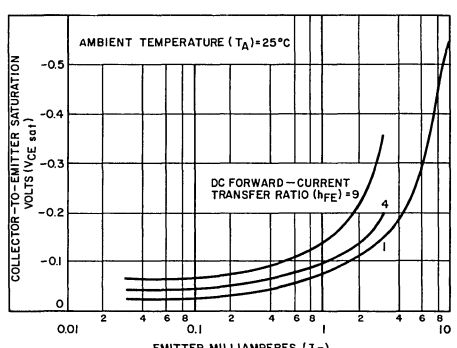


Fig.4– V_{CEsat} vs I_E.

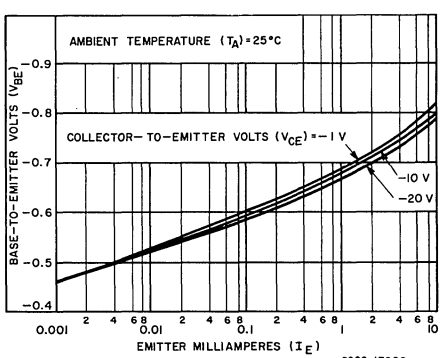


Fig.5– V_{BE} vs I_E.

STATIC CHARACTERISTICS FOR EACH TRANSISTOR

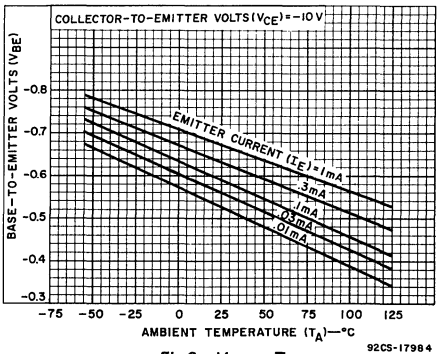


Fig.6– V_{BE} vs T_A.

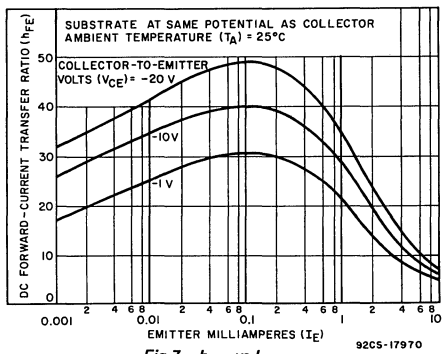


Fig.7– h_{FE} vs I_E.

STATIC CHARACTERISTICS FOR DIFFERENTIAL AMPLIFIER

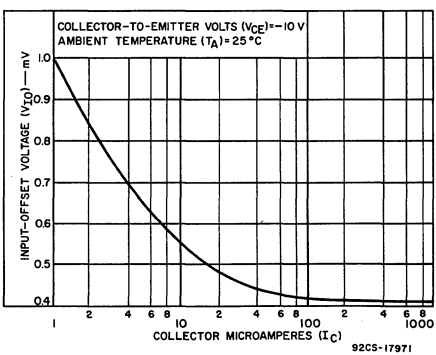


Fig.8–V_{IO} vs I_C, (transistors Q1 and Q2 as a differential amplifier).

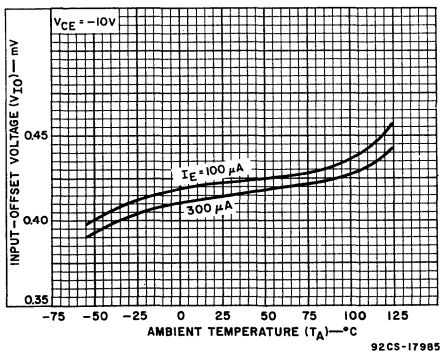


Fig.9–V_{IO} vs T_A (transistors Q1 and Q2 as a differential amplifier).

STATIC CHARACTERISTICS FOR CURRENT-MIRROR CONFIGURATION

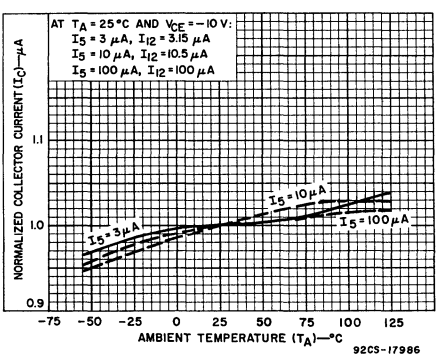


Fig.10–Normalized I_C vs T_A (transistors Q3 and Q4 in a current-mirror configuration.

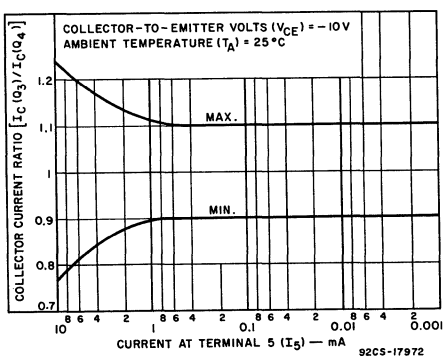


Fig.11–I_C ratio vs I_5 (transistors Q3 and Q4 in a current-mirror configuration.

STATIC CHARACTERISTICS FOR CURRENT-MIRROR CONFIGURATION

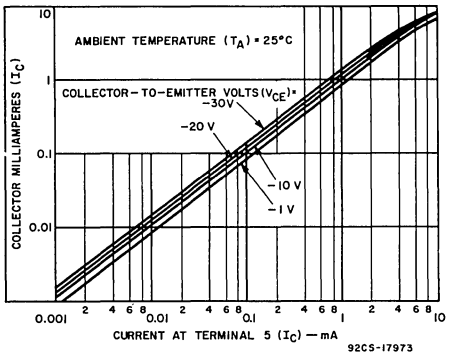


Fig.12 – I_C vs I_5 (transistors Q3 and Q4 in a current-mirror configuration).

STATIC CHARACTERISTICS FOR DARLINGTON CONFIGURATION

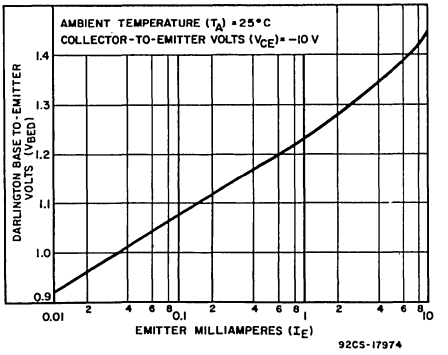


Fig.13 – V_{BE} vs I_E (transistors Q5 and Q6 in a darlingtron configuration).

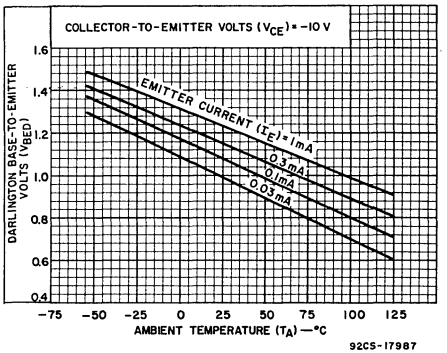


Fig.14 – V_{BE} vs T_A (transistors Q5 and Q6 in a darlington configuration).

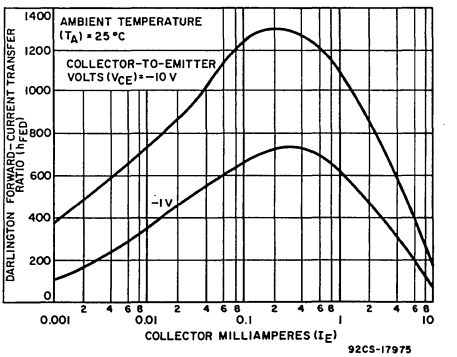


Fig.15 – h_{FE} vs. I_E (transistors Q5 and Q6 in a darlington configuration).

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR

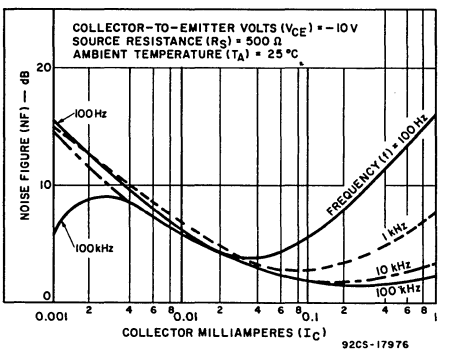


Fig.16 – NF vs I_C at R_S = 500 Ω

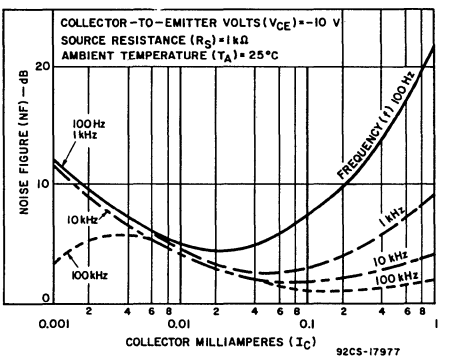


Fig.17 – NF bs I_C at R_S = 1 kΩ

DYNAMIC CHARACTERISTICS FOR EACH TRANSISTOR (Cont'd)

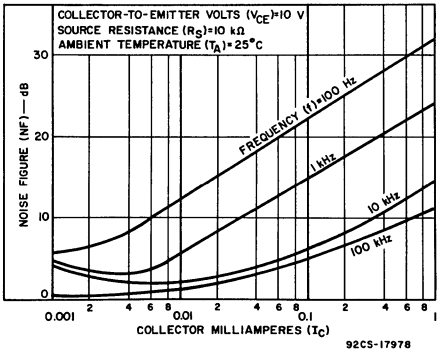


Fig.18– NF vs I_C at R_S = 10kΩ

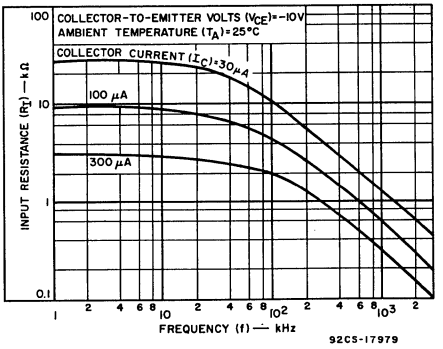


Fig.19– R_I vs f

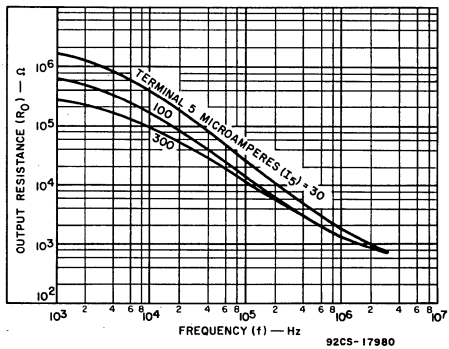


Fig.20– R_O vs f

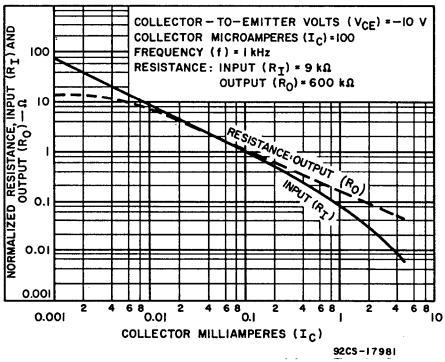


Fig.21– Normalized R_I and R_O vs I_C

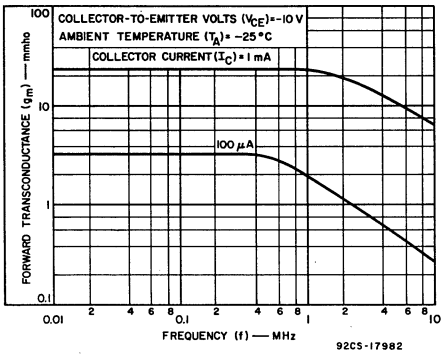


Fig.22– g_m vs f

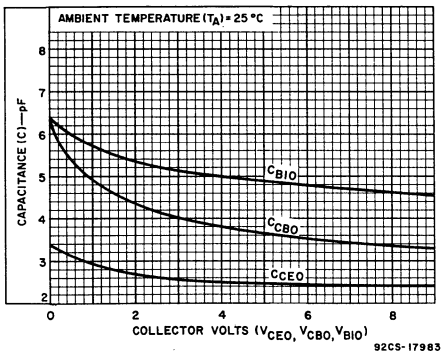
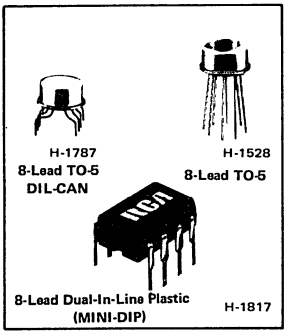


Fig.23– Transistor capacitances vs collector voltages (V_{CEO}, V_{CBO}, V_{CIO})

RCA Solid State Division

Linear Integrated Circuits

CA3085, CA3085A, CA3085B Types



Positive Voltage Regulators

For Regulated Voltages from 1.7 V to 46 V at Currents up to 100 mA

Features

- Up to 100 mA output current
- Input and output short-circuit protection
- Load and line regulation: 0.025%
- Pin compatible with LM100 Series
- Adjustable output voltage

Applications

- Shunt voltage regulator
- Current regulator
- Switching voltage regulator
- High-current voltage regulator
- Combination positive and negative voltage regulator
- Dual tracking regulator

RCA-CA3085, CA3085A, and CA3085B are silicon monolithic integrated circuits designed specifically for service as voltage regulators at output voltages ranging from 1.7 to 46 volts at currents up to 100 milliamperes.

A block diagram of the CA3085 Series is shown in Fig. 1. The diagram shows the connecting terminals that provide access to the regulator circuit components. The voltage regulators provide important features such as: frequency compensation, short-circuit protection, temperature-compensated reference voltage, current limiting, and booster input. These devices are useful in a wide range of applications for regulating high-current, switching, shunt, and positive and negative voltages. They are also applicable for current and dual-tracking regulation.

The CA3085A and CA3085B have output current capabilities up to 100 mA and the CA3085 up to 12 mA without the use of external pass transistors. However, all the devices can provide voltage regulation at load currents greater than 100 mA with the use of suitable external pass transistors. The CA3085 Series has an unregulated input voltage ranging from 7.5 to 30 V (CA3085), 7.5 to 40 V (CA3085A), and 7.5 to 50 V (CA3085B) and a minimum regulated output voltage of 26 V (CA3085), 36 V (CA3085A), and 46 V (CA3085B).

The CA3085A is unilaterally interchangeable with the CA3055.

These types are supplied in the 8-lead TO-5 style package (CA3085, CA3085A, CA3085B, and the 8-lead TO-5 with dual-in-line formed leads ("DIL-CAN", CA3085S, CA3085AS, CA3085BS). The CA3085 is also supplied in the 8-lead dual-in-line plastic package ("MINI-DIP", CA3085E), and in chip form (CA3085H).

Type	V_{IN} Range V	V_{OUT} Range V	Max. I_{OUT} mA	Max. Load Regulation % V_{OUT}
CA3085	7.5 to 30	1.8 to 26	12*	0.1
CA3085A	7.5 to 40	1.7 to 36	100	0.15
CA3085B	7.5 to 50	1.7 to 46	100	0.15

* This value may be extended to 100 mA; however, regulation is not specified beyond 12 mA.

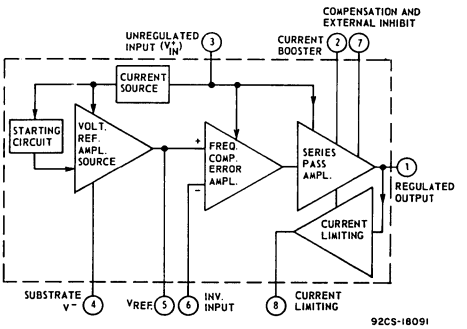


Fig. 1—Block diagram of CA3085 Series.

MAXIMUM RATINGS, ABSOLUTE-MAXIMUM VALUES at T_A = 25°C

POWER DISSIPATION: WITHOUT HEAT SINK	WITH HEAT SINK (TO-5 ONLY)
up to T_A = 55°C ..... 630 mW	up to T_C = 55°C .... 1.6 W
above T_A = 55°C derate linearly @6.67 mW/°C	above T_C = 55°C .... derate linearly at 16.7 mW/°C

TEMPERATURE RANGE:

Operating ..... −55 to +125°C
Storage ..... −65 to +150°C

UNREGULATED INPUT VOLTAGE:

CA3085 ..... 30 V
CA3085A ..... 40 V
CA3085B ..... 50 V

LEAD TEMPERATURE (DURING SOLDERING):

At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max. ..... +265°C

Maximum Voltage Ratings

The following chart gives the range of voltages which can be applied to the terminals listed vertically with respect to the terminals listed horizontally. For example, the voltage range between vertical Terminal No. 7 and horizontal Terminal No. 1 is +3 to −10 volts.

MAXIMUM VOLTAGE RATINGS

TERMINAL No.	5	6	7	8	1	2	3	4
5	–	+5 −5	•	•	•	•	•	+10 0
6	–	–	•	•	•	•	•	•
7	–	–	–	+3 −10	+3 −10	•	•	+‡ 0
8	–	–	–	–	+5 −1	•	•	•
1	–	–	–	–	–	+10 −‡	0 −‡	+‡ 0
2	–	–	–	–	–	–	0 –	+‡ 0
3	–	–	–	–	–	–	–	+‡ 0
4	–	–	–	–	–	–	–	Substrate & Case

*Voltages are not normally applied between these terminals; however, voltages appearing between these terminals are safe, if the specified voltage limits between all other terminals are not exceeded.

‡30V for CA3085
40V for CA3085A
50V for CA3085B

MAXIMUM CURRENT RATINGS

TERMINAL No.	I_{IN} mA	I_{OUT} mA
5	10	1.0
6	1.0	−0.1
7	1.0	−1.0
8	0.1	10
1	20	150
2	150	60
3	150	60
4	–	–

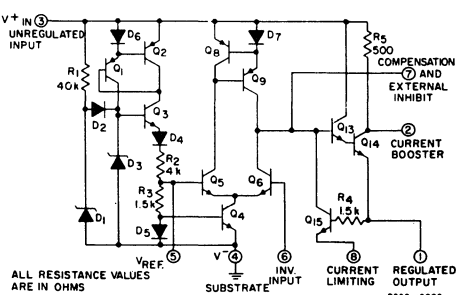


Fig.2—Schematic diagram of CA3085 Series.

ELECTRICAL CHARACTERISTICS

CHARACTERISTICS	SYMBOL	TEST CONDITIONS: Test Circuit Fig. No.	TEST CONDITIONS: T_A = 25°C [Unless indicated otherwise]	LIMITS: CA3085 MIN.	CA3085 TYP.	CA3085 MAX.	CA3085A MIN.	CA3085A TYP.	CA3085A MAX.	CA3085B MIN.	CA3085B TYP.	CA3085B MAX.	UNITS
Reference Voltage	V_{REF}	4	V^+_{IN} = 15V	1.4	1.6	1.8	1.5	1.6	1.7	1.5	1.6	1.7	V
Quiescent Regulator Current	$I_{quiescent}$	4	V^+_{IN} = 30V	–	3.3	4.5	—	–	–	–	–	–	mA
			V^+_{IN} = 40V	–	–	–	–	3.65	5	–	–	–	
			V^+_{IN} = 50V	–	–	–	–	–	–	–	4.05	7	
Input Voltage Range	$V_{IN(range)}$	–	–	7.5	–	30	7.5	–	40	7.5	–	50	V
Maximum Output Voltage	$V_{O(max.)}$	4	V^+_{IN} = 30,40,50V#; R_L = 365Ω; Term. No. 6 to Gnd.	26	27	–	36	37	–	46	47	–	V
Minimum Output Voltage	$V_{O(min.)}$	4	V^+_{IN} = 30V	–	1.6	1.8	–	1.6	1.7	–	1.6	1.7	V
Input-Output Voltage Differential	V_{IN}-V_{OUT}	–	–	4	–	28	4	–	38	3.5	–	48	V
Limiting Current	I_{LIM}	7	V^+_{IN} = 16V, V^+_{OUT} = 10V R_{SCP}* = 6Ω	–	96	120	–	96	120	–	96	120	mA
Load Regulation●	–	–	I_L = 1 to 100mA, R_{SCP} = 0	–	–	–	–	0.025	0.15	–	0.025	0.15	%V_{OUT}
		–	I_L = 1 to 100mA, R_{SCP} = 0 T_A = 0°C to +70°C	–	–	–	–	0.035	0.6	–	0.035	0.6	
		–	I_L = 1 to 12mA, R_{SCP} = 0	–	0.003	0.1	–	–	–	–	–	–	
Line Regulation▲	–	–	I_L = 1mA, R_{SCP} = 0	–	0.025	0.1	–	0.025	0.075	–	0.025	0.04	%/V
		–	I_L = 1mA, R_{SCP} = 0 T_A = 0°C to +70°C	–	0.04	0.15	–	0.04	0.1	–	0.04	0.08	
Equivalent Noise Output Voltage	V_{NOISE}	11	V^+_{IN} = 25V, C_{REF} = 0	–	0.5	–	–	0.5	–	–	0.5	–	mV p-p
			V^+_{IN} = 25V, C_{REF} = 0.22μF	–	0.3	–	–	0.3	–	–	0.3	–	
Ripple Rejection	–	12	V^+_{IN} = 25V, f = 1kHz, C_{REF} = 0	–	50	–	–	50	–	45	50	–	dB
			V^+_{IN} = 25V, f = 1kHz, C_{REF} = 2μF	–	56	–	–	56	–	50	56	–	
Output Resistance	r_o	12	V^+_{IN} = 25V, f = 1kHz	–	0.075	1.1	–	0.075	0.3	–	0.075	0.3	Ω
Temperature Coefficient of Reference and Output Voltages	ΔV_{REF}, ΔV_o	–	I_L = 0, V_{REF} = 1.6V	–	0.0035	–	–	0.0035	–	–	0.0035	–	%/°C
Load Transient Recovery Time: Turn On	t_{ON}	16	V^+_{IN} = 25V, +50mA Step	–	1	–	–	1	–	–	1	–	μs
Turn Off	t_{OFF}		V^+_{IN} = 25V, −50mA Step	–	3	–	–	3	–	–	3	–	μs
Line Transient Recovery Time: Turn On	t_{ON}	–	V^+_{IN} = 25V, f = 1kHz, 2V Step	–	0.8	–	–	0.8	–	–	0.8	–	μs
Turn Off	t_{OFF}	–		–	0.4	–	–	0.4	–	–	0.4	–	μs

30V (CA3085), 40V(CA3085A), 50V(CA3085B)

* RSCP: Short-circuit protection resistance

● Load Regulation = $\frac{\Delta V_{OUT}}{V_{OUT(initial)}} \times 100\%$

▲ Line Regulation = $\frac{(\Delta V_{OUT})}{[V_{OUT(initial)}]\ (\Delta V_{IN})} \times 100\%$

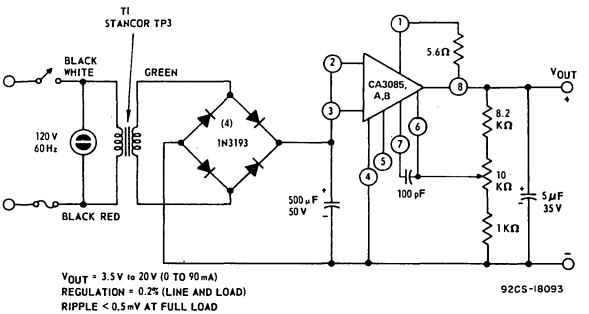
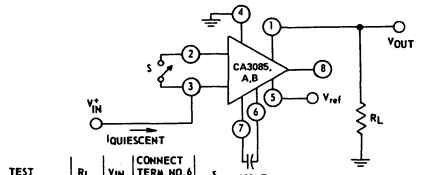


Fig.3—Application of the CA3085 Series in a typical power supply.

TEST CIRCUITS AND TYPICAL CHARACTERISTICS CURVES FOR CA3085 SERIES



TEST	R_L	V_{IN}	CONNECT TERM. NO. 6	S
$V_{REF.}$	∞	+1.6	OPEN	OPEN
$I_{QUIESCENT}$	∞	+40	OPEN	OPEN
V_{OUT} (MAX.)	365Ω	+40	GROUND	CLOSED
V_{OUT} (MIN.)	10 k	+30	TERM. NO. 1	OPEN

Fig.4—Test circuit for V_{REF}, $I_{quiescent}$, V_{OUT}(max.), V_{OUT}(min.).

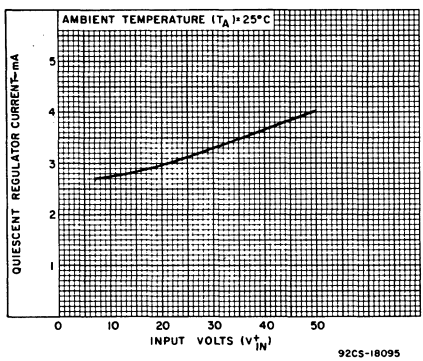


Fig.5—$I_{quiescent}$ vs. V^+_{IN}.

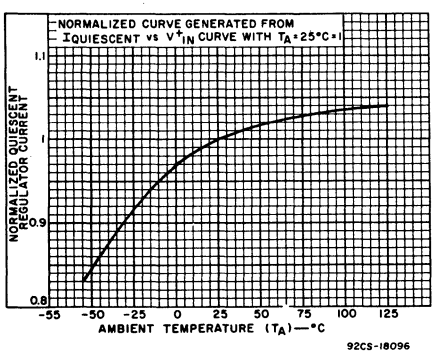


Fig.6—Normalized $I_{quiescent}$ vs. T_A.

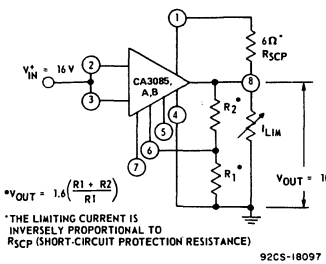


Fig.7—Test circuit for limiting current

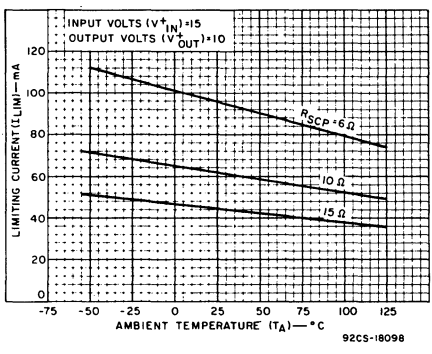


Fig.8—I_{LIM} vs. T_A.

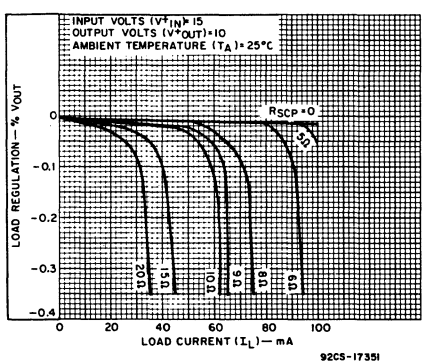


Fig.9—Load regulation characteristics.

TEST CIRCUITS AND TYPICAL CHARACTERISTICS CURVES FOR CA3085 SERIES

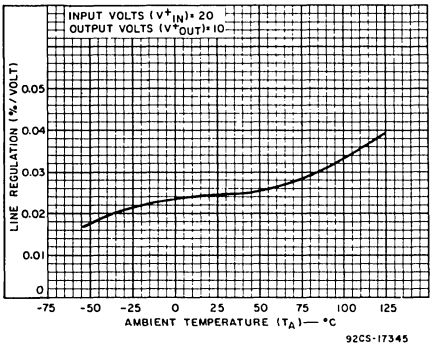


Fig.10—Line regulation temperature characteristics.

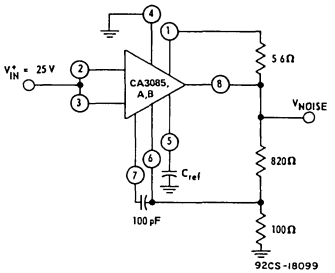


Fig.11—Test circuit for noise voltage.

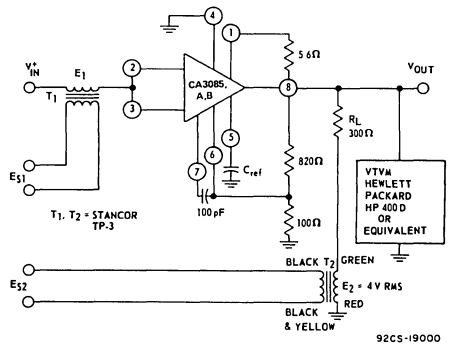


Fig.12—Test circuit for ripple rejection and output resistance.

TEST PROCEDURES FOR TEST CIRCUIT FOR RIPPLE REJECTION AND OUTPUT RESISTANCE

Output Resistance

Conditions:

1. V_{IN} = +25V, C_{REF} = 0, Short E_1
2. Set E_{S2} at 1 kHz so that E_2 = 4V rms
3. Read V_{OUT} on a VTVM, such as a Hewlett-Packard, HP400D or equivalent
4. Calculate R_{OUT} from $R_{OUT} = V_{OUT}(R_L/E_2)$

Ripple Rejection — I

Conditions:

1. V_{IN} = +25V, C_{REF} = 0, Short E_2
2. Set E_{S1} at 1 kHz so that E_1 = 3V rms
3. Read V_{OUT} on a VTVM, such as a Hewlett-Packard, HP400D or equivalent
4. Calculate Ripple Rejection from 20 log (E_1/V_{OUT})

Ripple Rejection — II

Conditions:

1. Repeat Ripple Rejection I with C_{REF} = 2 μF

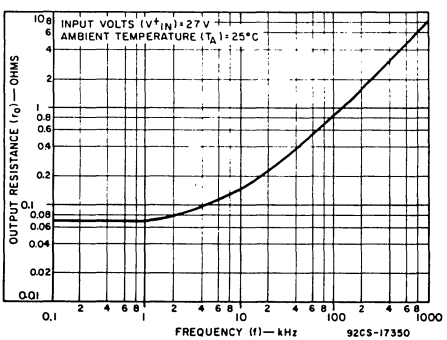


Fig.13—r_O vs. f.

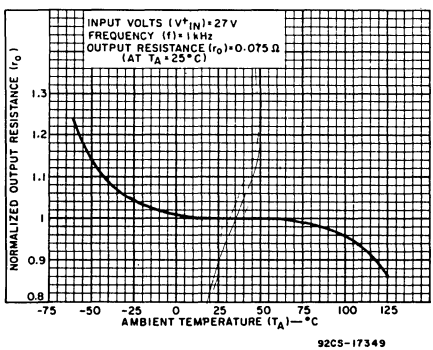


Fig.14—Normalized r_O vs. T_A.

TEST CIRCUIT AND TYPICAL CHARACTERISTICS CURVES FOR CA3085 SERIES

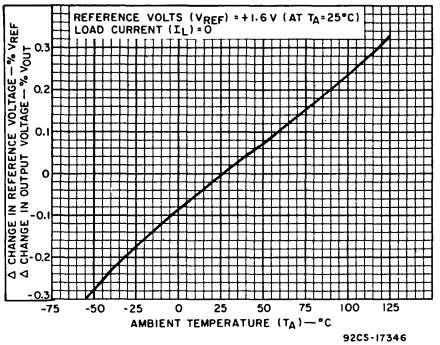


Fig.15—Temperature coefficient of V_{REF} and V_{OUT}.

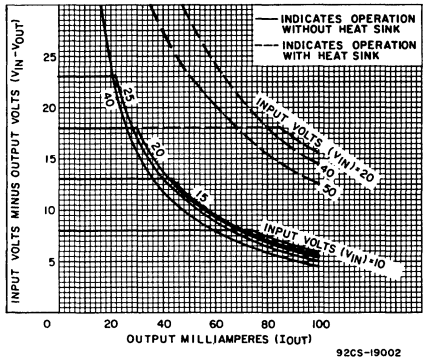


Fig.17—Dissipation limitation (V_{IN}—V_{OUT} vs. I_{OUT}).

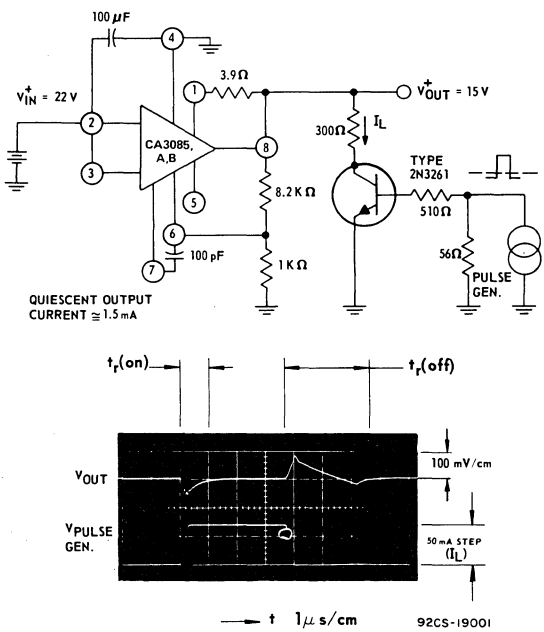


Fig.16—Turn-on and turn-off recovery time test circuit with associated waveforms.

TYPICAL REGULATOR CIRCUITS USING THE CA3085 SERIES

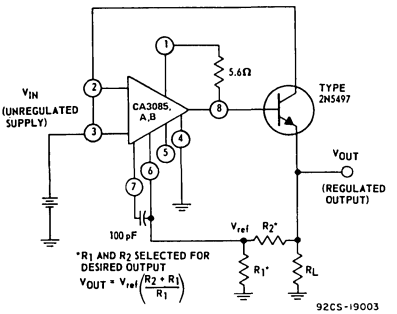


Fig.18—Typical high-current voltage regulator circuit.

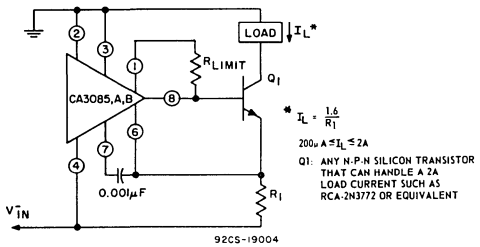


Fig.19—Typical current regulator circuit.

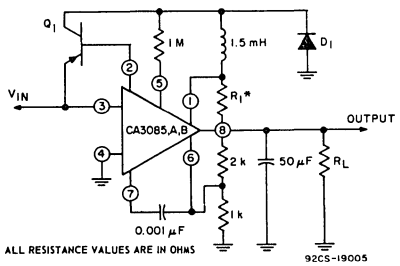
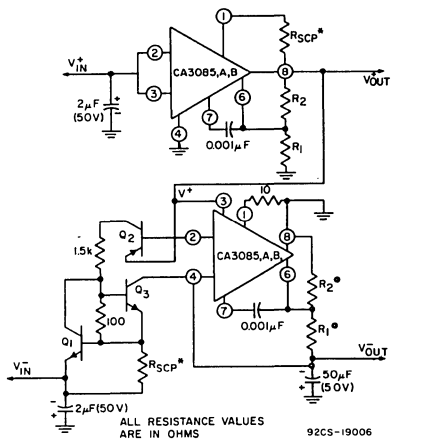


Fig.20—Typical switching regulator circuit.



ALL RESISTANCE VALUES ARE IN OHMS

Q1: RCA-2N2102 OR EQUIVALENT

Q2: ANY P-N-P SILICON TRANSISTOR (RCA-2N5322 OR EQUIVALENT)

Q3: ANY N-P-N SILICON TRANSISTOR THAT CAN HANDLE THE DESIRED LOAD CURRENT (RCA-2N3772 OR EQUIVALENT)

$*V_{OUT} = \left(\frac{R_1 + R_2}{R_1}\right)$

*RSCP: SHORT-CIRCUIT PROTECTION RESISTANCE

Fig.21—Combination positive and negative voltage regulator circuit.

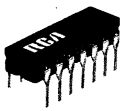
RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

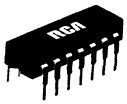
CA3086 CA3086F

14-Lead Dual-In-Line Frit-Seal (Hermetic) Package



H-1806

14-Lead Dual-In-Line Plastic Package



H-1517

General-Purpose N-P-N Transistor Array

Three Isolated Transistors and One Differentially–Connected Transistor Pair

For Low–Power Applications from DC to 120MHz

Applications

- **General-purpose use in signal processing systems operating in the DC to 120-MHz range**
- **Temperature compensated amplifiers**
- **See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Integrated-Circuit Transistor Array" for suggested applications.**

RCA-CA3086 consists of five general-purpose silicon n-p-n transistors on a common monolithic substrate. Two of the transistors are internally connected to form a differentially-connected pair.

The transistors of the CA3086 are well suited to a wide variety of applications in low-power systems at frequencies from DC to 120 MHz. They may be used as discrete transistors in conventional circuits. However, they also provide the very significant inherent advantages unique to integrated circuits, such as compactness, ease of physical handling and thermal matching.

The CA3086 is supplied in a 14-lead dual-in line plastic package. The CA3086F is supplied in a 14-lead dual-in-line hermetic (frit-seal) ceramic package.

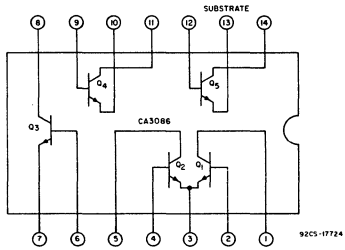


Fig.1 – Functional diagram of the CA3086.

MAXIMUM RATINGS, Absolute–Maximum Values at T_A = 25°C

DISSIPATION:		
Any one transistor	300	mW
Total package up to T_A = 55°C	750	mW
Above T_A = 55°C	derate linearly 6.67	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating	−55 to + 125	°C
Storage	−65 to + 150	°C
LEAD TEMPERATURE (During soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) From case for 10 seconds max.	+ 265	°C
The following ratings apply for each transistor in the device:		
COLLECTOR-TO-EMITTER VOLTAGE, V_{CEO}	15	V
COLLECTOR-TO-BASE VOLTAGE, V_{CBO}	20	V
COLLECTOR-TO-SUBSTRATE VOLTAGE, V_{CIO}*	20	V
EMITTER-TO-BASE VOLTAGE, V_{EBO}	5	V
COLLECTOR CURRENT, I_C	50	mA

* The collector of each transistor in the CA3086 is isolated from the substrate by an integral diode. The substrate (terminal 13) must be connected to the most negative point in the external circuit to maintain isolation between transistors and to provide for normal transistor action. To avoid undesirable coupling between transistors, the substrate (terminal 13) should be maintained at either DC or signal (AC) ground. A suitable bypass capacitor can be used to establish a signal ground.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOLS	TEST CONDITIONS		LIMITS			UNITS
			Typ. Characteristic Curves Fig. No.	Min.	Typ.	Max.	
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	–	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0	–	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_C = 10 μA, I_{CI} = 0	–	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	–	5	7	–	V
Collector-Cutoff Current	I_{CBO}	V_{CB} = 10V, I_E = 0	2	–	0.002	100	nA
Collector-Cutoff Current	I_{CEO}	V_{CE} = 10V, I_B = 0	3	—	See Curve	5	μA
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 1mA	4	40	100	–	

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

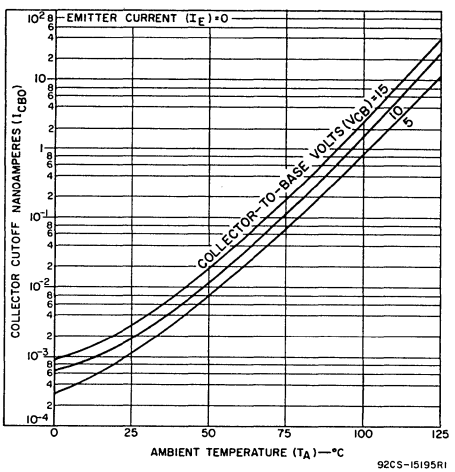


Fig.2– I_{CBO} vs T_A.

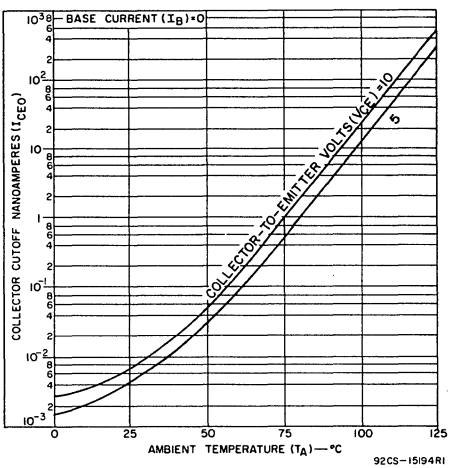


Fig.3– I_{CEO} vs T_A.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
Typical Values Intended Only for Design Guidance

CHARACTERISTICS	SYMBOL	TEST CONDITIONS		Typ. Characteristics Curves Fig. No.	TYPICAL VALUES	UNITS
DC Forward-Current Transfer Ratio	h_{FE}	V_{CE} = 3 V	I_C = 10 mA	4	100	
			I_C = 10 μA	4	54	
Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3 V	I_E = 1 mA	5	0.715	V
			I_E = 10 mA	5	0.800	V
V_{BE} Temperature Coefficient	$\Delta V_{BE}/\Delta T$	V_{CE} = 3V, I_C = 1mA		6	−1.9	mV/°C
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_B = 1mA, I_C = 10mA		–	0.23	V
Noise Figure (low frequency)	NF	f = 1kHz, V_{CE} = 3V, I_C = 100μA, R_S = 1kΩ		–	3.25	dB
Low-Frequency, Small-Signal Equivalent-Circuit Characteristics:						
Forward Current-Transfer Ratio	h_{fe}	f = 1kHz, V_{CE} = 3V, I_C = 1mA		7	100	–
Short-Circuit Input Impedance	h_{ie}			7	3.5	kΩ
Open-Circuit Output Impedance	h_{oe}			7	15.6	μmho
Open-Circuit Reverse-Voltage Transfer Ratio	h_{re}			7	1.8×10^{-4}	–
Admittance Characteristics:						
Forward Transfer Admittance	y_{fe}	f = 1MHz, V_{CE} = 3V, I_C = 1mA		8	31 − j1.5	mmho
Input Admittance	y_{ie}			9	0.3 + j0.04	mmho
Output Admittance	y_{oe}			10	0.001 + j0.03	mmho
Reverse Transfer Admittance	y_{re}			11	See Curve	–
Gain-Bandwidth Product	f_T	V_{CE} = 3V, I_C = 3mA		12	550	MHz
Emitter-to-Base Capacitance	C_{EBO}	V_{EB} = 3V, I_E = 0		–	0.6	pF
Collector-to-Base Capacitance	C_{CBO}	V_{CB} = 3V, I_C = 0		–	0.58	pF
Collector-to-Substrate Capacitance	C_{CIO}	V_{CI} = 3V, I_C = 0		—	2.8	pF

TYPICAL STATIC CHARACTERISTICS FOR EACH TRANSISTOR

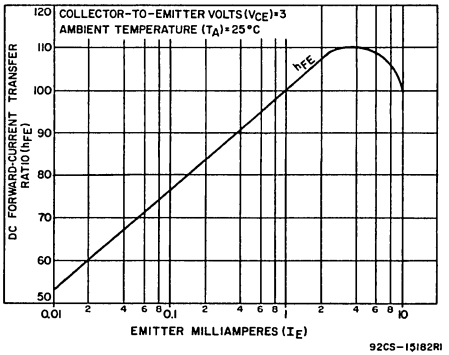


Fig.4– h_{FE} vs I_E.

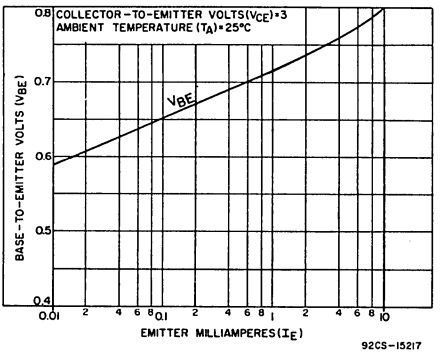


Fig.5– V_{BE} vs I_E.

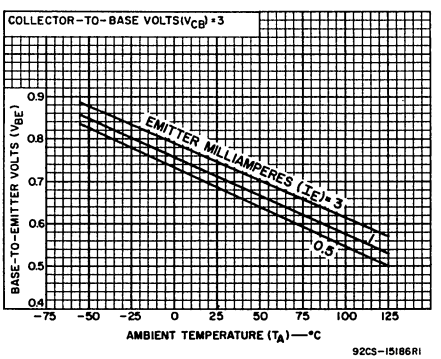


Fig.6– V_{BE} vs T_A.

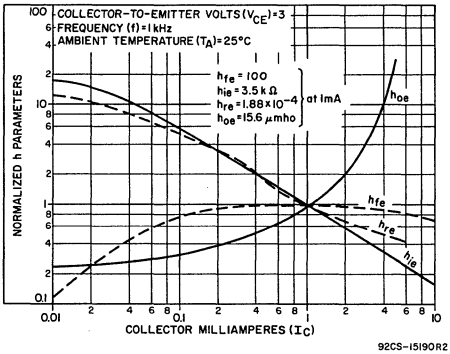


Fig.7 – Normalized h_{fe}, h_{ie}, h_{oe}, h_{re} vs I_C.

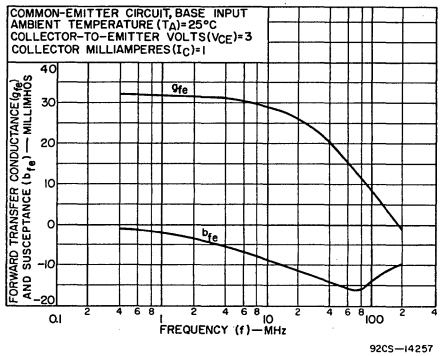


Fig.8 – y_{fe} vs f.

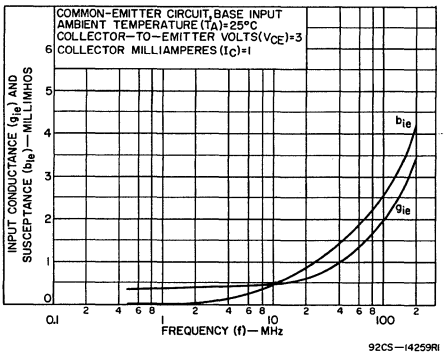


Fig.9 – y_{ie} vs f.

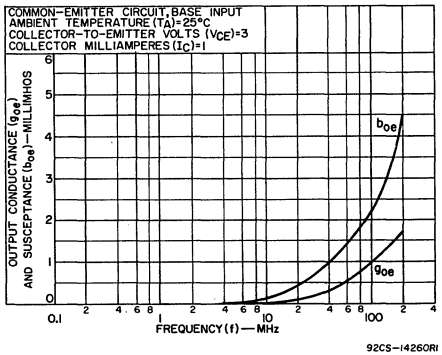


Fig.10 – y_{oe} vs f.

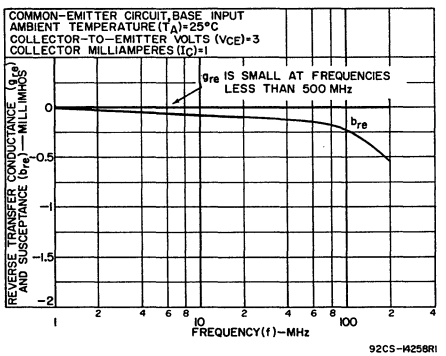


Fig.11 – y_{re} vs f.

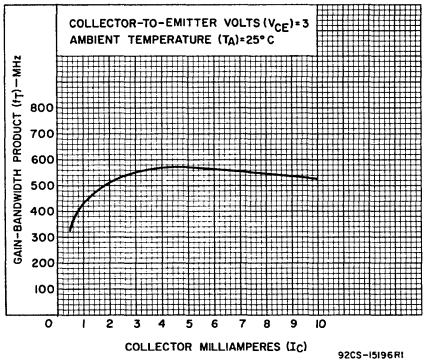


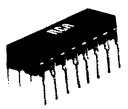
Fig.12 – f_T vs I_C.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3088E



H-1622

16-Lead Dual-In-Line Plastic Package

AM Receiver Subsystem

Includes: AM Converter, IF Amplifiers, Detector and Audio Preamplifier

For Applications in a Variety of AM Broadcast and Communications Receivers and Applications Requiring an Array of Amplifiers

Features:

- Excellent overload characteristics
- AGC for IF amplifier
- Buffered output signal for tuning meter
- Low harmonic distortion (THD)
- Delayed AGC for RF amplifier
- Terminals for optional inclusion of tone control
- Internal Zener diode provides voltage regulation
- Two IF amplifier stages
- Low-noise converter and first IF amplifier
- Operates from wide range of power supplies: V^+ = 6 to 16 volts
- Optional AC and/or DC feedback on wide-band amplifier
- Array of amplifiers for general-purpose applications
- Suitable for use with optional external RF stage, either MOS or bipolar

RCA-CA3088E*, a monolithic integrated circuit, is an AM subsystem that provides the converter, IF amplifier, detector, and audio preamplifier stages for an AM receiver.

The CA3088E also provides internal AGC for the first IF amplifier stage, delayed AGC for an optional external RF amplifier, a buffer stage to drive a tuning meter, and terminals facilitating the optional use of a tone control.

Fig. 2 is a functional diagram of the CA3088E. The signal from the low-noise converter is applied to the first IF amplifier and is then coupled to the second IF amplifier. This IF signal is then detected and externally filtered. The resultant audio signal is applied to an audio preamplifier. Optionally, a tone control circuit may be connected at the junction of the detector circuit and the audio preamplifier. The gain of the first IF amplifier stage is controlled by an internal AGC circuit. The CA3088E supplies a delayed AGC signal output for use with an external RF amplifier. A buffered output signal is also available for driving a tuning meter. A DC voltage, internally regulated by a Zener diode, supplies the second IF amplifier, the AGC and tuning meter circuits and may also be used with any other stage.

The CA3088E features four independent transistor amplifiers, each incorporating internal biasing for temperature tracking. These amplifiers are particularly useful in general-purpose amplifier, oscillator, and detector applications in a wide variety of equipment designs.

*Formerly Developmental Type TA5842.

MAXIMUM RATINGS, *Absolute Maximum Values, at T_A = 25°C*

DC SUPPLY VOLTAGE:		
Across Term. 5 and Terms. 3, 6, 13, 16, respectively	16	V
DC CURRENT:		
At Terms. 3, 6, 13, 16, respectively	10	mA
At Term. 10	30	mA
DEVICE DISSIPATION:		
Up to T_A = 50°C	760	mW
Above T_A = 50°C	derate linearly 7.6	mW/°C
AMBIENT TEMPERATURE RANGE:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
LEAD TEMPERATURE (During soldering):		
At distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C

TYPICAL ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS T_A = 25°C, V+ = 12 V	TEST CIRCUIT FIG. NO.	TYPICAL VALUES	UNITS
Static (DC) Characteristics					
DC Voltages:					
Terms. 1, 4, 9, 11	$V_{1, 4, 9, 11}$		1	0.7	V
Terms. 2, 7, 8	$V_{2, 7, 8}$			1.4	V
Term. 10	V_{10}			5.6	V
Term. 12	V_{12}			0	V
Term. 15	V_{15}			3.5	V
DC Current:					
Term. 3	I_3		1	0.35	mA
Term. 6	I_6			1.0	mA
Term. 10	I_{10}			20	mA
Term. 13	I_{13}			0	mA
Term. 16	I_{16}			1.2	mA
Dynamic Characteristics					
Detector Output		30% Modulation	4	75	mV RMS
Audio Amplifier Gain	A_{AF}	f = 1 kHz	4	30	dB
Audio Distortion		V_{OUT} = 100 mV	4	0.2	%
Sensitivity:		f_{IN} = 1 MHz			
At Converter Stage Input		Signal-to-Noise Ratio (S/N) = 20 dB	2	200	μV/m
At RF Stage Input			4	100	μV/m
Total Harmonic Distortion	THD	30% Modulation	4	1.0	%
Input Resistance:	R_I	No AGC, Input signal frequency (f_{IN}) = 1 MHz			
At Transistor Q1				3500	Ω
At Transistor Q5				2000	Ω
Input Capacitance:	C_I				
At Transistor Q1				12	pF
At Transistor Q5				17	pF
Feedback Capacitance:	C_{FB}				
At Transistor Q1				1.5	pF
At Transistor Q5				1.5	pF

The typical characteristics for the CA3088E are intended for guidance purposes in evaluating this device for equipment design.

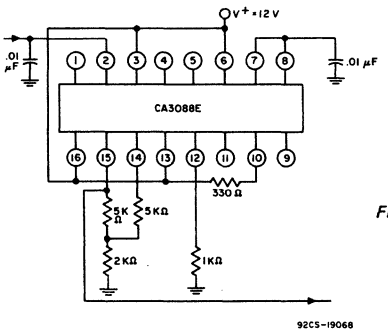


Fig.1—Test circuit for DC characteristics.

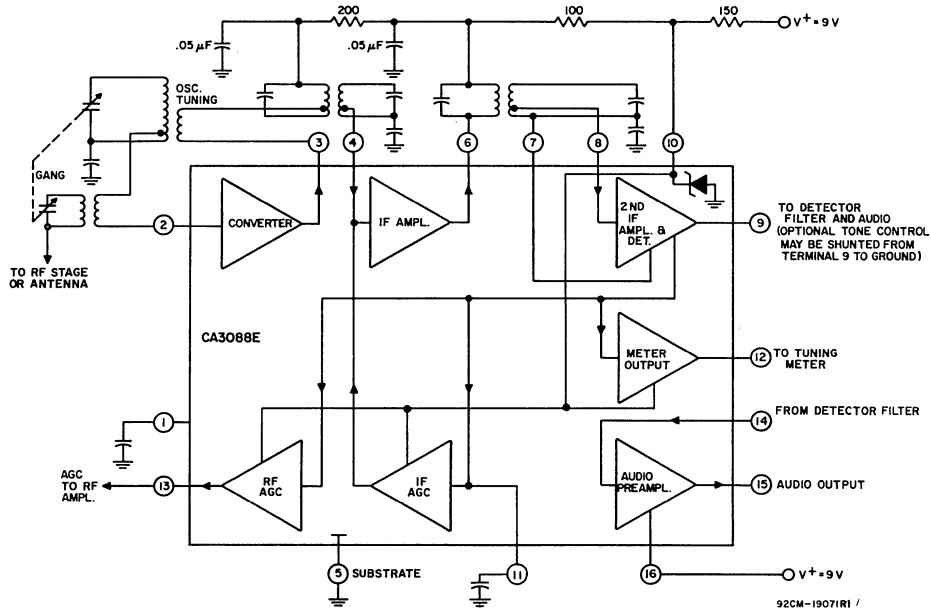


Fig.2—Functional block diagram of the CA3088E.

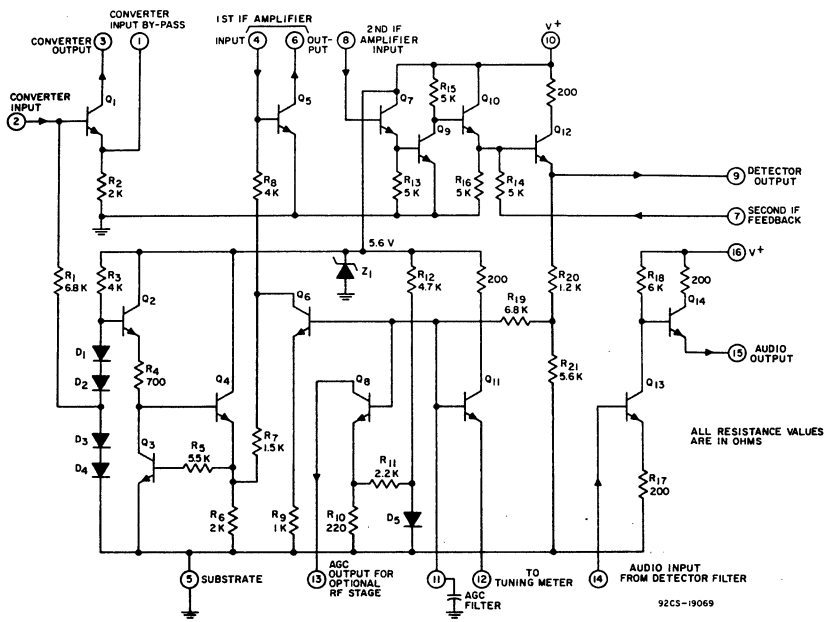


Fig.3—Schematic diagram of the CA3088E.

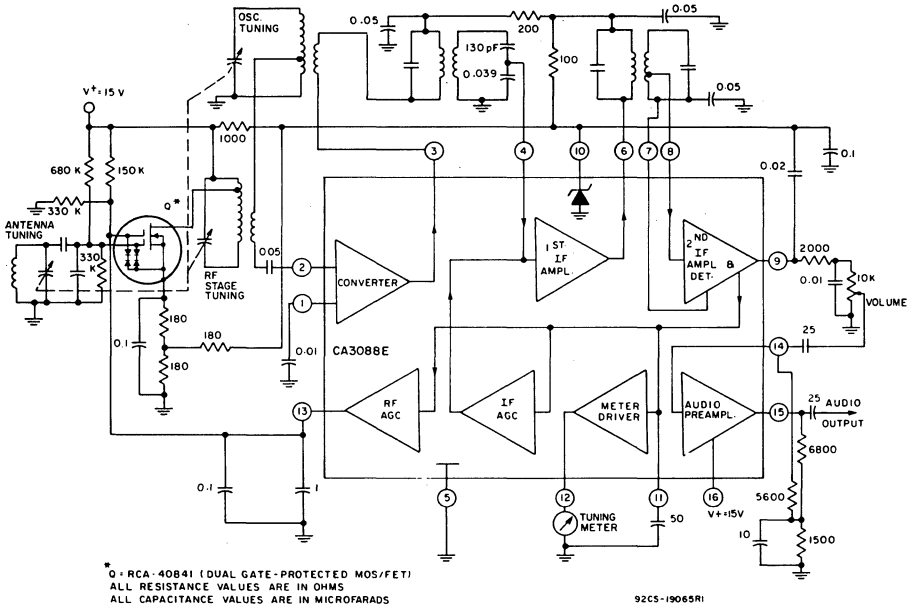


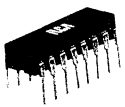
Fig.4—Typical AM broadcast receiver using the CA3088E with optional RF amplifier stage.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3089E



16-Lead Dual-in-Line
Plastic Package H-1622

FM IF System

Includes———IF Amplifier, Quadrature Detector, AF Preamplifier, and Specific Circuits for AGC, AFC, Muting (Squelch), and Tuning Meter

For FM IF Amplifier Applications in High-Fidelity, Automotive, and Communications Receivers

Features:

- **Exceptional limiting sensitivity: 12 μV typ. at −3 dB point**
- **Low distortion: 0.1% typ. (with double-tuned coil)**
- **Single-coil tuning capability**
- **High recovered audio: 400 mV typ.**
- **Provides specific signal for control of interchannel muting (squelch)**
- **Provides specific signal for direct drive of a tuning meter**
- **Provides delayed AGC voltage for RF amplifier**
- **Provides a specific circuit for flexible AFC**
- **Internal supply-voltage regulators**

RCA-CA3089E* is a monolithic integrated circuit that provides all the functions of a comprehensive FM-IF system. Fig. 1 is a block diagram showing the CA3089E features, which include a three-stage FM-IF amplifier/limiter configuration with level detectors for each stage, a doubly-balanced quadrature FM detector and an audio amplifier that features the optional use of a muting (squelch) circuit.

The advanced circuit design of the IF system includes desirable deluxe features such as delayed AGC for the RF tuner, an AFC drive circuit, and an output signal to drive a tuning meter and/or provide stereo switching logic. In addition, internal power supply regulators maintain a nearly constant current drain over the voltage supply range of +8.5 to +16 volts.

The CA3089E is ideal for high-fidelity operation. Distortion in a CA3089E FM-IF System is primarily a function of the phase linearity characteristic of the outboard detector coil.

* Formerly Developmental Type No. TA5628.

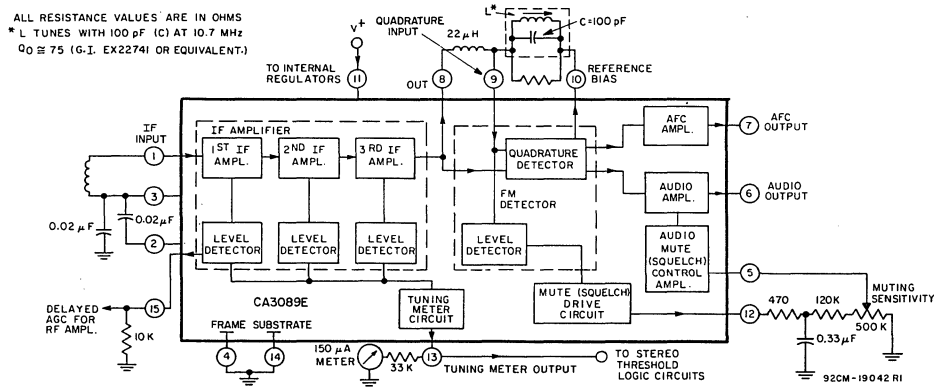


Fig.1-Block diagram of the CA3089E.

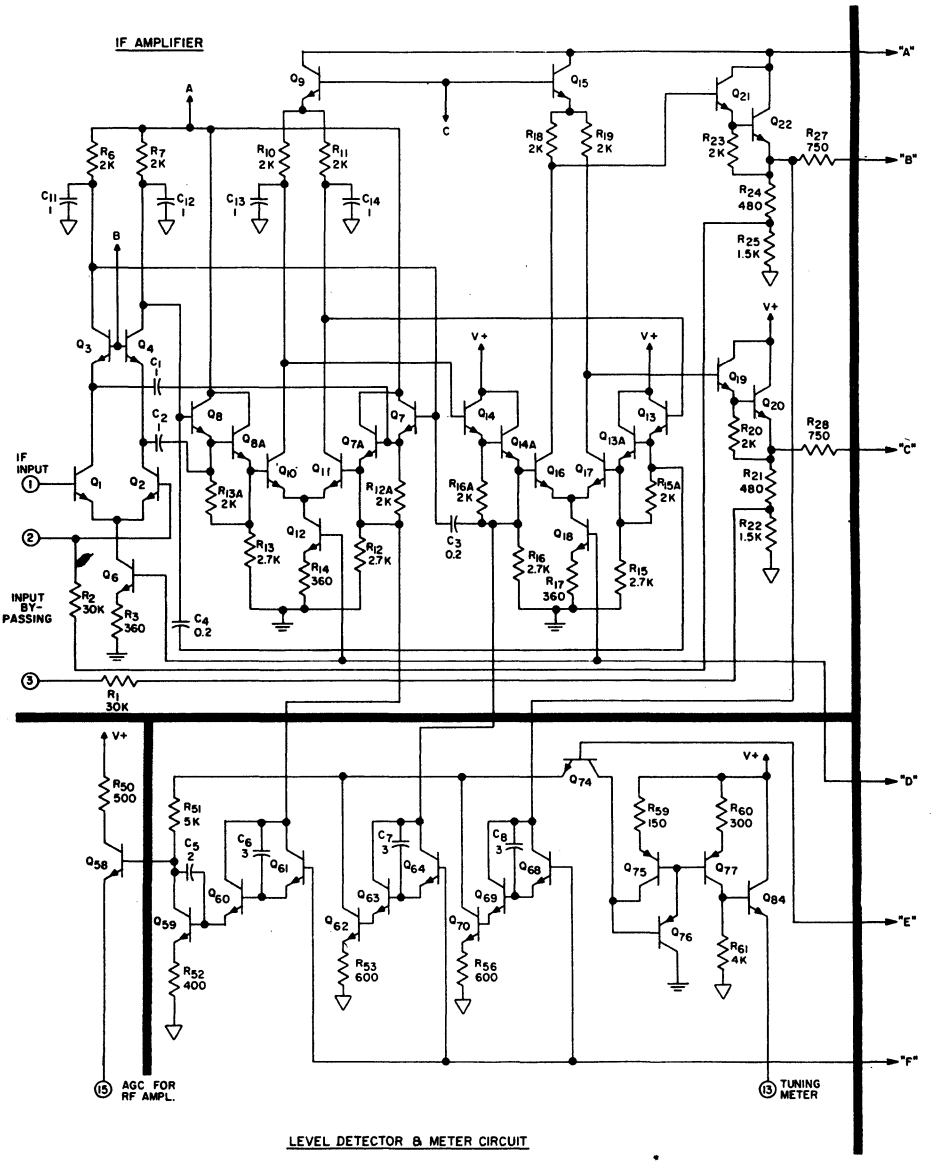


Fig.2-Schematic diagram of the CA3089E.

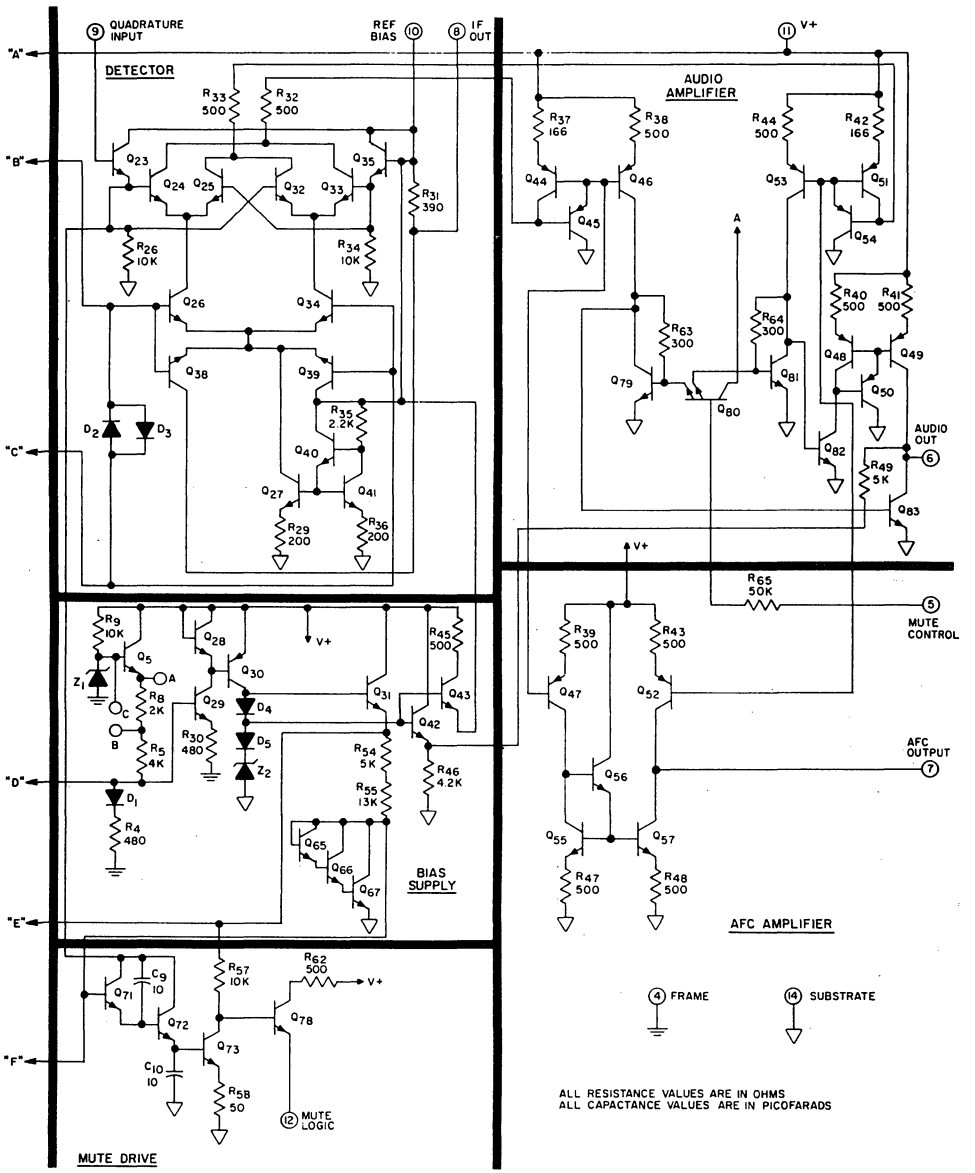


Fig.2-Schematic diagram of the CA3089E.

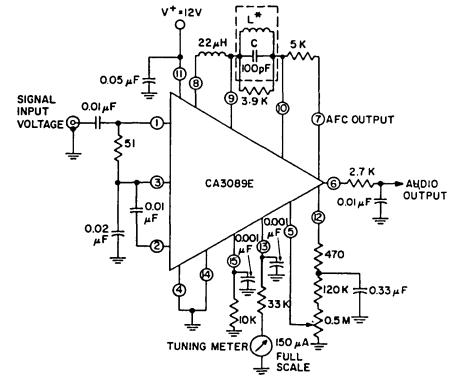
MAXIMUM RATINGS, *Absolute Maximum Values, at* $T_A = 25^oC$

DC Supply Voltage:		
Between Terminals 11 and 4	16	V
Between Terminals 11 and 14	16	V
DC Current (out of Terminal 15)	2	mA
Device Dissipation:		
Up to $T_A = 60^oC$	600	mW
Above $T_A = 60^oC$	derate linearly 6.7 mW/^oC	
Ambient Temperature Range:		
Operating	−55 to +125	^oC
Storage	-65 to +150	^oC
Lead Temperature (During Soldering):		
At distance not less than 1/32" (0.79mm) from case for 10 seconds max.	+265	^oC

ELECTRICAL CHARACTERISTICS, at $T_A = 25^oC$, $V^+ = 12$ Volts

CHARACTERISTIC	SYMBOL	TEST CONDITIONS		Circuit Fig. No.	LIMITS Min.	Typ.	Max.	UNITS
Static (DC) Characteristics								
Quiescent Circuit Current	I_{11}	No signal input, Non muted		3, 4	16	23	30	mA
DC Voltages:								
Terminal 1 (IF Input)	V_1				1.2	1.9	2.4	V
Terminal 2 (AC Return to Input)	V_2				1.2	1.9	2.4	V
Terminal 3 (DC Bias to Input)	V_3				1.2	1.9	2.4	V
Terminal 6 (Audio Output)	V_6				5.0	5.6	6.0	V
Terminal 10 (DC Reference)	V_{10}				5.0	5.6	6.0	V
Dynamic Characteristics								
Input Limiting Voltage (-3 dB point)	V_I(lim)	—		3, 4	—	12	25	μV
AM Rejection (Term. 6)	AMR	V_{IN} = 0.1V, AM Mod. = 30%	f_0 = 10.7 MHz,		45	55	—	dB
Recovered AF Voltage (Term. 6)	V_O(AF)				300	400	500	mV
Total Harmonic Distortion: *								
Single Tuned (Term. 6)	THD	V_{IN} = 0.1V	$f_{mod.}$ = 400 Hz,	3	—	0.5	1.0	%
Double Tuned (Term. 6)	THD		Deviation = ±75 kHz	4	—	0.1	—	%
Signal plus Noise to Noise Ratio (Term. 6)	S + N/N			3, 4	60	67	—	dB

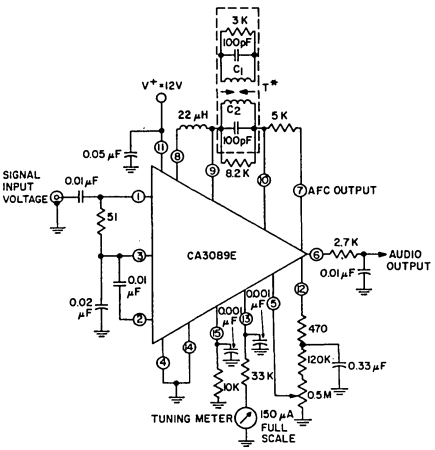
* THD characteristics are essentially a function of the phase characteristics of the network connected between terminals 8, 9, and 10.



ALL RESISTANCE VALUES ARE IN OHMS
*L TUNES WITH 100 pF (C) AT 10.7 MHz
Q_O(UNLOADED)≅75 (G.I. AUTOMATIC MFG. DIV. EX22741 OR EQUIVALENT)

92CM-19040R1

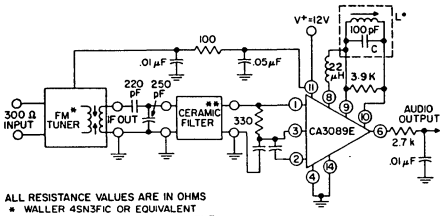
Fig.3-Test circuit for CA3089E using a single-tuned detector coil.



ALL RESISTANCE VALUES ARE IN OHMS
*T: PRI.-Q_O(UNLOADED)≅75 (TUNES WITH 100 pF (C1) 20 t OF 34e ON 7/32" DIA. FORM
SEC.-Q_O(UNLOADED)≅75 (TUNES WITH 100 pF (C2) 20 t OF 34e ON 7/32" DIA. FORM
kQ (PER CENT OF CRITICAL COUPLING)≅70%
(ADJUSTED FOR COIL VOLTAGE V_C)=150 mV
ABOVE VALUES PERMIT PROPER OPERATION OF MUTE (SQUELCH) CIRCUIT
"E" TYPE SLUGS, SPACING 4 mm

92CM-19041R1

Fig.4-Test circuit for CA3089E using a double-tuned detector coil.



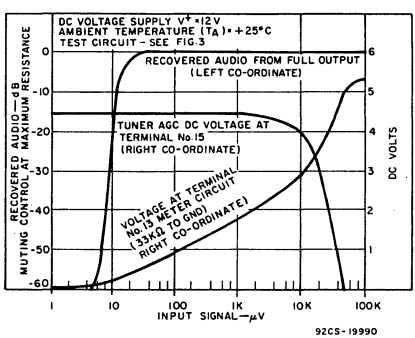
ALL RESISTANCE VALUES ARE IN OHMS
* WALLER 4SN3FIC OR EQUIVALENT
** MURATA SFG 10.7 MA OR EQUIVALENT
* L TUNES WITH 100 pF (C) AT 10.7 MHz
Q_O UNLOADED≅75 (G.I EX22741 OR EQUIVALENT)

92CS-19045

Performance data at f_o = 98 MHz, f_{MOD} = 400 Hz, Deviation = ±75 kHz:

-3dB Limiting Sensitivity	**2μV (Antenna Level)**
20dB Quieting Sensitivity	**1μV (Antenna Level)**
30dB Quieting Sensitivity	**1.5μV (Antenna Level)**

Fig.5-Typical FM tuner using the CA3089E with a single-tuned detector coil.



92CS-19990

Fig.6-Muting action, tuner AGC, and tuning meter output as a function of input signal voltage.

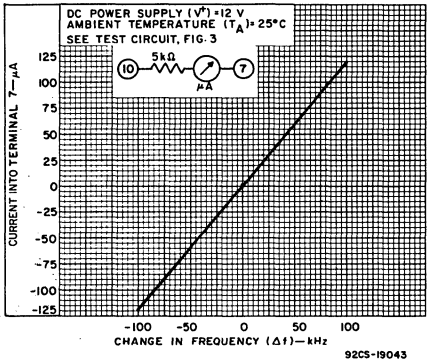
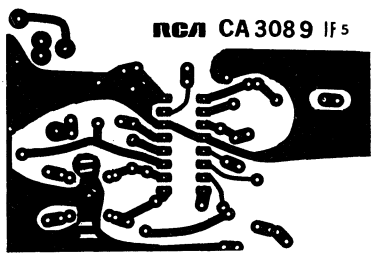
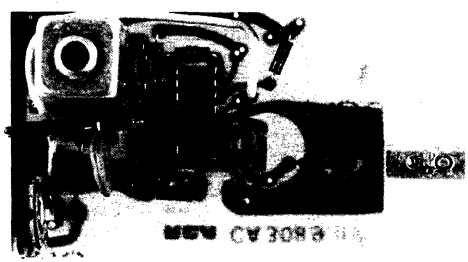


Fig.7-AFC characteristics (current at Term. 7 as a function of change in frequency).



a) Bottom view of printed-circuit board.



b) Component side — top view.

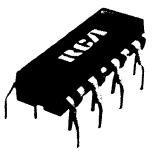
Fig.8-Actual size photographs of the CA3089E and outboard components mounted on a printed-circuit board.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3090AQ



16-Lead quad-in-line plastic package

H-1706

Stereo Multiplex Decoder

For FM Stereo Multiplex Systems

Features:

- **Requires the use of only one low-inductance tuning coil**
- **Automatic stereo switching**
- **Directly drives a stereo indicator lamp up to 100 mA**
- **Includes driver for stereo-lamp indicator**
- **Operates from a wide range of power supplies: 10 to 16 volts**
- **Requires only one adjustment for alignment**
- **Switching from monaural to stereo and stereo to monaural produces no audible thumps**
- **Low distortion: under 0.22% (typ.)**
- **Separate dc input permits stereo defeat or enable**
- **High signal output: directly drives audio amplifiers**
- **Excellent SCA (storecast) rejection: 55 dB typ.**
- **High audio channel separation: 40 dB typ.**

RCA-CA3090AQ*, a monolithic silicon integrated circuit, is a stereo multiplex decoder intended for FM multiplex systems.

The CA3090AQ is the successor to the CA3090Q; it offers three major advantages over the CA3090Q as follows:

1. Can directly drive a stereo indicator lamp with a current drain of up to 100 mA.
2. Stereo Defeat/Enable control-voltage specifications.
3. Capable of operation with lower distortion.

This stereo multiplex decoder requires only one low-inductance tuning coil (requires only one adjustment for complete alignment), provides automatic stereo switching, energizes a stereo indicator lamp, and operates from a wide range of voltage supplies.

Figure 1 shows the block diagram for the CA3090AQ. The input signal from the detector is amplified by a low-distortion preamplifier and simultaneously applied to both the 19-kHz and 38-kHz synchronous detectors. A 76-kHz signal, generated by a local voltage-controlled oscillator (VCO), is counted down by two frequency dividers to a 38-kHz signal and to two 19-kHz signals in phase quadrature. The 19-kHz pilot-tone supplied by the FM detector is compared to the locally generated 19-kHz signal in a synchronous detector. The resultant signal controls the voltage controlled oscillator (VCO) so that it produces an output signal to phase-lock the stereo decoder with the pilot tone. A second synchronous detector compares the locally generated 19-kHz signal with the 19-kHz pilot tone. If the pilot tone exceeds an externally adjustable threshold voltage, a Schmitt trigger circuit is energized. The signal from the Schmitt trigger lights the stereo indicator, enables the 38-kHz synchronous detector, and automatically switches the CA3090AQ from monaural to stereo operation. The output signal from the 38-kHz detector and the composite signal from the preamplifier are applied to a matrixing circuit from which emerge the resultant left and right channel audio signals. These signals are applied to their respective left and right post amplifiers for amplification to a level sufficient to drive most audio amplifiers.

The CA3090AQ may be used without the stereo defeat/enable function (see Fig. 6) if a control voltage for this function is not readily available. In this case, Terminal 4 should be grounded.

The CA3090AQ utilizes the 16-lead quad-in-line plastic package and operates over the ambient temperature range of −55°C to +125°C.

* Formerly Developmental Type No. TA6262G.

MAXIMUM RATINGS, *Absolute-Maximum Values at* $\bar{T}_A = 25^oC$

DC SUPPLY VOLTAGE	16 V
CURRENT AT TERM. 12	100 mA
INPUT SIGNAL VOLTAGE (COMPOSITE)■	400 mV
AMBIENT TEMPERATURE RANGE:	
Operating	−55 to +125°C
Storage	−65 to +150°C
LEAD TEMPERATURE (DURING SOLDERING):	
At distance not less than 1/32" (0.79 mm) from case for 10 s max.	+265°C

■ For stereo operation, a minimum input signal voltage (composite) of 40 mV is required.

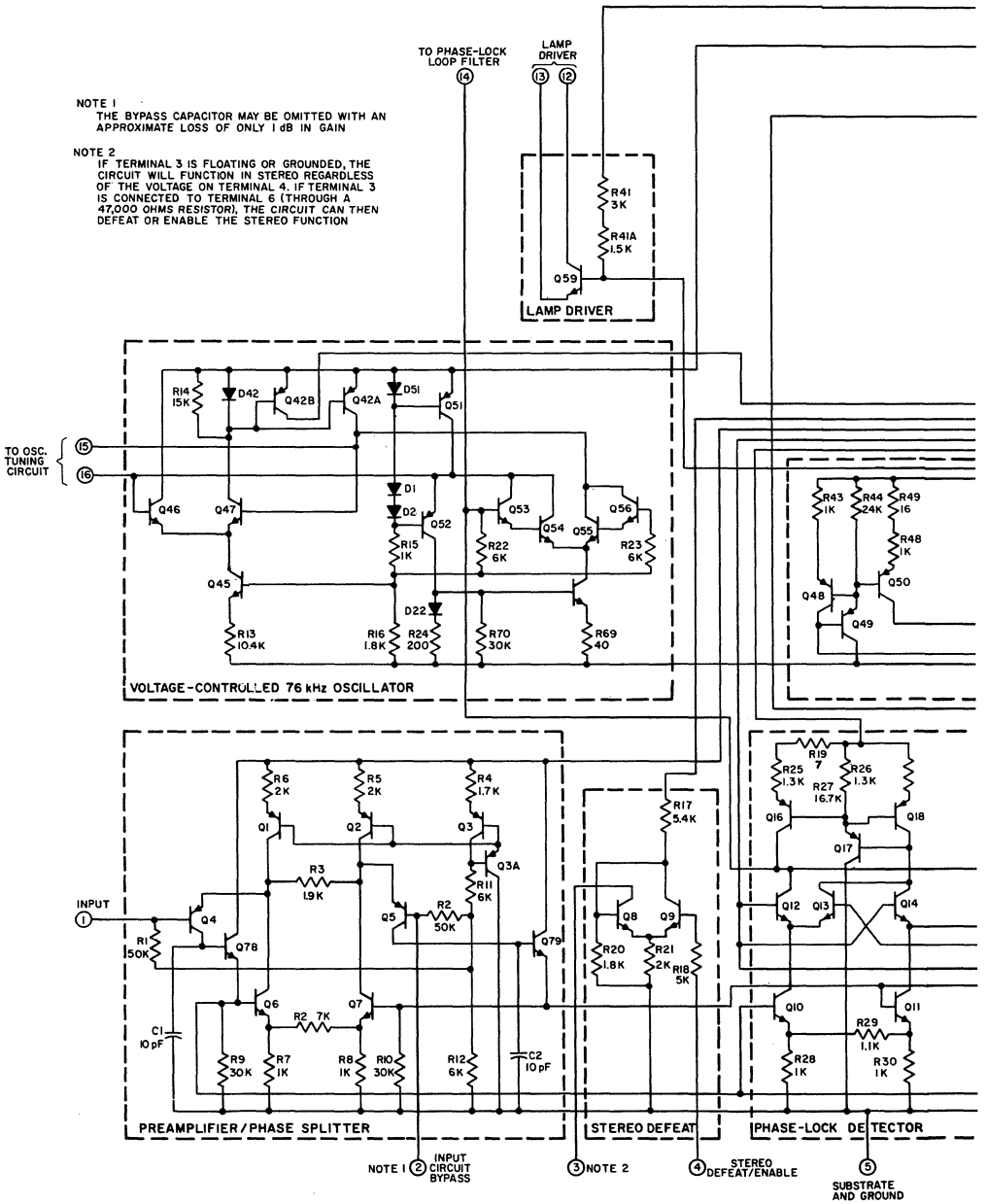


Fig. 1 — Schematic diagram of CA3090AQ.

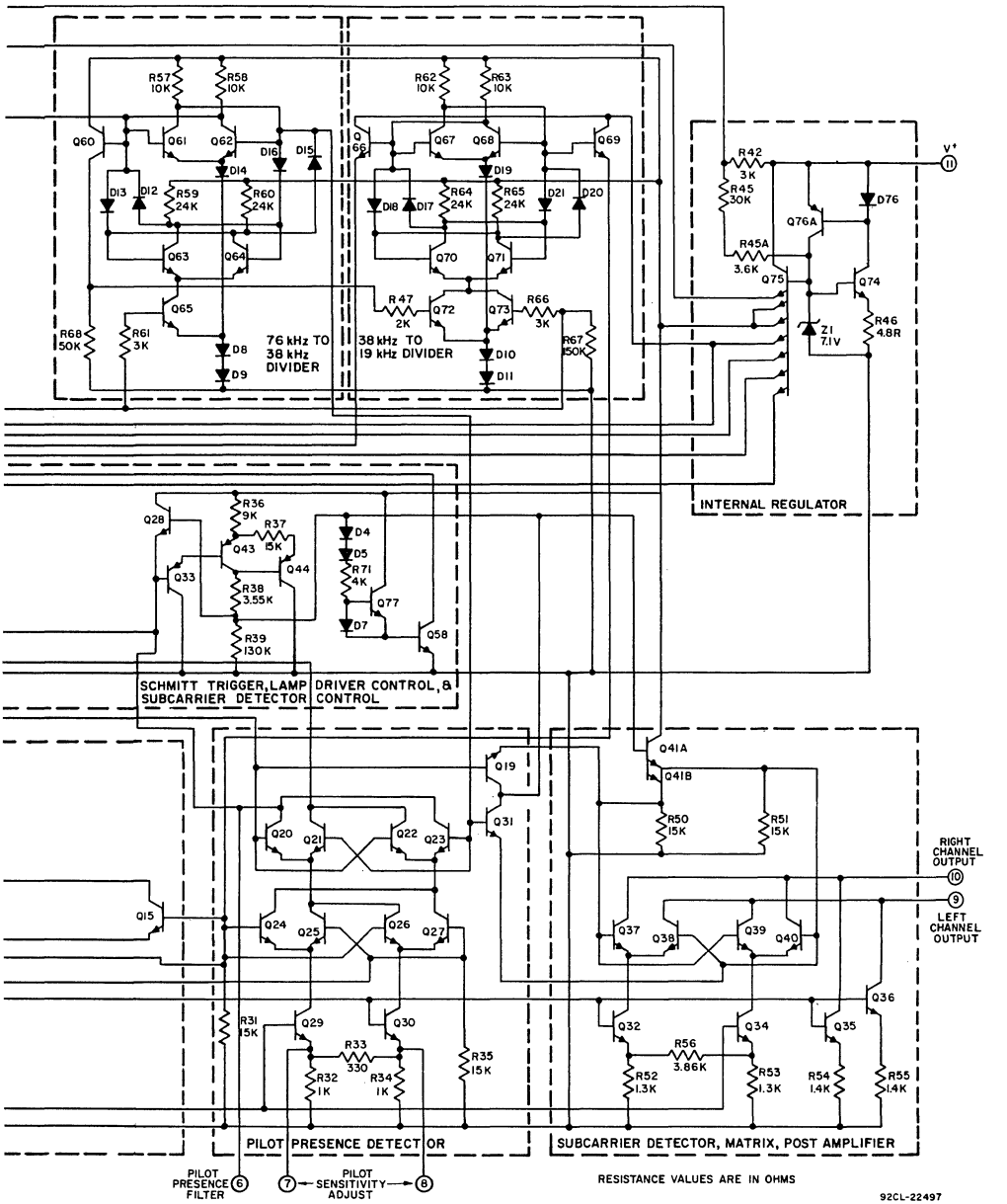


Fig.1 – Schematic diagram of CA3090AQ.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	TERMINAL MEASURED AND SYMBOL	TEST CONDITIONS $T_A = 25^\circ C$, V+ = 12 V (unless specified otherwise)	LIMITS Min.	Typ.	Max.	UNITS
Static Characteristics						
Total Current (Terms. 9, 10, 11)	I_{total}	Lamp OFF	–	22	27	mA
DC Voltage:						
Term. 1	V_1		1.6	2.3	3.1	V
Term. 6 (Indicator Lamp OFF)	V_6		–	2.1	3.6	V
Terms. 9 and 10	$V_{9\ \&\ 10}$		4.7	6.4	8.4	V
Term. 12 (Indicator Lamp OFF)	V_{12}	$V^+ = 16$ V	12.7	–	–	V
Voltage Differential (Term. 2–Term. 1)	$V_2 - V_1$		–	0	0.1	V
Current at Term. 12 (In actual use external circuit resistance (e.g. lamp should limit Term. 12 to the maximum rated value of 100 mA.)		V_{IN} (at f = 19 kHz) = 18 mV	75	100	–	mA
Dynamic Characteristics						
Input Impedance	Z_{IN}		–	50k	–	Ω
Channel Separation (L + R Reference)*		V_{IN} = 180 mV	25	40	–	dB
Channel Balance (Monaural)			–	0.3	3	dB
Monaural Gain			3	6	9	dB
Stereo/Monaural Gain Ratio*			–	±0.3	±3	dB
Indicator Lamp – Turn-ON Voltage		19-kHz pilot-tone @ Term. 1	–	4	–	mV
Capture Range (Deviation from 76-kHz center frequency)		19-kHz pilot-tone voltage = 18 mV	±6.6	±10	–	%
Distortion (75-μs de-emphasis):		V_{IN} = 240 mV				
2nd Harmonic			–	0.2	–	%
3rd, 4th, and 5th Harmonic			–	<0.1	–	%
19-kHz Rejection			–	35	–	dB
38-kHz Rejection			–	48	–	dB
SCA (storecast) Rejection			–	70	–	dB
Stereo Defeat Voltage (V_4)			–	1.2	<0.9	V
Stereo Enable Voltage (V_4)			>1.6	1.2	–	V

* For stereo operation, test conditions require a composite stereo input signal (modulated at 1 kHz) including a 19-kHz (18 mV) pilot-tone signal.

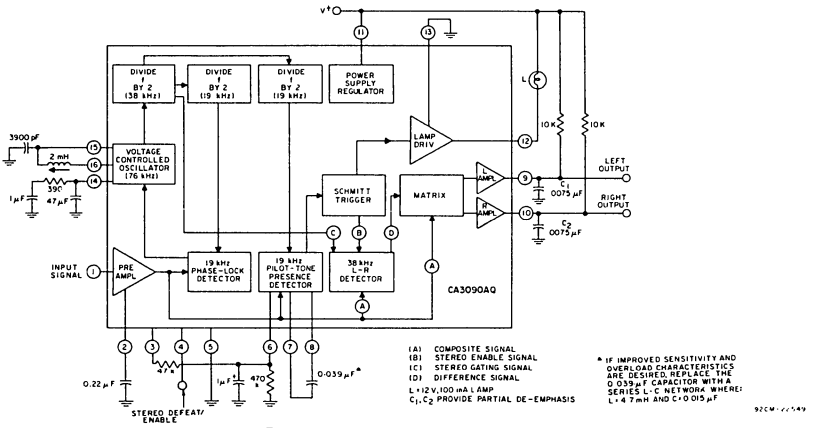


Fig.2 – Functional block diagram of the CA3090AQ.

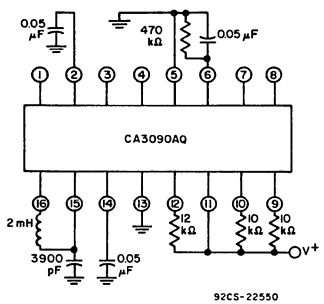


Fig. 3 – Test circuit for DC characteristics.

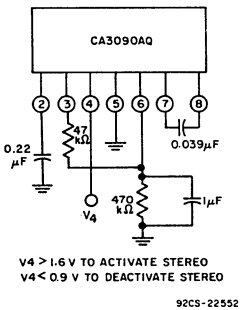


Fig. 5 – Test circuit for use with stereo defeat/enable.

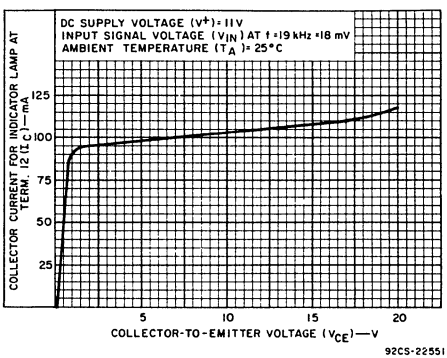


Fig. 4 – Indicator lamp characteristics (I_C vs. V_{CE}).

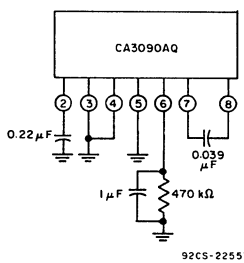


Fig. 6 – Test circuit for use without stereo defeat/enable.

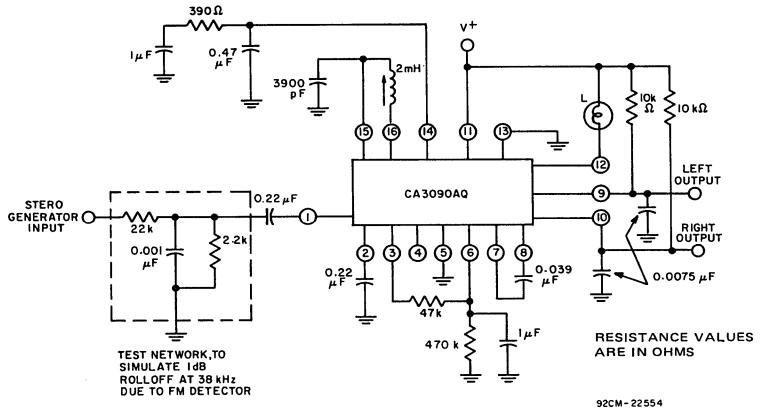


Fig. 7 — Test circuit for measurement of dynamic characteristics.

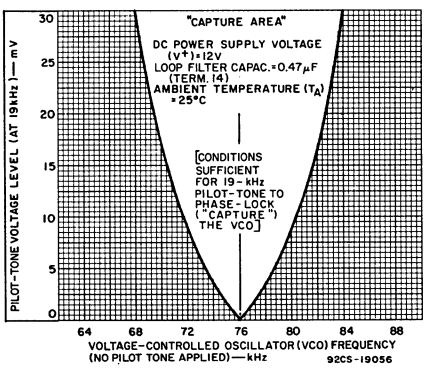


Fig. 8 — Pilot-tone voltage level vs. VCO frequency with no pilot-tone applied.

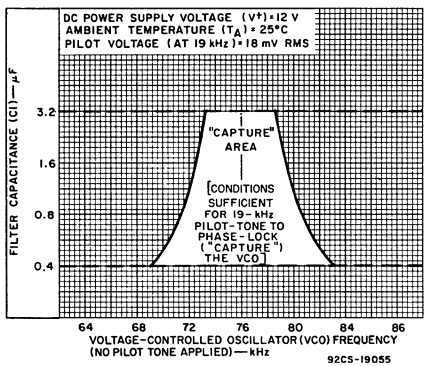
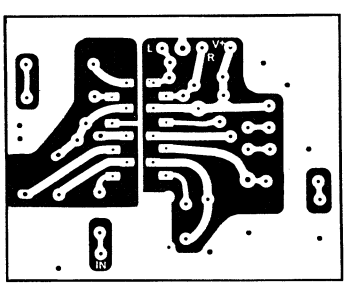
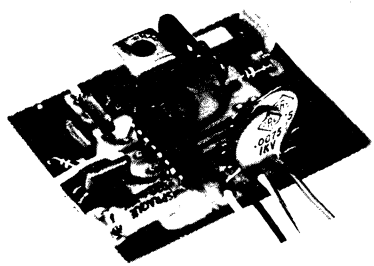


Fig. 9 — Filter capacitance vs. VCO frequency with no pilot-tone applied.



A—Foil side.



B—Component side.

Fig. 10 — Photographs of the CA3090AQ and outboard components mounted on a 2 X 2½-inch printed-circuit board to constitute a complete stereo multiplex decoder.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3091D



14-Lead Dual-In-Line
Ceramic Package H-1553

Four-Quadrant Multiplier

Applications:

- Multiplier ■ Divider ■ Squarer ■ Square Rooter
- Power-series approximator
- Full-wave rectifier
- Automatic level controller
- RMS converter
- Frequency discriminator
- Voltage-controlled filters and oscillators

RCA-CA3091D*, a monolithic silicon integrated circuit, is a four-quadrant multiplier that provides an output voltage that is the product of two input (x and y) voltages.

This device functions as a multiplier, divider, squarer, square rooter, and power-series approximator. In addition, this device is useful in applications such as ideal full-wave rectifiers, automatic level controllers, RMS converters, frequency discriminators, and voltage–controlled filters and oscillators.

The CA3091D comprises five basic circuits (See Fig. 1), including: a multiplier block, two linearity compensators, a current converter, a current source for biasing, and a regulator (reference voltage). A brief description of the operation, functions and typical applications is given in the section "Operating Considerations". In addition there is a separate section on "Symbols, Terms, and Definitions" that defines the terms and symbols used throughout the data bulletin.

The CA3091D is supplied in 14-lead dual-in-line ceramic package and operates over the full military temperature range of -55°C to +125°C.

* Formerly Developmental Type TA5855A.

Features: ●

- "Accuracy": ±4% (max.)
- "Linearity": 3.0% (max.)
- Feedthrough: 9 mV p-p (typ.)
- 3-db bandwidth: 4.4 MHz
- Low power operation capability: ±6.0 V, 4 mW drain
- Low power-supply sensitivity: 36 mV/V typ.
- Smooth overload characteristics — no foldback if full-scale input signal is exceeded
- Negligible warm-up drift
- Broadband operation capability (flat to 1 MHz) — both inputs have similar characteristics for reduced high-frequency phase shift between the inputs
- Low-level linearity correction circuitry minimizes low-level feedthrough for improved small-signal accuracy
- All multiplication is performed with wideband circuitry — this permits two signals of frequencies much higher than the -3 db frequency of the multiplier to produce a difference frequency that is within the multiplier's bandwidth
- High immunity to parasitic oscillation
- Essentially free from excess peaking — provides improved frequency response
- Requires no level shifting at the output — current-source operation at the output permits output signal to be referenced to ground or other levels within the output voltage swing capabilities of the multiplier
- Internal bias regulator

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

DC Supply Voltages:		
Between Terms. 12 and 1	+18	V
Between Terms. 4 and 1	−18	V
DC Supply Currents:		
At Term. 12 with DC Supply Voltage = +15 V	4	mA
At Term. 4 with DC Supply Voltage = −15 V	16	mA
Bias Current (At Term. 3)	1	mA
* Input Current	±1	mA
Output Short-Circuit Duration	No limitation	
Voltage Reference Current	10	mA
Linearity Correction Currents:		
At Terminals 7 and 8	10	mA
Device Dissipation (Up to 125°C)	200	mW
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−65 to +150	°C
Lead Temperature (during soldering):		
At distance not less than 1/32 inch (0.79 mm) from case for 10 seconds max.	+265	°C

* External resistance is required to limit the current to the indicated ±1 mA value.

ELECTRICAL CHARACTERISTICS, For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS: $T_A = 25^oC$, $I_{IB} = 0.5$ mA, $V^+ = 15$ V, $V^- = -15$ V	Circuit and/or Char. Curve	LIMITS: Min.	Typ.	Max.	UNITS
STATIC CHARACTERISTICS							
INPUT CIRCUIT							
Input Balance (Correction) Currents:							
At x Input	I_{IC}	x = 0	–	−20	−2.1	+20	μA
At y Input		y = 0	–	−20	−8.7	+20	μA
Feedthrough Linearity Balance (Correction) Current	I_{OC}		–	−34	−2.9	+34	μA
OUTPUT CIRCUIT							
Output Offset Current	I_{OO}	x & y = 0,	–	−10	−0.23	+10	μA
Output Offset Voltage	V_{OO}	I_{OO} thru R_L = 33kΩ	–	−0.330	−0.0076	+0.330	V
Output Peak Current Swing	$\|I_O\|$	Thru R_L = 24kΩ	3	0.41	0.45	–	mA
Output Peak Voltage Swing	$\|V_O\|$	Across R_L = 33kΩ	4	12	12.9	–	V
DC SUPPLIES & BIASING							
Current Drain (Idling):							
At Term. 4		V^- = -15 V	–	–	2.9	4.5	mA
At Term. 12		V^+ = +15 V	–	–	2.0	3.0	mA
Reference Voltage	V_{ref}	Measured across Terms. 6 & 4 at I = 1mA	–	5.5	6.1	6.7	V
DYNAMIC CHARACTERISTICS							
Output Current	I_O	With I = 0.2mA at each input	–	–	0.21	0.32	mA
Normalized k Factor $\left(k_N = \frac{k}{k_r}\right)$			11	0.69	1.0	1.7	
Accuracy		Worst case at 25°C	–	–	2.6	4.0	% of 10 V
Linearity				–	1.7	3.0	
Feedthrough Voltage:							
At y = 20V p-p, x = 0			–	–	9	20	mV p-p
At x = 20V p-p, y = 0			–	–	9	20	

ELECTRICAL CHARACTERISTICS, Typical Values Intended Only for Design Guidance

CHARACTERISTICS	SYMBOL	TEST CONDITIONS		TYPICAL VALUES	UNITS
		T_A = 25°C, I_{IB} = 0.5 mA V^+ = 15 V, V^- = −15 V	Circuit and/or Char. Curve		
STATIC CHARACTERISTICS					
INPUT CIRCUIT					
Input Resistance:					
At x Input	R_I	$\lvert I_x \rvert \leq 0.2$ mA	5	1.3	kΩ
At y Input		$\lvert I_y \rvert \leq 0.2$ mA		0.5	kΩ
Input Capacitance:					
At x Input	C_I	at 1 MHz	–	5.8	pF
At y Input				5.8	pF
OUTPUT CIRCUIT					
Output Resistance	R_O		6	1.0	MΩ
Output Capacitance:	C_O	at 1 MHz		4.0	pF
DC Supply Voltage Sensitivity:					
At Term. 4	$\frac{\Delta V_O}{\Delta V^-}$		11	26	mV/V
At Term. 12	$\frac{\Delta V_O}{\Delta V^+}$			36	mV/V
DYNAMIC CHARACTERISTICS					
Bandwidth (At -3dB point):					
Through x Input	BW		8, 10	4.8	MHz
Through y Input			8, 9	4.4	MHz
3° Error Frequency:					
Through x Input			–	360	kHz
Through y Input				310	kHz
Maximum Slew Rate	SR	7pF in parallel with 10 MΩ load	7	27	V/μs
Temperature Coefficients:					
Output Offset Current	$\Delta I_{OO}/\Delta T$	x & y = O	–	−0.021	μA/°C
x-Input Balance Current	$\Delta I_{IC}/\Delta T$	x = O	–	−0.063	μA/°C
y-Input Balance Current		y = O	–	−0.063	μA/°C
Normalized k Factor $\left(k_N = \frac{k}{k_r}\right)$	k_N		–	−0.76	%/%/°C
Accuracy			–	0.11	%/°C
Linearity			–	0.06	%/°C
Feedthrough:					
At x = O			–	5.6	mV/°C
At y = O				5.7	mV/°C

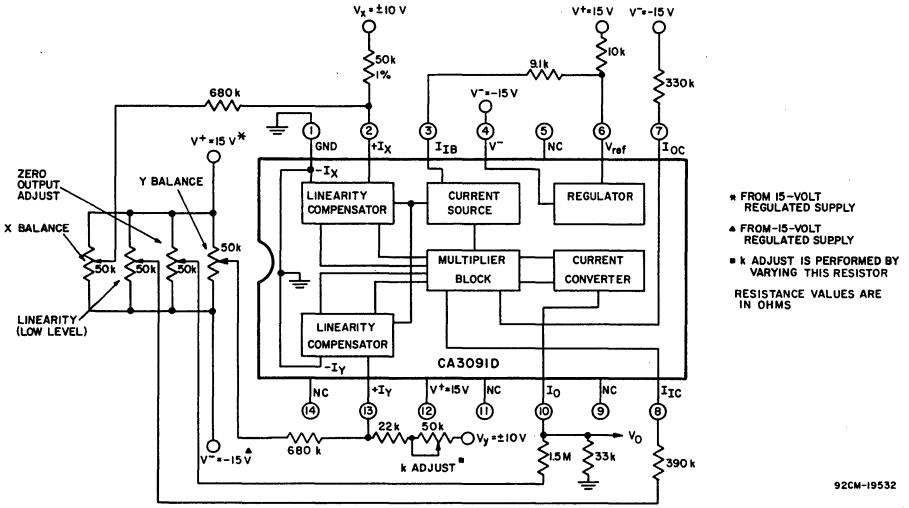


Fig.1—Functional block diagram of CA3091D with typical multiplier outboard(peripheral)circuitry.

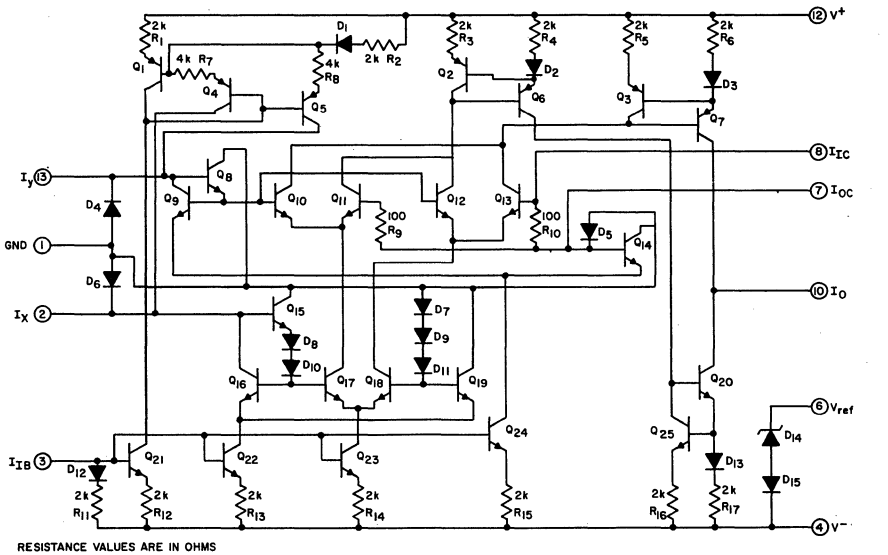


Fig.2—Schematic diagram of the CA3091D.

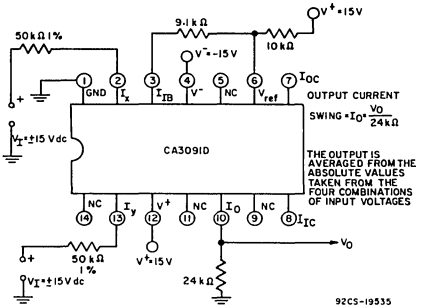


Fig.3—Test circuit for measurement of output current swing capability.

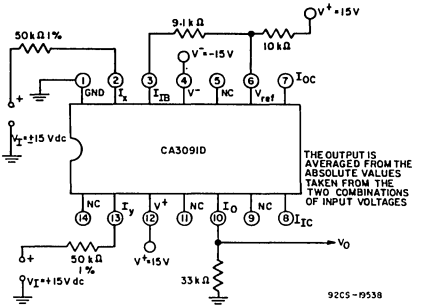


Fig.4—Test circuit for measurement of output voltage swing capability.

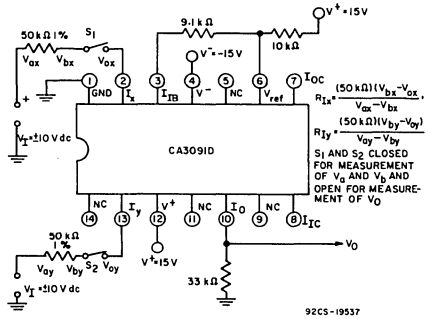


Fig.5—Test circuit for measurement of input resistance.

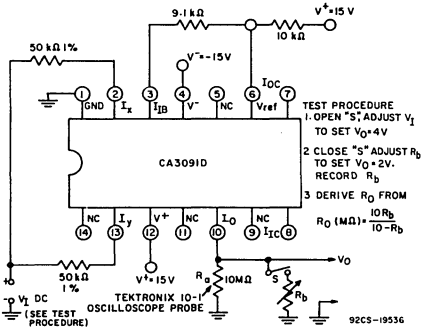


Fig.6—Test circuit for measurement of output resistance.

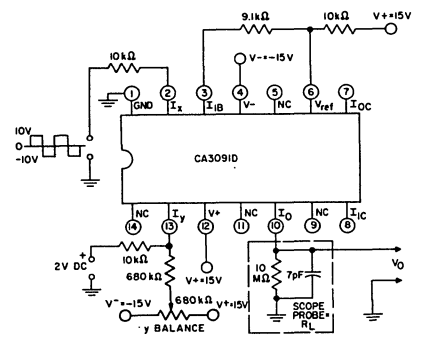


Fig.7—Test circuit for measurement of maximum slew rate.

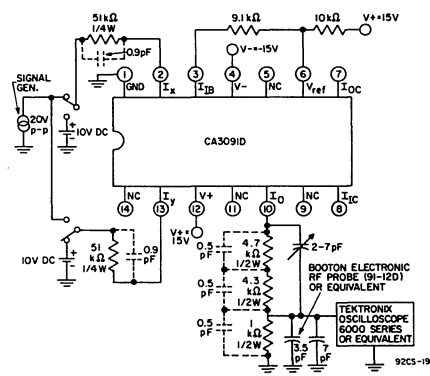


Fig.8—Test circuit for measurement of frequency response.

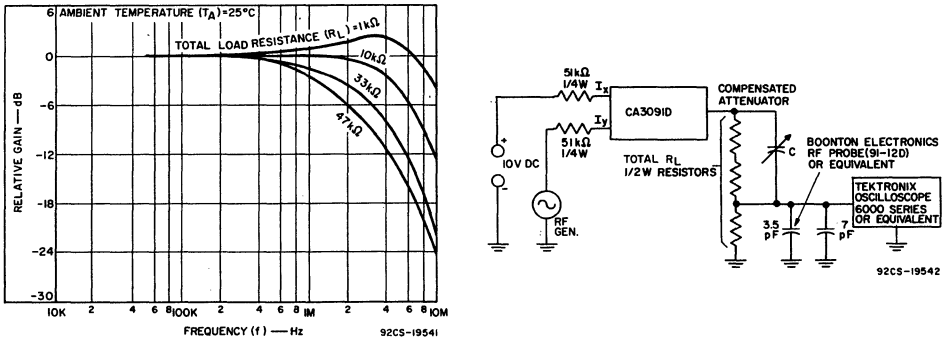


Fig.9- y—input frequency response characteristic curve with associated test circuit.

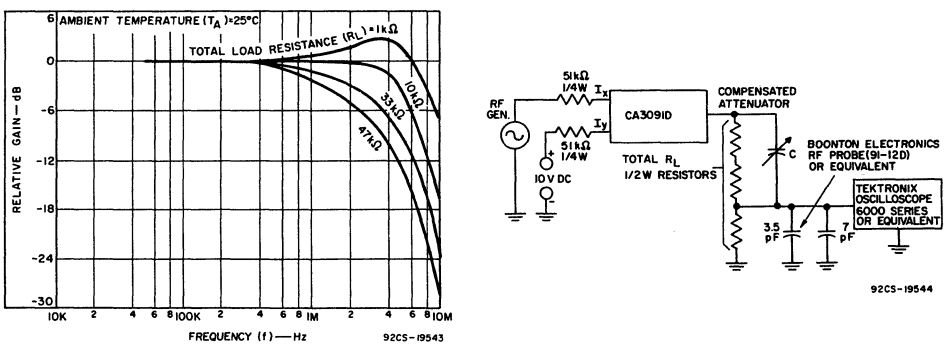


Fig.10- x—input frequency response characteristic curve with associated test circuit.

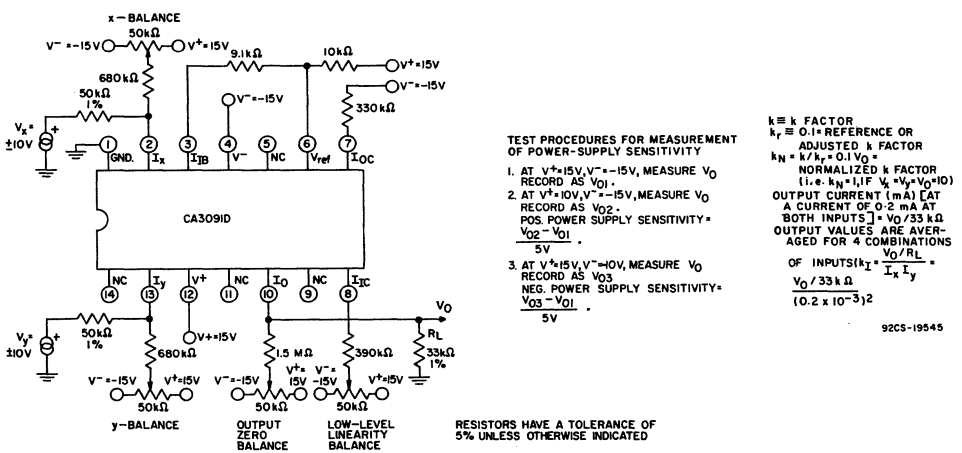
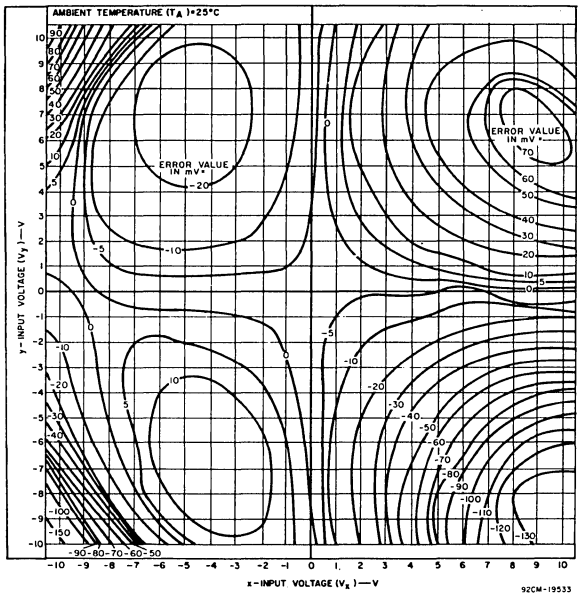


Fig.11—Test circuit for measurement of current gain and power-supply sensitivity.



Note: See "Contour Map" in "Symbols, Terms and Definitions" Section.

Fig.12—Contour mapping of multiplier accuracy (plotted on isomers) and linearity.

SYMBOLS, TERMS AND DEFINITIONS

Output Offset Current
The multiplier output current produced when both of the multiplier input signals are in the zero state.

Output Zero
Sets the output at the zero level when the x and y inputs are in the zero state. (It is implied that all other zeroing adjustments have been effected.)

R_I
Input Resistance — Converts the input voltage to an input current.

R_L
Output (Load) Resistance — Converts the output current to a voltage.

R_O
Output Resistance — See V_O and I_O for the equations associated with these properties.

Regulator Diode
A temperature compensated Zener diode, included in the multiplier circuit, to provide a stable I_{IB}.

Scale Factor or k factor (k)
Represents the basic gain of the multiplier as expressed in the equation $V_O = kV_XV_Y$

The equation indicates the ideal transfer function for the multiplier. The normalized k factor is expressed by $k_N = k/k_{ref}$ where k_{ref} is the ideal or reference k factor. The ideal factor, k_{ref} is the value at which the k factor is set when the k-factor adjust control is trimmed. Optimum operation of the CA3091D is achieved when the k-factor is 0.1.

V_{IM}
The maximum ac sine-wave voltage to be applied to the multiplier; a 20-volt p-p sine wave is the nominal maximum swing voltage recommended for use with 50-kilohm input resistors.

V_{MID}
An ac or dc voltage that approximately satisfies the equation

$$V_{MID} = V_{IM}/\sqrt{2}.$$

V_O
The output product voltage derived from the expression

$$(kV_XV_Y = V_O)$$

V_{ref}.
Temperature compensated zener connected to the −15 volt supply to provide a reference voltage as an aid in setting up a stable I_{IB}.

V_X, V_Y
The input voltages to be multiplied.

x-Balance Circuit
Sets the output to the zero level when the x-input is in the zero state.

y-Balance Circuit
Sets the output to the zero level when the y-input is in the zero state.

SYMBOLS, TERMS AND DEFINITIONS — continued

Accuracy
Accuracy defines the degree of error encountered in the operation of the multiplier. It is portrayed on a contour map by isomers (contour lines). Isomers with the highest values indicate "less-accurate" operation of the multiplier. (See illustrative Contour Map in Fig. 12.)

Contour Map
The contour map, shown in Fig. 12, is a graphical portrayal of the multiplier errors in the x, y input plane. Each contour line, termed "isomer", connects those points whose error values (in millivolts) are equal in magnitude. For example, a −20 mV contour line with points at $V_X = 5V$ and $V_Y = -3V$ indicates that the output voltage is 20 mV less than the theoretical output product (kV_XV_Y). This error voltage, presented in percent of full-scale input (±10 V), defines the "accuracy" of the device. Thus, a 20-mV error voltage represents an "accuracy" of 0.2% as derived from the equation:

$$\text{Accuracy} = 20\ \text{mV}/10 \times 100\% = 0.2\%.$$

A contour map provides a true indication of multiplier performance in each of the four quadrants. Each CA3091D is comprehensively tested and must provide the specified accuracy in the four quadrants.

Current Converter
This portion of the IC combines the multiplier's differential-amplifier output currents and converts them to a single-ended output current.

Current Sources
These circuits provide the biasing currents for the various circuits in the IC. The I_{IB} terminal provides the control current for the current-source circuit.

Feedthrough
Feedthrough occurs when an output signal is produced even though one of the input signals is zero. Consequently, feedthrough signal characteristics constitute a source of error in the operation of a multiplier. In the CA3091D, for example, the feedthrough signal output is specified to be less than 20 mV p-p when either terminal is set at 20 V p-p and the other terminal is set to zero.

I_{IB}
Circuit biasing control current.

I_{IC}
See I_{OC}.

I_O
Output product current ($k_I I_X I_Y = I_O$), where $k_I = kR_I^2 / R_L$

I_{OC}, I_{IC}
Compensatory input and output currents required to correct unlinearity along the x axis. (Optional for low-level signal use.)

I_X, I_Y
Input currents to be multiplied.

k
Voltage Scale Factor (determines the gain of the multiplier).

k_I
Current Scale Factor (k_I) = $(R_I^2 / R_L)k$.

k adjust
Scale-Factor Adjustment.

Linearity
"Linearity" indicates the degree of multiplier error (i.e. deviation from "straight-line" characteristics) along each of the four boundaries of the input x, y field. These boundaries are formed when one input is held at one of the two maximum values (10 volts or −10 volts) and the other input is swept through the voltage range. (See Contour Map for additional information.)

Linearity Adjust
An external circuit to provide vernier adjustment for optimum linearity. This control should be adjusted before adjusting the y-balance control.

Linearity Balance Circuit (Low-Level)
This circuit makes the multiplier's transfer function linear for low-level x-input signals.

Linearity Compensator
Internal circuitry that converts input current into a non-linear voltage, a requisite for producing a linear output in the differential amplifiers of the multiplier circuit.

Multiplier Circuitry
Provides the product of the two input voltages.

Multiplier Transfer Function
This function mathematically describes the interaction of the two inputs and the resulting output signal. The basic transfer function for a multiplier is

$$k(V_X + V_{Xe})(V_Y + V_{Ye}) = V_O + V_{Oe}$$

where: k = k factor and represents the basic gain of the multiplier

V_X, V_Y = the external inputs to be multiplied

V_O = the desired value of the product output signal

V_{Xe}, V_{Ye} = the "effective" errors that occur at the inputs of the multiplier and cause an output signal when either input is in a zero state.

V_{Oe} = the error voltage that develops at the output of the multiplier

DC correction factors are added to the multiplier inputs and output to compensate for the errors and offset variations. A complex linearity error term appears in the transfer function; however, this term is not included in the above equation for the purpose of clarity.

OPERATING CONSIDERATIONS

Operation of a Multiplier

A multiplier is, essentially, a gain-controlled amplifier (See Fig. 13) that multiplies the input signal (V_X) with the external gain controlling signal (V_Y) to produce the resultant output (V_O). The gain is externally adjustable by a coefficient (k). Stated simply, a multiplier produces an output voltage that is the linear product of two input voltages.

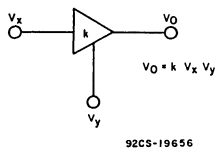
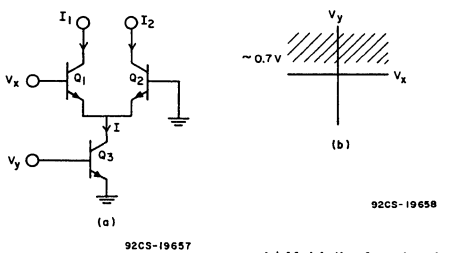


Fig.13—Gain-controlled amplifier.

The basic multiplier, shown in Fig. 14a, is a two-quadrant multiplier. The input signal (V_X) may have either a positive or negative polarity whereas, the external gain-controlling signal (V_Y) must be positive and greater than the base-to-emitter voltage (Fig. 14b). The output current ($I_1 - I_2$) of the differential amplifier, comprised of transistors Q1 and Q2, is related to both the input signal (V_X) and the current source (I). Since the current source (I) is related to the gain controlling signal (V_Y) the output current ($I_1 - I_2$), therefore, is related to both V_X and V_Y.



a) Basic circuit.

b) Multiplier functional only in shaded region.

Fig.14—Two-quadrant multiplier.

This relationship is essentially non-linear; thus an appropriate linearization circuit must be provided in the input stage to achieve the following linear relationship:

$$I_1 - I_2 = k' V_X V_Y \qquad \text{(Eq. 1)}$$

where k' is a constant

Figure 15 shows a typical arrangement of three differential amplifiers to form a four-quadrant multiplier. This arrangement incorporates the operating principles of the two-quadrant multiplier, but, in addition, it permits both of the input signals (V_X and V_Y) to have positive or negative polarities (or zero). When either input is zero, the output current ($I_1 - I_2$) must, theoretically, be zero as is shown by the following:

1. Assume $V_X = 0$,
   then $i_1 = i_2$ and $i_3 = i_4$
   therefore $i_1 + i_4 = i_2 + i_3$.
   Since $I_1 = i_1 + i_4$ and $I_2 = i_2 + i_3$,
   then $I_1 = I_2$.
   This equality is independent of V_Y
2. Now assume $V_Y = 0$,
   then $i_5 = i_6$.
   Sine $i_5 = i_1 + i_2$ and $i_6 = i_3 + i_4$,
   then $i_1 + i_2 = i_3 + i_4$.
   Since $i_1 = i_3$ and $i_2 = i_4$
   then $i_1 + i_4 = i_3 + i_2$.
   Therefore $I_1 = I_2$.
   This equality is independent of V_X.

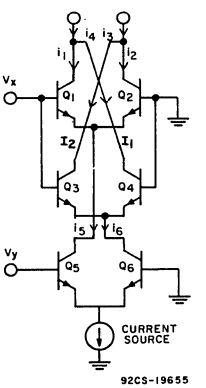


Fig.15—Basic four-quadrant multiplier.

The multiplying operation discussed in the previous section applies when neither V_X nor V_Y is zero. The output current ($I_1 - I_2$) then satisfies Equation 1,

$$I_1 - I_2 = k' V_X V_Y.$$

The multiplying action of the four-quadrant multiplier is dependent on current unbalance in the three differential amplifiers. Ideally, the multiplying operation should not occur if either V_X or V_Y is O. However, in practical applications slight current unbalances do exist. It is necessary, therefore, to null out such unbalances with external potentiometers prior to operation.

TYPICAL OPERATING CONSIDERATIONS

The RCA-CA3091D, shown in Fig. 2, is a four-quadrant multiplier that incorporates the basic multiplier principle, previously discussed in "Operation of a Multiplier". Because the design of this multiplier is based on the multiplication of two input currents to produce an output current it is necessary to convert the input voltages to input currents and the output current to an output voltage by inserting resistors at both input and output terminals. Fig. 1 shows the four-quadrant multiplier with its peripheral circuitry for nulling current unbalances.

The Bias Current (I_{IB}) at Term. 3 sets the operating current level for the entire multiplier circuit by means of a current-source circuit. Therefore, it is essential that this bias current level remain constant under all operating conditions. To maintain this steady state, a temperature compensated zener diode is provided on the chip and connected to the Reference Voltage (Term.6).

Linearity of the differential amplifier transconductance function is accomplished by linearity compensators as shown in Fig. 1. To correct low-level signal unbalances that may occur between Differential Amplifiers A and B, an external potentiometer is connected to Terminals 7 and 8 (See Fig. 1). The Current Converter circuit, which consists of a set of current mirrors, supplies the output current ($I_1 - I_2$). It is important that circuit unbalances be corrected prior to operation. Table I describes the alignment procedures for correcting these unbalances.

A multifunctional circuit board (Figs. 16 and 17) is available for performing the four basic applications, such as, multiplying, dividing, squaring and taking the square root.

When the CA3091D is used as a multiplier (Fig. 18) or as a squarer (Fig. 18) only the basic pheripheral circuitry on the multifunctional circuit board is utilized and the general-purpose operational amplifier (CA3741T) is disabled from operation. Follow the ac alignment procedures for these two applications before operating the circuit.

When the CA3091D is used as a divider (Fig. 20), the operational amplifier is required in order to provide the proper negative feedback. The limitations for operation as a divider are that $O < V_Y \leqslant 10V$ and $-10V \leqslant V_Z \leqslant 10V$. Note, the range of V_Y is limited to the positive polarity; if V_Y was permitted to go negative, the feedback loop would go positive and, thereby, create an unstable operating condition.

Alignment of the divider (Fig. 19) differs from multiplier and squarer alignment because of the additional variances introduced by the operational amplifier. A coupling capacitor is

Table I

AC Alignment Procedures For CA3091D, Four-Quadrant Multiplier

(Refer to Fig. 16, for circuit pertaining to following alignment procedures.)

Step No.	Voltage Setting		Control Adjust	Test Equipment Used	Measure	Notes
	V_X	V_Y				
1	–	–	–	–	–	Set all potentiometers to center of range.
2	O	V_{IM}	x Balance	AC VM	V_O	Adjust for a minimum reading.
3	O	V_{IM}	Linearity	AC VM	V_O	Adjust for a minimum reading.
4	–	–	–	–	–	Repeat Steps 1 and 2 until no further improvement is noted.
5	V_{IM}	O	y Balance	AC VM	V_O	Adjust for a minimum reading.
6	O	O	Zero Output	DC VM	V_O	Adjust for zero output.
7	V_{MID}	V_{MID}	R_k	AC/DC VM	V_O	Adjust for $V_{MID}^2/10$ at the output.
8	–	–	–	–	–	Check multiplier for alignment in all four quadrants.

V_{IM} – Is the maximum AC swing of the sine wave that will be applied to the multiplier. A 20-volt p-p value is the nominal maximum swing of the AC sine wave with input resistors of 50 kilohms.

V_{MID} – An AC or DC voltage that approximately satisfies the equation $V_{MID} = V_{IM}/\sqrt{2}$. For example, if a 50-kilohm resistor is used with a 7-volt input, then R_k should be adjusted for a 4.9-volt output.

provided at the output of the divider alignment circuit in order to separate the ac signal from the dc signal and, thus, avoid interaction between the calibrating potentiometers.

The alignment procedure for the square-rooter function (Fig. 21) is identical to the alignment procedure for the divider function. The input voltage range is limited to $0 < V_I \leqslant 10V$. This limitation is necessary in order to prevent the output voltage (V_O) from latching to the negative output saturation voltage of the operational amplifier. Table II describes the divider alignment procedure.

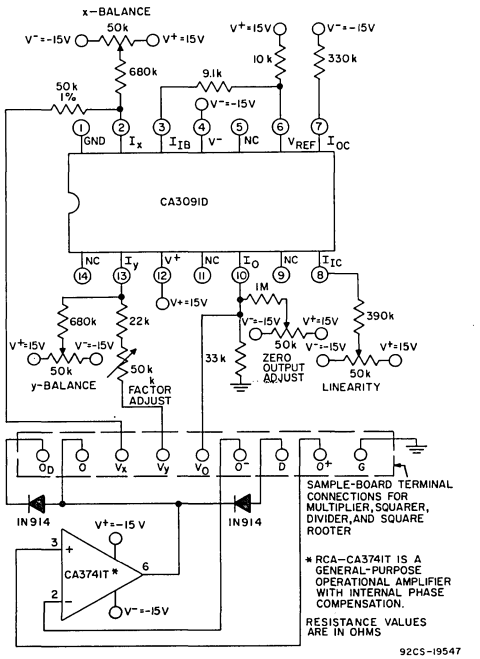
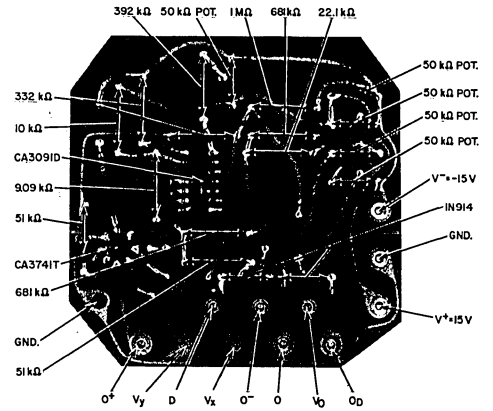
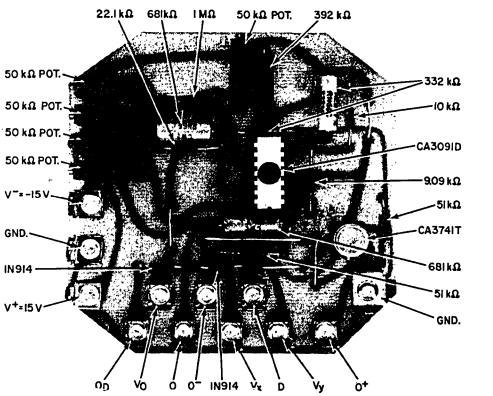


Fig.16—Typical multifunction circuit arrangement utilizing the CA3091D and CA3741T.



a) Foil side.

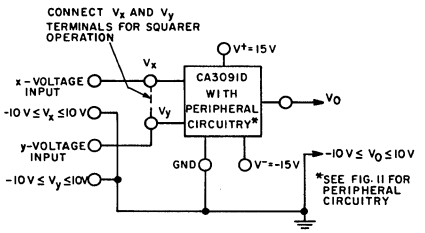


b) Component side.

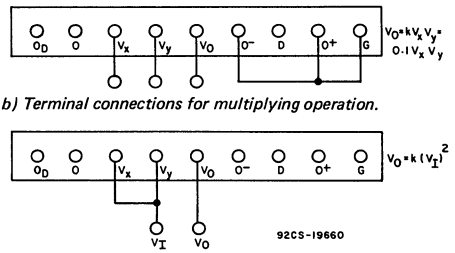
Fig.17—Photographs of a printed-circuit board for multifunction applications (multiplier, squarer, divider, square rooter) utilizing the CA3091D and CA3741T.

Table II — Divider Alignment Procedure

Step No.	Set V_z V	Set V_y V	Measure	Output Coupling	Test Equipment Used	Adjust	Notes
1	—	—	—	—	—	—	Set all potentiometers to center of range.
2	0	V_S	V_O	ac	ac — VM	O_{zero}	Adjust for minimum reading.
3	0	10V dc	V_O	dc	dc — VM	$x_{balance}$	Adjust for OV dc output.
4	V_S	V_S	V_O	ac	ac — VM	$y_{balance}$	Adjust for minimum reading.
5	5V dc	5V dc	V_O	dc	dc — VM	k_{adjust}	Adjust for 10V dc output.



a) Circuit arrangement for multiplier or squarer operation.



b) Terminal connections for multiplying operation.

c) Terminal connections for squarer operation.

Fig.18—Multifunction circuit-board arrangement with terminal connections for multiplier and squarer operation.

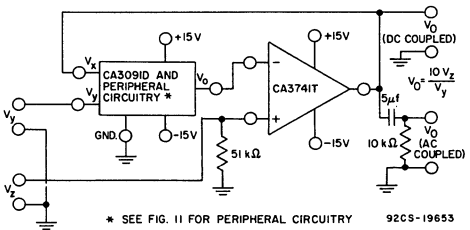


Fig.19—(a) Divider alignment circuit.

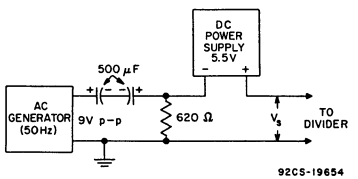
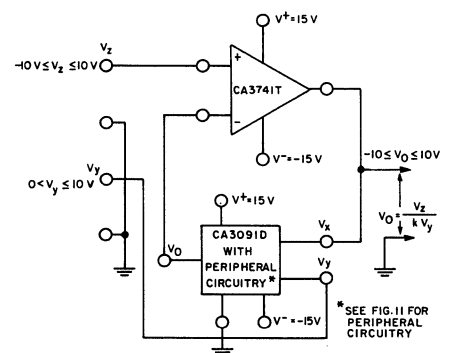
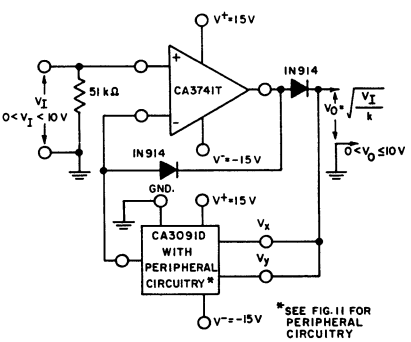


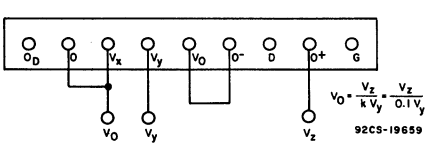
Fig.19—(b) Circuit to provide offset ac signal for use in divider alignment procedure.



a) Circuit arrangement for divider operation.

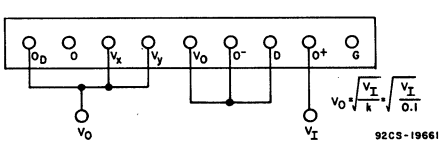


a) Circuit arrangement for square-rooter operation.



b) Terminal connections for divider operation.

Fig.20—Multifunction circuit-board arrangement with terminal connections for divider operation.



b) Terminal connections for square-rooter operation.

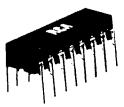
Fig.21—Multifunction circuit-board arrangement with terminal connections for square-rooter operation.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3093E



16-Lead Dual-In-Line Plastic Package

H1622

General-Purpose High-Current N-P-N Transistor-Zener Diode-Diode Array

Applications

- Signal processing and switching systems operating from DC to VHF
- Lamp and relay driver
- Differential amplifier
- Temperature-compensated amplifier
- Thyristor firing
- Temperature-compensated shunt regulator
- Temperature-compensated series regulator
- Level shifting
- Voltage-level clamping
- Current regulator
- Voltage clamping
- Simple off-line regulated supply
- See RCA Application Note, ICAN-5296 "Application of the RCA-CA3018 Circuit Transistor Array" for applications in addition to those given on pages 5 & 6 of this bulletin.

RCA CA3093E* is a versatile array of three high-current (to 100mA) NPN transistors, two 10%-tolerance Zener diodes and one conventional diode, all on a common monolithic substrate. Two of the transistors (Q_1 and Q_2) are matched at 1 mA for applications in which offset parameters are of special importance. The combination of positive Zener voltage temperature coefficients and negative forward base-emitter voltage temperature coefficients provides a unique temperature compensation capability.

Independent connections for each transistor and diode plus a separate terminal for the substrate permit maximum flexibility in circuit design.

*Formerly developmental type TA6119

#Z_1, Z_2 and D1 are transistors internally connected as shown below.

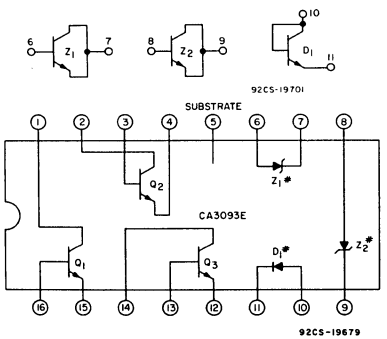


Fig. 1 — Functional diagram of the CA3093E (bottom view)

Features:

- **6 independent devices plus separate substrate connection**
- **Compensating temperature coefficients — V_{BE} and V_{D1} vs. V_Z**

Transistors

- **High I_C (100mA max)**
- **Matched pair (Q1 & Q_2)**
  V_{IO} = ± 5mV max
  I_{IO} = 2.5 μA max
  } at I_C = 1mA
  $\Delta V_{IO}/\Delta T$ = 5 μV/°C typ
- **h_{FE} = 40 min @ I_C = 10mA or 50mA**
- **Low V_{CE}sat . . . 0.7V max @ 50mA**

Zener Diodes

- **Two 1/4W Zeners**
- **V_Z = 7V ± 10%**
- **z_Z = 15Ω typ**

Diode

- **Close forward voltage match to V_{BE}'s of Q_1 and Q_2**
- **V_{PIV} = 5.5V min.**

MAXIMUM RATINGS, *Absolute-Maximum Values at T_A = 25°C*

Power Dissipation:		
Any one transistor	500	mW
Any one Zener Diode	250	mW
Total package	750	mW
Above 25°C ..... Derate linearly	6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−55 to +150	°C
Lead Temperature (During Soldering):		
At distance 1/16 ± 1/32 inch (1.59 ± 0.79mm) from case for 10 seconds max.	+265	°C
The following maximum ratings apply for each transistor		
Collector-to-Emitter Voltage (V_{CEO})	15	V
Collector-to-Base Voltage (V_{CBO})	20	V
Collector-to-Substrate Voltage (V_{CIO})*	20	V
Emitter-to-Base Voltage (V_{EBO})	5.5	V
Collector Current (I_C)	100	mA
Base Current (I_B)	35	mA
The following maximum ratings apply for each Zener Diode or Diode		
Zener Diode dc Current (I_Z)	35	mA
Zener Diode-to-Substrate Voltage (V_{ZIO})*	20	V
Diode (D1) Forward Current (I_{DF})	50	mA
Diode (D1) Reverse Voltage (V_{DR})	5.5	V
Diode (D1)-to-Substrate Voltage (V_{DIO})*	20	V

*The collector of each transistor, the cathode of each Zener diode, and the anode of the diode are isolated from the substrate by an internal diode. The substrate must be connected to a voltage which is more negative than any of these isolated terminals in order to maintain isolation between devices and provide normal transistor action. To avoid undesired coupling between devices, the substrate terminal (5) should be maintained at either dc or signal (ac) ground. A suitable bypass capacitor can be used to establish a signal ground.

ELECTRICAL CHARACTERISTICS at T_A = 25°C
For Equipment Design

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	LIMITS			UNITS
			Min.	Typ.	Max.	
For Each Transistor:						
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0	20	60	–	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1mA, I_B = 0	15	24	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0	20	60	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 500μA, I_C = 0	5.5	6.9	–	V
Collector-Cutoff-Current	I_{CEO}	V_{CE} = 10V, I_B = 0	–	–	10	μA
Collector-Cutoff-Current	I_{CBO}	V_{CB} = 10V, I_E = 0	–	–	1	μA
DC Forward Current Transfer Ratio	h_{FE}	V_{CE} = 3V, I_C = 10mA	40	76	–	
		V_{CE} = 3V, I_C = 50mA	40	75	–	
Forward Base-to-Emitter Voltage	V_{BE}	V_{CE} = 3V, I_C = 10mA	0.65	0.74	0.85	V
Collector-to-Emitter Saturation Voltage	V_{CEsat}	I_C = 50mA, I_B = 5mA	–	0.40	0.70	V
Forward Base-to-Emitter Temp. Coefficient	$\Delta V_{BE}/\Delta T$	I_E = 10mA	–	-1.9	–	mV/°C
For Transistors Q1 and Q2 (As a Differential Amplifier):						
Absolute Input Offset Voltage	$\lvert V_{IO} \rvert$	V_{CE} = 3V, I_C = 1mA	–	1.2	5	mV
Absolute Input Offset Current	$\lvert I_{IO} \rvert$		–	0.7	2.5	μA
Temp. Coefficient of Offset Voltage	$\lvert \Delta V_{IO}/\Delta T \rvert$	–	–	5	–	μV/°C
For Each Zener Diode						
Zener Voltage	V_Z	I_Z = 10mA	6.3	7	7.7	V
Zener Impedance	z_Z	I_Z = 10mA, f = 1 kHz	–	15	25	Ω
Zener Reverse Current	I_{ZR}	V_Z = +5V	–	–	1	μA
Zener Voltage Temp. Coefficient	$\Delta V_Z/\Delta T$	I_Z = 10mA	– i.e.	+3.6 +.05	–	mV/°C %/°C
Zener-to-Substrate Breakdown Voltage	$V_{(BR)ZIO}$	I_Z = 100μA (Terminals 7 & 9)	20	60	–	V
Dissipation		Refer to Example in Application "a"	–	–	250	mW
For Diode (D1)						
Diode Forward Voltage	V_{DF}	I_C = 10mA, V_{CE} = 3V	0.65	0.74	0.85	V
Diode Forward Current	I_{DF}		–	–	50	mA
Diode Reverse-Breakdown Voltage	$V_{(BR)DR}$	I_{DR} = 500μA	5.5	6.9	–	V
Diode-to-Substrate Breakdown Voltage	$V_{(BR)DIO}$	I_{Diode} = 100μA (Terminal 10)	20	60	–	V
Diode Forward-Voltage Temp. Coefficient	$\Delta V_{DF}/\Delta T$	I_{DF} = 5mA	–	-1.9	–	mV/°C

TYPICAL STATIC CHARACTERISTICS

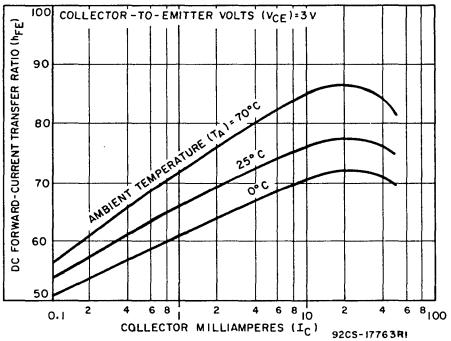


Fig. 2 – h_{FE} vs I_C

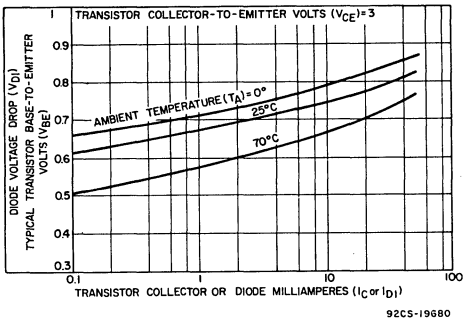


Fig. 3 – V_{BE} vs I_C and V_{D1} vs I_{D1}

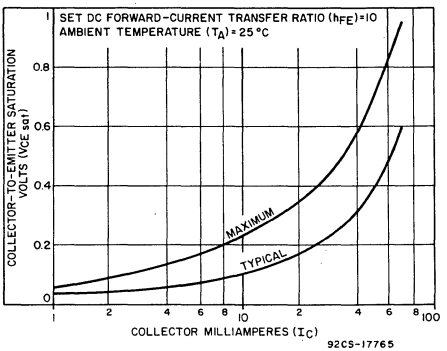


Fig. 4 – V_{CEsat} vs I_C at 25°C

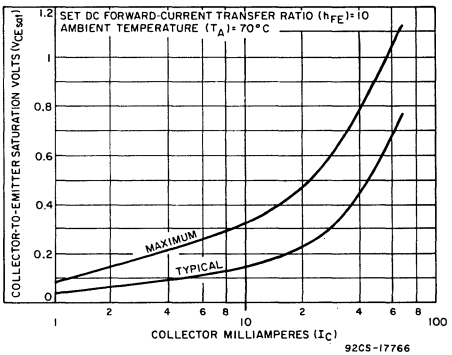


Fig. 5 – V_{CEsat} vs I_C at 70°C

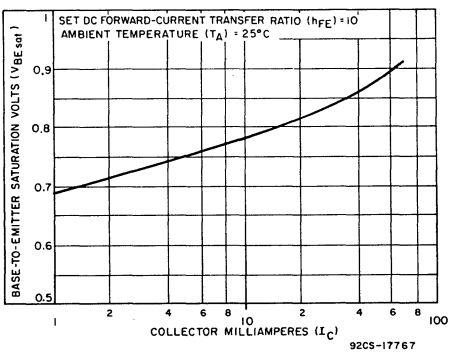


Fig. 6 – V_{BEsat} vs I_C

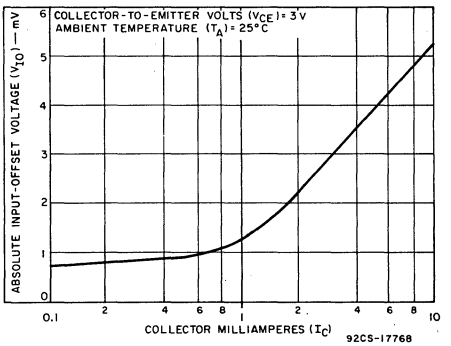


Fig. 7 – V_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier)

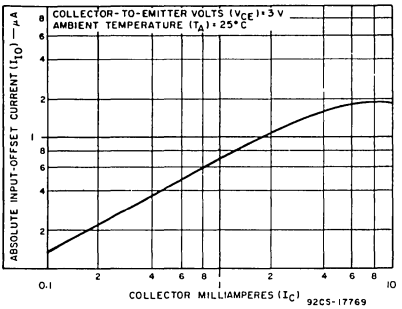


Fig. 8 – I_{IO} vs I_C (transistors Q1 and Q2 as a differential amplifier)

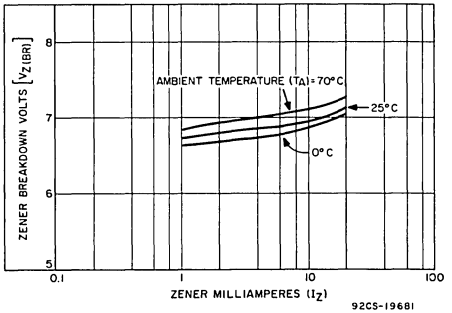


Fig. 9 – Typical Zener breakdown voltage vs current

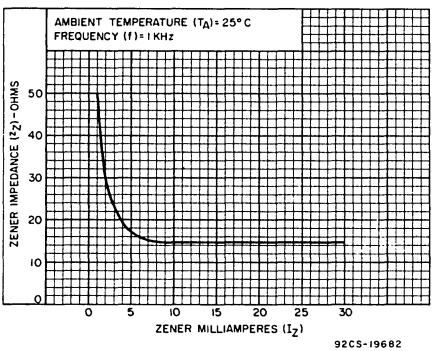
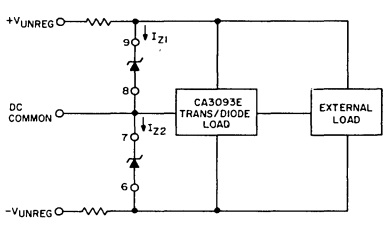


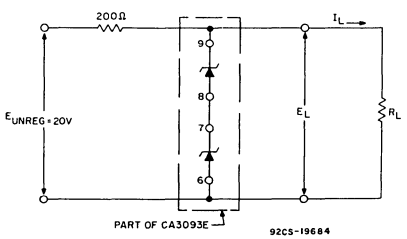
Fig. 10 – Typical Zener impedance vs current

TYPICAL APPLICATIONS

a) ±7V Regulator supplying CA3093E Transistors plus an external load.



b) 14V Regulator for Q1, Q2, Q3



Sample Computation for Determining Permissible Zener Dissipation at +25°C.

CA3093E Ratings at T_A = +25°C
Total Diss. Max = 750 mW (Derate @ 6.67 mW/°C above 25°C)
Each Zener Diss. Max = 250 mW
Max. Zener Current = 35 mA

Assume CA3093E Transistor/Diode Load Dissipation = 350 mW then max. total Zener Diss. ($P_{Z1} + P_{Z2}$) = 750 - 350 = 400 mW

$$(I_{Z1} + I_{Z2})\ max = \frac{400\ mW}{7V} = 57\ mA$$

(Note: Max. current rating on each Zener is 35 mA)

Typical Load Regulation for I_L = 0 to 25 mA

$\Delta E_L/E_L \times 100 \approx -6\%$ (no load to full load)

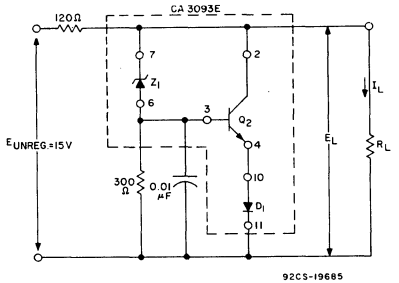
Typical Line Regulation

$$\frac{(\Delta E_L/E_L) \times 100}{\Delta E\ unreg.} \approx \pm 0.9\%/V$$

Typical Temperature Characteristic

$$\frac{\Delta E_L/E_L}{\Delta T} \times 100 = +0.05\%/°C$$

c) 8.6V Temp.-Compensated Shunt Regulator



Typical Temperature Characteristic @ R_L = 330Ω

$$\frac{\Delta E_L/E_L}{\Delta T} \times 100 = \pm\ 0.007\%/°C$$

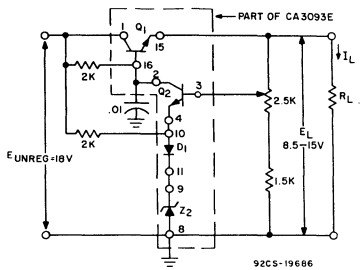
Typical Load Regulation I_L = 0 to 40 mA

$(\Delta E_L/E_L) \times 100 = -3\%$ (no load to full load)

Typical Line Regulation at R_L = 330Ω

$$\frac{\Delta E_L/E_L}{\Delta E \text{ unreg.}} \times 100 = \pm\ 0.55\%/V$$

d) Temp.-Compensated Series Voltage Regulator



Typical Temperature Characteristic @ E_L = 12V

$$\frac{\Delta E_L/E_L}{\Delta T} \times 100 = \pm\ 0.009\%/°C$$

Typical Load Regulation @ E_L = 12V

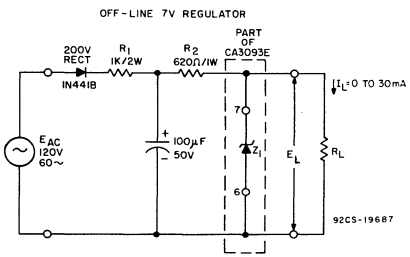
I_L = 0 to 40 mA

$$\frac{\Delta E_L}{E_L} \times 100 = \pm\ 0.4\% \text{ (no load to full load)}$$

Typical Line Regulation @ E_L = 12V

$$\frac{(\Delta E_L/E_L) \times 100}{\Delta E \text{ unreg.}} = \pm\ 0.45\%/V$$

e) Off-Line 7V Regulator



Typical E_L Ripple Voltage = 70 mV_{p-p}

Typical Load Regulation = $\frac{\Delta E_L}{E_L} \times 100 = -8.5\%$ (no load to full load)

I_L = 0 to 30 mA

Typical Line Regulation = $\frac{(\Delta E_L/E_L) \times 100}{\Delta E_{AC}} = \pm\ .075\%/V$

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3094, CA3094A, CA3094B Types



S-Suffix Types
8-LEAD TO-5
with Dual-In-Line
Formed Leads
"DIL-CAN"
H-1787



T-Suffix Types
8-LEAD TO-5
H-1528



CA3094E
8-LEAD DUAL-
IN-LINE PLASTIC
PACKAGE
"MINI-DIP" H-1817

Programmable Power Switch/ Amplifier

CA3094: For Operation Up to 24 Volts
CA3094A: For Operation Up to 36 Volts
CA3094B: For Operation Up to 44 Volts

For Control & General-Purpose Applications

Features:

- Designed for *single* or dual power supply
- Programmable: strobing, gating, squelching, AGC capabilities
- Can deliver 3 watts (avg.) or 10 W (peak) to external load (in switching mode)
- High-power, single-ended class A amplifier will deliver power output of 0.6 watt (1.6 W device dissipation)
- Total harmonic distortion (THD) @ 0.6 W in class A operation – 1.4% typ.
- High current-handling capability – 100 mA (avg.), 300 mA (peak)
- Sensitivity controlled by varying bias current
- Output: "sink" or "drive" capability

APPLICATIONS:

- Error-signal detector: temperature control with thermistor sensor; speed control for shunt wound dc motor
- Over-current, over-voltage, over-temperature protectors
- Dual-tracking power supply with RCA-CA3085
- Wide-frequency-range oscillator ■ Analog timer
- Level detector ■ Alarm systems ■ Voltage follower
- Ramp-voltage generator ■ High-power comparator
- Ground-fault interrupter (GFI) circuits

The CA3094 is a differential-input power-control switch/amplifier with auxiliary circuit features for ease of programmability. For example, an error or unbalance signal can be amplified by the CA3094 to provide an on-off signal or proportional-control output signal up to 100 mA. This signal is sufficient to directly drive high-current thyristors, relays, dc loads, or power transistors. The CA3094 has the generic characteristics of the RCA-CA3080 operational amplifier directly coupled to an integral Darlington power transistor capable of sinking or driving currents up to 100 mA.

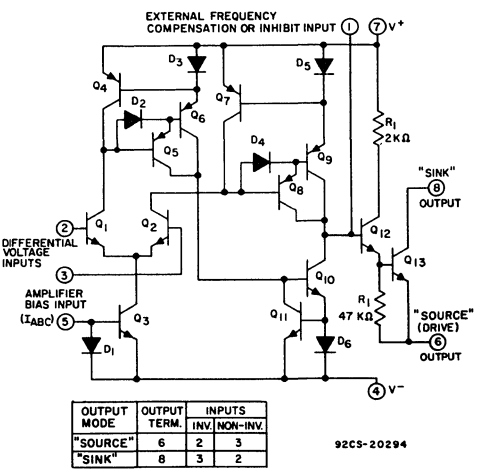
The gain of the differential input stage is proportional to the amplifier bias current (I_{ABC}), permitting programmable variation of the integrated circuit sensitivity with either digital and/or analog programming signals. For example, at an I_{ABC} of 100 μA, a one-millivolt change at the input will change the output from 0 to 100 mA (typical).

The CA3094 is intended for operation up to 24 volts and is especially useful for timing circuits, in automotive equipment, and in other applications where operation up to 24 volts is a primary design requirement (see Figs.27,28 and 29 in Applications Section). The CA3094A and CA3094B are like the CA3094 but are intended for operation up to 36 and 44 volts, respectively (single or dual supply).

Application Note ICAN-6668 describes the rudiments of Operational Transconductance Amplifiers (OTA's).

These types are available in 8-lead TO-5 style packages with standard leads ("T" suffix) and with dual-in-line formed leads "DIL-CAN" ("S" suffix). Type CA3094 is also available in an 8-lead dual-in-line plastic package "MINI-DIP" ("E" suffix), and in chip form ("H" suffix).

APPLICATION NOTE ICAN-6048 GIVES DETAILED APPLICATION INFORMATION FOR THE CA3094, CA3094A, AND CA3094B.



OUTPUT MODE	OUTPUT TERM.	INPUTS	
		INV.	NON-INV.
"SOURCE"	6	2	3
"SINK"	8	3	2

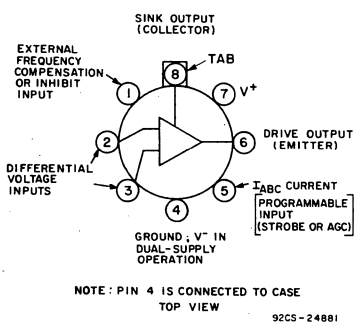
92CS-20294

Fig. 1—Schematic diagram of CA3094.

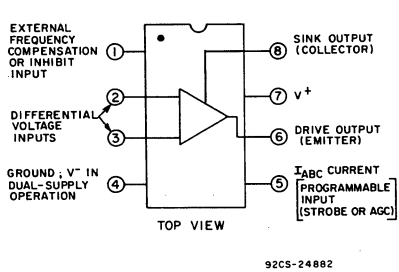
MAXIMUM RATINGS, *Absolute-Maximum Values:*

	CA3094	CA3094A	CA3094B	
DC SUPPLY VOLTAGE:				
Dual Supply	± 12 V	± 18 V	± 22 V	V
Single Supply	24 V	36 V	44 V	V
DC DIFFERENTIAL INPUT VOLTAGE (Terminals 2 and 3)	± 5*			V
DC COMMON-MODE INPUT VOLTAGE	Term. 4 ⩽ Term. 2 & 3 ⩽ Term. 7			
PEAK INPUT SIGNAL CURRENT (Terminals 2 and 3)	± 1			mA
PEAK AMPLIFIER BIAS CURRENT (Terminal 5)	2			mA
OUTPUT CURRENT:				
Peak	300			mA
Average	100			mA
DEVICE DISSIPATION:				
Up to T_A = 55°C:				
Without heat sink	630			mW
With heat sink	1.6			W
Above T_A = 55°C:				
Without heat sink derate linearly	6.67			mW/°C
With heat sink derate linearly	16.7			mW/°C
THERMAL RESISTANCE (Junction to Air)	140			°C/W
AMBIENT TEMPERATURE RANGE:				
Operating	−55 to +125			°C
Storage	−65 to +150			°C
LEAD TEMPERATURE (DURING SOLDERING): At distance 1/16 ± 1/32 in. (1.59 ± 0.79 mm) from case for 10 s max.	+ 300			°C

*Exceeding this voltage rating will not damage the device unless the peak input signal current (1 mA) is also exceeded.



TO-5 Style Package



Plastic Package

FUNCTIONAL DIAGRAMS

ELECTRICAL CHARACTERISTICS at $T_A = 25^\circ C$ For Equipment Design

CHARACTERISTIC	SYMBOL	TEST CONDITIONS			LIMITS			UNITS
		Test Circuit Fig. No.	Single Supply $V^+ = 30$ V Dual Supply $V^+ = 15$ V, $V^- = 15$ V $I_{ABC} = 100\ \mu A$ Unless Otherwise Specified	Char. Curves Fig. No.	Min.	Typ.	Max.	
INPUT PARAMETERS								
Input Offset Voltage	V_{IO}	17	$T_A = 25^\circ C$	2	–	0.4	5	mV
			$T_A = 0$ to $70^\circ C$		–	–	7	mV
Input-Offset-Voltage Change	$\lvert\Delta V_{IO}\rvert$		Change in V_{IO} Between $I_{ABC} = 100\ \mu A$ and $I_{ABC} = 5\ \mu A$		–	1	8	mV
Input Offset Current	I_{IO}	18	$T_A = 25^\circ C$	3	–	0.02	0.2	μA
			$T_A = 0$ to $70^\circ C$		–	–	0.3	μA
Input Bias Current	I_I	19	$T_A = 25^\circ C$	4	–	0.2	0.50	μA
			$T_A = 0$ to $70^\circ C$		–	–	0.70	μA
Device Dissipation	P_D	18	$I_{out} = 0$	5, 6	8	10	12	mW
Common-Mode Rejection Ratio	C_{MRR}	20			70	110	–	dB
Common-Mode Input–Voltage Range	V_{ICR}	20	$V^+ = 30$ V $\frac{\text{High}}{\text{Low}}$	7	27	28.8	–	V
					1.0	0.5	–	V
			$V^+ = 15$ V	7	+12	+13.8	–	V
			$V^- = 15$ V		–14	–14.5	–	V
Unity Gain-Bandwidth			$I_C = 7.5$ mA $V_{CE} = 15$ V $I_{ABC} = 500\ \mu A$		–	30	–	MHz
Open-Loop Bandwidth At –3 dB Point	BW_{OL}		$I_C = 7.5$ mA $V_{CE} = 15$ V $I_{ABC} = 500\ \mu A$	12	–	4	–	kHz
Total Harmonic Distortion (Class A Operation)	THD		$P_D = 220$ mW		–	0.4	–	%
			$P_D = 600$ mW		–	1.4	–	
Amplifier Bias Voltage (Terminal (No.5 to Terminal No.4)	V_{ABC}				–	0.68	–	V
Input Offset Voltage Temperature Coefficient	$\Delta V_{IO}/\Delta T$				–	4	–	μV/°C
Power-Supply Rejection	$\Delta V_{IO}/\Delta V$	17			–	15	150	μV/V
1/F Noise Voltage	E_N	21	$f = 10$ Hz $I_{ABC} = 50\ \mu A$	8	–	18	–	$\eta V/\sqrt{Hz}$
1/F Noise Current	I_N	21	$f = 10$ Hz $I_{ABC} = 50\ \mu A$	9	–	1.8	–	$pA/\sqrt{Hz}$
Differential Input Resistance	R_I		$I_{ABC} = 20\ \mu A$		0.50	1	–	MΩ
Differential Input Capacitance	C_I		$f = 1$ MHz $V^+ = 30$ V		–	2.6	–	pF

ELECTRICAL CHARACTERISTICS at T_A = 25°C For Equipment Design

CHARACTERISTIC	SYMBOL	TEST CONDITIONS			LIMITS			UNITS
		Test Circuit Fig. No.	Single Supply V^+ = 30 V Dual Supply V^+ = 15 V, V^- = 15 V I_{ABC} = 100 μA Unless Otherwise Specified	Char. Curves Fig. No.	Min.	Typ.	Max.	
OUTPUT PARAMETERS (Differential Input Voltage = 1V)								
Peak Output Voltage: (Terminal No. 6)			V^+ = 30 V					
With Q13 "ON"	V^+OM		R_L = 2 kΩ to ground		26	27	–	V
With Q13 "OFF"	V^-OM				–	0.01	0.05	V
Peak Output Voltage: (Terminal No. 6)			V^+ = +15 V, V^- = -15 V				0.5	
Positive	V^+OM		R_L = 2 kΩ to −15 V		+11	+12	–	V
Negative	V^-OM				–	−14.99	−14.95	V
Peak Output Voltage: (Terminal No. 8)			V^+ = 30 V					
With Q13 "ON"	V^+OM		R_L = 2 kΩ to 30 V		29.95	29.99	–	V
With Q13 "OFF"	V^-OM				–	0.040	–	V
Peak Output Voltage: (Terminal No. 8)			V^+ = 15 V, V^- = – 15 V					
Positive	V^+OM		R_L = 2 kΩ to + 15 V		+14.95	+14.99	–	V
Negative	V^-OM				–	14.96	–	V
Collector-to-Emitter Saturation Voltage (Terminal No. 8)	$V_{CE(sat)}$		V^+ = 30 V I_C = 50 mA Terminal No.6 grounded	10	–	0.17	0.80	V
Output Leakage Current (Terminal No. 6 to Terminal No. 4)			V^+ = 30 V		–	2	10	μA
Composite Small-Signal Current Transfer Ratio (Beta) (Q_{12} and Q_{13})	h_{fe}		V^+ = 30 V V_{CE} = 5 V I_C = 50 mA	11	16,000	100,000	–	
Output Capacitance:			f = 1 MHz					
Terminal No. 6	C_O		All Remaining		–	5.5	–	pF
Terminal No. 8			Terminals Tied to Terminal No. 4		–	17	–	pF
TRANSFER PARAMETERS								
Voltage Gain	A	22	V^+ = 30 V I_{ABC} = 100 μA ΔV_{out} = 20 V R_L = 2 kΩ	12	20,000 86	100,000 100	– –	V/V dB
Forward Transconductance To Terminal No. 1	g_m			13	1650	2200	2750	μmhos
Slew Rate:								
Open Loop:								
Positive Slope		23	I_{ABC} = 500 μA R_L = 2 kΩ	14	–	500	–	V/μs
Negative Slope					–	50	–	V/μs
Unity Gain (Non-Inverting, Compensated)		24	I_{ABC} = 500 μA R_L = 2 kΩ	15	–	0.7	–	V/μs

Typical Characteristics Curves

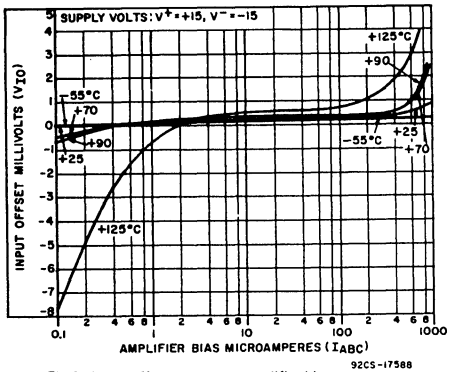


Fig.2—Input offset voltage vs. amplifier bias current (I_{ABC}, terminal No. 5).

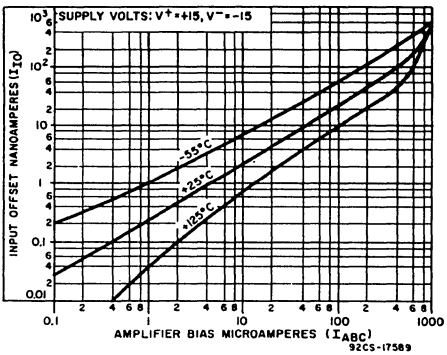


Fig.3—Input offset current vs. amplifier bias current (I_{ABC}, terminal No. 5).

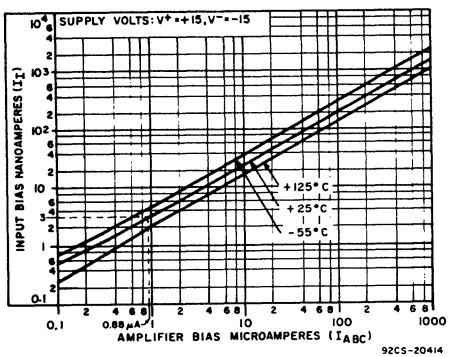


Fig.4—Input bias current vs. amplifier bias current (I_{ABC}, terminal No. 5).

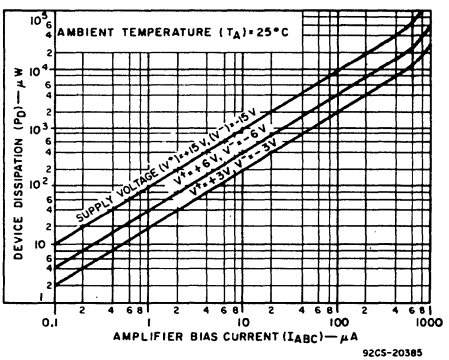


Fig.5—Device dissipation vs. amplifier bias current (I_{ABC}, terminal No. 5).

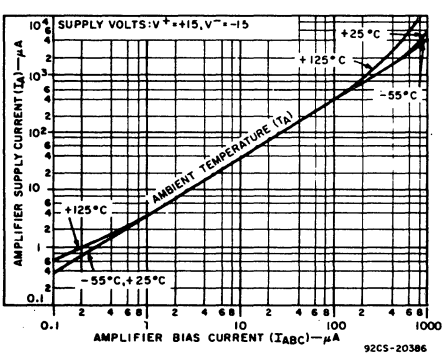


Fig.6—Amplifier supply current vs. amplifier bias current (I_{ABC}, terminal No. 5).

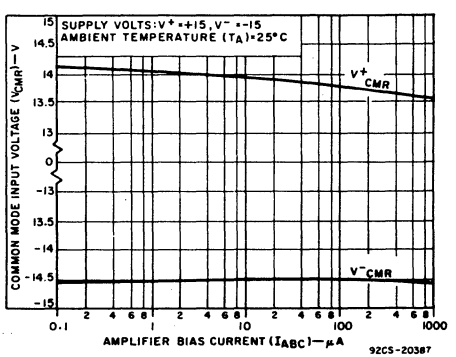


Fig.7—Common mode input voltage vs. amplifier bias current (I_{ABC}, terminal No. 5).

Typical Characteristics Curves

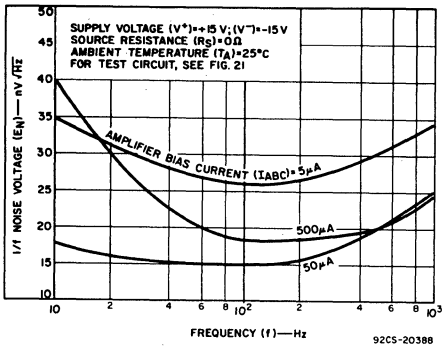


Fig.8—1/F Noise voltage vs. frequency.

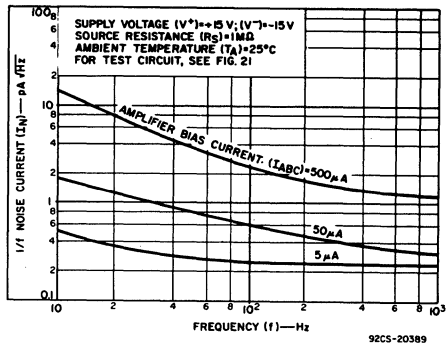


Fig.9—1/F Noise current vs. frequency.

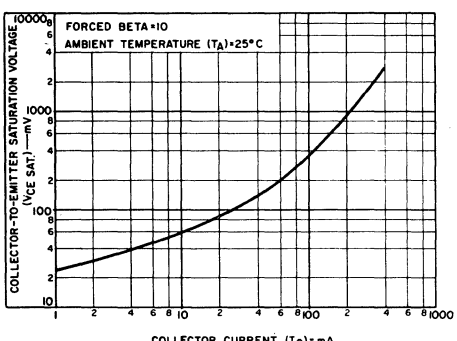


Fig.10—Collector-emitter saturation voltage vs. collector current of output transistor Q_{13}.

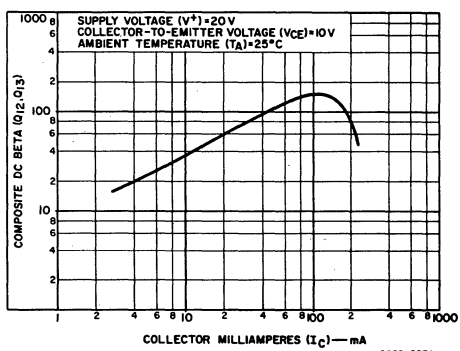


Fig.11—Composite dc beta vs. collector current of Darlington-connected output transistors Q_{12}, Q_{13}.

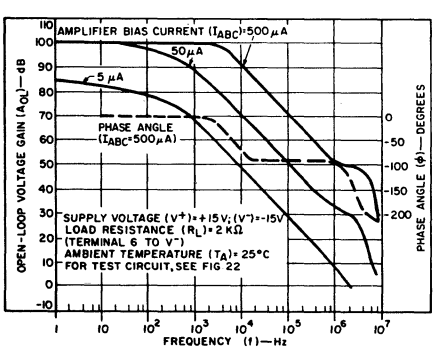


Fig.12—Open-loop voltage gain vs. frequency.

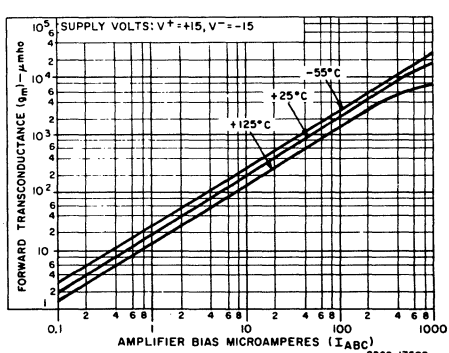


Fig.13—Forward transconductance vs. amplifier bias current.

Typical Characteristics Curves

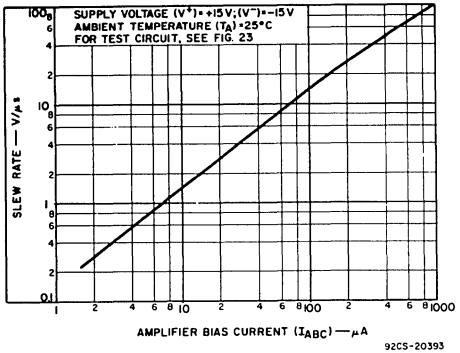


Fig. 14—Slew rate vs. amplifier bias current.

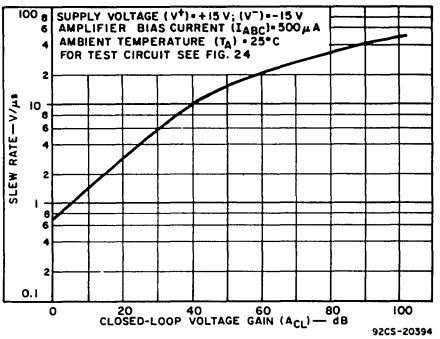


Fig. 15—Slew rate vs. closed-loop voltage gain.

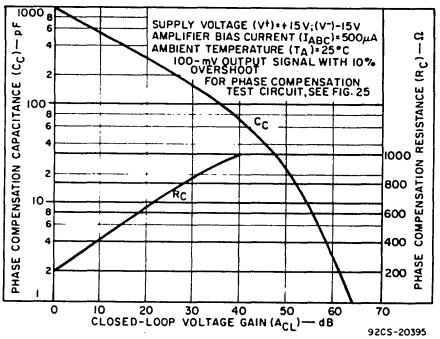


Fig. 16—Phase compensation capacitance and resistance vs. closed-loop voltage gain.

OPERATING CONSIDERATIONS

The "Sink" Output (terminal No. 8) and the "Drive" Output (terminal No. 6) of the CA3094 are not inherently current (or power) limited. Therefore, if a load is connected between terminal No. 6 and terminal No. 4 (V^- or ground), it is important to connect a current-limiting resistor between terminal 8 and terminal No. 7 (V^+) to protect transistor Q_{13} under shorted load conditions. Similarly, if a load is connected between terminal No. 8 and terminal No. 7, the current-limiting resistor should be connected between terminal 6 and terminal No. 4 or ground. In circuit applications where the emitter of the output transistor is not connected to the most negative potential in the system, it is recommended that a 100-ohm current-limiting resistor be inserted between terminal No. 7 and the V^+ supply.

TEST CIRCUITS

1/F Noise Measurement Circuit

When using the CA3094, A, or B audio amplifier circuits, it is frequently necessary to consider the noise performance of the device. Noise measurements are made in the circuit shown in Fig. 21. This circuit is a 30-dB, non-inverting amplifier with emitter-follower output and phase compensation from terminal No. 2 to ground. Source resistors (Rs) are set to 0.Ω or 1 MΩ for ^{E} noise and ^{I} noise measurements, respectively. These measurements are made at frequencies of 10 Hz, 100 Hz, and 1 kHz with a 1-Hz measurement bandwidth. Typical values for 1/f noise at 10 Hz and 50 μA I_{ABC} are $E_n = 18\ nV/\sqrt{HZ}$ and $I_N = 1.8\ pA/\sqrt{HZ}$.

Test Circuits

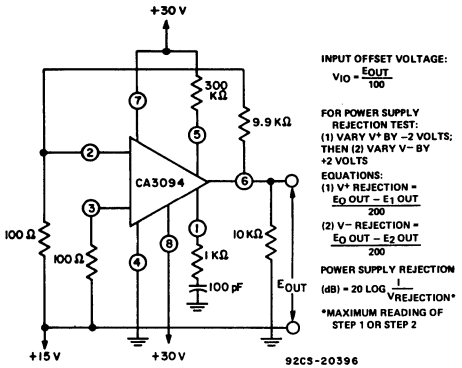


Fig. 17—Input offset voltage and power-supply rejection test circuit.

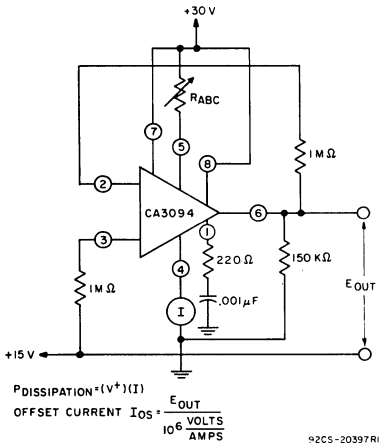


Fig. 18—Input offset current test circuit.

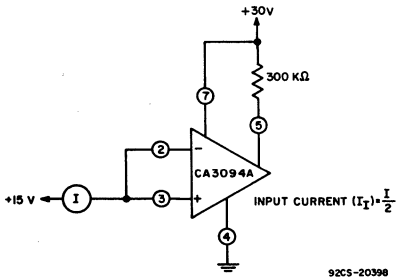


Fig. 19—Input bias current test circuit.

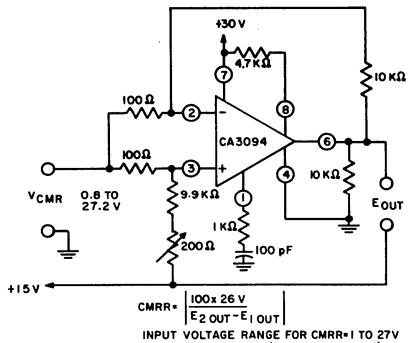
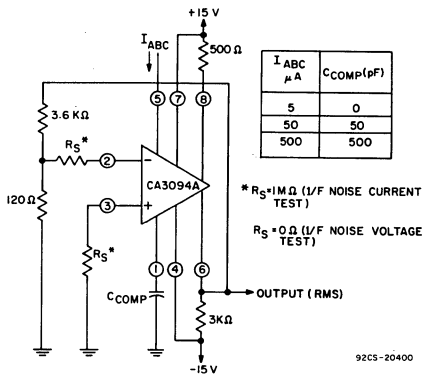
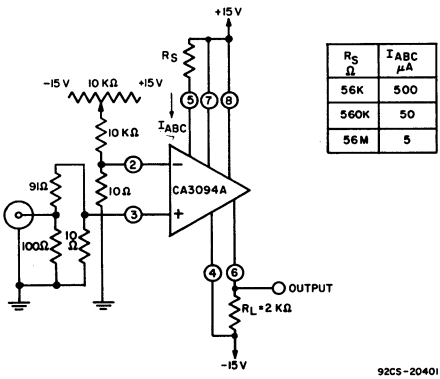


Fig. 20—Common-mode range and rejection ratio test circuit.



IABC μA	CCOMP (pF)
5	0
50	50
500	500

Fig. 21—1/F noise test circuit.



RS Ω	IABC μA
56K	500
560K	50
56M	5

Fig. 22—Open-loop gain vs. frequency test circuit.

Test Circuits (cont'd)

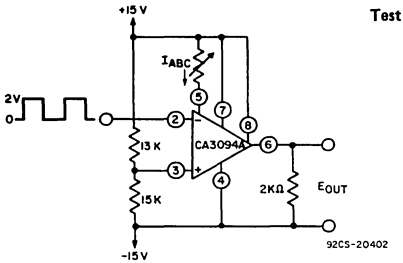


Fig.23—Open-loop slew rate vs. I_{ABC} test circuit.

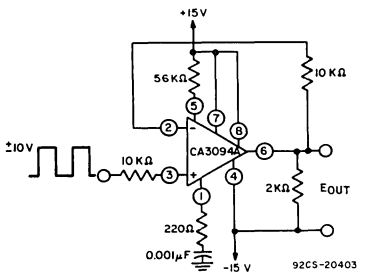
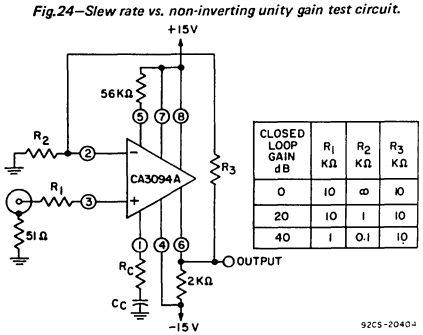
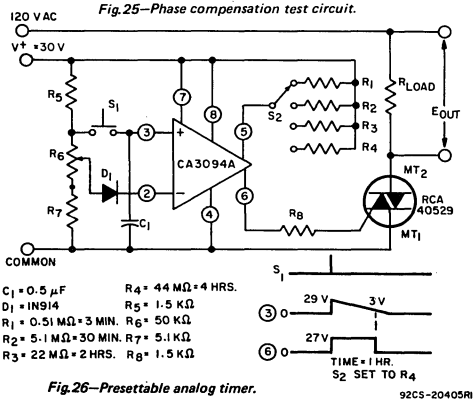


Fig.24—Slew rate vs. non-inverting unity gain test circuit.



CLOSED LOOP GAIN dB	R_1 KΩ	R_2 KΩ	R_3 KΩ
0	10	∞	10
20	10	1	10
40	1	0.1	10

Fig.25—Phase compensation test circuit.



C_1 = 0.5 μF
D_1 = IN914
R_1 = 0.51 MΩ = 3 MIN.
R_2 = 5.1 MΩ = 30 MIN.
R_3 = 22 MΩ = 2 HRS.
R_4 = 44 MΩ = 4 HRS.
R_5 = 1.5 KΩ
R_6 = 50 KΩ
R_7 = 5.1 KΩ
R_8 = 1.5 KΩ

Fig.26—Presettable analog timer.

TYPICAL APPLICATIONS

For Additional Application Information, refer to Application Note ICAN-6048 "Some Applications of a Programmable Power/Switch Amplifier IC".

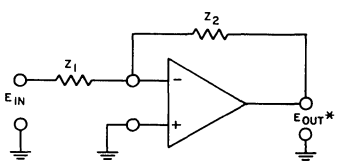
Design Considerations

The selection of the optimum amplifier bias current (I_{ABC}) depends on —

1. The Desired Sensitivity — the higher the I_{ABC}, the higher the sensitivity — i.e., a greater-drive current capability at the output for a specific voltage change at the input
2. Required Input Resistance — the lower the I_{ABC}, the higher the input resistance

If the desired sensitivity and required input resistance are not known and are to be experimentally determined, or the anticipated equipment design is sufficently flexible to tolerate a wide range of these parameters, it is recommended that the equipment designer begin his calculations with an I_{ABC} of 100 μA, since the CA3094 is characterized at this value of amplifier bias current.

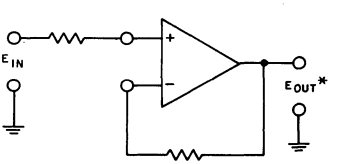
The CA3094 is extremely versatile and can be used in a wide variety of applications:



WHERE $\frac{E_{OUT}}{E_{IN}} = f\left(\frac{Z_2}{Z_1}\right)$ DEPENDS ON THE CHARACTERISTICS OF Z_1 AND Z_2

(a).

Fig. (a) As an inverting op-amp.



WHERE $E_{OUT} = E_{IN}$

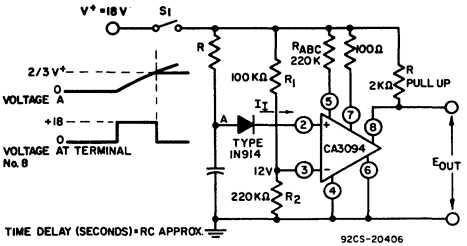
*IN SINGLE-ENDED OUTPUT OPERATION, THE CA3094 MAY REQUIRE A PULL UP OR PULL DOWN RESISTOR

(b)

92CS-20383

Fig. (b) In a non-inverting mode as a follower.

Typical Applications (cont'd)



Problem: To calculate the maximum value of R required to switch a 100-mA output current comparator

Given: $I_{ABC} = 5\ \mu A$, $R_{ABC} = 3.6\ M\Omega \approx \frac{18\ V}{5\ \mu A}$

$I_I = 500$ nA @ $I_{ABC} = 100\ \mu A$ (from Fig. 4)

$I_I = 5\ \mu A$ can be determined by drawing a line on Fig. 4 through $I_{ABC} = 100\ \mu A$ and $I_B = 500$ nA parallel to the typical $T_A = 25^oC$ curve.

Then: $I_I = 33$ nA @ $I_{ABC} = 5\ \mu A$

$$R_{max} = \frac{18-12 \text{ volts}}{33\ nA} = 180\ M\Omega \text{ @ } T_A = 25^oC$$

$R_{max} = 180\ M\Omega \times 2/3^* = 120\ M\Omega$ @ $T_A = -55^oC$

*Ratio of I_I at $T_A = +25^oC$ to I_I at $T_A = -55^oC$ for any given value of I_{ABC}.

Fig.27—RC timer.

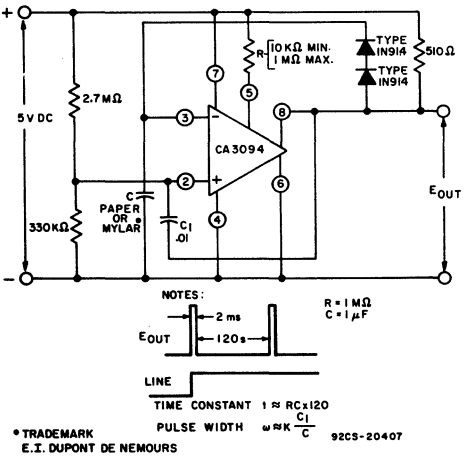
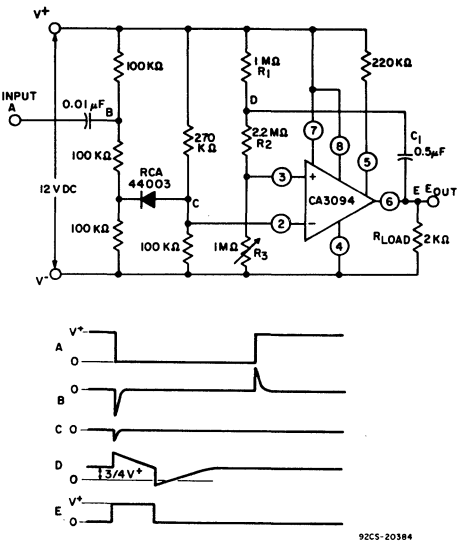


Fig.28—Free-running pulse generator.



On a negative-going transient at input (A), a negative pulse at C will turn "on" the CA3094, and the output (E) will go from a low to a high level.

At the end of the time constant determined by C_1, R_1, R_2, R_3, the CA3094 will return to the "off" state and the output will be pulled low by R_{LOAD}. This condition will be independent of the interval when input A returns to a high level.

Fig.29—RC timer triggered by external negative pulse.

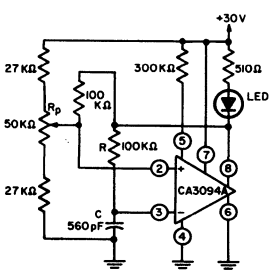


Fig.30—Single-supply astable multivibrator.

Typical Applications (cont'd)

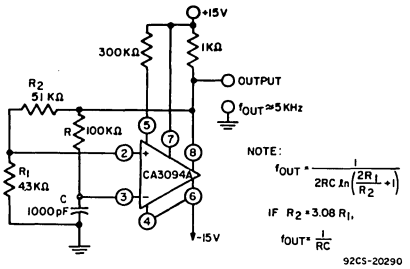
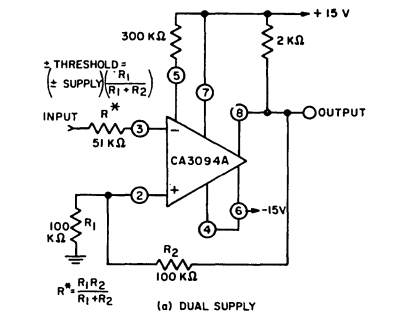


Fig.31—Op-amp astable multivibrator (dual-supply).



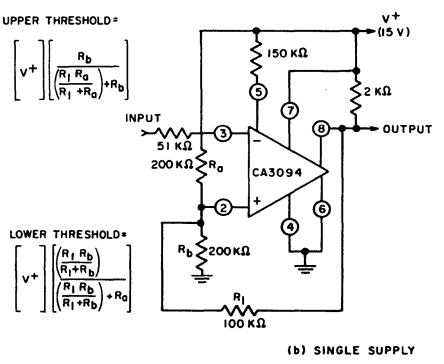


Fig.32—Comparator/threshold detector.

Typical Applications (cont'd)

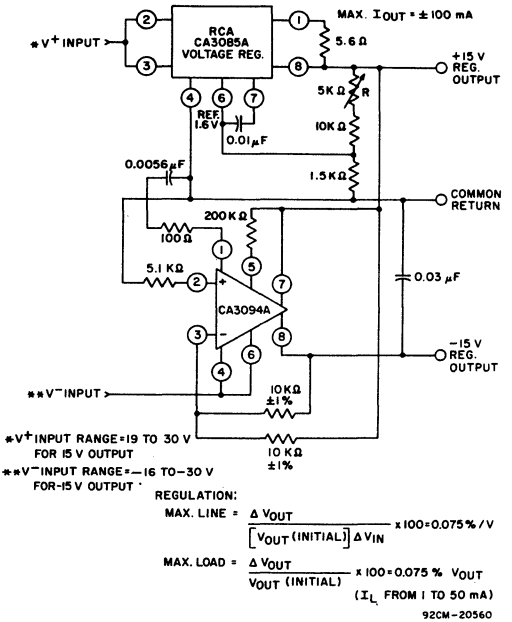


Fig.33—Dual tracking voltage regulator.

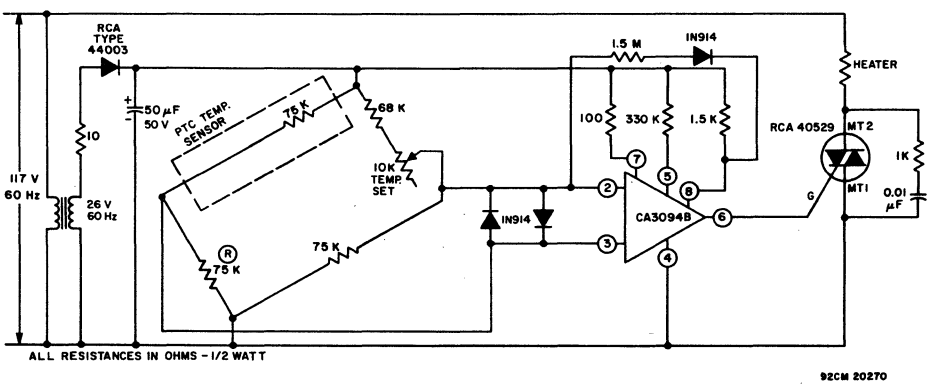
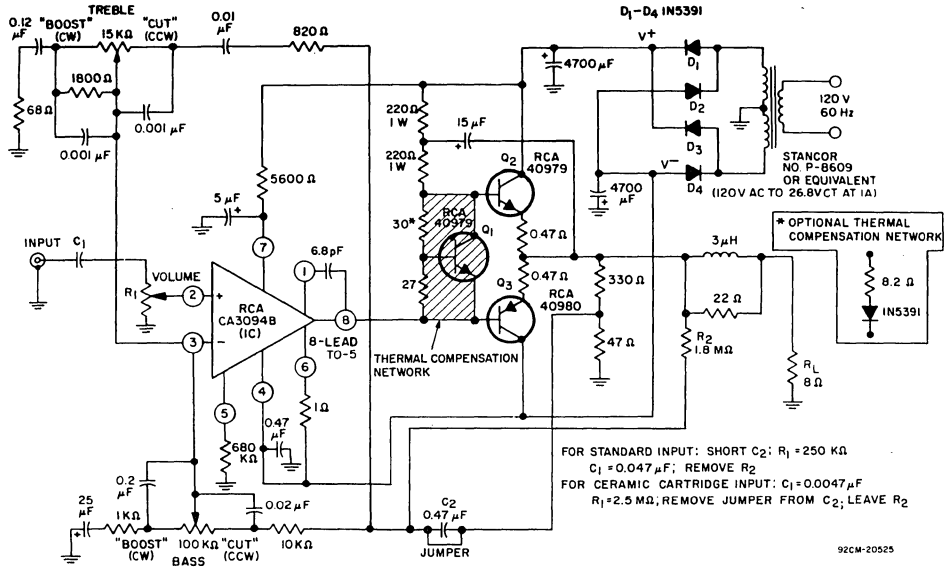


Fig.34—Temperature controller.



TYPICAL PERFORMANCE DATA — For 12-W Audio Amplifier Circuit

Power Output (8Ω load, Tone Control set at "Flat")		
Music (at 5% THD, regulated supply)	15	W
Continuous (at 0.2% IMD, 60 Hz & 2 kHz mixed in a 4:1 ratio, unregulated supply) See Fig. 8 In ICAN-6048	12	W
Total Harmonic Distoration		
At 1 W, unregulated supply	0.05	%
At 12 W, unregulated supply	0.57	%
Voltage Gain	40	dB
Hum and Noise (Below continuous Power Output)	83	dB
Input Resistance	250	kΩ
Tone Control Range	See Fig. 9 In ICAN-6048	

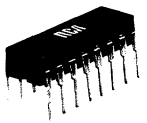
Fig.35—12-watt amplifier circuit featuring a true-complementary output stage with CA3094 in driver stage.

RCA
Solid State
Division

Linear Integrated Circuits

Monolithic Silicon

CA3095E



16-Lead Dual-In-Line
Plastic Package

Super-Beta Transistor Array

Differential Cascode Amplifier Plus 3 Independent Transistors

Applications

Differential Cascode Amplifier:

- Super-beta pre-amplifier for op-amp
- High-impedance dc meter amplifier
- Low-noise video amplifier
- Piezoelectric transducer amplifier
- Long-interval timer
- Long-duration one-shot multivibrator
- Comparator with high-input impedance
- Long-time-constant integrator
- Photocell amplifier
- Low-noise amplifier—for operation from high-source impedances

Independent Transistors:

- General use in signal processing systems in dc through vhf range

RCA-CA3095E* is a monolithic array of transistors connected as a super-beta differential cascode amplifier with three independent n-p-n transistors. (Refer to Fig. 1 for following description.)

The differential cascode amplifier incorporates two cascode amplifiers consisting of transistors Q1, Q3 and Q2, Q4, respectively, plus a voltage-limiting circuit, consisting of diodes D1, D2 and p-n-p transistor Q5. Two of these transistors, Q1 and Q2, are super-beta types that have an $h_{FE} > 1000$ and are capable of operating over a wide current range of 1 μA to 2 mA. Each of these types comprises the input section of its respective cascode amplifier. The output section of each cascode amplifier employs a conventional n-p-n transistor, Q3, Q4, respectively. The output signal is obtained at the collectors of these transistors. See Operating Considerations on page 8 for bias considerations of the differential cascode amplifier.

The exceptionally high-beta characteristics of Q1 and Q2, plus the large signal-voltage swing capability of Q3 and Q4, make the composite differential cascode amplifier an excellent choice for a broad range of small-signal, high-input-impedance amplifier applications including low-noise video amplifiers. This amplifier is also recommended for use in long-interval timers, oscillators, and long-duration one-shot applications.

The independent transistors, Q6, Q7 and Q8, are high-voltage silicon n-p-n conventional types for general use in signal processing systems in the frequency range from dc through vhf. Separate terminals for each of these transistors permit maximum flexibility in circuit design.

The CA3095E is supplied in a 16-lead dual-in-line plastic package and operates over the ambient temperature range of −55°C to +125°C

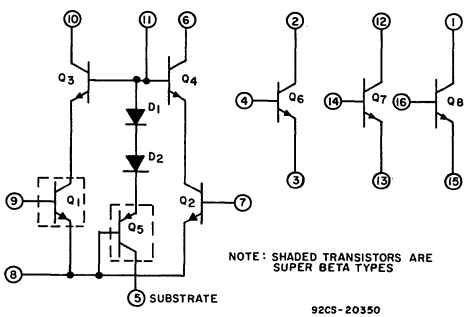
* Formerly developmental type TA6269X.

Features

- Two super-beta n-p-n transistors – $h_{FE} > 1000$
- Voltage-limiting circuitry (D1, D2, Q5)
- Operation possible at I_{IB} down to < 1 nA
- Matched pair (Q1 and Q2) –
  V_{IO} = 5 mV max. at I_C = 100 μA dc
  I_{IO} = 20 nA max. at I_C = 100 μA dc
- Wide current range – < 1 μA to 2 mA

Independent Transistors:

- h_{FE} = 300 typ. for each transistor
- Wide current range – < 1 μA to 10 mA
- Matched general-purpose transistors
- High voltage – V_{CBO} = 45 V max.



MAXIMUM RATINGS, *Absolute-Maximum Values at* $T_A = 25\ ^\circ C$

Power Dissipation:		
Any One Transistor	300	mW
Total Package–		
Up to 25 °C	750	mW
Above 25 °C ..... derate linearly	6.67	mW/°C
Ambient Temperature Range:		
Operating	−55 to +125	°C
Storage	−55 to +150	°C
Lead Temperature (During Soldering):		
At distance not less than 1/32" (0.79 mm) from case for 10 seconds max.	+265	°C
Voltage and Current Ratings Apply for Each Specified Transistor:		
Super-Beta Transistors (Q1, Q2)–		
Collector-to-Base Voltage (V_{CBO})	6	V
Emitter-to-Base Voltage (V_{EBO})	6	V
Collector-to-Substrate Voltage (V_{CIO})*	45	V
Collector Current (I_C)	50	mA
Base Current (I_B)	20	mA
Conventional N-P-N Transistors (Q3, Q4, Q6, Q7, Q8)–		
Collector-to-Base Voltage (V_{CBO})	45	V
Collector-to-Emitter Voltage (V_{CEO})	35	V
Emitter-to-Base Voltage (V_{EBO})	6	V
Collector-to-Substrate Voltage (V_{CIO})*	45	V
Collector Current (I_C)	50	mA
Base Current (I_B)	20	mA
Conventional P-N-P Transistor (Q5)–		
Collector-to-Base Voltage (V_{CBO})	−45	V
Collector-to-Emitter Voltage (V_{CEO})	−35	V
Limiting Circuit Current ($I_{Pin\ 11}$)	20	mA

* The collector of each transistor is isolated from the substrate by an integral diode. The substrate must be connected to a voltage which is more negative than any collector voltage in order to maintain isolation between transistors and provide normal transistor action. To avoid undesired coupling between transistors, the substrate terminal should be maintained at either dc or signal (ac) ground. A suitable bypass capacitor can be used to establish a signal ground.

STATIC CHARACTERISTICS

Characteristics	Symbol	Test Conditions $T_A = 25\ ^\circ C$		Limits Min.	Limits Typ.	Limits Max.	Units
Characteristics Apply for Each Super-Beta Cascode Amplifier Transistor Pair (Q1, Q3) and (Q2, Q4), Unless Indicated Otherwise							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10\ \mu A$, $I_E = 0$ See Note 1		6	–	–	V
Emitter-to-Base Breakdown Voltage (Applies only to Q1 & Q2)	$V_{(BR)EBO}$	$I_E = 100\ \mu A$, $I_C = 0$	Term. 9 to 8 or Term. 7 to 8	6	8	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_{CI} = 100\ \mu A$, $I_B = I_E = 0$		45	–	–	V
Collector Cutoff Current	I_{CER}	V_{6-8} or $V_{10-8} = 10$ V, $I_{11} = 100\ \mu A$ $R_{BE} = 100\ M\Omega$		–	–	100	nA
DC Forward-Current Transfer Ratio	h_{FE}	$V_{10-8} = 5$ V or $V_{6-8} = 5$ V	$I_C = 1$ mA	–	1500	–	
			$I_C = 100\ \mu A$	1000	2000	5000	
			$I_C = 10\ \mu A$	–	1500	–	
Base-to-Emitter Voltage (Applies only to Q1 & Q2)	V_{BE}	$I_C = 100\ \mu A$, V_{6-8} or $V_{10-8} = 5$ V		0.50	0.59	0.68	V
Saturation Voltage	V_{sat}	I_6 or $I_{10} = 1$ mA, $I_{11} = 100\ \mu A$, I_7 or $I_9 = 100\ \mu A$		–	0.22	0.7	V
For Cascode Amplifiers as a Differential Matched Pair							
Magnitude of Input-Offset Voltage	$\lvert I_{IO} \rvert$	$I_C = 100\ \mu A$		–	1	5	mV
Magnitude of Input-Offset Current	$\lvert I_{IO} \rvert$	$V_{6-8} = V_{10-8} = 5$ V		–	4	20	nA
Magnitude of Input-Offset Voltage Drift (Temp. Coeff.)	$\frac{\lvert \Delta V_{IO} \rvert}{\Delta T}$			–	3.3	–	μV/°C
Magnitude of Input-Offset Current Drift (Temp. Coeff.)	$\frac{\lvert \Delta I_{IO} \rvert}{\Delta T}$			–	0.05	–	nA/°C

Note 1: Terminal No. 9 to terminals 10 and 11 connected or terminal No. 7 to terminals 6 and 11 connected.

STATIC CHARACTERISTICS (Cont'd)

Characteristics	Symbol	Test Conditions $T_A = 25\ ^\circ C$		Limits Min.	Typ.	Max.	Units
For Each Conventional n-p-n Transistor (Q3, Q4, Q6, Q7, Q8)							
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	$I_C = 10\ \mu A$, $I_E = 0$		45	95	--	V
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	$I_C = 1$ mA, $I_B = 0$		35	50	—	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	$I_E = 100\ \mu A$, $I_C = 0$		6	8	—	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	$I_{CI} = 100\ \mu A$, $I_B = I_E = 0$		45	95	—	V
Collector Cutoff Current	I_{CEO}	$V_{CE} = 10$ V, $I_B = 0$		—	—	100	nA
Collector Cutoff Current	I_{CBO}	$V_{CB} = 10$ V, $I_E = 0$		—	—	10	nA
DC Forward-Current Transfer Ratio	h_{FE}	$V_{CE} = 5$ V	$I_C = 10$ mA	—	210	—	
			$I_C = 1$ mA	150	300	500	
			$I_C = 10\ \mu A$	—	180	—	
Base-to-Emitter Voltage	V_{BE}	$I_C = 1$ mA, $V_{CE} = 5$ V		0.60	0.69	0.78	V
Collector-to-Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C = 10$ mA, $I_B = 1$ mA		—	0.22	0.7	V

Dynamic Characteristics

Characteristics	Symbol	Test Conditions $T_A = 25\ ^\circ C$	Limits Min.	Typ.	Max.	Units
Characteristics Apply for Each Super-Beta Cascode Amplifier Transistor Pair (Q1, Q3), Unless Indicated Otherwise						
Gain-Bandwidth Product	f_T	$I_C = 100\ \mu A$, $V_{6-8} = V_{10-8} = 5$ V,	—	78	—	MHz
Noise Voltage (Referred to Input) For Differential Amplifier Operation	E_N	$I_C = 50\ \mu A$, f = 10 Hz	—	13	—	$nV/\sqrt{Hz}$
Noise Current (Referred to Input) For Differential Amplifier Operation	I_N	$I_C = 5\ \mu A$, f = 10 Hz	—	0.12	—	$pA/\sqrt{Hz}$
Collector-to-Base Capacitance	C_{CB}	$V_{6-7} = V_{10-9} = 5$ V, $I_E = 0$	—	0.3		pF
Collector-to-Substrate Capacitance	C_{CIO}	$V_{6-5} = V_{10-5} = 5$ V, $I_B = 0$	—	3.0	—	pF
For Each Conventional Transistor (Q3 through Q8)						
Gain-Bandwidth Product	f_T	$I_C = 100\ \mu A$, $V_{CE} = 5$ V	—	100	—	MHz
		$I_C = 3$ mA, $V_{CE} = 5$ V	—	320	—	
Noise Voltage (Referred to Input)	E_N	$I_C = 100\ \mu A$, $V_{CE} = 5$ V, f = 10 Hz	—	5	—	$nV/\sqrt{Hz}$
Noise Current (Referred to Input)	I_N	$I_C = 10\ \mu A$, $V_{CE} = 5$ V, f = 10 Hz	—	0.8	—	$pA/\sqrt{Hz}$
Collector-to-Base Capacitance	C_{CB}	$V_{CB} = 5$ V, $I_E = 0$	—	0.4	—	pF
Collector-to-Substrate Capacitance	C_{CIO}	$V_{CI} = 5$ V, $I_B = 0$	—	2	—	pF

* Curve plotted for I_{CEO} characteristic.

Test Circuits for Measurement of Super-Beta Cascode Amplifier Characteristics

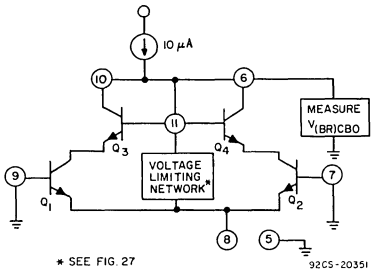


Fig.2–$V_{(BR)CBO}$ test circuit.

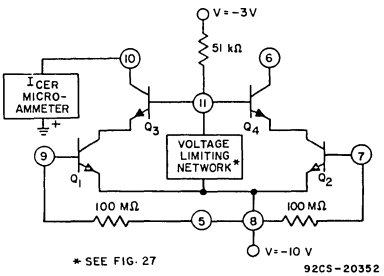


Fig.3–I_{CER} test circuit

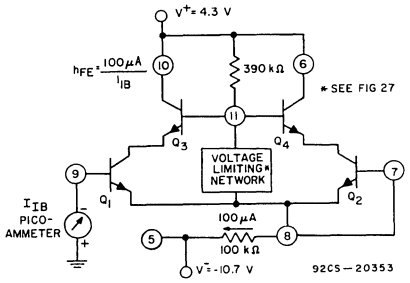


Fig.4–DC Beta (h_{FE}) test circuit.

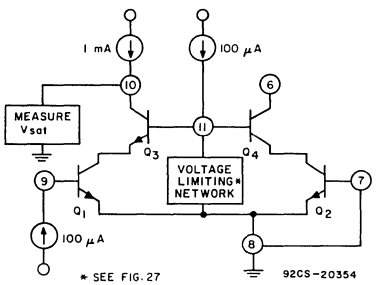


Fig.5–V_{sat} test circuit for super-beta cascode pairs.

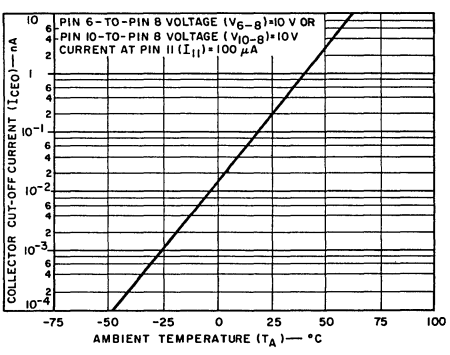


Fig.6–Collector cut-off current vs ambient temperature for super-beta cascode pairs.

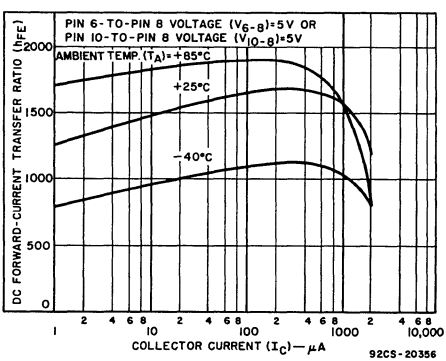


Fig.7–h_{FE} vs. I_C for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

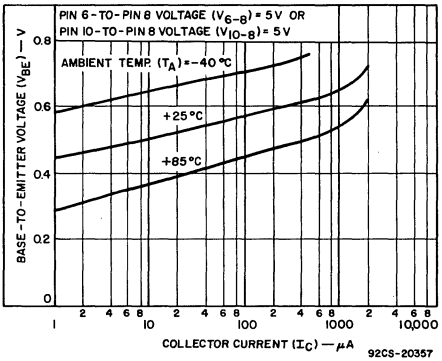


Fig.8–V_{BE} vs. I_C for each super-beta transistor (Q1 and Q2).

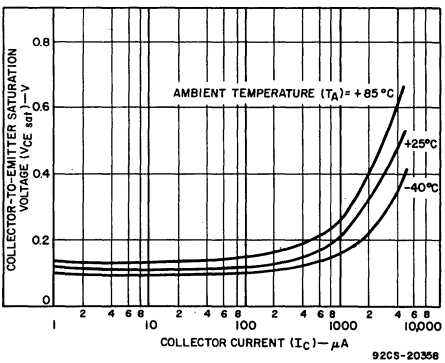


Fig.9–$V_{CE(sat)}$ vs. I_C for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

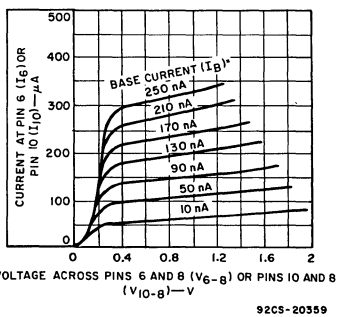


Fig.10–I-V characteristics for the super-beta cascode pairs.

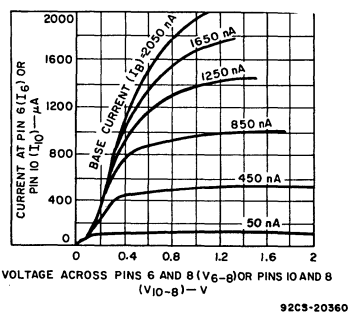


Fig.11–I-V characteristics for the super-beta cascode pairs.

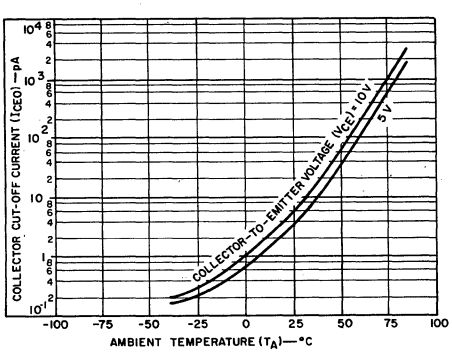


Fig.12–Collector cutoff current vs ambient temperature for the conventional transistors (V_{CE} = 5 V, 10 V).

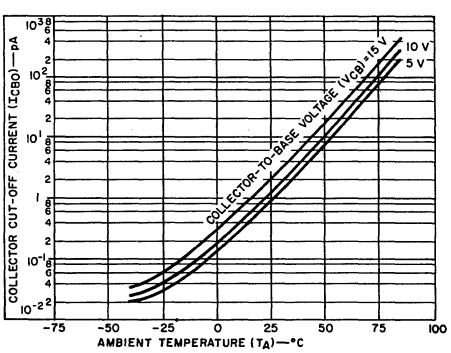


Fig.13–Collector cutoff current vs ambient temperature for the conventional transistors (V_{CB} = 5 V, 10 V, 15 V).

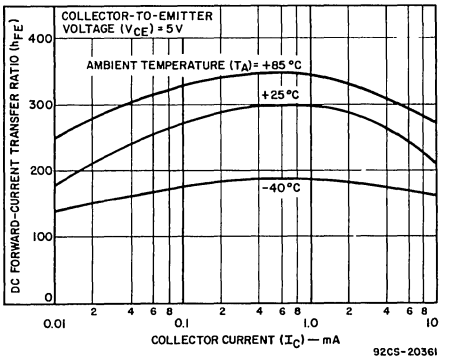


Fig. 14—h_{FE} vs. I_C for each conventional transistor (Q6, Q7, Q8).

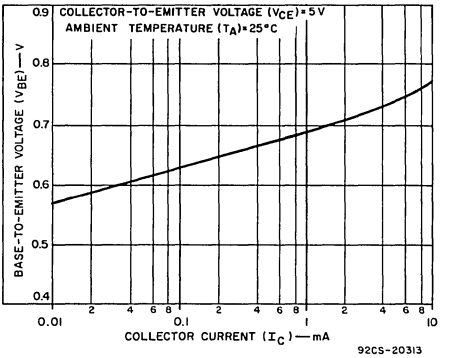


Fig. 15—V_{BE} as a function of collector current for the conventional transistors.

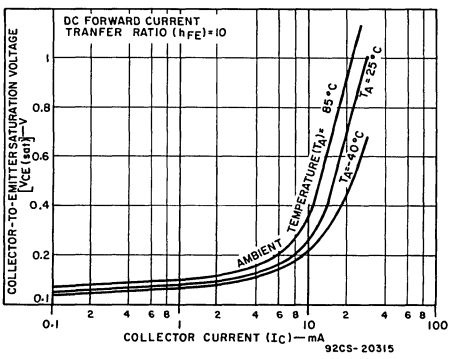


Fig. 16—$V_{CE(sat)}$ as a function of collector current for the conventional transistors.

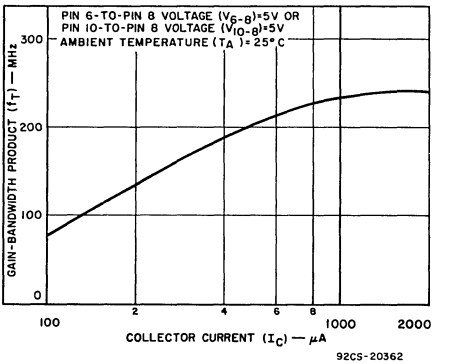


Fig. 17—Gain bandwidth product vs collector current for the super-beta cascode pairs.

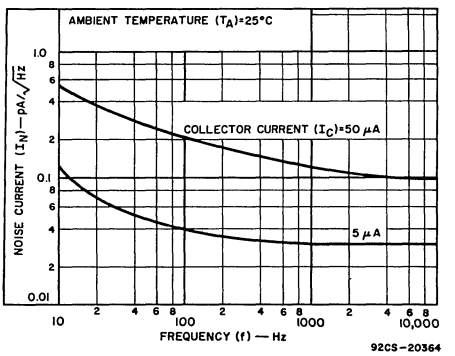


Fig. 18—I_N vs. f for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

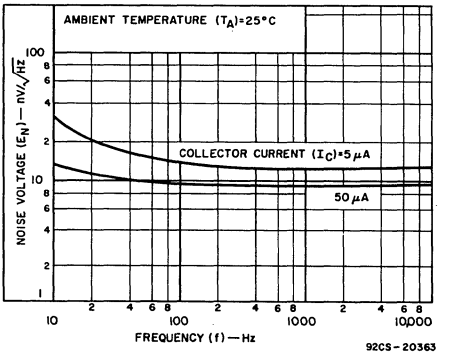


Fig. 19—E_N vs. f for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

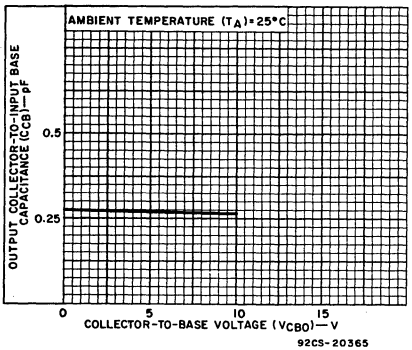


Fig.20—C_{CB} vs. V_{CBO} for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

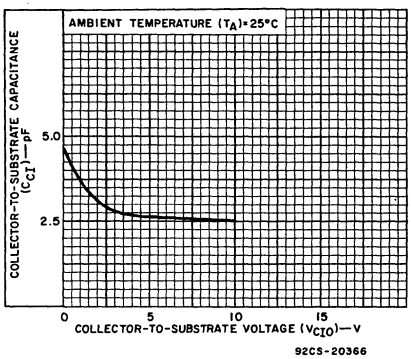


Fig.21—C_{CI} vs. V_{CIO} for each super-beta cascode amplifier transistor pair (Q1, Q3) and (Q2, Q4).

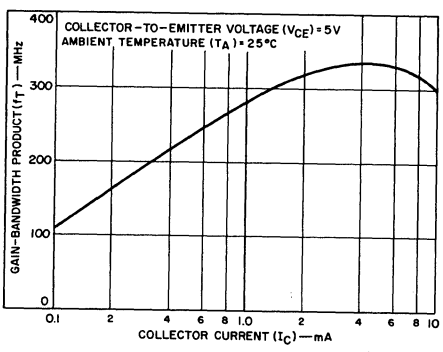


Fig.22—Gain bandwidth product vs collector current for the conventional transistors.

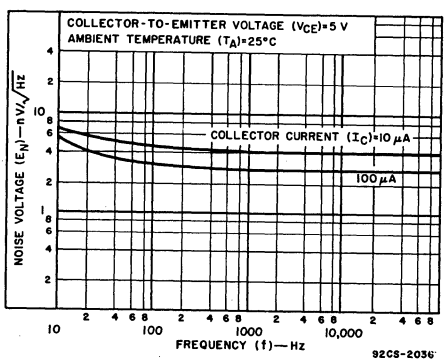


Fig.23—Noise voltage vs frequency for the conventional transistors.

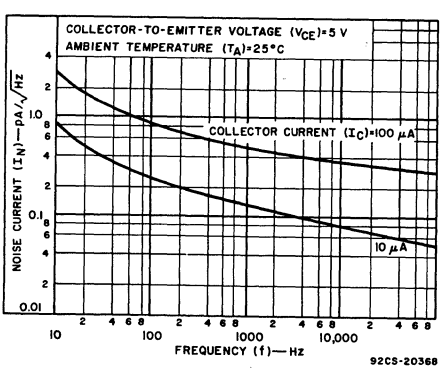


Fig.24—I_N vs. f for each conventional transistor (Q6, Q7, Q8).

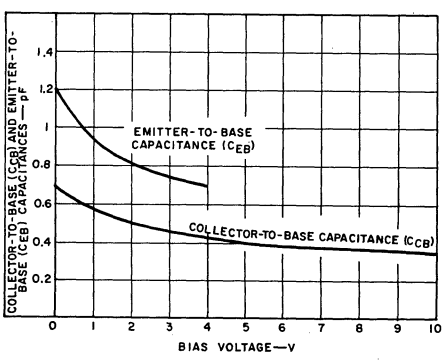


Fig.25—Collector-to-base and emitter-to-base capacitances vs bias voltage for the conventional transistors.

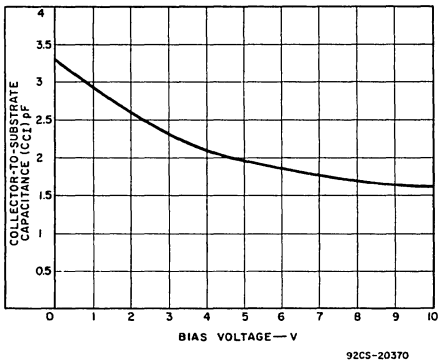


Fig.26—Collector-to-substrate capacitance vs bias voltage for the conventional transistors.

TYPICAL APPLICATIONS

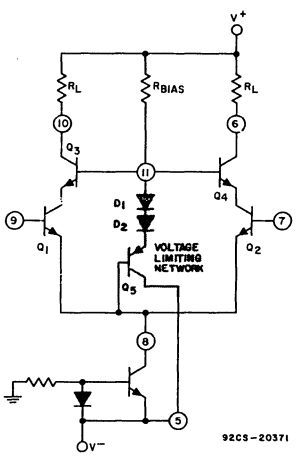


Fig.27—Bias arrangement for operation of the super-beta differential cascode amplifier.

Operating Considerations

Operation Considerations for the Super-Beta Differential Cascode Amplifier

An internal voltage-limiting network (diodes D1, D2 and p-n-p transistor Q5) incorporated in the differential cascode amplifier, assures that the applied collector-to-emitter voltage of each super-beta unit is maintained below two volts. Fig. 27 shows a typical bias arrangement of the super-beta differential cascode amplifier.

Bias current for this network must be supplied by an external source. This bias current can be obtained by simply connecting a resistor from Pin 11 to the positive supply of the differential amplifier. The return path for most of the bias current is through the substrate, Pin 5, rather than through the common emitter, Pin 8. This arrangement provides superior common-mode and power-supply rejection. As a general rule-of-thumb, the current supplied into Pin 11 should be approximately 0.04 to 0.1 times the value of the quiescent current of Pin 8.

TYPICAL APPLICATIONS (Cont'd)

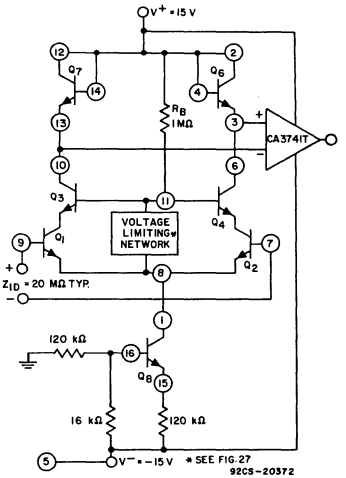


Fig.28—Super-beta Op-Amp with diode drive network.

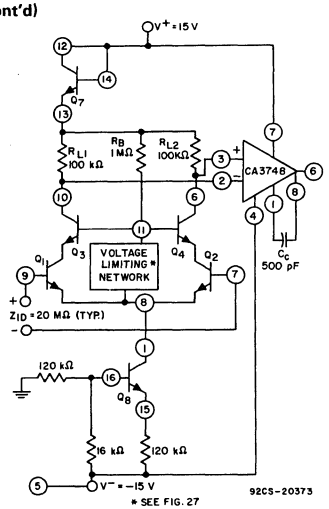


Fig.29—Super-beta Op-Amp with resistor drive network.

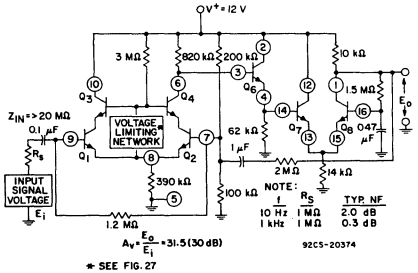


Fig.30—High-input-impedance, low-noise amplifier circuit.

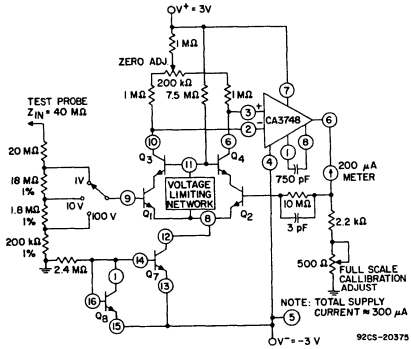


Fig.31—Typical high-input-impedance dc voltmeter circuit.

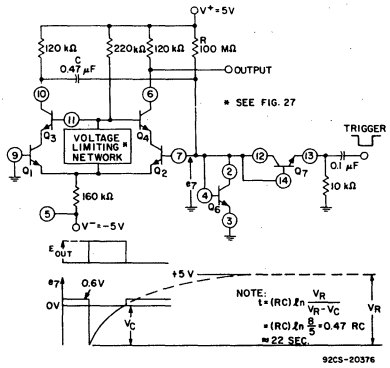


Fig.32—Long-delay monostable multivibrator circuit.

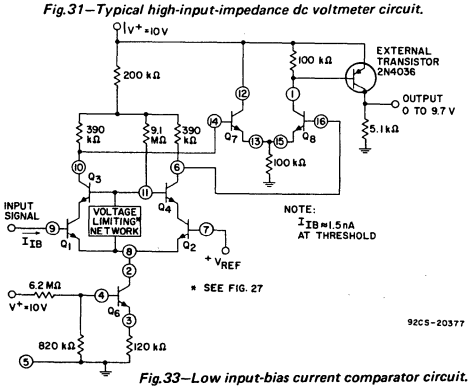


Fig.33—Low input-bias current comparator circuit.

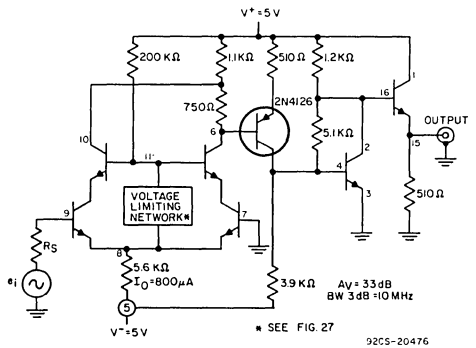


Fig.34—CA3095E wideband amplifier.

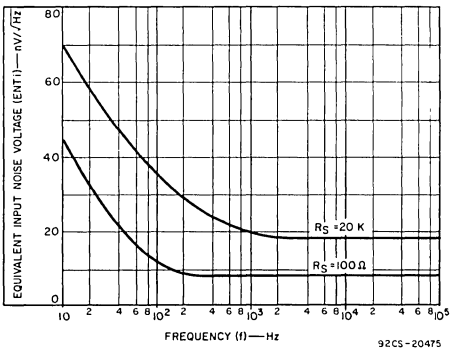


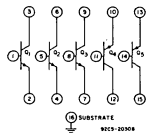
Fig.35—Equivalent input noise voltage vs. frequency for circuit of figure 34.

RCA
Solid State
Division

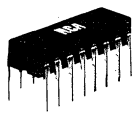
Linear Integrated Circuits

Monolithic Silicon

CA3096AE
CA3096E



Schematic Diagram



H-1622

N-P-N/P-N-P Transistor-Array IC

Features:

- Matched General-Purpose Transistors (CA3096AE Only)
- Input Offset Voltage ±5 mV
- Input Offset Current:
  - p-n-p Pair ± 250 nA max. @ I_C = −100 μA
  - n-p-n Pair ± 0.6 μA max. @ I_C = 1 mA
- High h_{FE}
  - n-p-n transistor: 150 min. @ I_C = 1 mA
  - p-n-p transistor: 40 min. @ I_C = 100 μA
- High Breakdown Voltages:
  - n-p-n transistor: $V_{(BR)CEO}$ = 35 V min; $V_{(BR)CBO}$ = 45 V min.
  - p-n-p transistor: $V_{(BR)CEO}$ = 40 V min; $V_{(BR)CBO}$ = 40 V min.
- Separate Substrate Connection
- Low Noise Figure:
  - n-p-n transistor: 2.2 dB typ. at 1 kHz
  - p-n-p transistor: 3 dB typ. at 1 kHz

RCA-CA3096E● and CA3096AE● are general-purpose high-voltage silicon transistor arrays. Each array consists of five independent transistors (two p-n-p and three n-p-n types) on a common substrate, which has a separate connection. Independent connections for each transistor permit maximum flexibility in circuit design.

Types CA3096AE and CA3096E are identical, except that the CA3096AE specifications include parameter matching and greater stringency in I_{CBO}, I_{CEO}, and $V_{CE(SAT)}$ (see Table I).

CA3096E and CA3096AE are supplied in 16-lead dual-in-line plastic packages.

Applications:

- Differential Amplifiers
- DC Amplifiers
- Sense Amplifiers
- Level Shifters
- Timers
- Lamp and Relay Drivers
- Thyristor Firing Circuits
- Temperature-Compensated Amplifiers
- Operational Amplifiers

● Formerly RCA Developmental No. TA6270.

MAXIMUM RATINGS, *Absolute Maximum Values at* T_A = 25°C

		Each n-p-n Transistor	Each p-n-p Transistor	
Collector-to-Emitter Voltage	V_{CEO}	35	−40	V
Collector-to-Base Voltage	V_{CBO}	45	−40	V
Collector-to-Substrate Voltage	V_{CIO}	45	−45	V
Emitter-to-Base Voltage	V_{EBO}	6	−40	V
Collector Current	I_C	50	−10	mA
Dissipation P_D:				
Up to T_A = 55 °C:				
Device (Total)			750	mW
Each Transistor			200	mW
Above T_A = 55 °C		Derate Linearly 6.67		mW/°C
Temperature Range:				
Operating		−55 to +125		°C
Storage		−65 to +150		°C
Lead Temperature (During Soldering) At distance 1/16 ± 1/32" (1.59 ± 0.79 mm) from case for 10 seconds max.		265		°C

TABLE I—CA3096AE AND CA3096E ESSENTIAL DIFFERENCES*

RCA TYPE	I_{CBO} (nA)		I_{CEO} (nA)		$V_{CE(SAT)}$ (V)		$\lvert V_{IO}\rvert$ (mV)		$\lvert I_{IO}\rvert$ (μA)	
	n-p-n	p-n-p	n-p-n	p-n-p	n-p-n	p-n-p	n-p-n	p-n-p	n-p-n	p-n-p
CA3096AE	40	−40	100	−100	0.7	0.4	5	5	0.6	0.25
CA3096E	100	−100	1000	−1000	1.0	0.7	–	–	–	–

* Maximum values.

STATIC ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$ (For Equipment Design)

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	CA3096AE, CA3096E LIMITS			UNITS
			Min.	Typ.	Max.	
For Each n-p-n Transistor:						
Collector-Cutoff Current (CA3096AE)	I_{CBO}	V_{CB} = 10 V, I_E = 0	–	0.0013	40	nA
Collector-Cutoff Current (CA3096AE)	I_{CEO}	V_{CE} = 10 V, I_B = 0	–	0.0055	100	nA
Collector-Cutoff Current (CA3096E)	I_{CBO}	V_{CB} = 10 V, I_E = 0	–	0.0013	100	nA
Collector-Cutoff Current (CA3096E)	I_{CEO}	V_{CE} = 10 V, I_B = 0	–	0.0055	1	μA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = 1 mA, I_B = 0	35	50	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = 10 μA, I_E = 0	45	100	–	V
Collector-to-Substrate Breakdown Voltage	$V_{(BR)CIO}$	I_{CI} = 10 μA, I_B = I_E = 0	45	100	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = 10 μA, I_C = 0	6	8	–	V
Emitter-to-Base Zener Voltage	V_Z	I_Z = 10 μA	6	7.9	9.8	V
Collector-to-Emitter Saturation Voltage (CA3096AE)	$V_{CE(SAT)}$	I_C = 10 mA, I_B = 1 mA	–	0.24	0.5	V
Collector-to-Emitter Saturation Voltage (CA3096E)	$V_{CE(SAT)}$	I_C = 10 mA, I_B = 1 mA	–	0.24	0.7	V
Base-to-Emitter Voltage	V_{BE}	I_C = 1 mA, V_{CE} = 5 V	0.6	0.69	0.78	V
DC Forward-Current Transfer Ratio	h_{FE}	I_C = 1 mA, V_{CE} = 5 V	150	390	500	
Magnitude of Temperature Coefficient: V_{BE} (for each transistor)	$\lvert\Delta V_{BE}/\Delta T\rvert$	I_C = 1 mA, V_{CE} = 5 V	–	−1.9	–	mV/°C
For Each p-n-p Transistor:						
Collector-Cutoff Current (CA3096AE)	I_{CBO}	V_{CB} = −10 V, I_E = 0	–	−0.055	40	nA
Collector-Cutoff Current (CA3096AE)	I_{CEO}	V_{CE} = −10 V, I_B = 0	–	−0.12	100	nA
Collector-Cutoff-Current (CA3096E)	I_{CEO}	V_{CE} = −10 V, I_B = 0	–	−0.12	1	μA
Collector-Cutoff-Current (CA3096E)	I_{CBO}	V_{CB} = −10 V, I_E = 0	–	−0.055	100	nA
Collector-to-Emitter Breakdown Voltage	$V_{(BR)CEO}$	I_C = −100 μA, I_B = 0	−40	−75	–	V
Collector-to-Base Breakdown Voltage	$V_{(BR)CBO}$	I_C = −10 μA, I_E = 0	−40	−80	–	V
Emitter-to-Base Breakdown Voltage	$V_{(BR)EBO}$	I_E = −10 μA, I_C = 0	−40	−100	–	V
Emitter-to-Base Zener Voltage	V_Z	I_Z = 10 μA	10	16	–	V
Emitter-to-Substrate Breakdown Voltage	$V_{(BR)EIO}$	I_{EI} = 10 μA, I_B = I_C = 0	−40	−100	–	V
Collector-to-Emitter Saturation Voltage	$V_{CE(SAT)}$	I_C = −1 mA, I_B = −100 μA	–	−0.16	−0.4	V
Base-to-Emitter Voltage	V_{BE}	I_C = −100 μA, V_{CE} = −5 V	−0.5	−0.6	−0.7	V
DC Forward-Current Transfer Ratio	h_{FE}	I_C = −100 μA, V_{CE} = −5 V	40	85	200	
		I_C = −1 mA, V_{CE} = −5 V	20	47	150	
Magnitude of Temperature Coefficient: V_{BE} (for each transistor)	$\lvert\Delta V_{BE}/\Delta T\rvert$	I_C = −100 μA, V_{CE} = −5 V	–	−2.2	–	mV/°C
For Transistors Q1 and Q2 (As a Differential Amplifier): CA3096AE ONLY						
Absolute Input Offset Voltage	$\lvert V_{IO}\rvert$	V_{CE} = 5 V, I_C = 1 mA	–	0.3	5	mV
Absolute Input Offset Current	$\lvert I_{IO}\rvert$		–	0.07	0.6	μA
Absolute Input Offset Voltage Temperature Coefficient	$\left\lvert\frac{\Delta V_{IO}}{\Delta T}\right\rvert$		–	1.1		μV/°C
For Transistors Q4 and Q5 (As a Differential Amplifier): CA3096AE ONLY						
Absolute Input Offset Voltage	$\lvert V_{IO}\rvert$	V_{CE} = −5 V, I_C = −100 μA, R_S = 0	–	0.15	5	mV
Absolute Input Offset Current	$\lvert I_{IO}\rvert$		–	2	250	nA
Absolute Input Offset Voltage Temperature Coefficient	$\left\lvert\frac{\Delta V_{IO}}{\Delta T}\right\rvert$		–	0.54	–	μV/°C

DYNAMIC ELECTRICAL CHARACTERISTICS at $T_A = 25^oC$
Typical Values Intended Only for Design Guidance

CHARACTERISTICS	SYMBOL	TEST CONDITIONS	TYPICAL VALUES	UNITS
For Each n-p-n Transistor				
Noise Figure (low frequency)	NF	f = 1 kHz, V_{CE} = 5 V, I_C = 1 mA, R_S = 1 kΩ	2.2	dB
Low-Frequency, Input Resistance	R_i	f = 1.0 kHz, V_{CE} = 5 V, I_C = 1 mA	10	kΩ
Low-Frequency Output Resistance	R_o		80	kΩ
Admittance Characteristics:		f = 1 MHz, V_{CE} = 5 V, I_C = 1 mA		
Forward Transfer Admittance	Y_{fe} $\frac{g_{fe}}{b_{fe}}$		7.5 −j13	mmho
Input Admittance	Y_{ie} $\frac{g_{ie}}{b_{ie}}$		2.2 j3.1	mmho
Output Admittance	Y_{oe} $\frac{g_{oe}}{b_{oe}}$		0.76 j2.4	mmho
Gain-Bandwidth Product	f_T	V_{CE} = 5 V, I_C = 1.0 mA	280	MHz
		V_{CE} = 5 V, I_C = 5 mA	335	
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = 3 V	0.75	pF
Collector-to-Base Capacitance	C_{CB}	V_{CB} = 3 V	0.46	pF
Collector-to-Substrate Capacitance	C_{CI}	V_{CI} = 3 V	3.2	pF
For Each p-n-p Transistor				
Noise Figure (low frequency)	NF	f = 1 kHz, I_C = 100 μA, R_S = 1 kΩ	3	dB
Low-Frequency Input Resistance	R_i	f = 1 kHz, V_{CE} = 5 V, I_C = 100 μA	27	kΩ
Low-Frequency Output Resistance	R_o		680	kΩ
Gain-Bandwidth Product	f_T	V_{CE} = 5 V, I_C = 100 μA	6.8	MHz
Emitter-to-Base Capacitance	C_{EB}	V_{EB} = −3 V	0.85	pF
Collector-to-Base Capacitance	C_{CB}	V_{CB} = −3 V	2.25	pF
Base-to-Substrate Capacitance	C_{BI}	V_{BI} = 3 V	3.05	pF

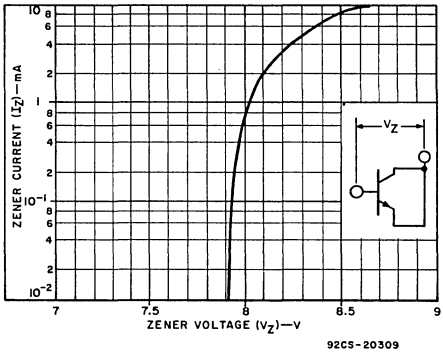


Fig.1—Base-to-emitter zener characteristic (n-p-n).

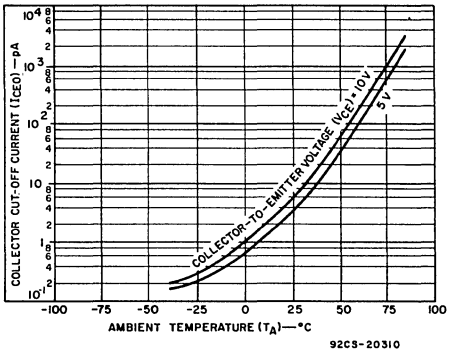


Fig.2—Collector cut-off current (I_{CEO}) as a function of temperature (n-p-n).

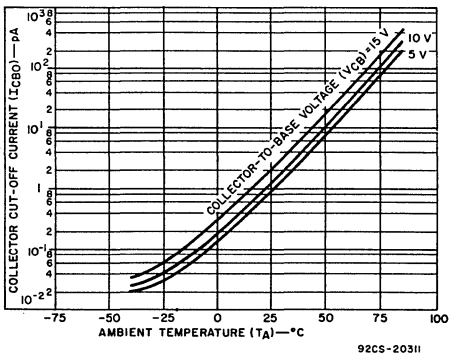


Fig.3—Collector cut-off current (I_{CBO}) as a function of temperature (n-p-n).

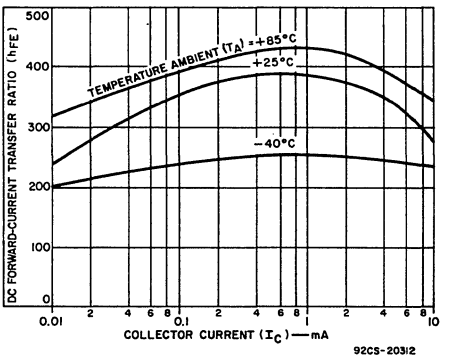


Fig.4—Transistor (n-p-n) h_{FE} as a function of collector current.

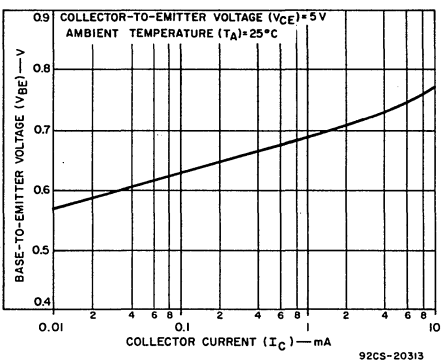


Fig.5—V_{BE} (n-p-n) as a function of collector current.

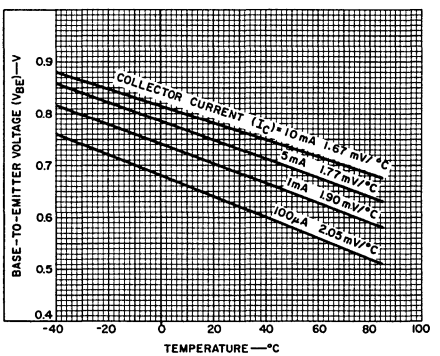


Fig.6—V_{BE} (n-p-n) as a function of temperature.

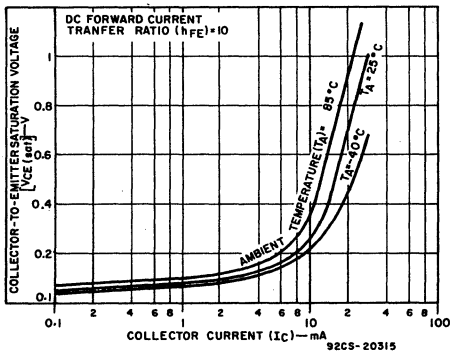


Fig.7—$V_{CE(SAT)}$(n-p-n) as a function of collector current.

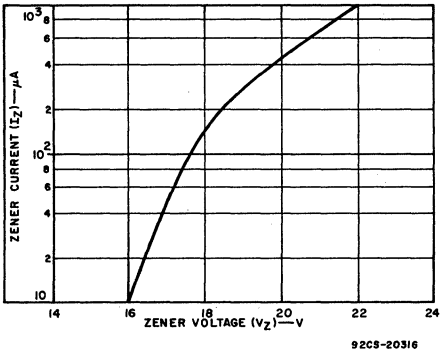


Fig.8—Base-to-emitter zener characteristic (p-n-p).

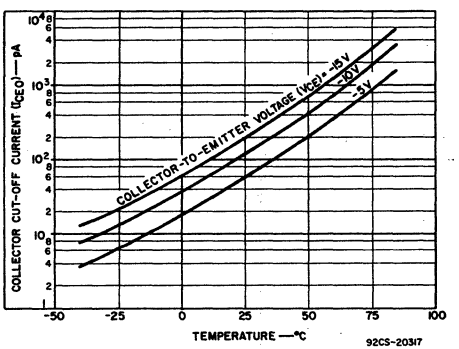


Fig.9—Collector cut-off current (I_{CEO}) as a function of temperature (p-n-p).

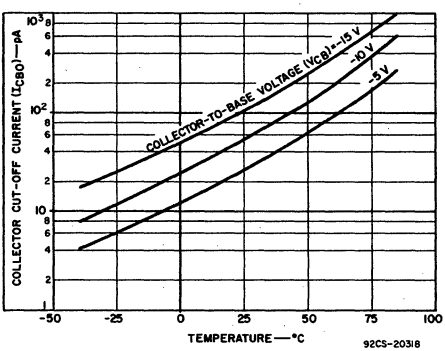


Fig.10—Collector cut-off current (I_{CBO}) as a function of temperature (p-n-p).

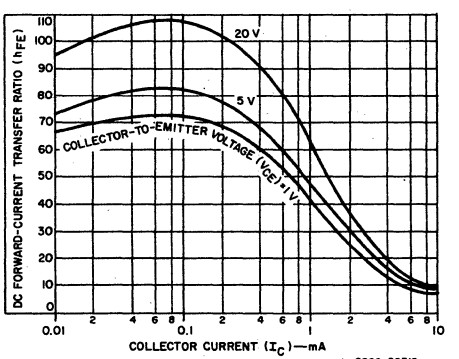


Fig.11—Transistor (p-n-p) h_{FE} as a function of collector current.

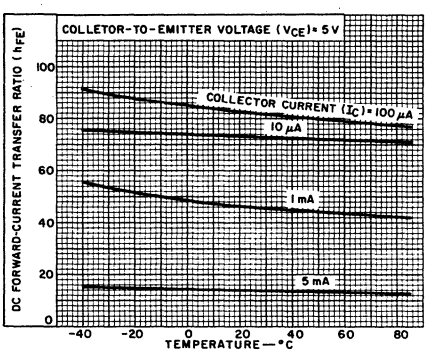


Fig.12—Transistor (p-n-p) h_{FE} as a function of temperature.

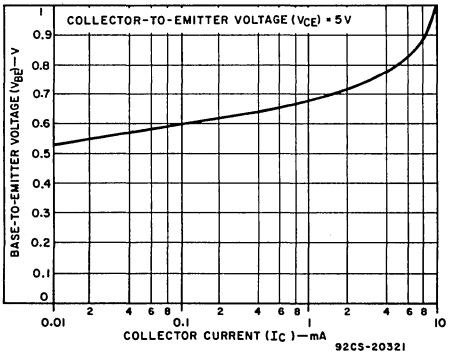


Fig.13—V_{BE} (p-n-p) as a function of collector current.

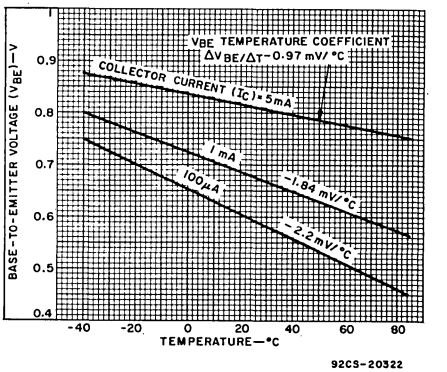


Fig.14—V_{BE} (p-n-p) as a function of temperature.

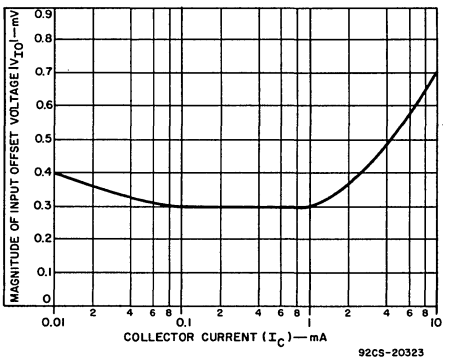


Fig.15—Magnitude of input offset voltage $|V_{IO}|$ as a function of collector current for n-p-n transistor Q_1–Q_2.

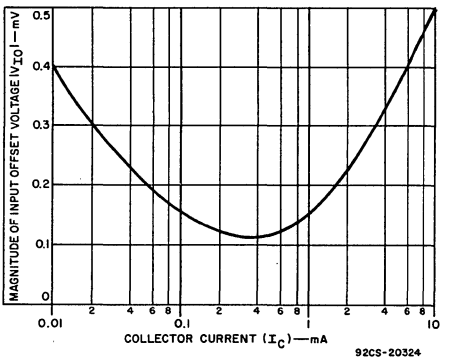


Fig.16—Magnitude of input offset voltage $|V_{IO}|$ as a function of collector current for p-n-p transistors Q_4–Q_5.

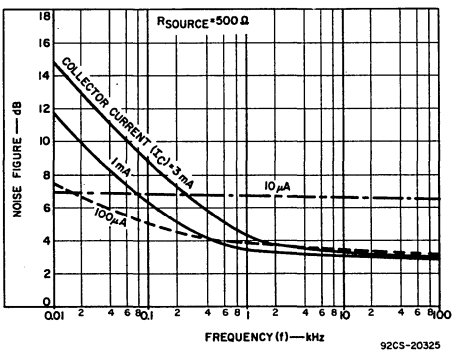


Fig.17—Noise figure as a function of frequency for n-p-n transistors.

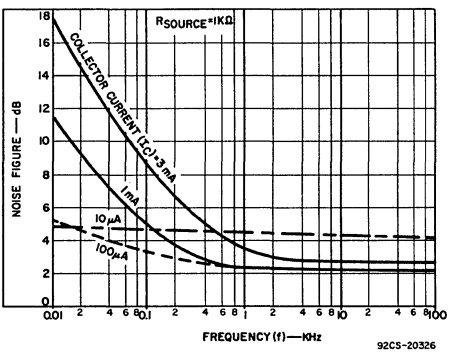


Fig.18—Noise figure as a function of frequency for n-p-n transistors.

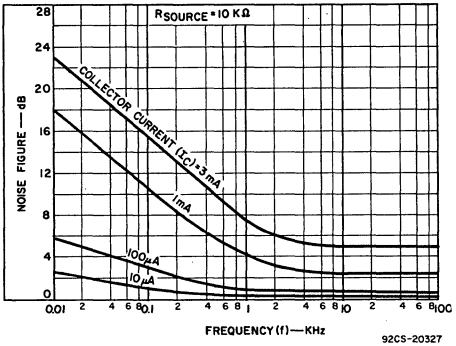


Fig.19—Noise as a function of frequency for n-p-n transistors.

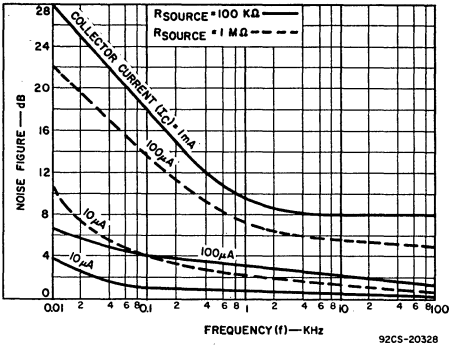


Fig.20—Noise figure as a function of frequency for n-p-n transistors.

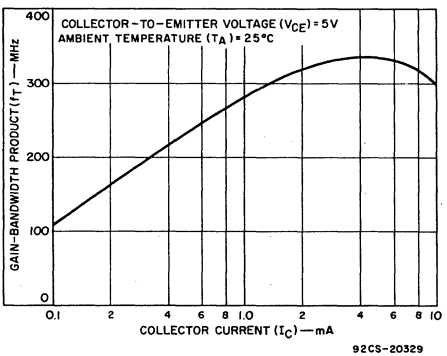


Fig.22—Capacitance as a function of bias voltage (n-p-n).

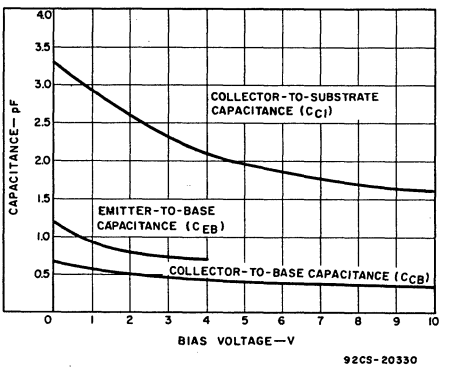


Fig.21—Gain-bandwidth product as a function of collector current (n-p-n).

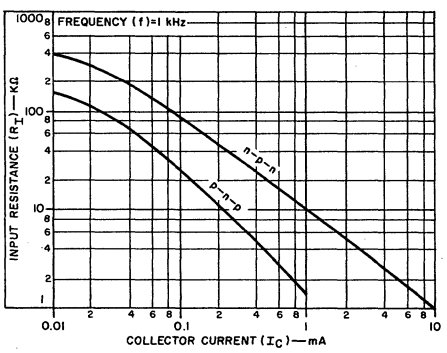


Fig.23—Input resistance as a function of collector current.

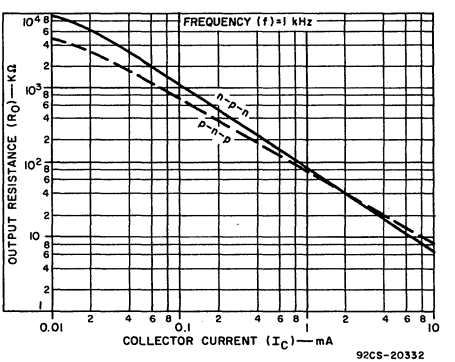


Fig.24—Output resistance as a function of collector current.

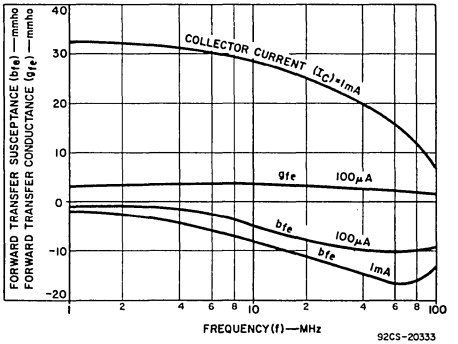


Fig.25—Forward transconductance as a function of frequency.

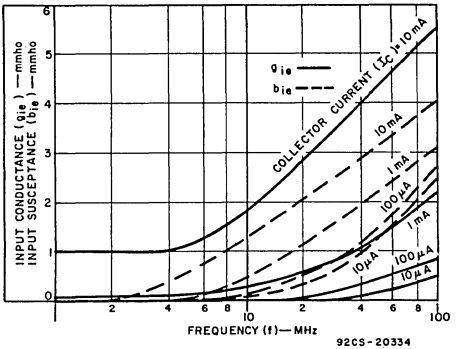


Fig.26—Input admittance as a function of frequency.

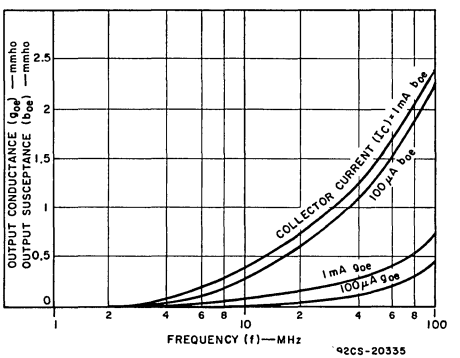


Fig.27—Output admittance as a function of frequency.

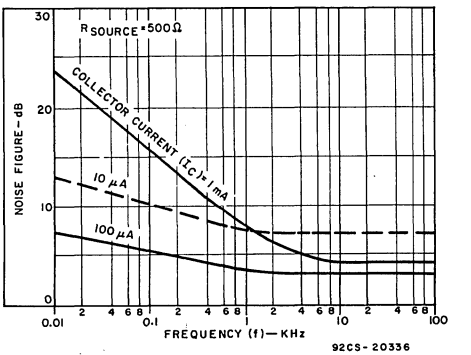


Fig.28—Noise figure as a function of frequency (p-n-p).

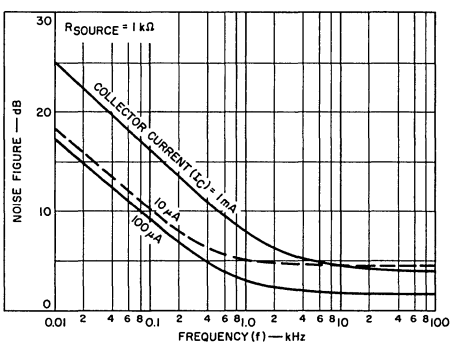


Fig.29—Noise figure as a function of frequency (p-n-p).

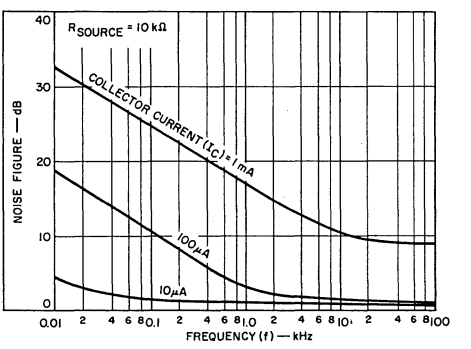


Fig.30—Noise figure as a function of frequency (p-n-p).

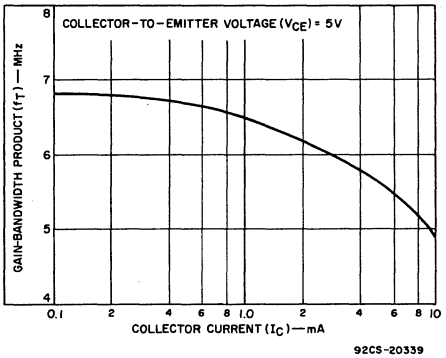


Fig.31—Gain-bandwidth product as a function of collector current (p-n-p).

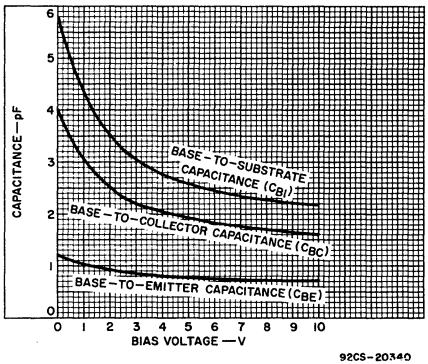


Fig.32—Capacitance as a function of bias voltage (p-n-p).

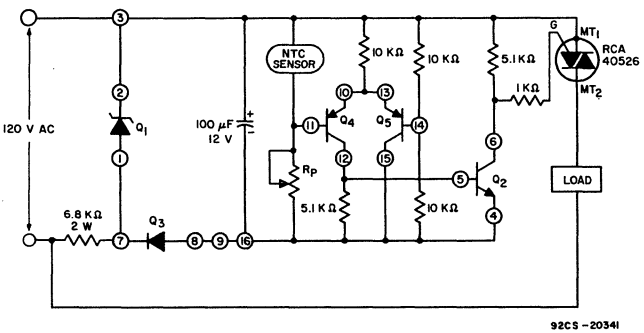


Fig.33—Line-operated level switch using CA3096AE or CA3096E.

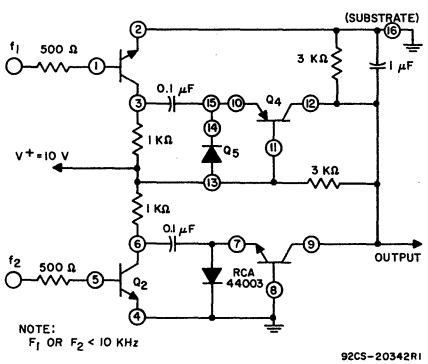


Fig.34a—Frequency comparator using CA3096E.

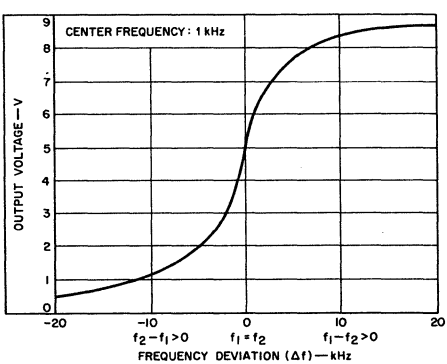


Fig.34b—Frequency comparator characteristics.

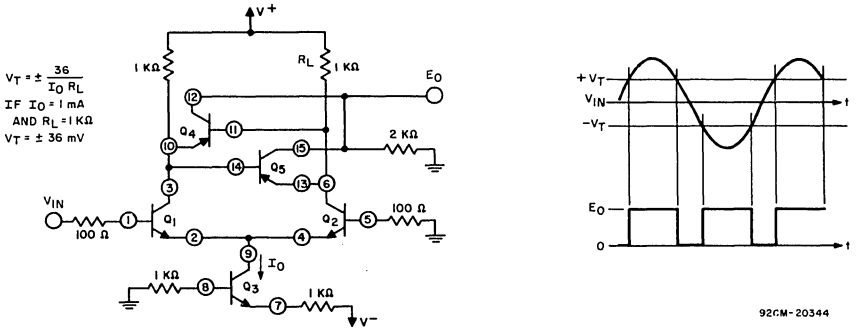


Fig.35—CA3096AE small-signal zero-voltage detector having noise immunity.

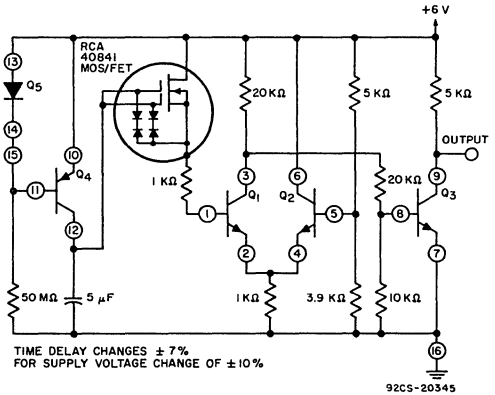


Fig.36a—One-minute timer using CA3096AE and a MOS/FET.

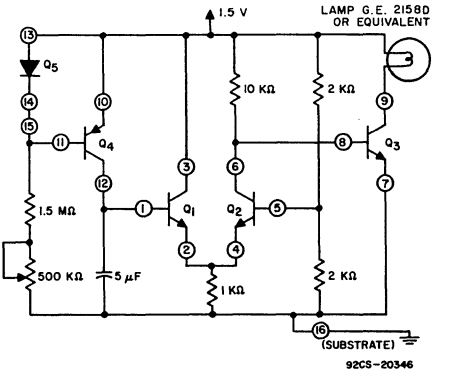
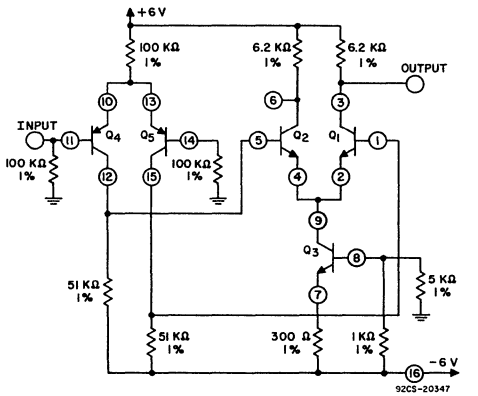


Fig.36b—Ten-second timer operated from 1.5-volt supply using CA3096E.



Features:

1. Can be operated with either dual supply or single supply.
2. Wide-input common-mode range +5 V to −5 V
3. Low bias current: <1 μA.

Fig.37a—Cascade of differential amplifiers using CA3096AE.

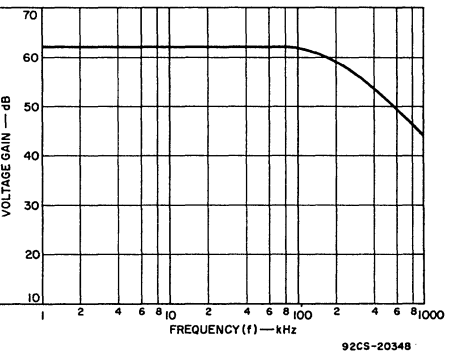


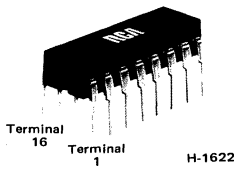
Fig.37b—Gain-frequency characteristics.

RCA Solid State Division

Linear Integrated Circuits

Monolithic Silicon

CA3097E



16-Lead Dual-In-Line Plastic Package

Thyristor/Transistor Array

For Military, Commercial, and Industrial Applications

Includes:

- Uncommitted n-p-n Transistor
- Sensitive-Gate Silicon Controlled Rectifier
- Programmable Unijunction Transistor (PUT)
- p-n-p/n-p-n Transistor Pair
- Zener Diode
- Separate Substrate Connection

RCA-CA3097E* Thyristor/Transistor Array is a monolithic integrated circuit that enables circuit designers to further integrate control systems. The CA3097E consists of five independent and completely isolated elements on one chip: an n-p-n transistor, a p-n-p/n-p-n transistor pair, a zener diode, a programmable unijunction transistor (PUT), and a sensitive-gate silicon controlled rectifier (SCR).

The CA3097 is supplied in either the 16-lead dual-in-line plastic package ("E" suffix) or the chip version ("H" suffix), and operates over the full military-temperature range of −55 to +125°C.

*Formerly Dev. No. TA6281

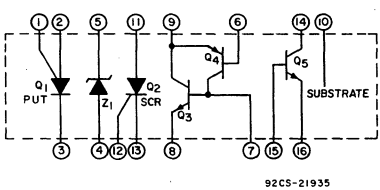


Fig. 1 – Schematic diagram of CA3097E.

Features:

- Complete isolation between elements
- n-p-n transistor – V_{CEO} = 30 V (min.) I_C = 100 mA (max.)
- p-n-p/n-p-n transistor pair – beta $\geq$ 8000 (typ.) @ I_C = 10 mA, individual p-n-p, n-p-n, or transistor pair operation
- Programmable unijunction transistor (PUT) – peak-point current = 15 nA (typ.) at R_G = 1 MΩ; V_{AK} = ±30 V
- (PUT) Extremely long RC time constants with low value of external capacitor
- Sensitive-gate silicon controlled rectifier (SCR) – 150 mA forward current (max.)
- Zener-diode impedance (Z_Z) = 15Ω (typ.) at 10 mA

Applications:

- Timers
- Light dimmers/motor controls
- Oscillators
- "One-shot" multivibrators
- Voltage regulators
- Comparators, Schmitt triggers
- Constant-current sources
- Amplifiers
- Logic circuits
- SCR triggering
- Pulse Circuits

MAXIMUM RATINGS, *Absolute-Maximum Values at $T_A = 25^oC$*

Isolation Voltage, any terminal to substrate*	+50 V
Dissipation, Total Package:	
Up to $T_A = 55^oC$	750 mW
Above $T_A = 55^oC$	derate linearly at 6.67 mW/°C
Ambient Temperature Range:	
Operating	−55 to +125°C
Storage	−65 to +150°C
Lead Temperature (During Soldering):	
At distance 1/16 ± 1/32 inch (1.59 ± 0.79 mm) from case for 10 seconds max.	+265 °C
Each n-p-n Transistor (Q3,Q5)	
The following ratings apply with terminals 6 & 9 connected together.	
Collector-to-Emitter Voltage (V_{CEO})	30 V
Collector-to-Base Voltage (V_{CBO})	50 V
Emitter-to-Base Voltage (V_{EBO})	5 V
Collector Current (I_C)	100 mA
Base Current (I_B)	20 mA
Dissipation (P_D)	500 mW
p-n-p Transistor (Q4)	
The following ratings apply with terminals 7 & 8 connected together.	
Collector-to-Emitter Voltage (V_{CEO})	−40 V
Collector-to-Base Voltage (V_{CBO})	−50 V
Emitter-to-Base Voltage (V_{EBO})	−40 V
Collector Current (I_C)	−10 mA
Base Current (I_B)	−3 mA
Dissipation (P_D)	200 mW
p-n-p/n-p-n Transistor Pair (Q3,Q4)	
Dissipation (P_D)	500 mW
Programmable Unijunction Transistor, PUT (Q1)	
Gate-to-Cathode Positive Voltage (V_{GK})	30 V
Gate-to-Cathode Negative Voltage (V_{GKR})	5 V
Gate-to-Anode Negative Voltage (V_{GA})	30 V
Anode-to-Cathode Voltage (V_{AK})	±30 V
DC Anode Current	150 mA
Peak Anode Non-Recurrent Forward (On-State) Current (10 μs pulse)	2 A
Total Average Dissipation	300 mW
Silicon Controlled Rectifier, SCR (Q2)	
Repetitive Peak Reverse Voltage (V_{RRXM}), R_{GK} = 1 KΩ	30 V
Repetitive Peak Off-State Voltage (V_{DRXM}), R_{GK} = 1 kΩ	30 V
DC On-State Current (I_{TDC})	150 mA
Peak Surge (Non-Repetitive) On-State Current (10 μs pulse)	2 A
Forward Peak Gate Current (I_{GFM})	20 mA
Peak Gate-to-Cathode Reverse Voltage (V_{GRM})	5 V
Total Average Dissipation	300 mW
Zener Diode, (Z1)	
DC Current (I_Z)	25 mA
Dissipation (P_D)	250 mW

* One or more of the terminals of each element of the CA3097E is isolated from the substrate by a junction diode. In order to maintain electrical isolation between elements, the substrate terminal must be connected to a voltage which is no more positive than that of any other terminal. To avoid undesirable coupling between elements, the substrate terminal (terminal 10) should be maintained at either dc or signal (ac) ground.

ELECTRICAL CHARACTERISTICS

CHARACTERISTIC	SYMBOL	TEST CONDITIONS Ambient Temperature (T_A) = 25°C Unless Otherwise Specified	FIG. NO.	LIMITS			UNITS
				Min.	Typ.	Max.	
n-p-n TRANSISTORS Q3, Q5 (TERMINALS 6 and 9 CONNECTED)							
COLLECTOR CUTOFF CURRENT	I_{CBO}	V_{CB} = 10 V, I_E = 0		–	–	1	μA
COLLECTOR CUTOFF CURRENT	I_{CEO}	V_{CE} = 10 V, I_B = 0		–	–	10	μA
COLLECTOR-TO-EMITTER BREAKDOWN VOLTAGE	$V_{(BR)CEO}$	I_C = 100μA, I_B = 0		30	–	–	V
COLLECTOR-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)CBO}$	I_C = 100μA, I_E = 0		50	–	–	V
COLLECTOR-TO-SUBSTRATE BREAKDOWN VOLTAGE	$V_{(BR)CIO}$	I_{CI} = 100μA, I_B = 0, I_E = 0		50	–	–	V
EMITTER-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)EBO}$	I_E = 100μA, I_C = 0		5	7.5	10	V
COLLECTOR-TO-EMITTER SATURATION VOLTAGE	$V_{CE(SAT)}$	I_C = 50mA, I_B = 5mA	5	–	–	0.65	V
		I_C = 10mA, I_B = 1 mA		–	0.10	–	
BASE-TO-EMITTER SATURATION VOLTAGE	$V_{BE(SAT)}$	I_C = 10mA, I_B = 1mA	2	–	0.76	–	V
BASE-TO-EMITTER VOLTAGE	V_{BE}	V_{CE} = 3V, I_C = 10mA	3	0.65	0.73	0.85	V
DC FORWARD-CURRENT TRANSFER RATIO	h_{FE}	V_{CE} = 3V, I_C = 10mA	4	100	130	–	
		V_{CE} = 3V, I_C = 50mA		80	120	–	
p-n-p TRANSISTOR Q4 (TERMINALS 7 and 8 CONNECTED)							
COLLECTOR CUTOFF CURRENT	I_{CBO}	V_{CB} = -10 V, I_E = 0		–	–	–1	μA
COLLECTOR CUTOFF CURRENT	I_{CEO}	V_{CE} = -10 V, I_B = 0		–	–	–10	μA
COLLECTOR-TO-EMITTER BREAKDOWN VOLTAGE	$V_{(BR)CEO}$	I_C = -100μA, I_B = 0		–40	–	–	V
COLLECTOR-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)CBO}$	I_C = -10μA, I_E = 0		–50	–	–	V
EMITTER-TO-SUBSTRATE BREAKDOWN VOLTAGE	$V_{(BR)EIO}$	I_{EI} = 10μA, I_B = 0, I_E = 0		–50	–	–	V
EMITTER-TO-BASE BREAKDOWN VOLTAGE	$V_{(BR)EBO}$	I_E = –10μA, I_C = 0		–40	–	–	V
COLLECTOR-TO-EMITTER SATURATION VOLTAGE	$V_{CE(SAT)}$	I_C = -1mA, I_B = -100μA	6	–	–	–0.33	V
BASE-TO-EMITTER SATURATION VOLTAGE	$V_{BE(SAT)}$	I_C = -1mA, I_B = –100μA-	7	–	–0.7	–	V
BASE-TO-EMITTER VOLTAGE	V_{BE}	V_{CE} = -3 V, I_C = –100μA	8	–0.5	–0.6	–0.7	V
DC FORWARD-CURRENT TRANSFER RATIO	h_{FE}	V_{CE} = – 3 V, I_C = –100μA	9	30	60	–	
		V_{CE} = –3 V, I_C = –1 mA		40	–	–	
n-p-n/p-n-p TRANSISTOR PAIR Q3, Q4							
DC FORWARD-CURRENT TRANSFER RATIO	h_{FE}	V_{CE} (n-p-n) = 3V, I_C = 10mA	10	–	8000	–	
		V_{CE} (n-p-n) = 3V, I_C = 50mA	10	–	6500	–	

ELECTRICAL CHARACTERISTICS (Cont'd.)

CHARACTERISTIC	SYMBOL	TEST CONDITIONS Ambient Temperature (T_A) = 25°C Unless Otherwise Specified	FIG. NO.	LIMITS Min.	LIMITS Typ.	LIMITS Max.	UNITS
PROGRAMMABLE UNIJUNCTION TRANSISTOR (PUT), Q1							
OFFSET VOLTAGE	V_T*	V_S = 10V, R_G = 10kΩ	11,22[a]	0.2	–	0.7	V
		V_S = 10V, R_G = 1MΩ		0.2	–	0.7	
ANODE-TO-CATHODE ON-STATE VOLTAGE	V_F	I_F = 50mA	12	–	0.90	1.5	V
		I_F = 100mA		–	1	–	
PEAK OUTPUT VOLTAGE	V_{OM}	C = 0.22μF Anode Supply Voltage = 20V	13,23	–	10	–	V
PEAK-POINT CURRENT	I_P	V_S = 10V, R_G = 10kΩ	14,22[a]	–	0.55	1	μA
		V_S = 10V, R_G = 1MΩ	–	–	0.015	0.15	
VALLEY-POINT CURRENT	I_V	V_S = 10V, R_G = 10kΩ	17,15	4	40	–	μA
		V_S = 10V, R_G = 1MΩ	16	–	–	25	
GATE REVERSE CURRENT	I_{GAO}	V_S = 30V	22[c]	–	0.02	–	nA
GATE REVERSE CURRENT	I_{GKS}	Anode-To-Cathode Short, V_S, = 30V	22[d]	–	0.2	–	nA
OUTPUT PULSE RISE TIME	t_r	Anode-Supply Voltage = 20V C = 0.22 μF	23	–	60	–	ns
SILICON CONTROLLED RECTIFIER (SCR), Q2							
PEAK OFF-STATE CURRENT: FORWARD	I_{DXM}	V_{DRXM} = 30V, R_{GK} = 1kΩ	24	–	–	2	μA
REVERSE	I_{RXM}	V_{RRXM} = 30V, R_{GK} = 1kΩ	24	–	–	2	
FORWARD DC VOLTAGE DROP	V_T	I_T = 50 mA	18	–	0.90	1.5	V
GATE-TO-SOURCE TRIGGER CURRENT	I_{GS}	T_A = 25°C	26	–	33	100	μA
		T_A = −55°C	26	–	50	–	
DC GATE-TRIGGER VOLTAGE	V_{GT}	V_L = 10V, R_L = 100Ω	19	–	0.55	0.75	V
HOLDING CURRENT	I_{HO}	R_{GK} = 1kΩ	20,24	–	1.2	–	mA
CRITICAL RATE-OF-RISE OF OFF-STATE VOLTAGE	dv/dt	EXPONENTIAL RISE, R_{GK} = 1kΩ, V_{DRXM} = 30V	25	–	150	–	V/μs
GATE-CONTROLLED TURN-ON TIME	t_{gt}	See Fig. 33	33	–	50	–	ns
CIRCUIT-COMMUTATED TURN-OFF TIME	t_q	See Fig. 33	33	–	10	–	μs
ZENER DIODE, Z1							
ZENER VOLTAGE	V_Z	I_Z = 10mA	21	7.2	8	8.8	V
ZENER IMPEDANCE	Z_Z	I_Z = 10mA, f = 1kHz		–	15	25	Ω
ZENER VOLTAGE TEMPERATURE COEFFICIENT	$(\Delta V_Z/V_Z)/\Delta T$	I_Z = 10mA		–	+0.05	–	%/°C
	$\Delta V_Z/\Delta T$			–	+4	–	mV/°C
ZENER-TO-SUBSTRATE BREAKDOWN VOLTAGE	$V_{(BR)ZIO}$	I_Z = 100μA TERM. 5 TO SUBSTRATE		50	80	– –	V

* $V_T = V_P - V_S$ (Fig. 22)

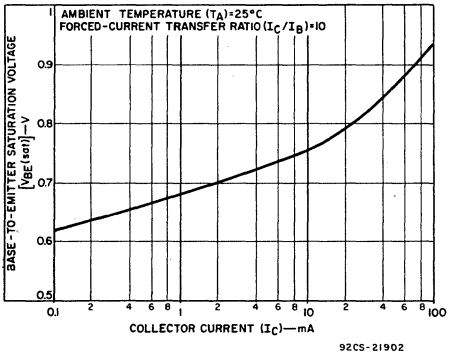


Fig.2 – Base-to-emitter saturation voltage vs. collector current for n-p-n transistors Q3 & Q5.

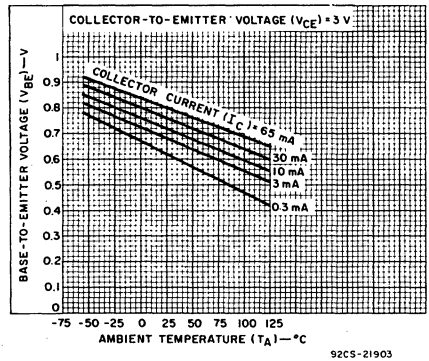


Fig.3 – Base-to-emitter voltage vs. ambient temperature for n-p-n transistors Q3 & Q5.

TYPICAL CHARACTERISTICS

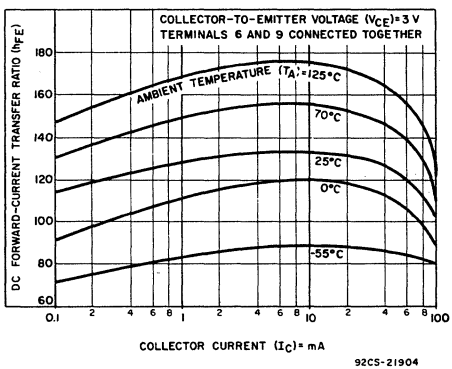


Fig.4 – DC forward-current transfer ratio vs. collector current for n-p-n transistors Q3 & Q5.

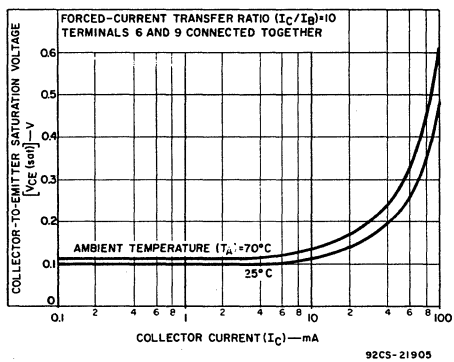


Fig.5 – Collector-to-emitter saturation voltage vs. collector for n-pn- transistors Q3 & Q5.

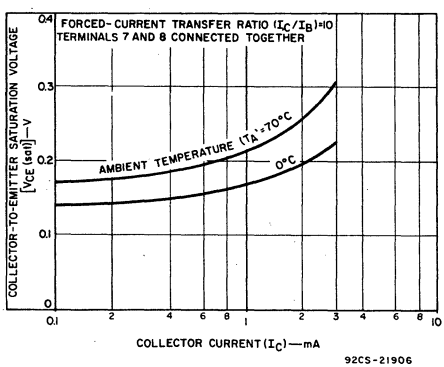


Fig.6 – Collector-to-emitter saturation voltage vs. collector current for p-n-p transistor Q4.